

Basics of Statistical Learning

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Preface

These are notes for STAT 432 Summer 2026 at UIUC. If you have any questions please contact me.

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Part I

Foundations

1 What is Statistical Learning?

1.1 Introduction

Statistical learning studies how to use data to understand relationships and make predictions.

In supervised learning, a common abstract model is

$$Y = f(X) + \varepsilon,$$

where:

- X represents features;
- Y represents the response;
- f is the underlying relationship;
- ε represents random noise or unexplained variation.

The main objective is to estimate the unknown function f using observed data.

More broadly, modern statistical learning is fundamentally about learning patterns that generalize well to unseen data.

Depending on the objective, statistical learning problems can be viewed from several different perspectives.

Task	Main Question
Prediction	What will happen?
Inference	Which variables matter?
Causality	What happens if we intervene?

- Prediction mainly focuses on predictive performance and generalization. A common formulation is

$$\hat{f}(x) \approx \mathbb{E}[Y \mid X = x].$$

- Inference focuses on understanding relationships between variables and often relies on tools such as confidence intervals and hypothesis testing.
- Causality studies intervention effects and is often written as

$$\mathbb{E}[Y \mid \text{do}(X = x)].$$

Important distinctions

1. Good training performance does not imply good generalization.
2. Good prediction does not imply valid inference.
3. Association does not imply causation.

Tip

In this course, we will primarily focus on **prediction** and **generalization** rather than formal statistical inference.

1.2 Types of Learning

Statistical learning problems can be divided into several major categories depending on the available data and learning objective.

We will mainly focus on supervised learning in this course, with some topics from unsupervised learning.

1.2.1 Supervised Learning

In supervised learning, we observe both predictors X and response Y .

The goal is to predict or explain the response, such as house price prediction, spam detection, medical diagnosis, etc..

Supervised learning is usually divided into two major tasks:

- **regression**, where the response is quantitative;
- **classification**, where the response is categorical.

Example 1.1.

- predicting salary is a regression problem;
- predicting whether a tumor is benign or malignant is a classification problem.

1.2.2 Unsupervised Learning

In unsupervised learning, we observe predictors X but no response Y .

The goal is to discover hidden structure in the data, such as clustering, dimensionality reduction, embedding learning, etc..

1.2.3 Reinforcement Learning

In reinforcement learning, an agent interacts with an environment and learns through rewards and penalties, such as robotics, game playing, recommendation systems, etc..

1.3 Types of Data

Machine learning methods are strongly connected to data structure, and different data types often require different modeling strategies.

This course mainly focuses on **tabular data**, which is the classical setting in statistics and applied machine learning.

1.3.1 Tabular Data

Tabular data is one of the most common data formats in statistical learning. It is typically represented as

$$X \in \mathbb{R}^{n \times p},$$

where:

- n is the number of observations;
- p is the number of predictors.

Example:

Age	Income	Balance	Student	Default
23	50000	1200	Yes	No
45	90000	3000	No	Yes

In tabular data:

- rows represent observations;
- columns represent variables or features.

For supervised learning, datasets usually contain:

- a feature matrix X ;
- a target variable y .

For unsupervised learning, we typically observe only the feature matrix X .

Since this course mainly focuses on tabular data, we will usually treat:

$$X \in \mathbb{R}^{n \times p}, \quad y \in \mathbb{R}^n.$$

 columns correspond to variables

The key feature of tabular data is NOT merely the matrix representation. More importantly, each column corresponds to a variable with its own semantic meaning, such as age, income, blood pressure, or transaction count.

 Tip

Tabular data is particularly well suited for methods such as linear models and tree-based models. Therefore, these model families will be the main focus of this course.

1.3.2 Other Types of Data

Besides tabular data, many machine learning problems involve other data formats. We briefly mention several common examples that appear in modern learning tasks.

i Image Data

Image data contains spatial structure, meaning nearby pixels are highly related.

Modern image models frequently use:

- convolutional neural networks,
- vision transformers.

A central goal is to extract meaningful visual patterns such as edges, textures, shapes, and objects.

i Text Data

Text data contains sequential and contextual structure. Modern NLP is largely based on transformers and large language models.

i Time Series Data

Time series data is ordered over time such as stock prices, weather data and sensor measurements.

i Graph Data

Graph data represents relationships between objects, such as social networks, citation networks, molecular structures. Modern graph learning often uses graph neural networks.

1.4 Generalization and Model Complexity

1.4.1 Training Error and Test Error

Classical regression focuses heavily on checking model assumptions, such as independence of errors, constant variance, approximate normality. These assumptions are especially important for formal inference, including confidence intervals and hypothesis testing.

In statistical learning, however, the primary goal is often **predictive performance** rather than formal inference. As a result, diagnostic tools still remain useful, but they usually play a secondary role. Their main purpose is often to detect serious issues such as strong nonlinearity, influential observations, data quality problems, severe model misspecification.

The central question becomes: How well does the model predict new observations?

Definition 1.1 (Training Error and Test Error).

- The training error measures how well a model fits the training data.
- The test error measures prediction error on new observations not used to fit the model.

For example, suppose we evaluate a regression model using mean squared error (MSE).

The **training MSE** measures fit on the observed training sample, while the **test MSE** measures prediction performance on new data drawn from the same population.

A model may achieve extremely small training error while still performing poorly on unseen data. This happens because the model begins fitting random noise rather than the underlying signal.

Definition 1.2 (Overfitting). Overfitting occurs when a model fits the training data very closely but fails to generalize well to new data. In this case, the model captures random noise or

idiosyncrasies in the training sample rather than the underlying relationship.

Model selection methods attempt to balance model complexity and prediction accuracy in order to avoid overfitting.

1.4.2 Bias–Variance Tradeoff

A central idea behind generalization is the **bias–variance tradeoff**.

Suppose

$$Y = f(X) + \varepsilon, \quad \text{Var}(\varepsilon) = \sigma^2.$$

For a fitted estimator $\hat{f}$, the test mean squared error at x_0 satisfies

$$\text{E} \left[(Y_0 - \hat{f}(x_0))^2 \right] = \text{Bias}^2(\hat{f}(x_0)) + \text{Var}(\hat{f}(x_0)) + \sigma^2.$$

The three terms represent:

- **bias**: systematic modeling error: $\text{Bias}(\hat{f}(x)) = \text{E}[\hat{f}(x)] - f(x)$. It measures the systematic difference between the average prediction and the true underlying function;
- **variance**: sensitivity to the training sample: $\text{Var}(\hat{f}(x))$. It measures how much the fitted model changes when the training data changes. Highly flexible models often have larger variance;
- **irreducible error**: σ^2 . This represents random noise or variability in the data generation process that cannot be explained or removed, even with the true model.

Roughly speaking, as model flexibility increases:

- training error usually decreases;
- test error often first decreases and then increases.

Model selection aims to balance these two effects in order to achieve good generalization performance.

i Information Criteria and Estimated Test Error

Classical regression often estimates expected test error indirectly using only the training data.

Some commonly used criteria are:

- AIC (Akaike Information Criterion),
- BIC (Bayesian Information Criterion),
- Mallows' C_p .

These criteria reward good fit while penalizing excessive model complexity.

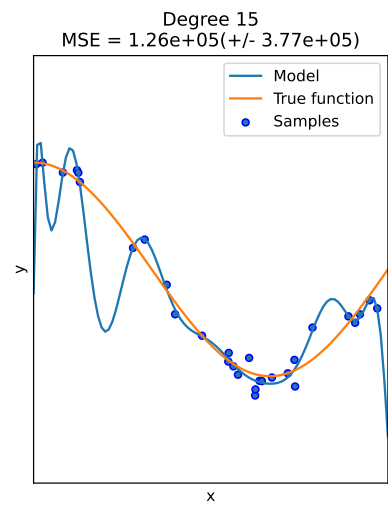
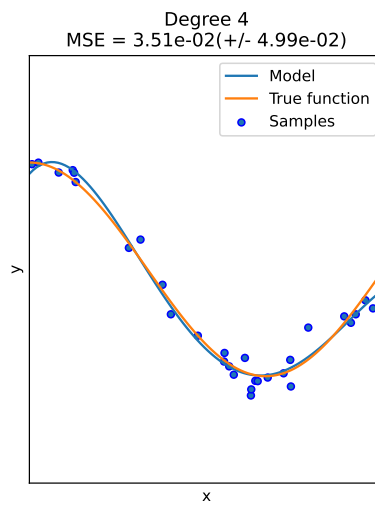
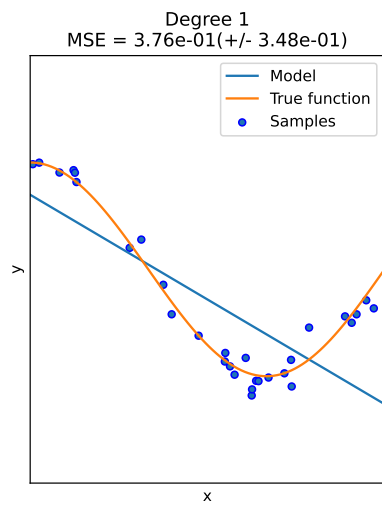
Although they are useful for model comparison, they still rely on approximations derived from the training sample.

1.4.3 Underfit vs Overfit

Bias and variance are closely related to underfit and overfit.

Roughly speaking, underfit means the model is not sufficient to fit the training samples, and overfit means that the models learns too many noise from the data. In many cases, high bias is related to underfit, and high variance is related to overfit.

The following example is from [the sklearn guide](#). Although it is a polynomial regression example, it grasps the key idea of underfit and overfit.



2 Classification Decisions and Evaluation

Unlike regression, where metrics such as Mean Squared Error (MSE) evaluate prediction error along a continuous scale, classification evaluation is fundamentally more nuanced.

A modern statistical learning classifier often produces a score or estimated probability rather than directly outputting a categorical decision. For example, a model may estimate

$$\hat{p}(x) = \Pr(Y = 1 \mid X = x),$$

and a separate decision rule then converts this soft prediction into a hard classification. As a result, classification performance can be evaluated from several different perspectives.

Broadly speaking, classification evaluation can be divided into three related but distinct goals.

Goal	Main question	Common tools
Decision quality	Are the final classification decisions good?	Accuracy, precision, recall, F1-score
Ranking quality	Are positive observations ranked above negative observations?	ROC curve, AUC, PR curve
Probability quality	Are predicted probabilities numerically trustworthy?	Calibration curve, Brier score, log loss

These goals are related but not equivalent. Accurate estimation of $\Pr(Y | X)$ is the most informative objective because it naturally supports both ranking and decision-making. In contrast, good performance on a threshold-based metric such as accuracy does not guarantee well-calibrated probabilities or reliable rankings.

In practice, however, decision quality is often easier to achieve and is frequently the primary objective in applied statistical learning. Many real-world applications ultimately require concrete actions, such as approving a loan, detecting fraud, or diagnosing disease. Consequently, threshold-based metrics such as accuracy, precision, recall, and F1-score often play the central role in applied machine learning.

Note

Classification evaluation can become quite sophisticated and may involve ranking performance, probability calibration, cost-sensitive learning, threshold optimization, and decision theory.

In this course, however, our primary goal is to understand the core ideas of statistical learning and predictive modeling rather than to develop a complete theory of statistical decision making. Therefore, we will mainly focus on accuracy and closely related threshold-based classification metrics.

2.1 Decision quality

Decision quality evaluates the final hard classifications after applying a threshold to predicted scores or probabilities.

This perspective focuses on whether the model makes useful decisions in practice. Metrics such as accuracy, precision, recall, and F1-score belong to this category.

2.1.1 Binary Classification Setup

Suppose the response is binary:

$$Y \in \{0, 1\}.$$

We call $Y = 1$ the positive class and $Y = 0$ the negative class.

A classifier produces a predicted label

$$\hat{Y} \in \{0, 1\}.$$

The four possible outcomes are summarized by the confusion matrix.

	Predicted positive	Predicted negative
True positive class	TP	FN
True negative class	FP	TN

Here, TP means true positive, FP means false positive, FN means false negative, TN means true negative.

The confusion matrix is the starting point for most threshold-based classification metrics.

2.1.2 Accuracy

Accuracy is the proportion of correctly classified observations:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}.$$

Accuracy is simple and useful when:

- the classes are reasonably balanced,
- false positives and false negatives have similar costs,
- the decision threshold is already fixed and meaningful.

However, accuracy can be misleading.

Suppose only 1% of observations are positive. A classifier that always predicts negative has 99% accuracy, but it never detects a positive case. In many applications, that classifier is useless.

Accuracy is not enough

Accuracy implicitly treats all mistakes as equally costly. When classes are imbalanced or error costs are asymmetric, accuracy can hide the most important behavior of the classifier.

2.1.3 Recall (sensitivity)

Recall is defined as

$$\text{Recall} = \frac{TP}{TP + FN}.$$

Recall answers the question:

Among all truly positive observations, how many did we detect?

In probability notation,

$$\text{Recall} = P(\hat{Y} = 1 \mid Y = 1).$$

Recall is also called sensitivity, or true positive rate (TPR).

High recall means few false negatives. Low recall means many true positive cases were missed.

Recall and Statistical Power

Recall has a direct connection to hypothesis testing. Suppose we think of the classification problem as testing

$$H_0 : Y = 0 \quad \text{versus} \quad H_1 : Y = 1.$$

Predicting positive is like rejecting H_0 . Predicting negative is like failing to reject H_0 .

Under this interpretation:

- a false positive is a Type I error,
- a false negative is a Type II error,

- recall plays a role analogous to statistical power.

Thus

$$\text{Recall} = 1 - \text{Type II error rate.}$$

Recall measures the ability to detect true signals.

2.1.4 Precision

Precision is defined as

$$\text{Precision} = \frac{TP}{TP + FP}.$$

Precision answers the question:

Among all observations predicted positive, how many are truly positive?

In probability notation,

$$\text{Precision} = P(Y = 1 \mid \hat{Y} = 1).$$

Precision is also called positive predictive value.

High precision means positive predictions are trustworthy. Low precision means many predicted positives are false alarms.

i Precision vs Recall

Precision and recall condition on different events.

Recall conditions on the truth:

$$P(\hat{Y} = 1 \mid Y = 1).$$

Precision conditions on the prediction:

$$P(Y = 1 \mid \hat{Y} = 1).$$

This distinction is very important.

Recall asks whether we find the positives. Precision asks whether our positive discoveries are reliable. Precision depends strongly on class prevalence.

2.1.5 F1 Score

The F1 score combines precision and recall:

$$F_1 = \frac{2PR}{P + R},$$

where P is precision and R is recall.

Equivalently,

$$F_1 = \frac{2TP}{2TP + FP + FN}.$$

The F1 score is the harmonic mean of precision and recall.

The harmonic mean strongly penalizes imbalance. If precision is high but recall is near zero, then F1 is near zero. If recall is high but precision is near zero, then F1 is also near zero.

F1 is useful when we want a single number that balances precision and recall. However, it assumes precision and recall are equally important. In applications where one type of error is more costly, a different metric or an explicit cost function may be better.

Note

F1 ignores true negatives entirely. This can be useful for rare event detection, where the number of true negatives may dominate the dataset and make accuracy misleading.

2.1.6 Recall vs Precision

2.1.6.1 Bayes' Rule and Class Imbalance

Precision and recall are connected by Bayes' rule.

Let $\pi = P(Y = 1)$ be the prevalence of the positive class.

Proposition 2.1.

$$\text{Precision} = P(Y = 1 \mid \hat{Y} = 1) = \frac{TPR \cdot \pi}{TPR \cdot \pi + FPR(1 - \pi)}.$$

This formula shows why precision can be low for rare events: when π is small, even a modest false positive rate can create many false positives.

Click to expand.

Let

$$TPR = P(\hat{Y} = 1 \mid Y = 1) = \text{Recall}$$

and

$$FPR = P(\hat{Y} = 1 \mid Y = 0).$$

Then by Bayes' Rule,

$$\begin{aligned} \text{Precision} &= P(Y = 1 \mid \hat{Y} = 1) \\ &= \frac{P(\hat{Y} = 1 \mid Y = 1)P(Y = 1)}{P(\hat{Y} = 1 \mid Y = 1)P(Y = 1) + P(\hat{Y} = 1 \mid Y = 0)P(Y = 0)} \\ &= \frac{TPR \cdot \pi}{TPR \cdot \pi + FPR \cdot (1 - \pi)}. \end{aligned}$$

Example 2.1 (Rare Event Example).

Click to expand.

Suppose there are 10,000 observations:

- 100 positives,
- 9,900 negatives.

Assume the classifier has:

$$TPR = 0.90, \quad FPR = 0.05.$$

Then:

$$TP = 90, \quad FP = 495.$$

The recall is high:

$$\text{Recall} = 0.90.$$

But precision is only

$$\text{Precision} = \frac{90}{90 + 495} \approx 0.154.$$

Most predicted positives are false positives. This is why precision-recall analysis is especially important for imbalanced data.

2.1.6.2 Classification Thresholds

Many classifiers first produce a score or estimated probability:

$$\hat{p}(x) = P(Y = 1 \mid X = x).$$

To make a hard classification, we choose a threshold c :

$$\hat{Y} = \begin{cases} 1, & \hat{p}(x) > c, \\ 0, & \hat{p}(x) \leq c. \end{cases}$$

The default decision threshold is often $c = 0.5$, but this is not always the optimal choice. Changing the threshold changes the confusion matrix, and therefore also changes metrics such as

accuracy, precision, recall, false positive rate (FPR), and F1 score.

To better understand the precision–recall tradeoff, let us examine how altering the decision threshold changes the behavior of a classifier.

Decision Threshold	TP	TN	FP	FN	Re- call	Preci- sion	Clas- sifier Be- hav- ior
Lower threshold	Non- decreasing	Non- increasing	Non- decreasing	Non- increasing	Non- decreasing	May decrease	
Higher threshold	Non- increasing	Non- decreasing	Non- increasing	Non- decreasing	Non- increasing	May decrease	

Lowering the threshold classifies more observations as positive. Therefore, both true positives (TP) and false positives (FP) may increase, while false negatives (FN) decrease.

Raising the threshold classifies more observations as negative. Therefore, both true negatives (TN) and false negatives (FN) may increase, while true positives (TP) decrease.

Example 2.2 (Example: Precision Is Not Strictly Monotone).

Click to expand.

Consider a small validation dataset with five observations. The model outputs predicted probabilities $\hat{P}(Y = 1 \mid X)$, ordered from highest to lowest.

Instance	Predicted Probability	True Label
A	0.95	1
B	0.90	0
C	0.85	1
D	0.80	1
E	0.70	0

As the threshold is raised, fewer observations are classified as positive.

Threshold	Predicted Positives	TP	FP	Precision
0.75	A, B, C, D	3	1	$3/4 = 0.75$
0.82	A, B, C	2	1	$2/3 \approx 0.67$
0.88	A, B	1	1	$1/2 = 0.50$
0.92	A	1	0	$1/1 = 1.00$

In this example, raising the threshold first decreases precision because some true positives are removed while the false positive remains. Precision increases only after the false positive is also removed.

This shows that precision can move up or down as the threshold changes, depending on the labels of the observations crossing the threshold.

i Thresholds are decisions

- The fitted model may estimate probabilities.
- The threshold determines the action.

Model fitting and decision making are related, but they are not the same thing.

2.1.6.3 Decision Theory and Error Costs

Suppose a false positive has cost C_{FP} and a false negative has cost C_{FN} .

If a model estimates

$$p(x) = P(Y = 1 \mid X = x),$$

then the expected cost of predicting positive is

$$\text{Cost}(1 \mid x) = C_{FP}P(Y = 0 \mid X = x) = C_{FP}(1 - p(x)).$$

The expected cost of predicting negative is

$$\text{Cost}(0 \mid x) = C_{FN}P(Y = 1 \mid X = x) = C_{FN}p(x).$$

We should predict positive when

$$C_{FP}(1 - p(x)) < C_{FN}p(x).$$

Solving this inequality gives

$$p(x) > \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

Thus the optimal threshold is

$$c^* = \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

This threshold minimizes the expected classification cost under the assumed probability model.

If false negatives are very costly, then C_{FN} becomes large, which lowers the threshold. In this case, the classifier predicts the positive class more aggressively in order to reduce missed detections.

Example 2.3 (Medical Screening). In medical screening, missing a serious disease may be much worse than sending a healthy patient for an additional test. Therefore false negatives are costly.

The threshold should often be lower, so the classifier prioritizes recall.

Example 2.4 (Spam Filtering). In spam filtering, putting an important legitimate email into spam may be very costly to the user. Therefore false positives may be more costly.

The threshold should often be higher, so the classifier prioritizes precision.

2.2 Ranking quality

Ranking quality evaluates whether positive observations tend to receive higher scores than negative observations.

This perspective is important when the classification threshold is not fixed or when the main goal is prioritization rather than direct classification. ROC curves, AUC, and precision–recall curves are commonly used to evaluate ranking performance.

2.2.1 ROC

ROC stands for receiver operating characteristic. An ROC curve plots:

$$TPR = \frac{TP}{TP + FN} \quad \text{against} \quad FPR = \frac{FP}{FP + TN}$$

as the probability threshold varies from 0 to 1.

2.2.2 ROC and Hypothesis Testing

ROC curves have a natural hypothesis testing interpretation. As discussed before,

- TPR is power,
- FPR is Type I error rate.

Therefore an ROC curve shows

power versus Type I error rate

across all possible thresholds.

The ideal classifier reaches the upper-left corner:

$$FPR = 0, \quad TPR = 1.$$

A classifier close to the diagonal line behaves almost like random guessing.

2.2.3 AUC

AUC refers to the Area Under the ROC Curve. It summarizes ranking performance across all possible classification thresholds.

Let $s(X^+)$ be the score for a randomly selected positive observation, and let $s(X^-)$ be the score for a randomly selected negative observation. Then

$$AUC = P(s(X^+) > s(X^-)),$$

up to small corrections for ties.

Thus AUC has a clean probabilistic interpretation:

AUC is the probability that a randomly selected positive observation receives a higher score than a randomly selected negative observation.

This explains why AUC is threshold-free. It evaluates ranking quality rather than the quality of a particular classification decision.

Importantly, AUC evaluates relative ordering rather than absolute probability accuracy.

AUC is not calibration

AUC can be excellent even when predicted probabilities are numerically inaccurate.

AUC only asks whether positive observations tend to receive higher scores than negative observations. It does not ask whether the predicted probabilities themselves are trustworthy or well calibrated.

2.2.4 Precision-Recall Curve

A precision-recall curve plots precision against recall as the classification threshold changes.

It is especially useful when the positive class is rare. In imbalanced datasets, an ROC curve may look strong because the false positive rate is divided by the large number of true negatives. However, precision directly measures how many predicted positives are actually positive.

Average precision summarizes the precision-recall curve as a single number. Like AUC, it evaluates ranking behavior across thresholds, but it focuses more directly on performance for the positive class.

2.3 Probability Quality

Probability quality asks whether predicted probabilities can be interpreted as actual probabilities.

For example, suppose a model predicts $\hat{p}(x) = 0.8$. If the model is well calibrated, then among many observations with predicted probability near 0.8, roughly 80% should truly belong to the positive class.

Thus probability quality concerns whether predicted probabilities are numerically trustworthy.

Unlike decision quality or ranking quality, probability quality focuses on the accuracy of the probability estimates themselves. Calibration curves, Brier score, and log loss are commonly used to evaluate probability quality.

Note

In many practical machine learning applications, probability quality is less important than classification accuracy or ranking performance.

Therefore, in this course, we will mainly focus on threshold-based classification metrics such as accuracy, precision, recall, and F1 score. This section is included primarily to provide a broader view of classification evaluation.

2.3.1 Calibration Curve

A calibration curve, also called a reliability diagram, is constructed as follows:

1. Divide observations into bins according to predicted probability.
2. For each bin, compute the average predicted probability.
3. Compute the observed positive proportion in each bin.
4. Plot observed proportion against average predicted probability.

For a perfectly calibrated model, the curve follows the diagonal line.

- If the curve lies below the diagonal, the model is overconfident.
- If the curve lies above the diagonal, the model is underconfident.

2.4 Model Tuning versus Threshold Tuning

In classification, model fitting and decision making should be viewed as separate problems.

First, the model estimates a score or probability:

$$\hat{p}(x) \approx \Pr(Y = 1 \mid X = x).$$

Second, we choose a threshold to convert this probability into an action.

Model tuning affects the quality of $\hat{p}(x)$ itself. This is where ideas such as bias, variance, regularization, and model complexity matter. A very flexible classifier may capture nonlinear structure, but its probability estimates may be unstable or poorly calibrated.

Threshold tuning does not change $\hat{p}(x)$. It only changes the decision rule built on top of the fitted probabilities. Lower thresholds usually increase recall and false positives; higher thresholds usually increase precision and reduce false positives.

Thus:

- model tuning controls the fitted scores or probabilities;
- threshold tuning controls the action taken from those scores;
- calibration checks whether the probabilities are numerically trustworthy.

A good classifier therefore involves not only learning accurate scores, but also choosing appropriate decision rules for the application.

2.5 Python Example: Breast Cancer Classification

We now use the [breast cancer dataset from sklearn](#). The goal is to classify tumors as malignant or benign. This example shows how to compute threshold-based metrics, ROC and PR curves, and calibration curves.

In this example, please mainly focus on the evaluation metrics. Other part like modeling, splitting, etc. will be introduced in later sections.

2.5.1 Load the dataset

First load the data and split it into training and test sets. By default, the dataset uses 0 for malignant and 1 for benign. Since many classification metrics interpret label 1 as the positive class, we redefine malignant as 1. Therefore we need to modify the y value.

```

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

cancer = load_breast_cancer(as_frame=True)

X = cancer.data
y_original = cancer.target
y = 1 - y_original

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, stratify=y, random_state=1
)

```

We may want to see the distribution of the test set.

```

import pandas as pd
pd.Series(y_test).value_counts().sort_index()

```

```

target
0      107
1       64
Name: count, dtype: int64

```

2.5.2 Fit models

We fit logistic regression model as an example, which is a typical strong baseline model.

```

from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

logit = make_pipeline(
    StandardScaler(),
    LogisticRegression(max_iter=5000),
)

logit.fit(X_train, y_train)

```



```
logit_prob = logit.predict_proba(X_test)[:, 1]
```

Note that `logit.predict_proba(X_test)` returns 2 columns:

- column 0: $\Pr(Y = 0 \mid X)$,
- column 1: $\Pr(Y = 1 \mid X)$.

Since we want the predicted probability for the positive class ($y=1$), we select the second column.

2.5.3 Confusion Matrix and Basic Metrics

All decision quality metrics can be computed using functions from `sklearn.metrics`. They are used to compare two list-like objects, while the true labels are passed first and the predicted labels second.

By default, `predict()` in `sklearn` uses threshold 0.5 for binary classification.

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

y_pred = logit.predict(X_test)

acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred)
rec = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
```

We can also compute the confusion matrix and extract the values of TP, TN, FP, and FN from it.

```
from sklearn.metrics import confusion_matrix

confusion_matrix(y_test, y_pred)
```

```
array([[106,  1],
       [ 6,  58]])
```

```
TN, FP, FN, TP = confusion_matrix(y_test, y_pred).ravel()
```

They are listed in this particular order since 0 stands for negative and 1 stands for positive in this example.

2.5.4 Modifying Thresholds

The default model use 0.5 as the threshold in `sklearn`. If you want to change the threshold, you may manually write the code.

```
threshold = 0.3
logit_prob = logit.predict_proba(X_test)[: , 1]
y_pred = (logit_prob >= threshold).astype(int)
```

You may also use `FixedThresholdClassifier` from `sklearn.model_selection` when you want the threshold to be part of the estimator workflow.

```
from sklearn.model_selection import FixedThresholdClassifier

threshold = 0.3
model_with_threshold = FixedThresholdClassifier(
    make_pipeline(
        StandardScaler(),
        LogisticRegression(max_iter=5000),
    ),
    threshold=threshold,
)

model_with_threshold.fit(X_train, y_train)
y_pred_wrapper = model_with_threshold.predict(X_test)
```

Note

`FixedThresholdClassifier` is useful when the threshold is fixed in advance.

However, there might be cases that you want to adjust the

threshold according to the dataset. In this case you may want to use `TunedThresholdClassifierCV`. We will talk about it in details later in the Logistic regression section.

2.5.5 Tuning Thresholds

In order to make the tuning process easier, we build an interface to quickly display the results.

```
def classification_summary(y_true, prob, threshold=0.5):
    pred = (prob >= threshold).astype(int)
    TN, FP, FN, TP = confusion_matrix(y_true, pred).ravel()

    return {
        "threshold": threshold,
        "TP": TP,
        "FP": FP,
        "FN": FN,
        "TN": TN,
        "accuracy": accuracy_score(y_true, pred),
        "precision": precision_score(y_true, pred, zero_division=0), ①
        "recall": recall_score(y_true, pred, zero_division=0),
        "f1": f1_score(y_true, pred, zero_division=0),
    }
```

① `zero_division=0` means that if the denominator is 0 the return value will be set to 0. This always happens in this case since some of the counts might be 0.

Now examine how metrics change as the threshold changes. To save space I only show the first few rows of the table.

```
import numpy as np
import pandas as pd

thresholds = np.linspace(0.05, 0.95, 19)
threshold_results = []

for threshold in thresholds:
```

```

    row = classification_summary(y_test, logit_prob, threshold=threshold)
    threshold_results.append(row)

threshold_df = pd.DataFrame(threshold_results)

threshold_df.head()

```

	threshold	TP	FP	FN	TN	accuracy	precision	recall	f1
0	0.05	62	8	2	99	0.941520	0.885714	0.968750	0.925373
1	0.10	61	7	3	100	0.941520	0.897059	0.953125	0.924242
2	0.15	60	5	4	102	0.947368	0.923077	0.937500	0.930233
3	0.20	60	5	4	102	0.947368	0.923077	0.937500	0.930233
4	0.25	60	4	4	103	0.953216	0.937500	0.937500	0.937500

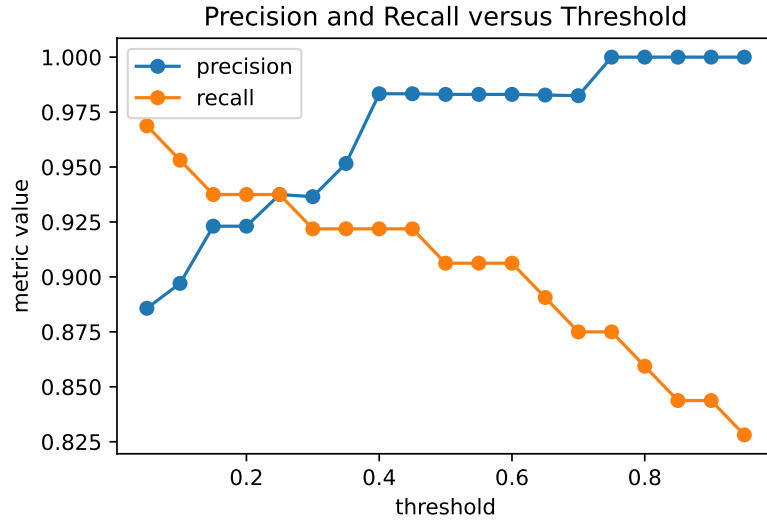
Plot precision and recall as functions of the threshold.

```

import matplotlib.pyplot as plt

plt.plot(threshold_df["threshold"], threshold_df["precision"], marker="o", label="precision")
plt.plot(threshold_df["threshold"], threshold_df["recall"], marker="o", label="recall")
plt.xlabel("threshold")
plt.ylabel("metric value")
plt.title("Precision and Recall versus Threshold")
plt.legend()

```



As the threshold increases, the classifier becomes more conservative in predicting the positive class. This often reduces false positives and increases precision, but may also increase false negatives and reduce recall.

2.5.6 Choosing a Threshold from Costs

In a medical setting, false negatives may be especially important because they correspond to missed malignant cases. Suppose a false negative is 5 times as costly as a false positive:

$$C_{FN} = 5, \quad C_{FP} = 1.$$

The decision-theoretic threshold is

$$c^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{1}{1 + 5} \approx 0.167.$$

Then the corresponding result is

```
cost_fp = 1
cost_fn = 5

cost_threshold = cost_fp / (cost_fp + cost_fn)
```

```
classification_summary(
    y_test,
    logit_prob,
    threshold=cost_threshold,
)
```

```
{'threshold': 0.16666666666666666,
 'TP': np.int64(60),
 'FP': np.int64(5),
 'FN': np.int64(4),
 'TN': np.int64(102),
 'accuracy': 0.9473684210526315,
 'precision': 0.9230769230769231,
 'recall': 0.9375,
 'f1': 0.9302325581395349}
```

This lower threshold prioritizes recall. It predicts malignant more aggressively because missing a malignant tumor is assumed to be more costly. Note that this threshold formula assumes that the predicted probabilities are reasonably calibrated so that they can be interpreted as approximate event probabilities.

2.5.7 ROC Curve and AUC

Unlike accuracy or precision, ROC analysis does not depend on a single fixed threshold. Instead, it studies model performance across all possible thresholds using the predicted probabilities or scores.

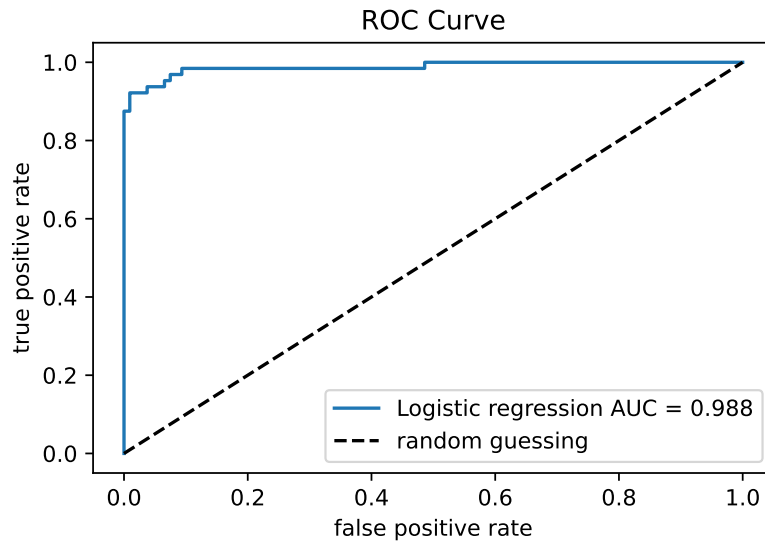
ROC curve and AUC can be computed using `roc_curve` and `roc_auc_score` from `sklearn.metrics`.

```
from sklearn.metrics import roc_curve, roc_auc_score

logit_fpr, logit_tpr, _ = roc_curve(y_test, logit_prob)
logit_auc = roc_auc_score(y_test, logit_prob)

plt.plot(logit_fpr, logit_tpr, label=f"Logistic regression AUC = {logit_auc:.3f}")
plt.plot([0, 1], [0, 1], "k--", label="random guessing")
```

```
plt.xlabel("false positive rate")
plt.ylabel("true positive rate")
plt.title("ROC Curve")
plt.legend()
```



We may fit another model and draw their ROC curve and AUC together in one plot to compare them.

```
from sklearn.ensemble import RandomForestClassifier

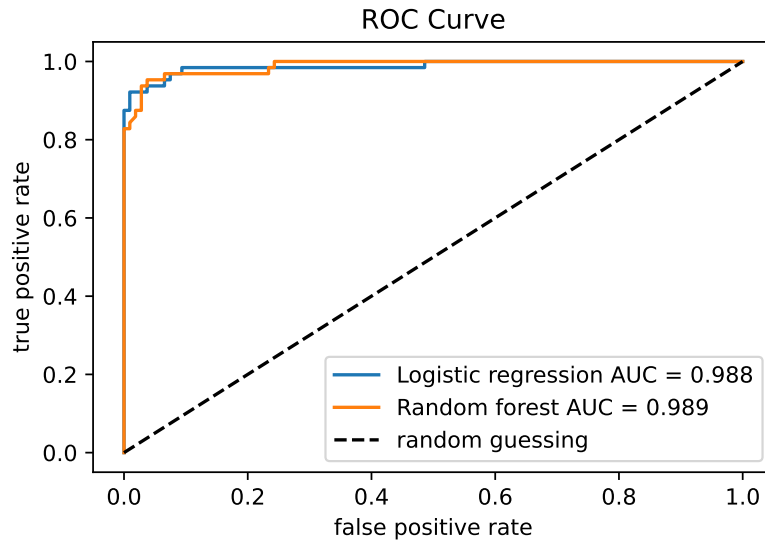
rf = RandomForestClassifier(
    n_estimators=500, max_features="sqrt", random_state=1, n_jobs=-1
)

rf.fit(X_train, y_train)
rf_prob = rf.predict_proba(X_test)[: , 1]

rf_fpr, rf_tpr, _ = roc_curve(y_test, rf_prob)
rf_auc = roc_auc_score(y_test, rf_prob)

plt.plot(logit_fpr, logit_tpr, label=f"Logistic regression AUC = {logit_auc:.3f}")
plt.plot(rf_fpr, rf_tpr, label=f"Random forest AUC = {rf_auc:.3f}")
plt.plot([0, 1], [0, 1], "k--", label="random guessing")
plt.xlabel("false positive rate")
```

```
plt.ylabel("true positive rate")
plt.title("ROC Curve")
plt.legend()
```



The ROC curve summarizes ranking performance across thresholds. AUC may be interpreted as the probability that a randomly chosen positive observation receives a higher score than a randomly chosen negative observation. Then a larger AUC means the model more often ranks malignant cases above benign cases.

2.5.8 Precision-Recall Curve

The precision-recall curve can be produced in a similar way.

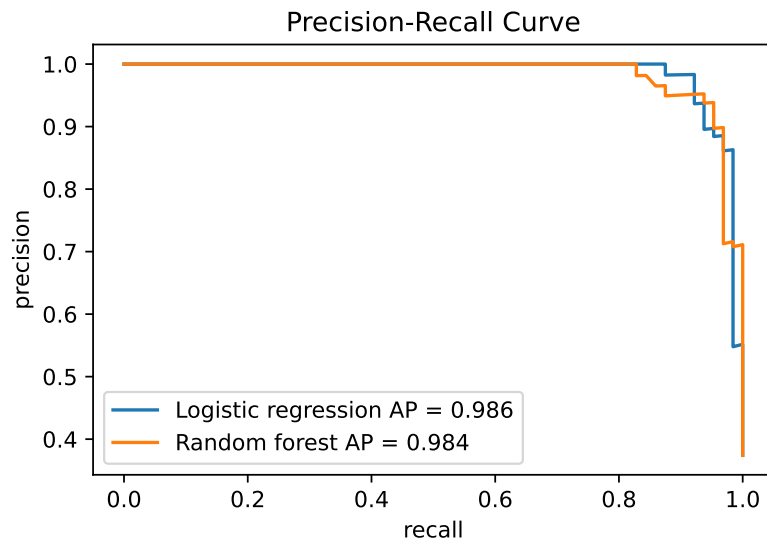
```
from sklearn.metrics import precision_recall_curve, average_precision_score

logit_precision, logit_recall, _ = precision_recall_curve(y_test, logit_prob)
rf_precision, rf_recall, _ = precision_recall_curve(y_test, rf_prob)

logit_ap = average_precision_score(y_test, logit_prob)
rf_ap = average_precision_score(y_test, rf_prob)
```



```
plt.plot(logit_recall, logit_precision, label=f"Logistic regression AP = {logit_ap:.3f}")
plt.plot(rf_recall, rf_precision, label=f"Random forest AP = {rf_ap:.3f}")
plt.xlabel("recall")
plt.ylabel("precision")
plt.title("Precision-Recall Curve")
plt.legend()
```



For highly imbalanced datasets, precision–recall curves are often more informative than ROC curves because ROC curves may appear overly optimistic when the negative class dominates.

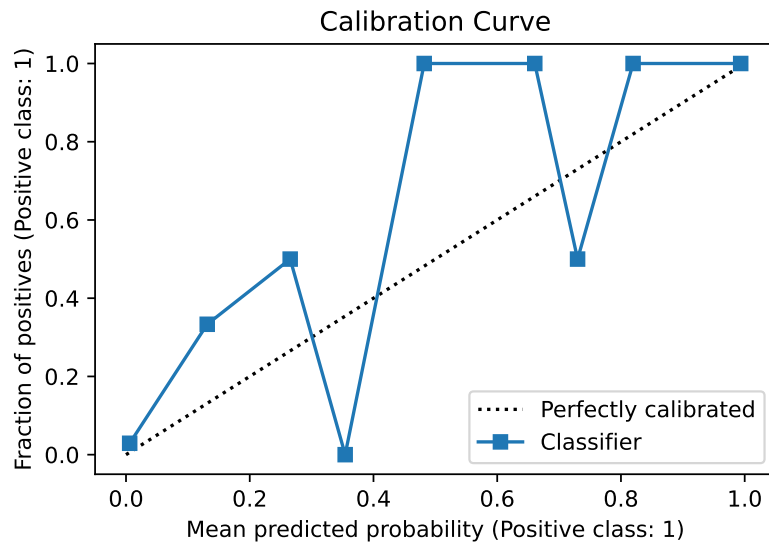
2.5.9 Optional: Calibration Curves

Click to expand.

Calibration focuses on probability quality rather than threshold-based decision quality or ranking quality. Calibration curves can be produced using `CalibrationDisplay` from `sklearn.calibration`.

```
from sklearn.calibration import CalibrationDisplay

CalibrationDisplay.from_predictions(y_test, logit_prob, n_bins=10)
plt.title("Calibration Curve");
```

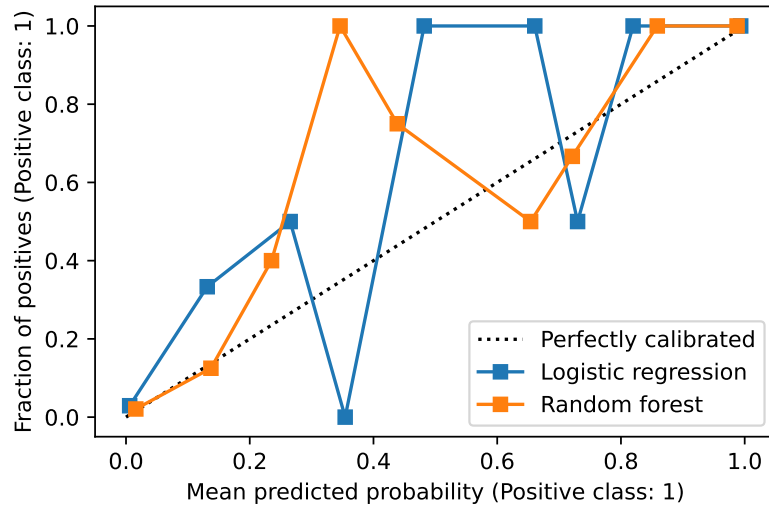


This is a packaged function provided by `sklearn`. If we want to plot multiple curves together in one plot, we can put the axis in the argument of this function.

```
fig, ax = plt.subplots()

CalibrationDisplay.from_predictions(
    y_test, logit_prob, n_bins=10, name="Logistic regression", ax=ax
)

CalibrationDisplay.from_predictions(
    y_test, rf_prob, n_bins=10, name="Random forest", ax=ax
)
```



The calibration curve compares predicted probabilities with observed frequencies.

Calibration curves diagnose probability quality; they do not by themselves recalibrate the model. Recalibration methods such as Platt scaling or isotonic regression can change the probability scale, but they do not necessarily improve AUC because AUC mainly depends on ranking.

Part II

**Parametric Supervised
Learning**

3 Linear Regression

This section offers a brief review of linear regression, with the main emphasis on its Python implementation through examples.

3.1 Quick Overview

Linear regression is the fundamental supervised learning method for a quantitative response. It models the conditional mean of the response as a linear function of the predictors:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \varepsilon.$$

In matrix notation, this can be written as

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon.$$

Here, $\mathbf{X}$ is the design matrix, whose first column is a column of ones representing the intercept.

3.2 The Ordinary Least Squares Estimator

The most common fitting method is ordinary least squares (OLS). It chooses the coefficient vector β that minimizes the residual sum of squares (RSS):

$$\text{RSS}(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \|\mathbf{y} - \mathbf{X}\beta\|_2^2.$$

Theorem 3.1 (OLS Estimator). *The coefficient vector that minimizes RSS is*

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y},$$

provided that $\mathbf{X}^\top \mathbf{X}$ is invertible. This estimator is called the ordinary least squares (OLS) estimator.

Once the coefficients are estimated, the fitted value for observation i is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}.$$

Interpretability is one of the main advantages of linear regression compared with many more complex models.

1. The coefficient $\hat{\beta}_j$ describes how the predicted response changes when x_j increases by one unit, while the other predictors are held fixed.
2. When interaction terms are present, a main effect cannot be interpreted in isolation. The effect of a predictor depends on the values of the variables with which it interacts.
3. If predictors are centered so that 0 corresponds to the average of the original variable, the intercept represents the predicted response at the average predictor levels.

i Note

Classical regression primarily focuses on inference and model assumptions, such as normality, constant variance, and independence of errors.

As discussed earlier in the statistical learning basics section, statistical learning places greater emphasis on predictive performance and generalization to unseen data. As a result, estimated test error and model validation often play a more central role than formal inference procedures.

3.3 Linear Regression in Python

There are many Python libraries for fitting linear regression models. In this course, we use `scikit-learn` as the primary modeling tool, with support from `pandas` and `numpy` for data manipulation, and `matplotlib` and `seaborn` for visualization.

To demonstrate the code, we generate a dataset with 2 predictors and 1 response variable, where $\beta_0 = 1$, $\beta_1 = 2$, and $\beta_2 = 3$.

```
import numpy as np
import pandas as pd

n = 100

rng = np.random.default_rng(1)

x1 = rng.normal(1, 0.1, size=n)
x2 = rng.normal(0, 1, size=n)
y = 1 + 2 * x1 + 3 * x2 + rng.normal(0, 0.5, size=n)

X = pd.DataFrame({'x1': x1, 'x2': x2})
```

When using `sklearn`, a dataset is typically represented by features `X` and a target variable `y`.

- `X` usually has 2 dimensions (number of samples $\times$ features).
- `y` is usually 1-dimensional, though it may also be represented as a 2-dimensional array with a single column.

Both `X` and `y` can be stored as either `numpy` arrays or `pandas` objects.

In this example we construct `X` as a `DataFrame`, since column names make it easier to modify variables during the model-building stage, e.g. when adding interactions or transformations.

3.3.1 Linear regression models

We fit a linear regression model using `LinearRegression`.

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X, y)

print(f"intercept: {model.intercept_}")
print(f"coefficients: {model.coef_}")
```

```
intercept: 0.9846240766122039
coefficients: [1.93387446 2.99292344]
```

We can then use the fitted model to generate predictions and compute residuals.

```
y_pred = model.predict(X)
res = y - y_pred
```

3.3.2 Measure model performance

`sklearn` provides many performance measures, most of which can be found in `sklearn.metrics`. A common syntax is `metric(y_true, y_pred)`.

```
from sklearn.metrics import mean_squared_error, r2_score, root_mean_squared_error

y_pred = model.predict(X)

print(f"MSE: {mean_squared_error(y, y_pred)}")
print(f"R2: {r2_score(y, y_pred)}")
print(f"RMSE: {root_mean_squared_error(y, y_pred)}")
```

Each model also provides a built-in scoring method. For a specific model, you can check the official documentation to see which metric is used, but the general idea is:

- regressors use `r2_score`, and
- classifiers use `accuracy`

```
model.score(X, y)
```

0.9769128696597689

i Different Definitions Across Software

Some evaluation metrics may be defined slightly differently across software packages and textbooks.

For example, mean squared error (MSE) in statistics is defined as $\frac{RSS}{n-(p+1)}$ where $(p+1)$ is the number of estimated parameters. This adjustment accounts for the degrees of freedom used in fitting the model.

In contrast, MSE in machine learning (and `sklearn`) are commonly defined as $\frac{RSS}{n}$ which is simply the average squared prediction error over the observations.

These differences are usually small when the sample size is large, but they can affect numerical comparisons between software packages or published benchmarks. Therefore, it is important to verify the exact definition of any metric before making comparisons.

For example, according to the `sklearn` [documentation](#), `mean_squared_error` is computed by averaging over the number of samples n .

3.3.3 Model Diagnostics

In traditional regression, diagnostic plots are powerful tools for detecting potential problems in a fitted model. However, statistical learning primarily focuses on prediction performance rather than model assumptions. Therefore, diagnostics are not our primary focus in this course, and this section is optional.

[Click for more details.](#)

In Python there is no single function that automatically produces all diagnostic plots. Instead, we usually combine tools from

several libraries. In these notes we focus on basic tools from `statsmodels`, `scipy`, `seaborn`, and `matplotlib`.

We can also use `seaborn`'s theme to improve the appearance of plots produced by `matplotlib`-based tools.

```
import seaborn as sns
sns.set_theme()
```

We compute several quantities needed for residual analysis. Although our primary modeling tool is `sklearn`, residual diagnostics are easier with `statsmodels`. Therefore we construct a `statsmodels` model that holds the same data.

Since we already have the feature matrix and the target vector, the fastest way to use `statsmodels` is through `statsmodels.api` (as opposed to the R-style interface `statsmodels.formula.api`).

```
import statsmodels.api as sm
X_sm = sm.add_constant(X)
model = sm.OLS(y, X_sm).fit()
model.params
```

```
const    0.984624
x1        1.933874
x2        2.992923
dtype: float64
```

The fitted values, residuals, leverage, Cook's distance, and other quantities can be extracted directly from this `model` object. We wrap them into `DataFrames` for easier plotting.

```
y_pred = model.fittedvalues
res = model.resid

model_output = pd.DataFrame({
    "y_true": y,
    "y_pred": y_pred,
    "res": res
})
```

```
influence = model.get_influence()

std_resid = influence.resid_studentized_internal
sqrt_std_resid = np.sqrt(np.abs(std_resid))
leverage = influence.hat_matrix_diag
cooks = influence.cooks_distance[0]

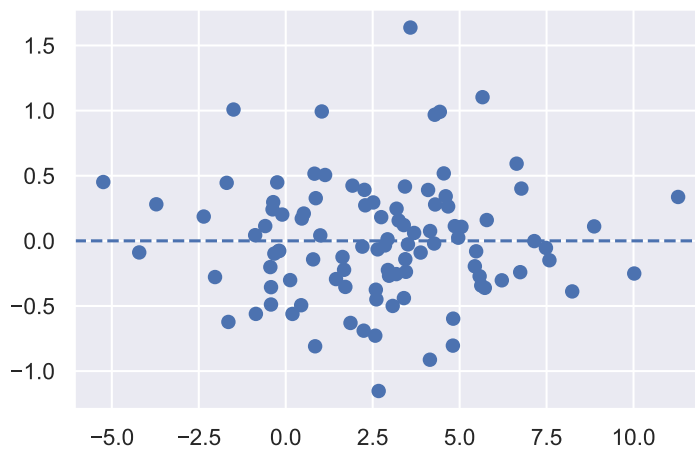
df_diag = pd.DataFrame({
    "leverage": leverage,
    "std_resid": std_resid,
    "sqrt_std_resid": sqrt_std_resid,
    "cooks": cooks
})
```

i Residual plots

The most basic residual plot is simply residuals versus fitted values.

```
import matplotlib.pyplot as plt

plt.scatter(y_pred, res)
plt.axhline(0, linestyle="--")
```

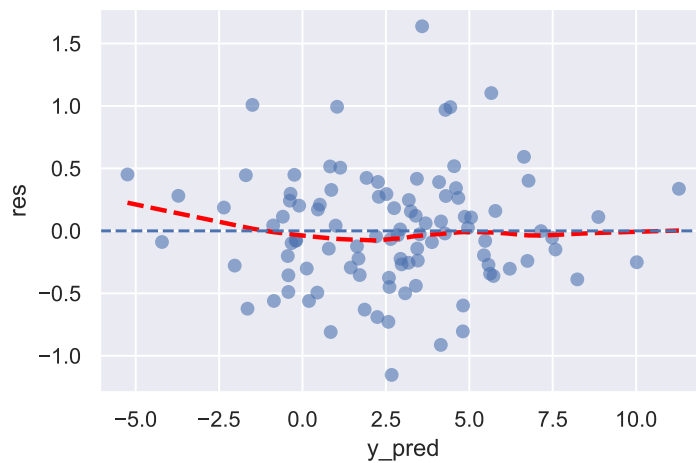


If we use **seaborn**, additional features are computed automatically. For example, setting **lowess=True** will display a smooth trend curve.

```
import seaborn as sns

sns.regplot(
    data=model_output,
    x="y_pred",
    y="res",
    lowess=True,
    scatter_kws={"alpha": 0.6},
    line_kws={"color": "red", "linestyle": "--"},
)

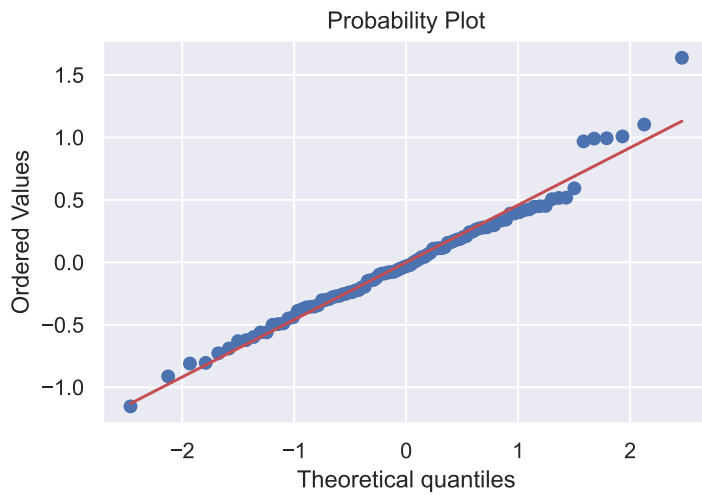
plt.axhline(0, linestyle="--")
```



i Q-Q plot

```
import scipy.stats as stats

stats.probplot(res, dist="norm", plot=plt);
```



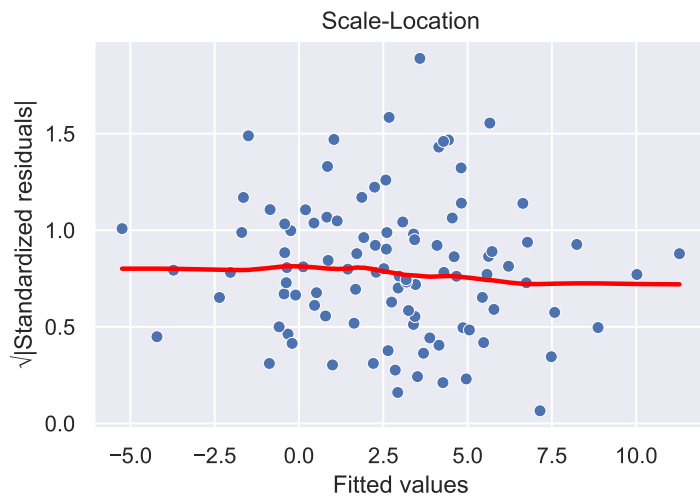
i Scale-location plot

```
import numpy as np

sns.scatterplot(x=y_pred, y=sqrt_std_resid, data=df_diag)

sns.regplot(
    x=y_pred,
    y=sqrt_std_resid,
    data=df_diag,
    lowess=True,
    scatter=False,
    line_kws={"color": "red"}
)

plt.xlabel("Fitted values")
plt.ylabel(" $\sqrt{|\text{Standardized residuals}|}$ ")
plt.title("Scale-Location");
```

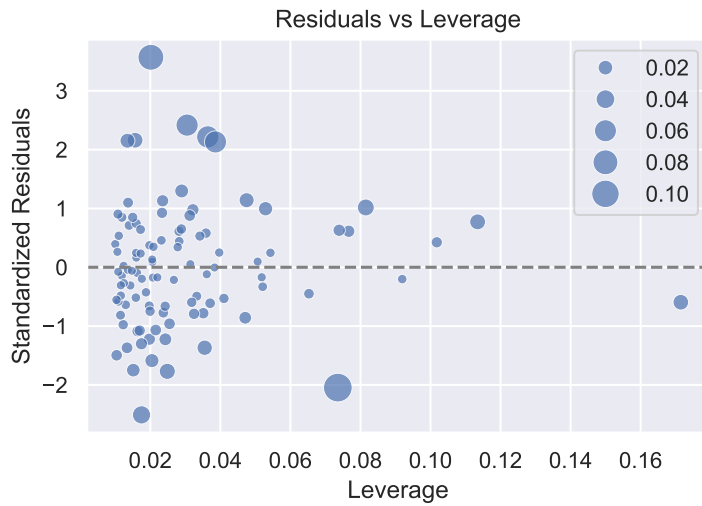


i Leverage with Cook's distance

```
sns.scatterplot(
    x=leverage,
    y=std_resid,
    data=df_diag,
    size=cooks,
    sizes=(20, 200),
    alpha=0.7
)

plt.axhline(0, linestyle="--", color="gray")

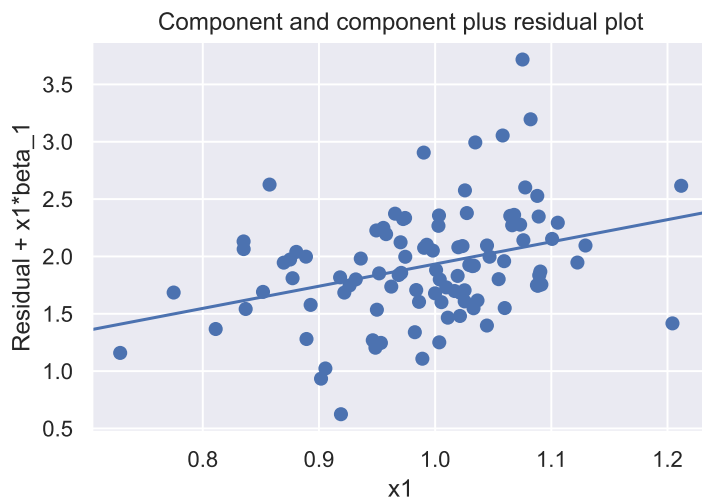
plt.xlabel("Leverage")
plt.ylabel("Standardized Residuals")
plt.title("Residuals vs Leverage");
```



i Partial residual plots

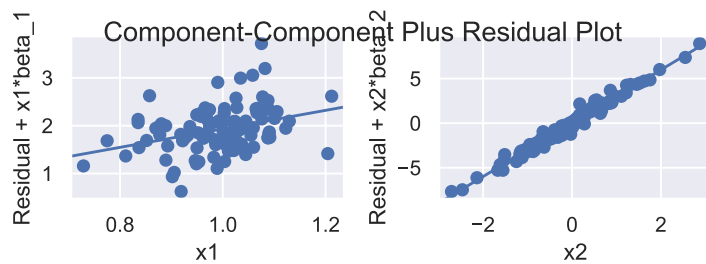
Partial residual plots help visualize the relationship between a predictor and the response after adjusting for the effects of other predictors. `plot_ccpr()` from `sm.graphics` can be used to generate partial residual plots.

```
sm.graphics.plot_ccpr(model, "x1");
```



To generate plots for all predictors at once, we can use

```
sm.graphics.plot_ccpr_grid(model);
```



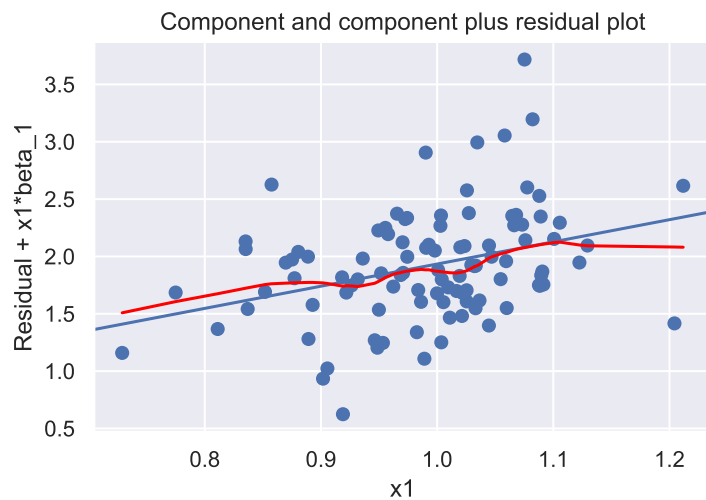
The default `plot_ccpr()` does not include a smooth curve. If a smooth trend line is desired, we can compute it using the `lowess()` function and add it manually.

```
from statsmodels.nonparametric.smoothers_lowess import lowess

fig = sm.graphics.plot_ccpr(model, "x1")

ax = fig.axes[0]
x = ax.lines[0].get_xdata()
y = ax.lines[0].get_ydata()

smooth = lowess(y, x, frac=0.5)
ax.plot(smooth[:, 0], smooth[:, 1], color="red")
```

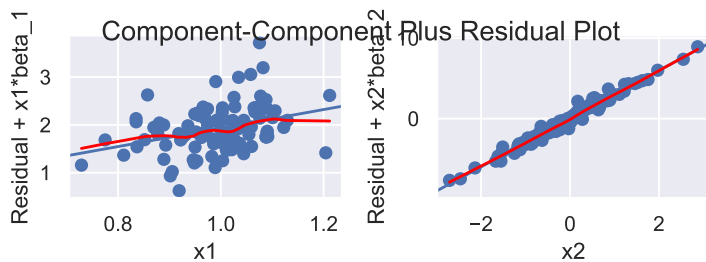


The same idea can be applied to the grid version.


```
fig = sm.graphics.plot_ccpr_grid(model)

for ax in fig.axes:
    if len(ax.lines) != 0:
        x = ax.lines[0].get_xdata()
        y = ax.lines[0].get_ydata()

        smooth = lowess(y, x, frac=0.5)
        ax.plot(smooth[:, 0], smooth[:, 1], color="red")
```



3.4 Model evaluation

3.4.1 Train-Test Split

Instead of estimating test performance indirectly from the training data, we can evaluate a model more directly by using data that were not used to fit the model. This leads to the idea of data splitting:

The dataset is divided into two parts:

- Training set: used to fit the model.
- Test set: used to evaluate prediction performance.

After fitting the model on the training set, predictions are made on the test set, and performance measures such as MSE are computed. This provides a more realistic estimate of how the model performs on new data.

In many situations we also need to choose between several competing models or tuning parameters. To avoid using the test data for model selection, a third subset is often introduced.

- Training set: used to fit models.
- Validation set: used to compare models and select tuning parameters.
- Test set: used only once at the end to estimate the final model's performance.

This separation helps prevent overly optimistic performance estimates.

In `sklearn`, the function `train_test_split()` from `sklearn.model_selection` is commonly used to split the data into a training set and a test set.

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split

n = 10

rng = np.random.default_rng(1)
x1 = rng.normal(loc=1, scale=0.1, size=n)
x2 = rng.normal(loc=0, scale=1, size=n)
y = 1 + 2 * x1 + 3 * x2 + rng.normal(0, 0.5, size=n)

X = pd.DataFrame({'x1': x1, 'x2': x2})

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)
```

The argument `test_size=0.2` means that 20% of the data are used as the test set, while the remaining 80% are used as the training set. The argument `random_state=0` ensures reproducibility.

3.4.2 Cross-Validation

When the dataset is not large, splitting the data into separate training, validation, and test sets may leave too little data for model fitting. In such cases, cross-validation is commonly used to estimate validation performance more efficiently.

The most common form is k -fold cross-validation.

1. The available training data are divided into k roughly equal parts (called folds).
2. For each fold:
 - The model is trained on $k - 1$ folds.
 - The remaining fold is used as a validation set.
3. This process is repeated k times so that each fold serves once as the validation set.
4. The validation errors from all folds are averaged to estimate the model's expected test error.

Cross-validation is primarily used to estimate test error or to compare competing models. When it is used for model selection, two additional steps are typically performed. Note that cross-validation uses only the training data and does not involve the test set.

5. After the best model configuration has been selected, the model is retrained on the entire training data.
6. The final model is then evaluated once on the test set, which serves as an unbiased estimate of its performance on new data.

In `sklearn`, there are several ways to perform k -fold cross-validation. From `KFold` to `cross_validate` and then `cross_val_score`, the interface becomes easier to use, but also less flexible.

The three approaches perform essentially the same task, but at different levels of abstraction.

KFold

KFold from `sklearn.model_selection` is used to split the dataset into k groups, called folds. In each iteration, one fold is used as the validation set and the remaining folds are used as the training set.

```
from sklearn.model_selection import KFold

kf = KFold(n_splits=5)

for train_idx, val_idx in kf.split(range(10)):
    print(train_idx, val_idx)
```

```
[2 3 4 5 6 7 8 9] [0 1]
[0 1 4 5 6 7 8 9] [2 3]
[0 1 2 3 6 7 8 9] [4 5]
[0 1 2 3 4 5 8 9] [6 7]
[0 1 2 3 4 5 6 7] [8 9]
```

- Here I use `range(10)` inside `kf.split()` because only the indices are needed. If a dataset is passed instead, the output is still the indices for the training set and validation set.
- If you want the split to be randomized, set `shuffle=True` when creating KFold. In that case, you may also specify `random_state` to make the result reproducible.

Let us now apply KFold to our simulated dataset.

```

from sklearn.model_selection import KFold
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

kf = KFold(n_splits=5, shuffle=True, random_state=1)

cv_scores = []
for train_idx, val_idx in kf.split(X):
    X_train, X_val = X.iloc[train_idx], X.iloc[val_idx]
    y_train, y_val = y[train_idx], y[val_idx]

    model = LinearRegression()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_val)
    score = r2_score(y_val, y_pred)

    cv_scores.append(score)

print(cv_scores)
print(sum(cv_scores) / len(cv_scores))

[-1.473916411368759, 0.3859310045149441, -
0.7579033227389274, 0.7043154482687257, 0.8824788076538627]
-0.051818894734030785

```

This gives the validation score for each fold, together with their average.

i cross_validate

Using `KFold` directly is somewhat manual. We may use `cross_validate()` to automate the cross-validation process. Depending on the arguments supplied, `cross_validate()` may internally use `KFold`.

You may specify which scoring methods using `scoring` argument. The default choice for `scoring` depends on the model. In `LinearRegression` case, the default choice is `r2`. Note that this `scoring` can accept a list of scoring

methods, or a single scoring method.

```
from sklearn.model_selection import cross_validate

model = LinearRegression()
cv_result = cross_validate(model, X, y, cv=5, scoring=["r2", "neg_mean_squared_error"])

cv_result

{'fit_time': array([0.00148249, 0.00122499, 0.00104594, 0.00093675, 0.00085187]),
 'score_time': array([0.00161695, 0.00092888, 0.00085115, 0.00079632, 0.00077248]),
 'test_r2': array([-0.39366753,  0.10843993,  0.60326202,  0.73431811,  0.91493437]),
 'test_neg_mean_squared_error': array([-0.79253928, -0.34840152, -1.25856101, -0.05339861, -0.05188008])}
```

If you are only interested in a certain validation scores, you may extract them directly.

```
cv_result['test_r2']

array([-0.39366753,  0.10843993,  0.60326202,  0.73431811,  0.91493437])
```

- More info on different scoring methods can be found in [the document](#). You can also print it from `sklearn.metrics.get_scorer_names()`.
- Here the negative MSE values are negative because `sklearn` follows the convention that larger scores are better.
- If `cv=5`, then `KFold(5, shuffle=False)` is used by default. If you want randomized folds, you may pass a `KFold` object explicitly.

```
cv_result = cross_validate(
    model, X, y, cv=KFold(5, shuffle=True, random_state=1)
)

cv_result["test_score"]

array([-1.47391641,  0.385931, -0.75790332,  0.70431545,  0.88247881])
```

You may compare these scores with the ones obtained in the KFold section. Usually, the overall cross-validation score is taken to be the mean of the fold scores. Note that in this case the `scoring` is not given. So it will use the default R^2 .

```
cv_result['test_score'].mean()
```

```
np.float64(-0.051818894734030764)
```

i cross_val_score

`cross_val_score()` is a shorter way to obtain `cv_result["test_score"]` from the `cross_validate()` section. The arguments `cv` and `scoring` work in almost the same way, except that `scoring` only accept a single scorer instead of a list in this case.

```
from sklearn.model_selection import cross_val_score
```

```
cv_scores = cross_val_score(  
    model, X, y, cv=KFold(5, shuffle=True, random_state=1)  
)
```

```
cv_scores
```

```
array([-1.47391641,  0.385931  , -  
0.75790332,  0.70431545,  0.88247881])
```

```
cv_scores.mean()
```

```
np.float64(-0.051818894734030764)
```

So `cross_val_score()` is convenient when you only need the validation scores and do not need the extra information returned by `cross_validate()`.

Cross-validation is often used to compare competing models. One important application is hyperparameter tuning. In this setting, we compare models from the same family under different choices of hyperparameters. In `sklearn`, this process is commonly carried out with `GridSearchCV`.

The CV procedure is summarized (again) as follows:

1. Split the training data into several folds.
2. For each hyperparameter combination, hold out one fold as the validation set and use the remaining folds as the training set.
3. Fit the model on the training folds and evaluate its performance on the validation fold.
4. Repeat this process so that each fold serves once as the validation set. The average validation score is the cross-validation score for that hyperparameter combination.
5. Repeat Steps 2–4 for all hyperparameter combinations, and select the combination with the best average cross-validation score.
6. **Final step:** After the best hyperparameter combination is identified, the model is refit on the entire training set using these hyperparameters.

`GridSearchCV` automates this process. To illustrate the idea, suppose we want to decide whether a linear regression model should be fit with or without an intercept.

First, specify the hyperparameters to be tuned:

```
params = {"fit_intercept": [True, False]}
```

! Pipeline and ColumnTransformer

If the model is wrapped inside a `Pipeline` or a `ColumnTransformer` (discussed later), hyperparameters inside a step must be referred to using the syntax `step__param` with two underscores.

For example, suppose a pipeline contains a step named

linear, and that step has a hyperparameter called `fit_intercept`. Then the parameter grid should be written as

```
params = {"linear__fit_intercept": [True, False]}
```

Next, construct the grid search.

- The argument `cv` is specified in the same way as in ordinary cross-validation.
- The argument `refit=True` controls whether the model is refit on the entire training set using the best hyperparameters. The default value is `True`.

```
from sklearn.model_selection import GridSearchCV

gs = GridSearchCV(model, param_grid=params, cv=5, refit=True)
gs.fit(X, y)
```

<code>estimator</code>	<code>LinearRegression()</code>
<code>estimator:</code>	
<code>estimator object</code>	

This is assumed to implement the scikit-learn estimator interface. Either estimator needs to provide a “score” function, or “scoring” must be passed.

<code>param_grid</code>	<code>{'fit_intercept':</code>
<code>param_grid: dict</code>	<code>[True, False]}</code>
or list of	
dictionaries	

Dictionary with parameters names ('str') as keys and lists of parameter settings to try as values, or a list of such dictionaries, in which case the grids spanned by each dictionary in the list are explored. This enables searching over any sequence of parameter settings.

scoring scoring: str, None
callable, list, tuple
or dict,
default=None

Strategy to
evaluate the
performance of the
cross-validated
model on
the test set.

If ‘scoring’
represents a single
score, one can use:

- a single
string (see :ref:‘scoring_string_names’);
- a callable (see
:ref:‘scoring_callable’) that
returns a single
value;
- ‘None’, the
‘estimator’s
:ref:‘default
evaluation criterion
‘ is used.

If ‘scoring’
represents multiple
scores, one can use:

- a list or tuple of
unique strings;
- a callable
returning a
dictionary where
the keys are the
metric
names and the
values are the
metric scores;
- a dictionary with
metric names as
keys and callables
as values.

See :ref:‘multimet-
ric_grid_search’

`n_jobs n_jobs: int, None`
`default=None`

Number of jobs to
run in parallel.

“None“ means 1

unless in a

:obj:‘joblib.parallel_backend‘

context.

“-1“ means using

all processors. See

:term:‘Glossary ‘

for more details.

.. versionchanged::

v0.20

‘n_jobs‘ default

changed from 1 to

None

refit refit: bool, str, True
or callable,
default=True

Refit an estimator
using the best
found parameters
on the whole
dataset.

For multiple metric
evaluation, this
needs to be a 'str'
denoting the
scorer that would
be used to find the
best parameters for
refitting
the estimator at
the end.

Where there are
considerations
other than
maximum score in
choosing a best
estimator, "refit"
can be set to a
function which
returns the selected
"best_index_"
given
"cv_results_". In
that
case, the
"best_estimator_"
and
"best_params_"
will be set
according to the
returned
"best_index_"
while the
"best_score_"
attribute will not
be available.

The refitted
estimator is made
available at the
"best_estimator_"

`cv`: int,
cross-validation
generator or an
iterable,
`default=None`

Determines the
cross-validation
splitting strategy.
Possible inputs for
`cv` are:

- None, to use the
default 5-fold cross
validation,
- integer, to specify
the number of folds
in a
'(Stratified)KFold',
- `:term:'CV
splitter'`,
- An iterable
yielding (train,
test) splits as
arrays of indices.

For integer/None
inputs, if the
estimator is a
classifier and “y” is
either binary or
multiclass,
`:class:'StratifiedK-
Fold'` is used. In all
other cases,
`:class:'KFold'` is
used. These
splitters are
instantiated
with `'shuffle=False'`
so the splits will be
the same across
calls.

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Refer `:ref:'User
Guide '` for the
various
cross-validation
strategies that can
be used here.

verbose verbose: 0
int

Controls the
verbosity: the
higher, the more
messages.

- >1 : the
computation time
for each fold and
parameter
candidate is
displayed;
- >2 : the score is
also displayed;
- >3 : the fold and
candidate
parameter indexes
are also displayed
together with the
starting time of the
computation.

```
pre_dispatch          '2*n_jobs'
pre_dispatch: int,
or str,
default='2*n_jobs'
```

Controls the number of jobs that get dispatched during parallel execution.

Reducing this number can be useful to avoid an explosion of memory consumption when more jobs get dispatched than CPUs can process. This parameter can be:

- None, in which case all the jobs are immediately created and spawned. Use this for lightweight and fast-running jobs, to avoid delays due to on-demand spawning of the jobs
- An int, giving the exact number of total jobs that are spawned
- A str, giving an expression as a function of `n_jobs`, as in `'2*n_jobs'`

<code>error_score</code>	<code>nan</code>
<code>error_score: 'raise'</code> or numeric, <code>default=np.nan</code>	

Value to assign to the score if an error occurs in estimator fitting. If set to 'raise', the error is raised. If a numeric value is given, `FitFailedWarning` is raised. This parameter does not affect the refit step, which will always raise the error.

`return_train_score` False
re-
`turn_train_score`:
bool, default=False

If “False“, the
“`cv_results_`“
attribute will not
include training
scores.

Computing
training scores is
used to get insights
on how different
parameter settings
impact the overfit-
ting/underfitting
trade-off.

However
computing the
scores on the
training set can be
computationally
expensive and is
not strictly
required to select
the parameters
that
yield the best
generalization
performance.

.. `versionadded::`
0.19

.. `versionchanged::`
0.21

Default value was
changed from
“True“ to “False“

<code>fit_intercept</code>	True
<code>fit_intercept: bool,</code> <code>default=True</code>	

Whether to calculate the intercept for this model. If set to False, no intercept will be used in calculations (i.e. data is expected to be centered).

<code>copy_X copy_X:</code>	True
<code>bool, default=True</code>	

If True, X will be copied; else, it may be overwritten.

tol tol: float, 1e-06
default=1e-6

The precision of the solution ('coef_') is determined by 'tol' which specifies a different convergence criterion for the 'lsqr' solver. 'tol' is set as 'atol' and 'btol' of :func:'scipy.sparse.linalg.lsqr' when fitting on sparse training data. This parameter has no effect when fitting on dense data.

.. versionadded::
1.7

`n_jobs` `n_jobs`: int, None
default=None

The number of jobs to use for the computation. This will only provide speedup in case of sufficiently large problems, that is if firstly `'n_targets > 1'` and secondly `'X'` is sparse or if `'positive'` is set to `'True'`. `"None"` means 1 unless in a `:obj:'joblib.parallel_backend'` context. `"-1"` means using all processors. See `:term:'Glossary'` for more details.

positive positive: False
bool, default=False

When set to
“True“, forces the
coefficients to be
positive. This
option is only
supported for
dense arrays.

For a comparison
between a linear
regression model
with positive
constraints
on the regression
coefficients and a
linear regression
without such
constraints,
see
[:ref:‘sphinx_glr_auto_examples_linear_model_plot_nnls.py‘](#).

.. versionadded::
0.24

After fitting, the best estimator and other information about the search can be obtained.

```
gs.best_estimator_
```

<code>fit_intercept</code>	True
<code>fit_intercept: bool,</code> <code>default=True</code>	

Whether to calculate the intercept for this model. If set to False, no intercept will be used in calculations (i.e. data is expected to be centered).

<code>copy_X copy_X:</code>	True
<code>bool, default=True</code>	

If True, X will be copied; else, it may be overwritten.

tol tol: float, 1e-06
default=1e-6

The precision of the solution ('coef_') is determined by 'tol' which specifies a different convergence criterion for the 'lsqr' solver. 'tol' is set as 'atol' and 'btol' of :func:'scipy.sparse.linalg.lsqr' when fitting on sparse training data. This parameter has no effect when fitting on dense data.

.. versionadded::
1.7

`n_jobs` `n_jobs`: int, None
default=None

The number of jobs to use for the computation. This will only provide speedup in case of sufficiently large problems, that is if firstly `'n_targets > 1'` and secondly `'X'` is sparse or if `'positive'` is set to `'True'`. `"None"` means 1 unless in a `:obj:'joblib.parallel_backend'` context. `"-1"` means using all processors. See `:term:'Glossary'` for more details.

positive positive: False
bool, default=False

When set to “True”, forces the coefficients to be positive. This option is only supported for dense arrays.

For a comparison between a linear regression model with positive constraints on the regression coefficients and a linear regression without such constraints, see [:ref:‘sphinx_glr_auto_examples_linear_model_plot_nnls.py’](#).

.. versionadded::
0.24

```
gs.best_params_
```

```
{'fit_intercept': True}
```

```
gs.best_score_
```

```
np.float64(0.3934573792950554)
```

`gs.cv_results_` stores the results for all folds. Itself is a `dict`. Usually we convert it into a `DataFrame`.

```
import pandas as pd

df_result = pd.DataFrame(gs.cv_results_)
df_result
```

	mean_fit_time	std_fit_time	mean_score_time	std_score_time	param_fit_intercept	params
0	0.001264	0.000358	0.000937	0.000373	True	{'fit_intercep
1	0.001417	0.000631	0.000819	0.000146	False	{'fit_intercep

Then we could get access to the mean test score or other metrics to see how scores are changed along the parameters.

```
df_result["mean_test_score"]
```

```
0    0.393457
1    0.366843
Name: mean_test_score, dtype: float64
```

i n_jobs

You may assign `n_jobs` for parallel computing. You may leave it `None` right now. We will come back to it later when we really need it.

3.4.3 Modern Hyperparameter Tuning

Historically, one of the most common approaches to hyperparameter tuning was exhaustive grid search, which is introduced above. This approach is conceptually simple and works well when the number of hyperparameters is small. As a result, `GridSearchCV` became one of the standard tools in early machine learning workflows.

However, as models became more complicated, researchers observed that exhaustive grid search often wastes substantial computation. In many problems, only a few hyperparameters

strongly affect performance, while others have relatively small influence.

A particularly influential paper [1] by Bergstra and Bengio showed that random search can often outperform grid search in high-dimensional hyperparameter spaces. The main idea is that random search explores more distinct values along important hyperparameter directions instead of spending excessive computation on unimportant dimensions.

This led to the widespread use of methods such as:

- `RandomizedSearchCV`
- Bayesian optimization
- TPE (Tree-structured Parzen Estimator)
- Optuna
- AutoML frameworks

Modern frameworks such as `Optuna` use adaptive search strategies. Instead of testing hyperparameters independently, they use information from previous trials to guide future searches toward more promising regions of the hyperparameter space. These methods have become increasingly important for large and computationally expensive models.

Nevertheless, in this course we will mainly use ordinary grid search because it is simple, transparent, and easy to understand. For most models in this course, naive grid search is already sufficient.

Only for more computationally expensive models such as XGBoost will we occasionally use more practical workflows such as:

- random search followed by refined grid search
- stagewise tuning
- early stopping

3.5 Feature Engineering and Preprocessing

Feature engineering refers to modifying or creating predictors in order to better represent the relationship between the predictors

and the response. Here we mainly cover the **sklearn** way to perform feature engineering.

i Modeling Perspective

In statistical learning (commonly implemented using **sklearn**), the primary goal is predictive performance. Models are therefore typically constructed by building a feature matrix, and modifying a model usually means modifying the feature matrix rather than changing a model formula.

In contrast, traditional regression analysis (commonly implemented using **statsmodels**) allows explicit specification of statistical models through a formula language. This approach is convenient when precise control of model terms and statistical inference are required.

3.5.1 Creating Features Manually

Before introducing automated tools, we can always create new predictors directly by modifying the original dataset.

```
import pandas as pd

X = pd.DataFrame({'x1': [0, 2, 4], 'x2': [1, 3, 5]})
X
```

	x1	x2
0	0	1
1	2	3
2	4	5

For example, we can add the squared term of x_2 and the interaction term x_1x_2 to form a new feature matrix X2.

```
X2 = X.copy()
X2["x2_sq"] = X["x2"] ** 2
X2["x1_x2"] = X["x1"] * X["x2"]
X2
```

	x1	x2	x2_sq	x1_x2
0	0	1	1	0
1	2	3	9	6
2	4	5	25	20

This corresponds to fitting a model of the form

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2 + \beta_4 x_1 x_2 + \varepsilon.$$

Tip

Although the model is called linear regression, it can represent nonlinear relationships by including transformed predictors such as x^2 , $\log x$, or $\sqrt{x}$. The model remains linear because it is linear in the coefficients β .

3.5.2 Polynomial Features

Click to expand.

When many predictors are present, manually creating polynomial and interaction terms can become tedious. In such cases we can use `PolynomialFeatures` from `sklearn.preprocessing`, which automatically generates polynomial and interaction terms up to the specified degree.

```
from sklearn.preprocessing import PolynomialFeatures

poly = PolynomialFeatures(degree=2, interaction_only=False, include_bias=True)
poly.fit_transform(X)
```

```
array([[ 1.,  0.,  1.,  0.,  0.,  1.],
       [ 1.,  2.,  3.,  4.,  6.,  9.],
       [ 1.,  4.,  5., 16., 20., 25.]])
```

The argument `interaction_only=False` means that both squared terms and interaction terms are included. For example,

if the original variables are x_1 and x_2 , then the transformed matrix contains terms such as

$$1, x_1, x_2, x_1^2, x_1x_2, x_2^2.$$

```
poly2 = PolynomialFeatures(degree=2, interaction_only=True, include_bias=False)
poly2.fit_transform(X)
```

```
array([[ 0.,  1.,  0.],
       [ 2.,  3.,  6.],
       [ 4.,  5., 20.]])
```

Here `interaction_only=True` means that only interaction terms are added, while powers such as x_1^2 and x_2^2 are excluded. In addition, the argument `include_bias=False` removes the constant column of ones.

3.5.3 Categorical Variables

Dummy Variable Trap

Including all dummy variables together with an intercept creates perfect multicollinearity because the dummy variables sum to one. Therefore, when creating dummy variables with k categories, we typically include only $k - 1$ variables.

pandas version

We could use `pd.get_dummies()` to directly modify the dataset.

```
df = pd.DataFrame({"color": ["red", "blue", "green", "red"]})
pd.get_dummies(df, drop_first=True)
```

	color_green	color_red
0	False	True

	color_green	color_red
1	False	False
2	True	False
3	False	True

The argument `drop_first=True` removes the base category column to avoid the dummy variable trap described above. If you want to keep the base column, you may set it to `False` (which is the default).

`pd.get_dummies()` automatically detects categorical variables and leaves numerical variables unchanged.

```
X = pd.DataFrame({"x": [1, 2, 3, 4, 5, 6], "group": ["A", "A", "B", "B", "C", "C"]})
pd.get_dummies(X, drop_first=True)
```

	x	group_B	group_C
0	1	False	False
1	2	False	False
2	3	True	False
3	4	True	False
4	5	False	True
5	6	False	True

If you want to convert numeric variables into dummy variables, you may manually cast them to categorical variables.

```
X["x"] = X["x"].astype("category")
pd.get_dummies(X, drop_first=True)
```

	x_2	x_3	x_4	x_5	x_6	group_B	group_C
0	False	False	False	False	False	False	False
1	True	False	False	False	False	False	False
2	False	True	False	False	False	True	False
3	False	False	True	False	False	True	False

	x_2	x_3	x_4	x_5	x_6	group_B	group_C
4	False	False	False	True	False	False	True
5	False	False	False	False	True	False	True

`sklearn` version

`OneHotEncoder` from `sklearn` is another tool for converting categorical variables into dummy variables. It follows the standard `sklearn` API. Compared with `pd.get_dummies()`, which is convenient for quick data analysis, `OneHotEncoder` is better suited for statistical learning pipelines because it can be fitted on training data and then applied consistently to new data.

```
from sklearn.preprocessing import OneHotEncoder

df = pd.DataFrame({"color": ["red", "blue", "green", "red"]})
encoder = OneHotEncoder(drop="first", sparse_output=False)

encoder.fit_transform(df)
```

```
array([[0., 1.],
       [0., 0.],
       [1., 0.],
       [0., 1.]])
```

The argument `sparse_output=False` controls the output format. If `False`, the output is a regular matrix. If `True`, the output is a sparse matrix.

```
from sklearn.preprocessing import OneHotEncoder

X = pd.DataFrame({"x": [1, 2, 3, 4, 5, 6], "group": ["A", "A", "B", "B", "C", "C"]})
encoder = OneHotEncoder(drop="first", sparse_output=False)

encoder.fit_transform(X)
```

```
array([[0., 0., 0., 0., 0., 0., 0.],
       [1., 0., 0., 0., 0., 0., 0.]])
```

```
[0., 1., 0., 0., 0., 1., 0.],  
[0., 0., 1., 0., 0., 1., 0.],  
[0., 0., 0., 1., 0., 0., 1.],  
[0., 0., 0., 0., 1., 0., 1.]])
```

When applied to a `DataFrame` with multiple columns, unlike the `pandas` version, `OneHotEncoder` converts each column into dummy variables. In the example above, the first five columns correspond to `x` and the last two correspond to `group`.

If you want to apply `OneHotEncoder` only to certain columns, you need to use a `ColumnTransformer`, which will be introduced later.

3.5.4 Scaling Variables

Centralization, standardization and normalization

In regression and statistical learning, these transformations are typically applied to the predictors X to control their location and scale. They are rarely applied to the response y , unless there is a specific modeling reason.

- Centralization: Subtract the sample mean so that the variable has mean 0.
- Standardization: Subtract the sample mean and divide by the sample standard deviation so that the variable has mean 0 and standard deviation 1.
- Normalization: Rescale the variable so that its values lie between 0 and 1.

Standardization is especially useful when predictors are measured on very different scales. It is commonly used in statistical learning, particularly for methods such as ridge regression and lasso.

```
import numpy as np  
import pandas as pd  
  
rng = np.random.default_rng(1)
```

```
X = pd.DataFrame({"x1": rng.normal(10, 2, size=8), "x2": rng.normal(100, 20, size=8)})
X
```

	x1	x2
0	10.691168	107.291448
1	11.643236	105.882650
2	10.660874	100.568445
3	7.393686	110.934260
4	11.810712	85.270918
5	10.892749	96.741801
6	8.926094	90.357614
7	11.162236	111.976924

Standardization:

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
scaler.fit_transform(X)
```

```
array([[ 0.20941102,  0.67782481],
       [ 0.88853621,  0.52289215],
       [ 0.18780166, -0.06153794],
       [-2.14273599,  1.07844331],
       [ 1.00799911, -1.74388454],
       [ 0.35320178, -0.48237337],
       [-1.04964497, -1.18447483],
       [ 0.54543119,  1.19311041]])
```

Centralization (standardization without scaling):

```
scaler = StandardScaler(with_std=False)
scaler.fit_transform(X)
```

```
array([[ 0.29357401,  6.16344046],
       [ 1.24564191,  4.75464247],
       [ 0.26327978, -0.55956264],
```

```
[ -3.00390884,   9.80625227],
[  1.41311736, -15.8570892 ],
[  0.49515477,  -4.38620642],
[ -1.47150084, -10.77039372],
[  0.76464184,  10.84891679]])
```

Normalization:

```
from sklearn.preprocessing import MinMaxScaler

minmax = MinMaxScaler()
minmax.fit_transform(X)
```

```
array([[0.74653912, 0.82455346],
       [0.96208412, 0.77180136],
       [0.73968061, 0.57281222],
       [0.          , 0.96095768],
       [1.          , 0.          ],
       [0.79217633, 0.42952446],
       [0.34693206, 0.19047009],
       [0.8531873  , 1.          ]])
```

Note that scaling is not required for ordinary least squares (OLS). However, it is essential for regularization methods such as ridge and lasso because the penalty depends on the scale of the coefficients.

3.5.5 ColumnTransformer

Click to expand.

To apply different transformations to different columns, we can use `ColumnTransformer`. `ColumnTransformer` consists of multiple transformations. Each transformation is specified by a triple containing:

- a name (identifier),
- the transformer,
- the columns to which it applies.

```

from sklearn.compose import ColumnTransformer

prep = ColumnTransformer(
    transformers=[
        ("cat", OneHotEncoder(drop="first", sparse_output=False), ["group"]),
        ("x", "passthrough", ["x"]),
        ("x_std", StandardScaler(), ["x"]),
    ]
)

X = pd.DataFrame({"x": [1, 2, 3, 4, 5, 6], "group": ["A", "A", "B", "B", "C", "C"]})
prep.fit_transform(X)

```

```

array([[ 0.          ,  0.          ,  1.          , -
1.46385011],
       [ 0.          ,  0.          ,  2.          , -
0.87831007],
       [ 1.          ,  0.          ,  3.          , -
0.29277002],
       [ 1.          ,  0.          ,  4.          ,  0.29277002],
       [ 0.          ,  1.          ,  5.          ,  0.87831007],
       [ 0.          ,  1.          ,  6.          ,  1.46385011]])

```

You may use the id to get into each transformer.

```
prep.named_transformers_
```

```

{'cat': OneHotEncoder(drop='first', sparse_output=False),
 'x': FunctionTransformer(accept_sparse=True, check_inverse=False,
                          feature_names_out='one-to-one'),
 'x_std': StandardScaler()}

```

```
prep.named_transformers_['x_std']
```

`copy`: bool,
`default=True`

True

If False, try to
avoid a copy and
do inplace scaling
instead.

This is not
guaranteed to
always work
inplace; e.g. if the
data is
not a NumPy array
or `scipy.sparse`
CSR matrix, a
copy may still be
returned.

`with_mean`
`with_mean`: bool,
`default=True`

True

If True, center the
data before scaling.
This does not work
(and will raise an
exception) when
attempted on
sparse matrices,
because centering
them entails
building a dense
matrix which in
common use cases
is likely to be too
large to fit in
memory.

`with_std with_std: True`
`bool, default=True`

If True, scale the data to unit variance (or equivalently, unit standard deviation).

💡 Estimator, Predictor and Transformer

In `sklearn`, all models are called estimators. An estimator is any object that can be trained using the `.fit()` method. Once an estimator is fitted,

- if it can transform input data using `.transform()`, it is called a transformer;
- if it can generate predictions using `.predict()`, it is called a predictor.

Typical examples include

- Transformers: `StandardScaler`, `MinMaxScaler`, `PolynomialFeatures`, `OneHotEncoder`, etc.
- Predictors: `LinearRegression` and most other machine learning models.

Usually a model is applied in two steps:

1. `.fit()` learns the parameters from the training data.
2. `.transform()` or `.predict()` applies the model to data.

For transformers, the combined method `.fit_transform()` performs both steps in one call.

3.6 Pipelines and transformations of y

In many modeling workflows, several steps must be performed before fitting the model. These steps may include

- feature transformations,
- scaling variables,
- generating polynomial features.

A **pipeline** combines these steps into a single object. Using pipelines ensures that the same transformations are applied consistently during training, cross-validation, and prediction, and helps prevent data leakage.

Here is a typical **pipeline**. Each step is indicated by a pair consisting of

- a name (identifier) of the step
- the estimator

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression

steps = [
    ("scaler", StandardScaler()),
    ("linear_model", LinearRegression()),
]
pipe = Pipeline(steps=steps)
```

This **pipe** object behaves like a **model** object. You may use the step names to access the internal components.

```
pipe.named_steps
```

```
{'scaler': StandardScaler(), 'linear_model': LinearRegression()}
```

```
pipe.named_steps['scaler']
```

`copy`: bool,
`default=True`

True

If False, try to
avoid a copy and
do inplace scaling
instead.

This is not
guaranteed to
always work
inplace; e.g. if the
data is
not a NumPy array
or `scipy.sparse`
CSR matrix, a
copy may still be
returned.

`with_mean`
`with_mean`: bool,
`default=True`

True

If True, center the
data before scaling.
This does not work
(and will raise an
exception) when
attempted on
sparse matrices,
because centering
them entails
building a dense
matrix which in
common use cases
is likely to be too
large to fit in
memory.

`with_std with_std: True`
`bool, default=True`

If True, scale the
data to unit
variance (or
equivalently,
unit standard
deviation).

In a pipeline, you normally need to manually name all components. To make this easier, you may use `make_pipeline()`, which automatically generates default names for each step.

In addition, `make_pipeline()` does not require a list of (name, step) pairs. Instead, you simply provide the steps one by one.

```
from sklearn.pipeline import make_pipeline

pipe_easy = make_pipeline(
    StandardScaler(),
    StandardScaler(),
    LinearRegression(),
)
```

It will automatically name each step by its class name.

```
pipe_easy.named_steps
```

```
{'standardscaler-1': StandardScaler(),
 'standardscaler-2': StandardScaler(),
 'linearregression': LinearRegression()}
```

Note that in most cases we won't stack two `StandardScaler()` together. Here is just an example to show how the steps are automatically named.

i Note

Pipelines are particularly useful when combined with cross-validation and hyperparameter tuning, since they ensure that preprocessing steps are applied within each training fold.

3.6.1 TransformedTargetRegressor

`TransformedTargetRegressor` from `sklearn.compose` is a wrapper that applies a transformation to the response variable y during model fitting and automatically applies the inverse transformation when making predictions. In other words, the regressor is trained on the transformed response, while predictions are converted back to the original scale.

Here is a simple example of a `TransformedTargetRegressor`. The transformation of y is `centralizer`, which is essentially `standardizer` without rescaling.

```
from sklearn.compose import TransformedTargetRegressor

model = TransformedTargetRegressor(
    regressor=LinearRegression(),
    transformer=StandardScaler(with_std=False)
)
```

`TransformedTargetRegressor` must know how to reverse the transformation applied to the response variable. If the transformer has an `.inverse_transform()` method (e.g., `StandardScaler()`), the inverse transformation is handled automatically. Otherwise, you must specify both the transformation and its inverse using the `func=` and `inverse_func=` arguments.

```
model = TransformedTargetRegressor(
    regressor=LinearRegression(),
    func=np.log,
    inverse_func=np.exp
)
```

3.6.2 Full Example

Click to expand.

In this example we build a more complicated pipeline.

```
import numpy as np
import pandas as pd

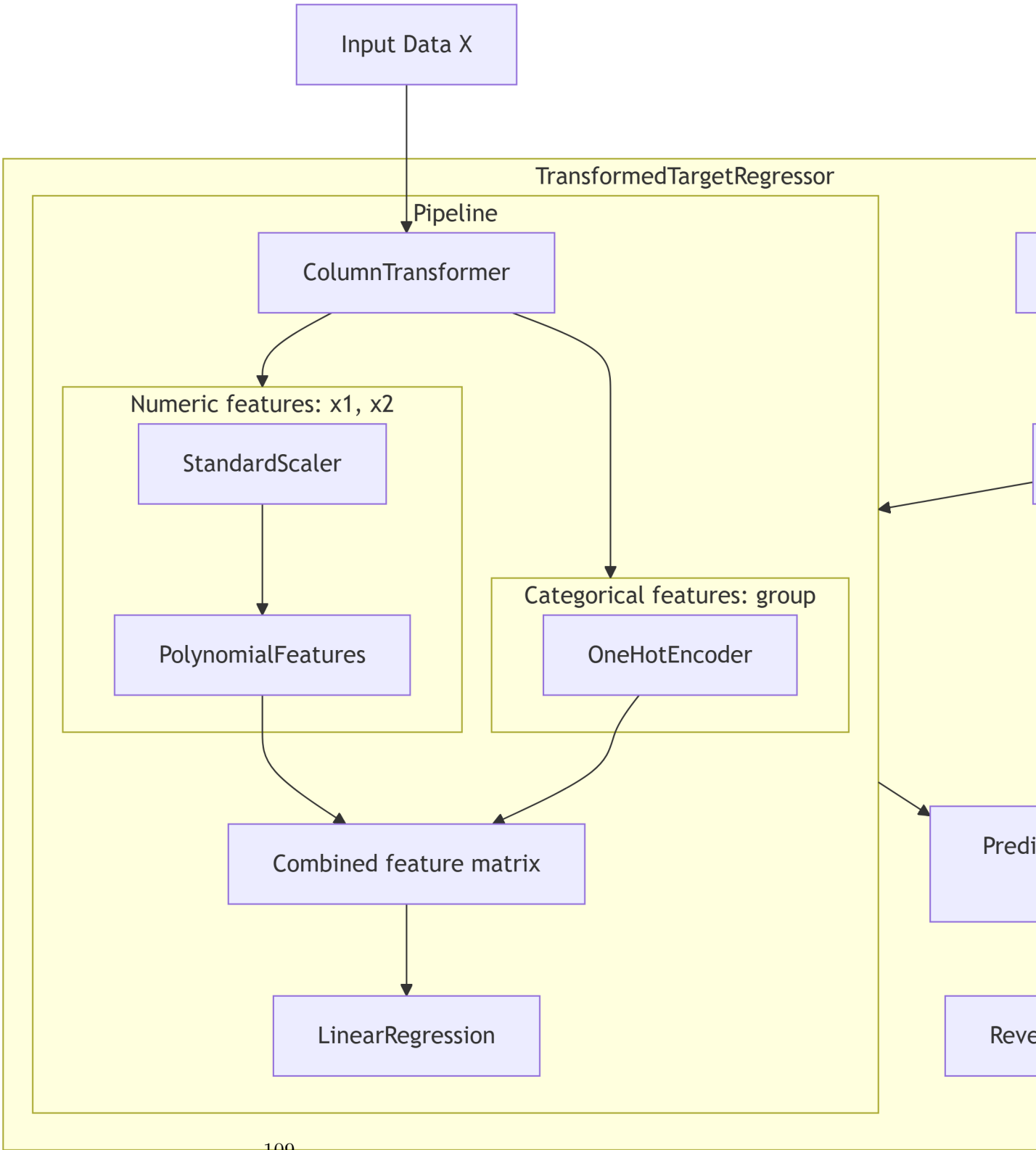
n = 100

rng = np.random.default_rng(1)
X = pd.DataFrame(
    {
        "x1": rng.normal(0, 1, n),
        "x2": rng.normal(5, 2, n),
        "group": rng.choice(["A", "B", "C"], size=n),
    }
)

y = 1 + 2 * X["x1"] - 1.5 * X["x2"] + 3 * (X["group"] == "B") - 2 * (X["group"] == "C") + rng.randn(n)
```

The pipeline is illustrated as follows.

1. `ColumnTransformer` separates numeric and categorical features.
2. Numeric features go through: `StandardScaler` → `PolynomialFeatures`.
3. Categorical features go through `OneHotEncoder`.
4. The transformed features are combined into a single feature matrix.
5. The model `LinearRegression` is fitted to the transformed data.
6. The response variable y is centralized.
7. The fitted model can then be used to generate predictions, while the transformation of y is automatically reversed.



```

from sklearn.compose import ColumnTransformer, TransformedTargetRegressor
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder, PolynomialFeatures
from sklearn.linear_model import LinearRegression

numeric_features = ["x1", "x2"]
categorical_features = ["group"]

numeric_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("poly", PolynomialFeatures(degree=2, include_bias=False)),
    ]
)

preprocessor = ColumnTransformer(
    [
        ("num", numeric_pipe, numeric_features),
        ("cat", OneHotEncoder(drop="first"), categorical_features),
    ]
)

pipe = Pipeline(
    [
        ("preprocess", preprocessor),
        ("model", LinearRegression()),
    ]
)

model = TransformedTargetRegressor(
    regressor=pipe,
    transformer=StandardScaler(with_std=False),
)

```

The entire pipeline is trained on the training set and then evaluated on the validation set, where the training and validation sets are determined by cross-validation.

```

from sklearn.model_selection import cross_validate

```

```
cv_result = cross_validate(  
    model,  
    X,  
    y,  
    cv=KFold(5, shuffle=True, random_state=1),  
    scoring=["r2", "neg_mean_squared_error"],  
)  
  
print(f"test R2: {cv_result['test_r2'].mean()}")  
print(f"test neg mse: {cv_result['test_neg_mean_squared_error'].mean()}")
```

```
test R2: 0.9158850679164349  
test neg mse: -1.0901713224790424
```

4 Regularization

4.1 Motivation

The OLS estimator is attractive because, under the Gauss–Markov assumptions, it is BLUE: the best linear unbiased estimator. However, when some predictors are nearly collinear, the matrix $X^\top X$ becomes nearly singular. In that case, the variance of $\hat{\beta}_{\text{OLS}}$ can be very large, so the estimated coefficients may become unstable and unreliable.

This leads to the bias-variance trade-off. By allowing a small amount of bias, we may substantially reduce the variance and improve predictive performance. A common way to do this is through regularization.

Two of the most important regularization methods are ridge regression and lasso. Both methods tend to shrink the coefficients toward zero, so they are also called shrinkage methods.

4.2 Ridge Regression

4.2.1 OLS estimator review

Consider a linear model

$$y = X\beta + \varepsilon.$$

Given the dataset X and y , the OLS estimator $\hat{\beta}_{\text{OLS}}$ finds the coefficient vector β that minimizes the residual sum of squares

$$\text{RSS}(\beta) = \|y - X\beta\|_2^2.$$

In other words,

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta} \{ \|y - X\beta\|_2^2 \}.$$

When $X^\top X$ is invertible, the solution is

$$\hat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top y.$$

4.2.2 Ridge estimator

Ridge regression modifies the OLS loss function by adding a quadratic penalty.

Definition 4.1 (Ridge Regression). For a response vector $y \in \mathbb{R}^n$ and a design matrix $X \in \mathbb{R}^{n \times p}$, the Ridge estimator $\hat{\beta}_{\text{ridge}}$ is defined as:

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \}$$

where

- $\lambda \geq 0$ is the regularization parameter that controls the trade-off between the fit to the data and the penalty on the coefficient size.
- $\|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2$ is the L_2 norm of the coefficient vector

The penalty shrinks the coefficients toward zero. Larger values of λ produce stronger shrinkage. However, ridge regression does not perform variable selection, since it rarely sets any coefficient exactly to zero.

Similar to the OLS estimator, the Ridge estimator can be solved analytically.

Theorem 4.1 (The Closed-Form of Ridge estimator). *The closed-form solution of $\hat{\beta}_{\text{ridge}}$ is*

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

Click for proof.

Consider the ridge regression objective

$$Q(\beta) = (y - X\beta)^\top (y - X\beta) + \lambda \beta^\top \beta, \quad \text{where } \lambda \geq 0.$$

We derive the ridge estimator by minimizing $Q(\beta)$ with respect to β .

First expand the quadratic form:

$$\begin{aligned} Q(\beta) &= (y^\top - \beta^\top X^\top)(y - X\beta) + \lambda \beta^\top \beta \\ &= y^\top y - y^\top X\beta - \beta^\top X^\top y + \beta^\top X^\top X\beta + \lambda \beta^\top \beta. \end{aligned}$$

Since $y^\top X\beta$ is a scalar, it equals its transpose:

$$y^\top X\beta = \beta^\top X^\top y.$$

Thus

$$Q(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta + \lambda \beta^\top \beta.$$

Combine the last two quadratic terms:

$$Q(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top (X^\top X + \lambda I)\beta.$$

Now differentiate with respect to β :

$$\nabla_\beta Q(\beta) = -2X^\top y + 2(X^\top X + \lambda I)\beta.$$

Set the gradient equal to zero:

$$-2X^\top y + 2(X^\top X + \lambda I)\beta = 0.$$

So

$$(X^\top X + \lambda I)\beta = X^\top y.$$

Assuming $X^\top X + \lambda I$ is invertible, we obtain

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

Therefore the ridge estimator is

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

In this formulation, the term λI (where I is the identity matrix) is added to the cross-product matrix $X^\top X$ before inversion. When $\lambda > 0$, this ensures that the matrix is non-singular and improves its conditioning, even in the presence of multicollinearity or when $p > n$.

4.2.3 About intercept

Linear regression models are commonly written in the form

$$y = X\beta + \varepsilon.$$

This notation can represent two different model structures depending on whether an intercept is included.

- Model with intercept: $\beta = [\beta_0, \beta_1, \dots, \beta_p]^\top$ where β_0 is the intercept. The design matrix X contains a leading column of ones representing the constant term.
- Model without intercept: $\beta = [\beta_1, \dots, \beta_p]^\top$ and the design matrix contains only the predictor variables. In this case the fitted regression surface is forced to pass through the origin.

In most practical applications, models with an intercept are preferred because they allow the regression function to shift vertically rather than being constrained to go through the origin.

i Centering and the No-Intercept Form

A model with an intercept can be rewritten as a no-intercept model by centering both the response and the predictors:

$$y^c = y - \bar{y}, \quad X_j^c = X_j - \bar{X}_j.$$

After centering, all variables have mean zero. In ordinary least squares, the intercept estimate is

$$\hat{\beta}_0 = \bar{y} - \sum_{j=1}^p \hat{\beta}_j \bar{X}_j.$$

When the data are centered, this quantity becomes zero automatically. As a result, fitting a regression without an intercept on centered data produces the same fitted model as fitting a regression with an intercept on the original data.

For OLS, the intercept formulation and the centered no-intercept formulation are mathematically equivalent. However, the situation is slightly different in ridge regression. The ridge estimator introduces the penalty

$$\lambda \sum_{j=1}^p \beta_j^2,$$

which is intended to shrink the slope coefficients. In practice the intercept is not penalized, because penalizing it would shrink the overall level of the model toward zero and make the result depend on the arbitrary origin of the response variable.

For this reason ridge regression is usually formulated in the no-intercept form after centering the data, so that the penalty naturally applies only to the slope coefficients.

i Standardization in Practice

Because the ridge penalty depends on the sizes of the coefficients, ridge regression is usually implemented after standardizing the predictors. This puts the predictors on a common scale, so that the penalty treats them fairly. In practice, the predictors are typically standardized, whereas the response is usually only centered rather than standardized.

For details see Section 3.5.4.

i Implementation in `scikit-learn`

In `scikit-learn`, ridge and lasso regression automatically center the data internally when fitting the model (while leaving the final predictions on the original scale). The intercept is therefore handled separately and excluded from the penalty.

Because of this internal preprocessing, users typically do not need to manually center the response when using `Ridge` or `Lasso`. However, it is still common to standardize the predictors explicitly using tools such as `StandardScaler`, especially when building modeling pipelines.

4.2.4 Alternative Formulation

By the theory of Lagrange multipliers, ridge regression can also be written in a constrained optimization form:

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \|y - X\beta\|_2^2 \quad \text{subject to} \quad \|\beta\|_2^2 \leq s.$$

Here $s \geq 0$ controls the size of the coefficient vector and is directly related to the penalty parameter λ . The two parameters have an inverse relationship:

- Large s corresponds to small λ . The constraint is weak, and β approaches the OLS estimator.
- Small s corresponds to large λ . The constraint is strong, and β shrinks toward 0.

This formulation highlights that ridge regression limits the magnitude of the coefficient vector rather than directly penalizing prediction error. We will later use this interpretation to explain the difference between ridge regression and lasso.

i The Constrained Optimization Problem

Ridge regression can be formulated as a constrained optimization problem, where we minimize RSS subject to a

constraint s on the squared L_2 norm of the coefficients:

$$\min_{\beta} \|y - X\beta\|_2^2 \quad \text{subject to} \quad \|\beta\|_2^2 \leq s.$$

To solve this, we define the Lagrangian function $\mathcal{L}(\beta, \lambda)$:

$$\mathcal{L}(\beta, \lambda) = \|y - X\beta\|_2^2 + \lambda(\|\beta\|_2^2 - s)$$

where $\lambda \geq 0$ is the Lagrange multiplier. Taking the partial derivative with respect to β and setting it to zero gives:

$$\frac{\partial \mathcal{L}}{\partial \beta} = -2X^\top(y - X\beta) + 2\lambda\beta = 0$$

Rearranging the terms yields the standard ridge estimator:

$$(X^\top X + \lambda I)\beta = X^\top y \implies \hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

This shows that for every $\lambda > 0$, there exists an equivalent constrained problem with some constraint s .

i Relation Between λ and s

The connection between the penalty parameter λ and the constraint s can be understood through the constrained formulation

$$\min_{\beta} \|y - X\beta\|_2^2 \quad \text{subject to} \quad \|\beta\|_2^2 \leq s.$$

There are two possible scenarios:

- **Non-binding constraint:** If the OLS solution already satisfies $\|\hat{\beta}_{\text{ols}}\|_2^2 \leq s$, then the OLS solution remains optimal for the constrained problem. Therefore, $\hat{\beta}_{\text{ridge}} = \hat{\beta}_{\text{ols}}$. The constraint has no effect, so the corresponding penalized formulation uses $\lambda = 0$.
- **Binding constraint:** If $\|\hat{\beta}_{\text{ols}}\|_2^2 > s$, then the OLS solution lies outside the constraint region. The constrained optimum must therefore lie on the boundary $\|\hat{\beta}_{\text{ridge}}\|_2^2 = s$. Consequently, the corresponding penalized formulation requires $\lambda > 0$.

Therefore, when $\lambda > 0$, the parameters λ and s are related through $\|(X^\top X + \lambda I)^{-1} X^\top y\|_2^2 = s$.

The parameter λ can be viewed as the “price” paid for enforcing the constraint. As s decreases, the corresponding value of λ increases, producing greater shrinkage toward the origin. Since the norm of the ridge estimator decreases as λ increases, smaller values of s correspond to larger values of λ .

Consequently, for every $s < \|\hat{\beta}_{\text{ols}}\|_2^2$, there exists a unique $\lambda > 0$ such that $\|\hat{\beta}_{\text{ridge}}\|_2^2 = s$.

This relationship is consistent with the Karush–Kuhn–Tucker (KKT) complementary slackness condition in optimization theory,

$$\lambda (\|\beta\|_2^2 - s) = 0.$$

4.2.5 SVD and Tikhonov Regularization

Recall the OLS estimator

$$\hat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top y.$$

Let the singular value decomposition (SVD) of X be

$$X = UDV^\top,$$

where

- U is an $n \times n$ orthogonal matrix,
- V is a $p \times p$ orthogonal matrix,
- D is an $n \times p$ diagonal matrix whose nonzero entries $d_1, \dots, d_r$ are the singular values of X .

i Thin SVD

Let $X = UDV^\top$ be the (full) singular value decomposition of X , and let $r = \text{rank}(X)$. If $r < \min(n, p)$, then $D^\top D$

contains zero eigenvalues and is therefore singular.

In statistical learning, we are usually interested in expressions involving D^{-1} or $(D^\top D)^{-1}$ rather than the complete orthogonal basis structure of U and V . Therefore, it is often more convenient to work with the **thin (reduced) SVD** $X = U_r D_r V_r^\top$, where U_r and V_r are obtained by removing the columns corresponding to zero singular values, and D_r is obtained by deleting the associated zero rows and columns of D . Consequently, $D_r = \text{diag}(\sigma_1, \dots, \sigma_r)$ is square and invertible.

By a slight abuse of notation, statistical learning texts often write $X = UDV^\top$ while actually referring to the thin SVD. We follow this convention throughout this section. Unless otherwise stated, all SVDs appearing below should be interpreted as thin SVDs.

Using the SVD,

$$X^\top X = V D^\top D V^\top.$$

Since $D^\top D = \text{diag}(d_1^2, \dots, d_r^2)$, the eigenvalues of $X^\top X$ are d_i^2 . Hence the OLS estimator can be written as

$$\hat{\beta}_{\text{OLS}} = V(D^\top D)^{-1} D^\top U^\top y.$$

Since $D^\top D = \text{diag}(d_1^2, \dots, d_r^2)$, we have

$$(D^\top D)^{-1} = \text{diag}\left(\frac{1}{d_1^2}, \dots, \frac{1}{d_r^2}\right).$$

Let u_i and v_i denote the columns of U and V . Then the estimator can be written as

$$\hat{\beta}_{\text{OLS}} = \sum_{i=1}^r \frac{1}{d_i} v_i u_i^\top y.$$

If some singular values d_i are very small, the factors $1/d_i$ become very large, which leads to unstable coefficient estimates. Therefore this expression reveals the instability of OLS.

Now turn to Ridge regression. We replace the OLS estimator with

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

Substituting the SVD gives

$$\hat{\beta}_{\text{ridge}} = V(D^\top D + \lambda I)^{-1} D^\top U^\top y.$$

Because $D^\top D$ is diagonal, the estimator can be written componentwise as

$$\hat{\beta}_{\text{ridge}} = \sum_{i=1}^r \frac{d_i}{d_i^2 + \lambda} v_i u_i^\top y.$$

This representation shows that ridge regression applies a shrinkage factor

$$\frac{d_i^2}{d_i^2 + \lambda}$$

to each principal component direction.

- When d_i is large, the factor is close to 1, so the component is almost unchanged.
- When d_i is small, the factor becomes small, shrinking the unstable directions.

Thus ridge regression stabilizes the estimator by shrinking the contributions of directions associated with small singular values.



Tikhonov regularization

Tikhonov regularization is a method used to solve ill-posed problems and handle multicollinearity by adding a penalty term, $\alpha \|L\beta\|_2^2$. When the Tikhonov matrix L is the identity matrix I , it simplifies to ridge regression. Therefore, ridge regression is a special case of Tikhonov regularization [2, 3].

4.2.6 Bias-Variance Tradeoff

One of the main motivations for regularization methods is the bias-variance tradeoff. A more flexible model can fit the training data very well, but it may also be highly sensitive to random noise in the sample. In contrast, a more constrained model is usually more stable, but may systematically miss part of the true signal.

This tradeoff can be described mathematically through a decomposition of the expected prediction error.

Suppose the true model is

$$Y = f(x) + \varepsilon, \quad \mathbb{E}(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2.$$

Let $\hat{f}(x)$ be an estimator of $f(x)$ constructed from a training sample. Since the training sample is random, $\hat{f}(x)$ is also random. We study the expected prediction error at a fixed point x :

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 \right].$$

Here, the expectation is taken over both the randomness in the training data and the randomness in the new response Y .

Definition 4.2 (Bias and Variance).

- $\text{Bias}(\hat{f}(x)) = \mathbb{E}[\hat{f}(x)] - f(x)$.
- $\text{Var}(\hat{f}(x)) = \mathbb{E} \left[\left(\hat{f}(x) - \mathbb{E}[\hat{f}(x)] \right)^2 \right]$.

Theorem 4.2 (Bias-Variance Decomposition).

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 \right] = \text{Bias}(\hat{f}(x))^2 + \text{Var}(\hat{f}(x)) + \sigma^2.$$

Click to expand.

Since $Y = f(x) + \varepsilon$,

$$\begin{aligned} \mathbb{E} \left[(Y - \hat{f}(x))^2 \right] &= \mathbb{E} \left[(f(x) + \varepsilon - \hat{f}(x))^2 \right] \\ &= \mathbb{E} \left[(f(x) - \hat{f}(x))^2 + 2\varepsilon(f(x) - \hat{f}(x)) + \varepsilon^2 \right] \\ &= \mathbb{E} \left[(f(x) - \hat{f}(x))^2 \right] + 2\mathbb{E} \left[\varepsilon(f(x) - \hat{f}(x)) \right] + \mathbb{E}(\varepsilon^2). \end{aligned}$$

Since the new noise ε is independent of the training data and has mean 0,

$$\mathbb{E} [\varepsilon(f(x) - \hat{f}(x))] = 0.$$

Also,

$$\mathbb{E}(\varepsilon^2) = \text{Var}(\varepsilon) = \sigma^2.$$

Therefore,

$$\mathbb{E} [(Y - \hat{f}(x))^2] = \mathbb{E} [(f(x) - \hat{f}(x))^2] + \sigma^2.$$

Thus the prediction error is the sum of:

- the estimation error $\mathbb{E}[(f(x) - \hat{f}(x))^2]$,
- the irreducible error σ^2 .

Now we further decompose

$$\mathbb{E} [(f(x) - \hat{f}(x))^2].$$

Let

$$m(x) = \mathbb{E}[\hat{f}(x)],$$

where the expectation is taken over the randomness in the training sample. Then

$$f(x) - \hat{f}(x) = (f(x) - m(x)) + (m(x) - \hat{f}(x)).$$

Squaring,

$$(f(x) - \hat{f}(x))^2 = (f(x) - m(x))^2 + (m(x) - \hat{f}(x))^2 + 2(f(x) - m(x))(m(x) - \hat{f}(x)).$$

Taking expectation,

$$\mathbb{E} [(f(x) - \hat{f}(x))^2] = (f(x) - m(x))^2 + \mathbb{E} [(m(x) - \hat{f}(x))^2] + 2(f(x) - m(x)) \mathbb{E}[m(x) - \hat{f}(x)].$$

But

$$\mathbb{E}[m(x) - \hat{f}(x)] = m(x) - \mathbb{E}[\hat{f}(x)] = 0.$$

So the cross term vanishes, and we obtain

$$\mathbb{E} [(f(x) - \hat{f}(x))^2] = (f(x) - \mathbb{E}[\hat{f}(x)])^2 + \mathbb{E} [(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2].$$

The first term is the squared bias:

$$\text{Bias}(\hat{f}(x))^2 = \left(\mathbb{E}[\hat{f}(x)] - f(x) \right)^2.$$

The second term is the variance:

$$\text{Var}(\hat{f}(x)) = \mathbb{E} \left[\left(\hat{f}(x) - \mathbb{E}[\hat{f}(x)] \right)^2 \right].$$

Therefore,

$$\mathbb{E} \left[(f(x) - \hat{f}(x))^2 \right] = \text{Bias}(\hat{f}(x))^2 + \text{Var}(\hat{f}(x)).$$

Combining this with the earlier decomposition gives

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 \right] = \text{Bias}(\hat{f}(x))^2 + \text{Var}(\hat{f}(x)) + \sigma^2.$$

This formula shows that the expected prediction error has three parts:

- Squared bias: error caused by systematic deviation of the estimator from the truth.
- Variance: error caused by sensitivity of the estimator to the particular training sample.
- Irreducible error: the noise level σ^2 , which cannot be removed by any method.

A very flexible model often has small bias but large variance. A more constrained model often has larger bias but smaller variance. The goal of regularization is not necessarily to minimize bias or variance alone, but to balance them so that the total prediction error is as small as possible.

Note

Ordinary least squares is often unbiased, but its variance can be very large when predictors are highly correlated or when the model is too flexible. Ridge and lasso (which will be discussed later) introduce bias by shrinking the coefficients toward 0, but this often substantially reduces variance.

As a result, although ridge or lasso may fit the training data

less perfectly than OLS, they can achieve better prediction performance on new data. This is the central idea behind the bias-variance tradeoff and one of the main reasons regularization methods are useful in statistical learning.

4.3 Ridge Regression in Python

We use the following simulated dataset to demonstrate ridge regression.

The covariance matrix has a very high correlation, so the two predictors x_1 and x_2 are highly correlated.

```
import numpy as np

rng = np.random.default_rng(2)
n = 30

Sigma = np.array([[1, 0.99], [0.99, 1]])
mean = np.array([0, 0])

X = rng.multivariate_normal(mean, Sigma, size=n)
y = rng.normal(loc=X[:, 0] + X[:, 1], scale=1)
```

The data set is generated from a model with $E(y \mid x_1, x_2) = x_1 + x_2$. Therefore the true coefficients are $\beta_1 = \beta_2 = 1$.

4.3.1 Ridge estimator

We use `Ridge` from `sklearn.linear_model` to fit a ridge regression model. Its API is very similar to that of OLS. The argument `alpha` is the regularization parameter λ .

For simplicity, we use a no-intercept model, so we set `fit_intercept=False`.

```
from sklearn.linear_model import Ridge

ridge = Ridge(alpha=5, fit_intercept=False).fit(X, y)
print(f"Ridge beta = {ridge.coef_}")
```

```
Ridge beta = [0.85769874 0.84964704]
```

For comparison, we also fit the OLS estimator.

```
from sklearn.linear_model import LinearRegression

ols = LinearRegression(fit_intercept=False).fit(X, y)
print(f"OLS beta = {ols.coef_}")
```

```
OLS beta = [0.99640827 0.85969191]
```

In this example, you may notice two things:

- The ridge coefficients are smaller in magnitude than the OLS coefficients.
- The OLS coefficients may happen to be closer to the true values in this single sample.

The second point is due to randomness in the simulated data. To see the general pattern, we repeat the simulation many times.

```
import pandas as pd

res = []
for _ in range(100):
    X = rng.multivariate_normal(mean, Sigma, size=n)
    y = rng.normal(loc=X[:, 0] + X[:, 1], scale=1)
    ols = LinearRegression(fit_intercept=False).fit(X, y)
    ridge = Ridge(alpha=5, fit_intercept=False).fit(X, y)
    res.append(
        {
            "ols_b1": ols.coef_[0],
            "ols_b2": ols.coef_[1],
            "ridge_b1": ridge.coef_[0],
```

```

        "ridge_b2": ridge.coef_[1],
    }
)
res = pd.DataFrame(res)

```

The first five results are shown below. We can see that the OLS estimates vary substantially from sample to sample.

```
res.head(5)
```

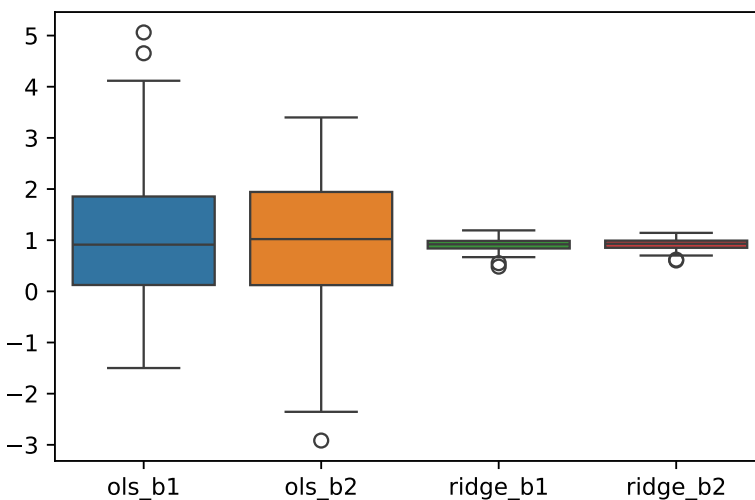
	ols_b1	ols_b2	ridge_b1	ridge_b2
0	1.642753	0.343583	0.908249	0.883823
1	1.105216	0.774858	0.885627	0.858433
2	-0.978268	3.091366	0.797745	1.092525
3	1.301118	0.325411	0.781313	0.728571
4	-0.626684	2.741871	0.926826	1.008491

This becomes even clearer in the boxplot below, which shows that the ridge estimator has much smaller variance than the OLS estimator.

```

import seaborn as sns
sns.boxplot(res)

```



4.3.2 Contour map

The variance of both $\hat{\beta}_1$ and $\hat{\beta}_2$ is quite large under OLS. This is because the variance of $\hat{\beta}$ is $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$. Since the columns of $\mathbf{X}$ are highly correlated, the smallest eigenvalue of $\mathbf{X}^\top \mathbf{X}$ is close to zero, making the largest eigenvalue of $(\mathbf{X}^\top \mathbf{X})^{-1}$ very large.

To see things in more details, we first build the RSS function. For each β_1 and β_2 in the range, we form a model $\hat{y}_\beta = \beta_1 x_1 + \beta_2 x_2$ and use it to compute $\text{RSS} = \sum (y - \hat{y})^2$.

```
beta1_vals = np.linspace(-1, 3, 200)
beta2_vals = np.linspace(-1, 3, 200)
B1, B2 = np.meshgrid(beta1_vals, beta2_vals)

betas = np.column_stack([B1.ravel(), B2.ravel()])
preds = X @ betas.T
rss_term = np.sum((y.reshape(-1, 1) - preds) ** 2, axis=0)
rss_value0 = rss_term.reshape(B1.shape)
```

We then construct the contour map for the RSS surface, plotting the isolines that correspond to the 1%, 2.5%, 5%, 20%, 50%, and 75% quantiles of the distribution. The true coefficient ($\beta_1 = 1, \beta_2 = 1$) is displayed as the red dot, as well as the estimated β as a blue square in the contour map.

```
import matplotlib.pyplot as plt

quant_levels = np.quantile(rss_value0, [0.01, 0.025, 0.05, 0.2, 0.5, 0.75])

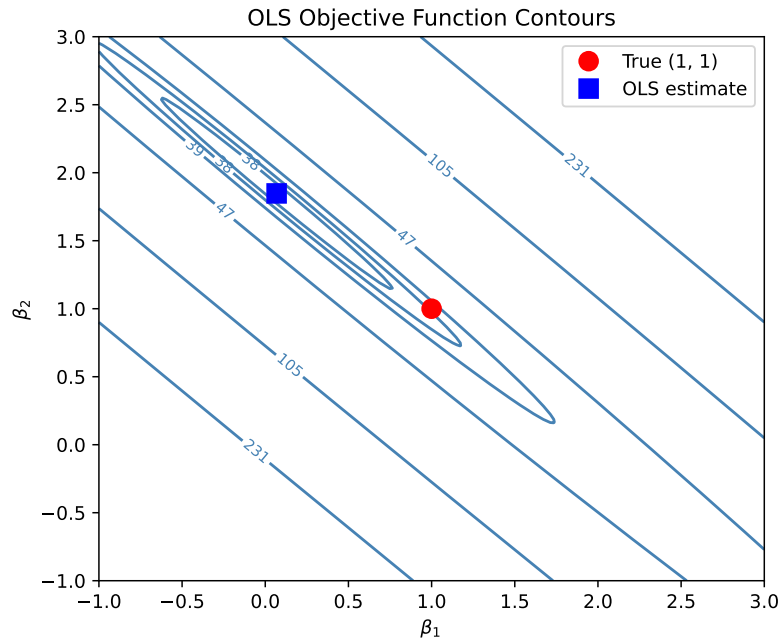
fig, ax = plt.subplots(figsize=(6, 5))
cs = ax.contour(B1, B2, rss_value0, levels=quant_levels, colors="steelblue")
ax.clabel(cs, inline=True, fontsize=8, fmt="%.0f")

ax.plot(1, 1, "ro", markersize=10, label="True (1, 1)")
ax.plot(ols.coef_[0], ols.coef_[1], "bs", markersize=10, label="OLS estimate")

ax.set_xlabel(r"$\beta_1$")
ax.set_ylabel(r"$\beta_2$")
ax.set_title("OLS Objective Function Contours")
```



```
ax.legend()
plt.tight_layout()
```



The contour map reveals a narrow, elongated valley in the RSS objective function. While the surface is highly sensitive to movement perpendicular to the valley, it remains relatively flat along the valley floor. Because the surface is particularly flat in this longitudinal direction, numerous parameter combinations yield nearly identical RSS values. Consequently, the data provides insufficient information to distinguish between points along this axis, making the RSS minimum highly unstable. This lack of identifiability is the fundamental cause of high variance in the estimated coefficients.

When ridge regression is applied, the RSS objective function is modified by adding a penalty term. As a result, the objective surface no longer exhibits a long, flat valley floor. Instead, the minimum becomes more well-defined, and the estimate is pulled toward the origin. Although the true parameter may no longer lie at the bottom of this modified valley, the ridge estimate has substantially lower variance. In this way, ridge regression stabilizes the estimation by controlling the region

along the valley floor where the OLS solution would otherwise vary widely.

Here we show the RSS objective function contours for $\lambda = 0.1, 1, 10$ and 100 .

```
from sklearn.linear_model import Ridge

beta1_vals = np.linspace(-1, 3, 200)
beta2_vals = np.linspace(-1, 3, 200)
B1, B2 = np.meshgrid(beta1_vals, beta2_vals)

lambda_vals = [0.1, 1, 10, 100]

fig, axes = plt.subplots(2, 2, figsize=(10, 8))

for ax, reg_lambda in zip(axes.ravel(), lambda_vals):
    betas = np.column_stack([B1.ravel(), B2.ravel()])
    preds = X @ betas.T
    rss_term = np.sum((y.reshape(-1, 1) - preds) ** 2, axis=0)
    penalty_term = reg_lambda * np.sum(betas**2, axis=1)
    rss_value = (rss_term + penalty_term).reshape(B1.shape)

    quant_levels = np.quantile(rss_value, [0.01, 0.025, 0.05, 0.2, 0.5, 0.75])

    cs = ax.contour(B1, B2, rss_value, levels=quant_levels, colors="steelblue")
    ax.clabel(cs, inline=True, fontsize=7, fmt="%.0f")

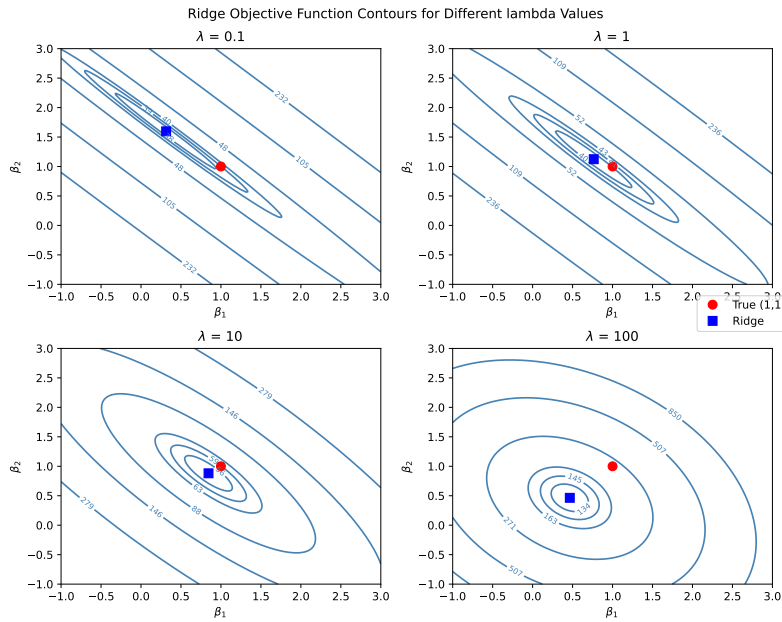
    ridge = Ridge(alpha=reg_lambda, fit_intercept=False).fit(X, y)

    ax.plot(1, 1, "ro", markersize=8, label="True (1,1)")
    ax.plot(ridge.coef_[0], ridge.coef_[1], "bs", markersize=8, label="Ridge")

    ax.set_title(r"$\lambda$ = {}".format(reg_lambda))
    ax.set_xlabel(r"$\beta_1$")
    ax.set_ylabel(r"$\beta_2$")

handles, labels = axes[0, 0].get_legend_handles_labels()
fig.suptitle("Ridge Objective Function Contours for Different lambda Values")
fig.legend(handles, labels, loc="right")
```

```
plt.tight_layout()
```



As λ increases, the bottom of the valley gradually becomes more rounded, and the objective surface rises more steeply as the coefficients move away from the origin. This additional curvature removes the long flat valley floor and makes the minimum more clearly defined. At the same time, the penalty pulls the estimated coefficients toward the origin, so the ridge estimate is effectively dragged toward zero.

Geometrically, this shrinkage prevents the estimator from drifting freely along the nearly flat valley direction that appears in the OLS objective. As a result, the ridge estimator is much more stable and typically has substantially lower variance than the OLS estimator, although this improvement comes at the cost of introducing some bias.

i Gram matrix and Covariance matrix

The Gram matrix $X^T X$ represents the raw inner products of predictors. In the context of RSS map, it is the direct source of the surface geometry.

The Covariance matrix is essentially the Gram matrix of the centered data, usually scaled by $n - 1$.

$$\mathbf{S} = \frac{1}{n-1} \sum (x_i - \bar{x})(x_i - \bar{x})^top.$$

Therefore if the data is already centered, the Gram matrix and the covariance matrix are directly related: $X^T X = (n - 1) \text{Cov}$.

Mathematically, these two directions (the longitudinal axis and the perpendicular axis) correspond to the eigenvectors of the $\mathbf{X}^T \mathbf{X}$ matrix (or, equivalently, the principal components of the predictor space). The flatness or steepness in those directions is governed by their corresponding eigenvalues.

```
XTX = X.T @ X
eigenvalues, eigenvectors = np.linalg.eigh(XTX)
for i in range(len(eigenvalues)):
    print(f'eigenvalue: {eigenvalues[i]: .2f}, eigenbasis: {eigenvectors[i]}')
```

```
eigenvalue:  0.26, eigenbasis: [ 0.70320986 -
0.71098234]
eigenvalue:  95.19, eigenbasis: [-0.71098234 -
0.70320986]
```

Then mark the eigenvectors in the contour map.

```

fig, ax = plt.subplots(figsize=(6, 5))

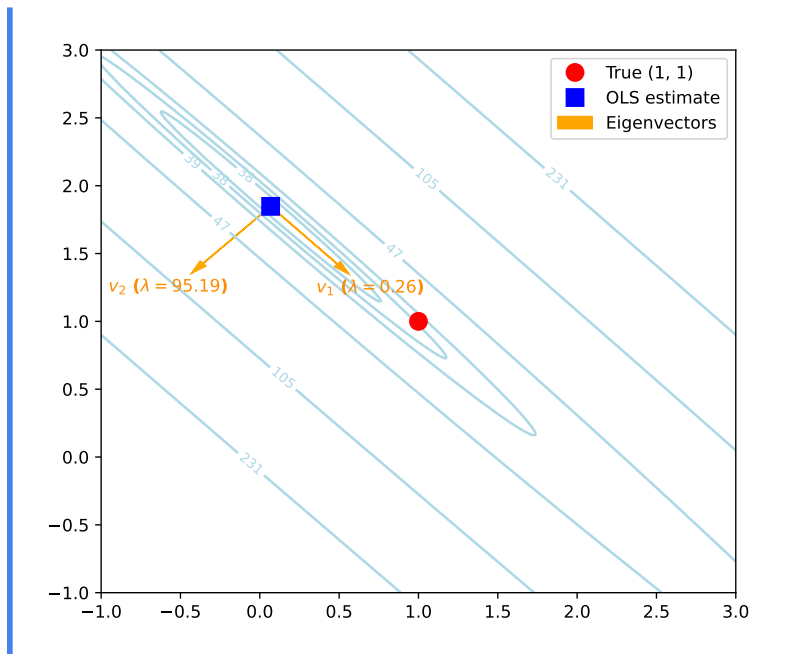
quant_levels = np.quantile(rss_value0, [0.01, 0.025, 0.05, 0.2, 0.5, 0.75])
cs = ax.contour(B1, B2, rss_value0, levels=quant_levels, colors="lightblue")
ax.clabel(cs, inline=True, fontsize=8, fmt="%.0f")

ax.plot(1, 1, "ro", markersize=10, label="True (1, 1)")
ax.plot(ols.coef_[0], ols.coef_[1], "bs", markersize=10, label="OLS estimate")

scale = 0.6
for i in range(len(eigenvalues)):
    v = eigenvectors[:, i]
    ax.arrow(
        ols.coef_[0],
        ols.coef_[1],
        v[0] * scale,
        v[1] * scale,
        head_width=0.05,
        head_length=0.1,
        fc="orange",
        ec="orange",
        label="Eigenvectors" if i == 0 else "",
    )

    ax.text(
        ols.coef_[0] + v[0] * scale * 1.5,
        ols.coef_[1] + v[1] * scale * 1.5,
        r"$v_{\text{}}$ ($\lambda={:.2f}$)".format(i + 1, eigenvalues[i]),
        color="darkorange",
        fontweight="bold",
        ha="center",
    )
ax.legend()
plt.tight_layout()

```



4.3.3 Choosing λ

The regularization parameter λ determines how much shrinkage is applied.

In practice, the modern standard is to choose λ by cross-validation. Similar to the model selection procedure discussed in the linear regression case, the process works as follows:

1. Fix a value of λ . Split the dataset into k folds. Each time, choose one fold as the validation set and use the remaining folds as the training set to fit the model and compute the validation score.
2. Repeat this process so that each fold serves once as the validation set. The average validation score is the cross-validation score for this λ .
3. Repeat the above steps for each candidate value of λ , and select the λ with the best cross-validation score.
4. Finally, refit the model on the entire training set using the selected λ .

Since this procedure is very common in ridge regression, `sklearn` provides `RidgeCV`, which has built-in cross-validation. Alternatively, the same task can be performed using `GridSearchCV`.

i Leave-one-out cross-validation

`GridSearchCV` typically uses k -fold cross-validation, where the training set is split into k folds and each fold is used once as the validation set.

If each observation is treated as its own fold (that is, the validation set contains exactly one observation), the procedure is called leave-one-out cross-validation (LOOCV). In other words, LOOCV is equivalent to N -fold cross-validation, where N is the number of observations.

`RidgeCV` uses leave-one-out cross-validation by default.

The arguments of `RidgeCV` are similar to those of `Ridge`. The main differences are:

- `alphas` should be a list of candidate values for α . In the example below, α is selected from values ranging from $1e-4$ to $1e4$.
- `cv` works in the same way as in ordinary cross-validation and specifies the number of folds used in the cross-validation procedure. If it is not set, LOOCV will be used.

```
from sklearn.linear_model import RidgeCV
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

alphas = np.logspace(-4, 4, 100)

ridge_cv_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("model", RidgeCV(alphas=alphas)),
    ]
)

ridge_cv_pipe.fit(X, y)
```

```
best_alpha_ridge = ridge_cv_pipe.named_steps["model"].alpha_
best_alpha_ridge
```

```
1.3219411484660315
```

4.3.4 Validation curve and Coefficient paths

RidgeCV returns only the final fitted model at the selected value of `alpha`, rather than the entire validation curve or coefficient path. If we want to examine how the validation score changes with `alpha`, we may fit ordinary Ridge models over a grid of `alpha` values and then add a vertical line at the value selected by RidgeCV.

sklearn provides `validation_curve()` to streamline the process.

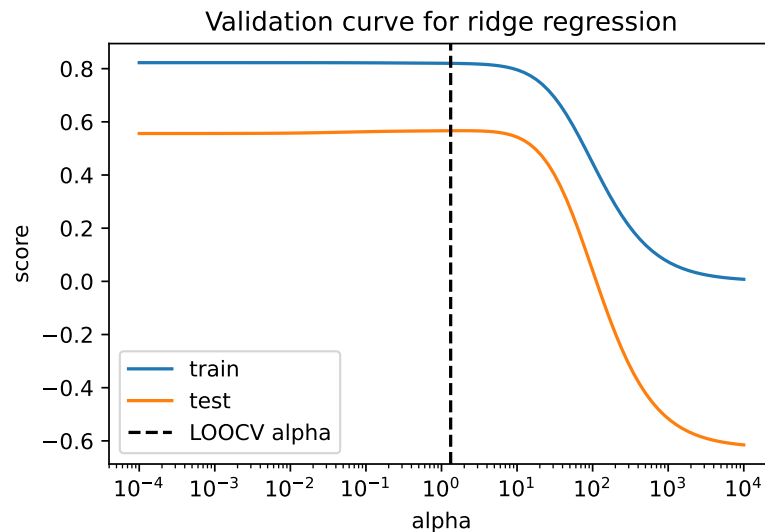
```
from sklearn.model_selection import validation_curve

alphas = np.logspace(-4, 4, 100)

ridge_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("model", Ridge()),
    ]
)
train_scores, test_scores = validation_curve(
    ridge_pipe, X, y, param_name="model__alpha", param_range=alphas
) ①

plt.plot(alphas, train_scores.mean(axis=1), label="train")
plt.plot(alphas, test_scores.mean(axis=1), label="test")
plt.axvline(best_alpha_ridge, color="black", linestyle="--", label="LOOCV alpha") ②
plt.xscale("log")
plt.xlabel("alpha")
plt.ylabel("score")
plt.title("Validation curve for ridge regression")
plt.legend()
```


- ① The validation curve can also be drawn using `GridSearchCV.cv_results_`.
There is no major difference between the two methods.
- ② This best `alpha` here is based on the LOOCV formula. So it is slightly different from the best `alpha` under 5-fold cross-validation.



Similarly, if we want to examine how the coefficients change with `alpha`, we fit ordinary Ridge repeatedly over a grid of `alpha` values and record the fitted coefficients. In this case, cross-validation is not needed, since the goal is only to visualize the coefficient path rather than to compare models.

```
from sklearn.linear_model import Ridge
import matplotlib.pyplot as plt

alphas = np.logspace(-4, 4, 100)
coefs = []
for a in alphas:
    pipe = Pipeline(
        [
            ("scaler", StandardScaler()),
            ("model", Ridge(alpha=a)),
        ]
    )
    pipe.fit(X, y)
```

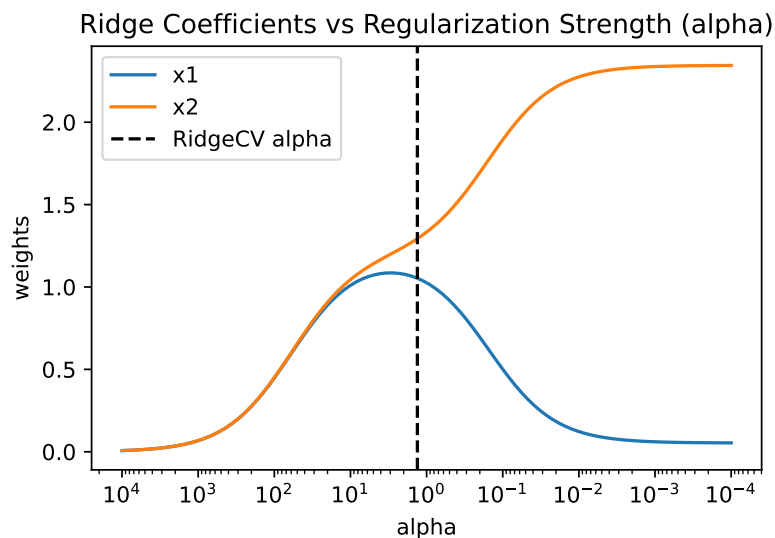
```

    coefs.append(pipe.named_steps["model"].coef_)
coefs = np.array(coefs)
for j in range(coefs.shape[1]):
    plt.plot(alphas, coefs[:, j], label=f"x{j + 1}")
plt.axvline(best_alpha_ridge, linestyle="--", color="black", label="RidgeCV alpha")

ax = plt.gca()
ax.set_xlim(ax.get_xlim()[::-1]) ①
plt.xscale("log")
plt.xlabel("alpha")
plt.ylabel("weights")
plt.title("Ridge Coefficients vs Regularization Strength (alpha)")
plt.axis("tight")
plt.legend()

```

- ① In the coefficient path plot, the horizontal axis is shown in reverse order so that the coefficients move from heavily regularized models on the left to weakly regularized models on the right.



4.3.5 LOOCV

Naively, LOOCV would require fitting the model N times. Each time one observation is removed and the model is refit. However, ridge regression admits an analytic formula that allows the LOOCV error to be computed without refitting the model repeatedly. This is one reason ridge regression works very well with LOOCV. `RidgeCV` from `sklearn` is also implemented using LOOCV formula instead of refitting the model multiple times.

Theorem 4.3 (LOOCV). *The LOOCV error is*

$$CV(\lambda) = \frac{1}{N} \sum_{i=1}^N \left(\frac{y_i - \hat{y}_i}{1 - H_{\lambda,ii}} \right)^2.$$

Click for proof.

Recall that the ridge estimator is

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

The fitted values can be written as

$$\hat{y} = X\hat{\beta}_\lambda = H_\lambda y,$$

where

$$H_\lambda = X(X^\top X + \lambda I)^{-1} X^\top$$

is the ridge hat matrix.

Let

$$r_i = y_i - \hat{y}_i$$

be the ordinary residual from the ridge regression fitted using the entire dataset.

Suppose we remove observation i and refit the ridge regression model. Let $\hat{y}_{(i)}$ denote the predicted value for y_i from this leave-one-out model. The corresponding leave-one-out residual is

$$r_{(i)} = y_i - \hat{y}_{(i)}.$$

By the PRESS residual formula [4], this residual can be computed directly from the full-data fit:

$$r_{(i)} = \frac{r_i}{1 - H_{\lambda,ii}},$$

where $H_{\lambda,ii}$ is the i -th diagonal element of the ridge hat matrix.

Therefore the LOOCV error is

$$CV(\lambda) = \frac{1}{N} \sum_{i=1}^N \left(\frac{y_i - \hat{y}_i}{1 - H_{\lambda,ii}} \right)^2.$$

i Note

This formula means that the leave-one-out residuals can be obtained by simply adjusting the ordinary residuals using the leverage term $H_{\lambda,ii}$. Consequently, the LOOCV error can be computed without refitting the model N times.

4.4 Lasso Regression

4.4.1 Definition

Lasso regression (short for least absolute shrinkage and selection operator) modifies the OLS objective by adding an L_1 penalty.

Definition 4.3 (Lasso Regression). For a response vector $y \in \mathbb{R}^n$ and a design matrix $X \in \mathbb{R}^{n \times p}$, the lasso estimator $\hat{\beta}_{\text{lasso}}$ is defined by

$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}$$

where

- $\lambda \geq 0$ is the regularization parameter,
- $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$ is the L_1 norm of the coefficient vector.

i Solving lasso

Lasso is solved using numerical optimization algorithms. The most widely used methods include

- coordinate descent
- LARS (Least Angle Regression)
- proximal gradient methods

These algorithms iteratively update the coefficients until convergence.

Unlike OLS or ridge regression, the lasso estimator does not admit a closed-form solution because the L_1 penalty is not differentiable at zero.

The lasso penalty also shrinks the coefficients toward zero, and therefore behaves similarly to ridge regression. In practice, the penalty is typically applied only to the slope coefficients, so the intercept is not penalized. For this reason, it is common to standardize the predictors and center the response variable before fitting the model.

However, lasso has an important difference from ridge regression, which we discuss in the next section.

4.4.2 Geometry and sparsity

Similar to ridge regression, by the theory of Lagrange multipliers, lasso can also be written in constrained form:

$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta} \|y - X\beta\|_2^2 \quad \text{subject to} \quad \|\beta\|_1 \leq s.$$

Here $s \geq 0$ controls the size of the coefficient vector. This constrained formulation is useful because it makes the geometry of lasso easier to visualize.

In $\mathbb{R}^2$, the ridge constraint region is based on the L_2 norm and has a circular boundary, whereas the lasso constraint region is based on the L_1 norm and has a diamond-shaped boundary. The solution is obtained where the smallest residual RSS contour first touches the constraint region.

For ridge regression, the circular boundary is smooth, so the point of tangency typically occurs away from the coordinate axes. As a result, ridge usually shrinks coefficients continuously toward zero, but does not make them exactly zero.

For lasso, the diamond-shaped boundary has sharp corners lying on the coordinate axes. Because of these corners, the point of tangency is much more likely to occur at a location where one coordinate is zero. Consequently, lasso can produce solutions with some coefficients exactly equal to zero.

Therefore, ridge is mainly a shrinkage method, whereas lasso is both a shrinkage method and a variable-selection method.

```
import numpy as np
import matplotlib.pyplot as plt

mu = np.array([2.0, 0.35])

theta = np.deg2rad(32)
R = np.array([[np.cos(theta), -np.sin(theta)], [np.sin(theta), np.cos(theta)]])

D = np.diag([7.0, 0.45])
A = R @ D @ R.T

def rss(beta1, beta2):
    B = np.stack([beta1 - mu[0], beta2 - mu[1]], axis=-1)
    return np.einsum("...i,ij,...j->...", B, A, B)

x = np.linspace(-0.5, 3.2, 600)
y = np.linspace(-2.0, 2.0, 600)
X, Y = np.meshgrid(x, y)
Z = rss(X, Y)

levels = [0.15, 0.4, 0.8, 1.4, 2.4, 3.8, 5.8]

ridge_r = 1.7
lasso_t = 2.0

fig, axs = plt.subplots(1, 2, figsize=(10, 5))
```

```

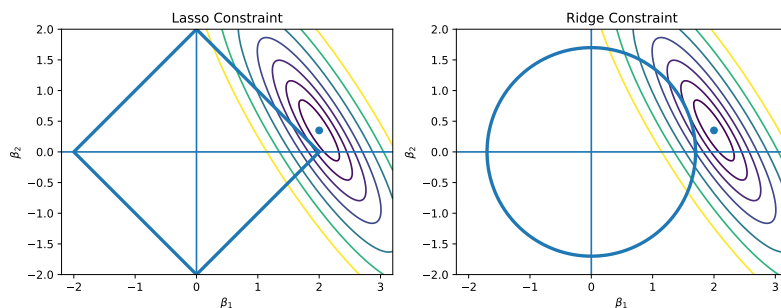
axs[0].contour(X, Y, Z, levels=levels)
diamond = np.array(
    [[lasso_t, 0], [0, lasso_t], [-lasso_t, 0], [0, -lasso_t], [lasso_t, 0]]
)
axs[0].plot(diamond[:, 0], diamond[:, 1], linewidth=3)
axs[0].scatter(*mu)
axs[0].axhline(0)
axs[0].axvline(0)
axs[0].set_title("Lasso Constraint")

axs[1].contour(X, Y, Z, levels=levels)
t = np.linspace(0, 2 * np.pi, 400)
axs[1].plot(ridge_r * np.cos(t), ridge_r * np.sin(t), linewidth=3)
axs[1].scatter(*mu)
axs[1].axhline(0)
axs[1].axvline(0)
axs[1].set_title("Ridge Constraint")

for ax in axs:
    ax.set_xlim(-2.2, 3.2)
    ax.set_ylim(-2, 2)
    ax.set_aspect("equal")
    ax.set_xlabel(r"$\beta_1$")
    ax.set_ylabel(r"$\beta_2$")

plt.tight_layout()

```



4.4.3 One-variable lasso and soft-thresholding

To understand how lasso sets coefficients to 0, it is useful to first look at the one-variable case.

Suppose we want to solve

$$\min_{\beta} \left\{ \|y - X\beta\|_2^2 + \lambda|\beta| \right\}.$$

Note that

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta} \left\{ \|y - X\beta\|_2^2 \right\}$$

denotes the OLS solution for this one-variable problem.

Theorem 4.4 (Soft-thresholding Rule).

$$\hat{\beta}_{\text{lasso}} = \begin{cases} \hat{\beta}_{\text{OLS}} + \frac{\lambda}{2a} & \text{if } \hat{\beta}_{\text{OLS}} < -\frac{\lambda}{2a}, \\ 0 & \text{if } |\hat{\beta}_{\text{OLS}}| \leq \frac{\lambda}{2a}, \\ \hat{\beta}_{\text{OLS}} - \frac{\lambda}{2a} & \text{if } \hat{\beta}_{\text{OLS}} > \frac{\lambda}{2a}, \end{cases}$$

Click to expand.

Since there is only one β ,

$$\text{RSS}(\beta) = \|y - X\beta\|_2^2 + \lambda|\beta| = (\sum x_i^2)\beta^2 - 2(\sum x_i y_i)\beta + \sum y_i^2 + \lambda|\beta|.$$

Let $a = \sum x_i^2$, $c = \sum x_i y_i$. Then

$$\text{RSS}(\beta) = a\beta^2 - 2c\beta + \lambda|\beta| + \text{constant}.$$

Note that

$$\hat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top y = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{c}{a}.$$

Now due to absolute value we split into different cases.

If $\beta \geq 0$: Then $|\beta| = \beta$. Therefore $\text{RSS}(\beta) = a\beta^2 - 2c\beta + \lambda\beta + \text{constant}$. Let $\frac{d\text{RSS}(\beta)}{d\beta} = 2a\beta - 2c + \lambda = 0$. So $\beta = \frac{c}{a} - \frac{\lambda}{2a} = \hat{\beta}_{\text{OLS}} - \frac{\lambda}{2a}$ is the critical point.

In this case since $\beta \geq 0$,

- if $\hat{\beta}_{\text{OLS}} - \frac{\lambda}{2a} > 0$, then the minimizer is $\hat{\beta}_{\text{OLS}} - \frac{\lambda}{2a}$.
- Otherwise, the minimum over the region $\beta \geq 0$ is attained at 0.

Therefore

$$\hat{\beta}_{\text{lasso}} = \begin{cases} \hat{\beta}_{\text{OLS}} - \frac{\lambda}{2a} & \text{if } \hat{\beta}_{\text{OLS}} > \frac{\lambda}{2a}, \\ 0 & \text{if } \hat{\beta}_{\text{OLS}} \leq \frac{\lambda}{2a}. \end{cases}$$

If $\beta \leq 0$: Then $|\beta| = -\beta$. Therefore $\text{RSS}(\beta) = a\beta^2 - 2c\beta - \lambda\beta + \text{constant}$. Let $\frac{d\text{RSS}(\beta)}{d\beta} = 2a\beta - 2c - \lambda = 0$. So $\beta = \frac{c}{a} + \frac{\lambda}{2a} = \hat{\beta}_{\text{OLS}} + \frac{\lambda}{2a}$ is the critical point.

In this case since $\beta \leq 0$,

- if $\hat{\beta}_{\text{OLS}} + \frac{\lambda}{2a} < 0$, then the minimizer is $\hat{\beta}_{\text{OLS}} + \frac{\lambda}{2a}$.
- Otherwise, the minimum over the region $\beta \leq 0$ is attained at 0.

In other words,

$$\hat{\beta}_{\text{lasso}} = \begin{cases} \hat{\beta}_{\text{OLS}} + \frac{\lambda}{2a} & \text{if } \hat{\beta}_{\text{OLS}} < -\frac{\lambda}{2a}, \\ 0 & \text{if } \hat{\beta}_{\text{OLS}} \geq -\frac{\lambda}{2a} \end{cases}$$

To sum up,

$$\hat{\beta}_{\text{lasso}} = \begin{cases} \hat{\beta}_{\text{OLS}} + \frac{\lambda}{2a} & \text{if } \hat{\beta}_{\text{OLS}} < -\frac{\lambda}{2a}, \\ 0 & \text{if } |\hat{\beta}_{\text{OLS}}| \leq \frac{\lambda}{2a}, \\ \hat{\beta}_{\text{OLS}} - \frac{\lambda}{2a} & \text{if } \hat{\beta}_{\text{OLS}} > \frac{\lambda}{2a}, \end{cases}$$

This formula shows the essential behavior of lasso:

- if the OLS coefficient is large enough in magnitude, it is shrunk toward zero;
- if the OLS coefficient is small enough, it is shrunk all the way to zero.

i Orthogonal design

If the design matrix satisfies $X^\top X = nI$, then the lasso problem separates into p one-variable problems. In that case, each lasso coefficient is obtained by applying soft-thresholding to the corresponding OLS coefficient.

This is the clearest way to see why lasso performs variable selection.

4.5 Lasso in Python

In `scikit-learn`, `Lasso` and `LassoCV` from `linear_model` are used in a way very similar to `Ridge` and `RidgeCV`.

- The argument `alpha` is the regularization parameter.
- `fit_intercept=True` means the intercept is handled automatically.
- If `fit_intercept=False`, the data is expected to be centered beforehand.
- It is recommended to standardize the features X .

A key computational difference from ridge is that lasso does not have a closed-form estimator. Likewise, unlike ridge regression, there is no closed-form LOOCV formula available for selecting the tuning parameter. Therefore, `LassoCV` uses ordinary K -fold cross-validation to choose the best value of `alpha`.

Note that `LassoCV` is very tricky here and you should be very careful when you use it. We will come back to it later.

4.5.1 A simulated example

To illustrate lasso and ridge in code, we start with a simulated dataset. The data are constructed so that:

- there are 100 predictors and 150 observations,
- many predictors are almost collinear because they are obtained from a rank 12 latent structure,
- but only 10 predictors truly contribute to the response.

```
import numpy as np

rng = np.random.default_rng(0)

n = 150
p = 100
k = 12

Z = rng.normal(size=(n, k))
A = rng.normal(size=(k, p))
X = Z @ A + 0.8 * rng.normal(size=(n, p))

beta = np.zeros(p)
beta[:10] = [8, 7, 6, 5, 4, 3, 2, 1.5, 1, 0.5]

y = X @ beta + 12 * rng.normal(size=n)
```

We split the data into training and test sets.

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=0
)
```

4.5.2 Initial model fitting

Before tuning the regularization parameter, we first fit OLS, ridge, and lasso with simple default or manually chosen settings. This gives a first look at how the methods behave on the same dataset.

```

from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

ols_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("model", LinearRegression()),
    ]
)

ols_pipe.fit(X_train, y_train)
y_pred_ols = ols_pipe.predict(X_test)

print("OLS")
print(f"Test MSE: {mean_squared_error(y_test, y_pred_ols)}")
print(f"Test R^2 : {r2_score(y_test, y_pred_ols)}")

```

OLS

Test MSE: 1212.515485649412

Test R² : 0.49121698288112325

```

from sklearn.linear_model import Ridge

ridge_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("model", Ridge(alpha=1.0)),
    ]
)

ridge_pipe.fit(X_train, y_train)
y_pred_ridge = ridge_pipe.predict(X_test)

print("Ridge")
print(f"Test MSE: {mean_squared_error(y_test, y_pred_ridge)}")
print(f"Test R^2 : {r2_score(y_test, y_pred_ridge)}")

```

Ridge

Test MSE: 249.13745912332723
Test R² : 0.895459552549109

```
from sklearn.linear_model import Lasso

lasso_pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("model", Lasso(alpha=0.1, max_iter=10000)),
])
lasso_pipe.fit(X_train, y_train)
y_pred_lasso = lasso_pipe.predict(X_test)

print("\nLasso")
print(f"Test MSE: {mean_squared_error(y_test, y_pred_lasso)}")
print(f"Test R^2 : {r2_score(y_test, y_pred_lasso)}")
```

Lasso
Test MSE: 204.53280796906878
Test R² : 0.9141760906397303

One important difference between ridge and lasso is that lasso can set some coefficients exactly to zero, while ridge typically only shrinks coefficients toward zero without eliminating them.

```
ridge_coef = ridge_pipe.named_steps["model"].coef_
lasso_coef = lasso_pipe.named_steps["model"].coef_

print("Number of nonzero ridge coefficients:", np.sum(ridge_coef != 0))
print("Number of nonzero lasso coefficients:", np.sum(lasso_coef != 0))
```

Number of nonzero ridge coefficients: 100
Number of nonzero lasso coefficients: 64

You can see that in the lasso case, many coefficients are exactly 0. This is one reason lasso is often viewed as performing both regularization and variable selection.

4.5.3 Coefficient paths

To better understand how the regularization parameter affects the estimates, we now plot the coefficient paths.

- In the lasso plot, some coefficient curves hit exactly 0 as `alpha` increases.
- In the ridge plot, all coefficients are shrunk continuously toward 0, but usually remain nonzero.

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import lasso_path, Ridge
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)

fig, axs = plt.subplots(1, 2, figsize=(10, 5))

alphas_lasso, coefs_lasso, _ = lasso_path(X_train_scaled, y_train)

axs[0].plot(alphas_lasso, coefs_lasso.T)
axs[0].set_title("Lasso coefficient paths")

alphas_ridge = np.logspace(-4, 4, 100)
coefs_ridge = []
for a in alphas_ridge:
    ridge = Ridge(alpha=a)
    ridge.fit(X_train_scaled, y_train)
    coefs_ridge.append(ridge.coef_)
coefs_ridge = np.array(coefs_ridge)

axs[1].plot(alphas_ridge, coefs_ridge)
axs[1].set_title("Ridge coefficient paths")

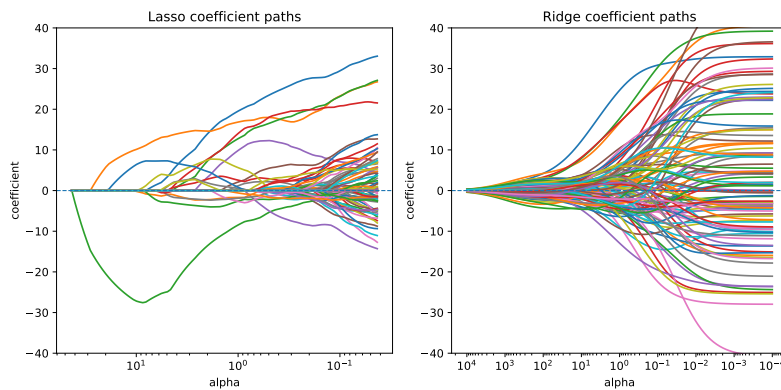
for ax in axs:
    ax.set_xscale("log")
    ax.axhline(0, linestyle="--", linewidth=1)
    ax.invert_xaxis()
```

```

ax.set_ylim(-40, 40)
ax.set_xlabel("alpha")
ax.set_ylabel("coefficient")

plt.tight_layout()

```



4.5.4 Tuning with cross-validation

The choice of **alpha** strongly affects the fitted model. Therefore in practice we usually tune it by cross-validation.

We first use `GridSearchCV` to select the best value of **alpha**.

```

from sklearn.linear_model import Lasso
from sklearn.model_selection import GridSearchCV

alphas = np.logspace(-2, 4, 100)

lasso_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("model", Lasso(max_iter=100000, tol=1e-3, random_state=0)),
    ]
)

lasso_cv_pipe = GridSearchCV(lasso_pipe, param_grid={"model__alpha": alphas})

lasso_cv_pipe.fit(X_train, y_train)

```

```

y_pred_lasso_cv = lasso_cv_pipe.predict(X_test)

best_alpha_lasso = lasso_cv_pipe.best_params_["model__alpha"]

print("Best lasso alpha:", best_alpha_lasso)
print("Lasso Test MSE:", mean_squared_error(y_test, y_pred_lasso_cv))
print("Lasso Test R^2:", r2_score(y_test, y_pred_lasso_cv))

```

```

Best lasso alpha: 0.49770235643321115
Lasso Test MSE: 221.63625982655304
Lasso Test R^2: 0.9069993197512844

```

For comparison, we tune ridge regression using the same cross-validation procedure and alpha grid.

```

ridge_cv_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("model", Ridge()),
    ]
)

ridge_cv_pipe = GridSearchCV(ridge_cv_pipe, param_grid={"model__alpha": alphas})

ridge_cv_pipe.fit(X_train, y_train)
y_pred_ridge_cv = ridge_cv_pipe.predict(X_test)

best_alpha_ridge = ridge_cv_pipe.best_params_["model__alpha"]

print("Best ridge alpha:", best_alpha_ridge)
print("Ridge Test MSE:", mean_squared_error(y_test, y_pred_ridge_cv))
print("Ridge Test R^2:", r2_score(y_test, y_pred_ridge_cv))

```

```

Best ridge alpha: 12.32846739442066
Ridge Test MSE: 280.8056227406724
Ridge Test R^2: 0.8821712929419413

```

After obtaining the best regularization parameters, we mark them on the coefficient path plots. This helps connect the selected model to the whole regularization path.


```

import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import lasso_path, Ridge
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)

fig, axs = plt.subplots(1, 2, figsize=(10, 5))

alphas_lasso, coefs_lasso, _ = lasso_path(X_train_scaled, y_train)

axs[0].plot(alphas_lasso, coefs_lasso.T)
axs[0].axvline(
    best_alpha_lasso, linestyle="--", color="black", label="selected alpha"
)
axs[0].set_title("Lasso coefficient paths")

alphas_ridge = np.logspace(-4, 4, 100)
coefs_ridge = []

for a in alphas_ridge:
    ridge = Ridge(alpha=a)
    ridge.fit(X_train_scaled, y_train)
    coefs_ridge.append(ridge.coef_)

coefs_ridge = np.array(coefs_ridge)

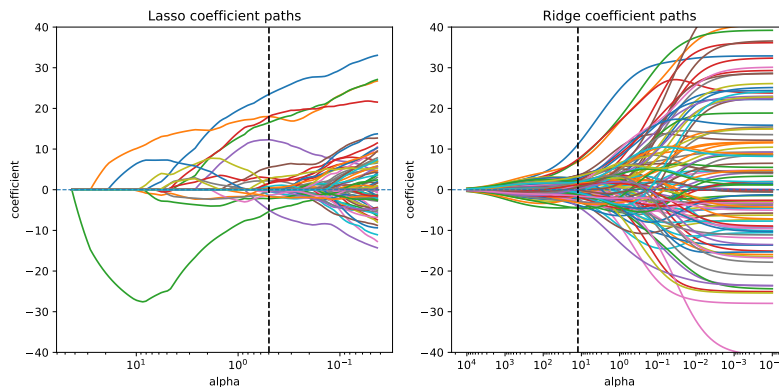
axs[1].plot(alphas_ridge, coefs_ridge)
axs[1].axvline(
    best_alpha_ridge, linestyle="--", color="black", label="selected alpha"
)
axs[1].set_title("Ridge coefficient paths")

for ax in axs:
    ax.set_xscale("log")
    ax.axhline(0, linestyle="--", linewidth=1)
    ax.invert_xaxis()
    ax.set_ylim(-40, 40)

```

```
ax.set_xlabel("alpha")
ax.set_ylabel("coefficient")

plt.tight_layout()
```



🔥 Data leaks in LassoCV

When using a pipeline such as `StandardScaler->LassoCV`, the `StandardScaler` is fitted once using the entire training set before `LassoCV` performs its internal cross-validation. As a result, the validation folds inside `LassoCV` are evaluated on data that were standardized using information from all training observations, including observations outside the current fold.

This introduces a small amount of data leakage and may make the cross-validation estimate slightly optimistic.

A safer approach is to place the entire pipeline inside `GridSearchCV` using `Lasso` instead of `LassoCV`. In this setup, the scaler is refit separately within each training fold, preventing leakage.

`LassoCV` is still useful when no preprocessing pipeline is involved, and it is often faster than `GridSearchCV` because of its specialized implementation.

Similarly, `RidgeCV` may suffer from the same issue when placed after preprocessing steps in a pipeline. However, when `cv=None`, `RidgeCV` uses formulas to estimate the LOOCV. In practice, the resulting bias is often small, so many users accept this shortcut for convenience and speed.

4.5.5 Validation curves

Another way to visualize tuning is through the validation curve. Here we plot the mean training and validation scores across the same grid of `alpha` values used above. The score is negative MSE, so higher values are better.

- When `alpha` is too small, the model may overfit.
- When `alpha` is too large, the model may underfit.
- A good value of `alpha` balances these two effects.

```
from sklearn.model_selection import validation_curve
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge, Lasso
import numpy as np
import matplotlib.pyplot as plt

fig, axs = plt.subplots(1, 2, figsize=(10, 4.5))

lasso_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("model", Lasso(max_iter=100000, tol=1e-3, random_state=0)),
    ]
)

train_scores_lasso, test_scores_lasso = validation_curve(
    lasso_pipe, X_train, y_train, param_name="model__alpha", param_range=alphas
)

axs[0].plot(alphas, train_scores_lasso.mean(axis=1), label="train")
axs[0].plot(alphas, test_scores_lasso.mean(axis=1), label="test")
axs[0].axvline(best_alpha_lasso, color="black", linestyle="--", label="best alpha")
axs[0].set_title("Validation curve for lasso")

ridge_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("model", Ridge()),
    ]
)
```

```

    ]
)

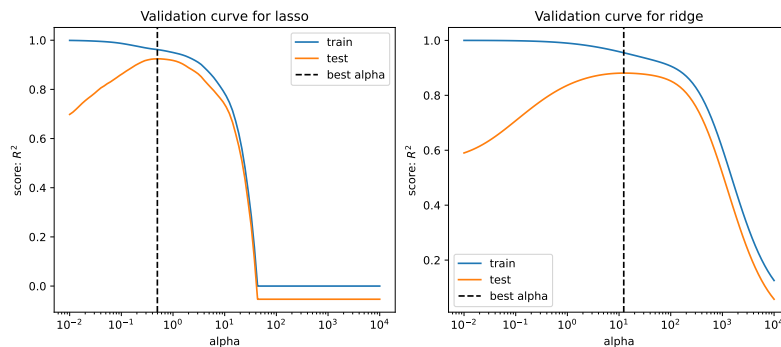
train_scores_ridge, test_scores_ridge = validation_curve(
    ridge_pipe, X_train, y_train, param_name="model__alpha", param_range=alphas
)

axs[1].plot(alphas, train_scores_ridge.mean(axis=1), label="train")
axs[1].plot(alphas, test_scores_ridge.mean(axis=1), label="test")
axs[1].axvline(best_alpha_ridge, color="black", linestyle="--", label="best alpha")
axs[1].set_title("Validation curve for ridge")

for ax in axs:
    ax.set_xscale("log")
    ax.set_xlabel("alpha")
    ax.set_ylabel(r"score:  $R^2$ ")
    ax.legend()

plt.tight_layout()

```



4.6 Elastic net

Elastic net combines the L_1 penalty of lasso and the L_2 penalty of ridge regression. It is useful when we want a sparse model, but the predictors are strongly correlated.

Definition 4.4 (Elastic net). The elastic net estimator is defined by

$$\hat{\beta}_{\text{EN}} = \arg \min_{\beta} \left\{ \|y - X\beta\|_2^2 + \lambda [\alpha \|\beta\|_1 + (1 - \alpha) \|\beta\|_2^2] \right\},$$

where $\lambda \geq 0$ controls the overall regularization strength and $0 \leq \alpha \leq 1$ controls the mixture of the two penalties.

- When $\alpha = 1$, elastic net reduces to lasso.
- When $\alpha = 0$, elastic net reduces to ridge regression.
- The L_1 penalty can set coefficients exactly to zero, while the L_2 penalty improves stability when predictors are strongly correlated.

Grouping effect

An important feature of elastic net is the **grouping effect**. When several predictors are strongly correlated, lasso may select only one of them, and the selected predictor may change substantially across different samples. In contrast, the L_2 component of elastic net encourages correlated predictors to have similar coefficients, so they tend to enter or leave the model together.

Elastic net therefore combines variable selection with more stable coefficient estimates when predictors are strongly correlated.

4.6.1 Elastic net in Python

In `sklearn`, elastic net is implemented by `ElasticNet`. The parameter names are slightly confusing:

- `alpha` corresponds to λ , the overall regularization strength.
- `l1_ratio` corresponds to α , the mixture of the L_1 and L_2 penalties.

Thus, `l1_ratio=1` gives lasso, while values closer to 0 behave more like ridge regression.

```

from sklearn.linear_model import ElasticNet

en = ElasticNet(alpha=0.1, l1_ratio=0.5, max_iter=10000)
en.fit(X_train, y_train)
y_pred_en = en.predict(X_test)

```

4.6.2 Choosing α and λ

Both parameters can be selected using `ElasticNetCV`.

```

from sklearn.linear_model import ElasticNetCV

en_cv = ElasticNetCV(
    l1_ratio=[0.1, 0.5, 0.9, 1.0],
    alphas=np.logspace(-4, 4, 100),
    cv=5,
    max_iter=10000,
).fit(X_train, y_train)

```

As noted in the earlier caution, preprocessing should be performed separately within each fold. Therefore, we use `GridSearchCV` with `ElasticNet` instead of a `StandardScaler-ElasticNetCV` pipeline.

```

import numpy as np
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

en_pipe = make_pipeline(
    StandardScaler(),
    ElasticNet(max_iter=100000, tol=1e-3),
)

alphas = np.logspace(-2, 4, 100)

en_grid = GridSearchCV(
    en_pipe,
    {

```

```

        "elasticnet__alpha": alphas,
        "elasticnet__l1_ratio": [0.1, 0.5, 0.9, 1.0],
    },
)
en_grid.fit(X_train, y_train)

y_pred_en = en_grid.predict(X_test)

print("Best lambda:", en_grid.best_params_["elasticnet__alpha"])
print("Best alpha :", en_grid.best_params_["elasticnet__l1_ratio"])
print("Test MSE    :", mean_squared_error(y_test, y_pred_en))
print("Test R^2    :", r2_score(y_test, y_pred_en))

```

```

Best lambda: 0.49770235643321115
Best alpha  : 1.0
Test MSE    : 221.63625982655304
Test R^2    : 0.9069993197512844

```

4.6.3 Summary

Finally, we summarize the test performance and the number of nonzero coefficients for each model.

```

import pandas as pd

results = []

models = {
    "OLS": ols_pipe,
    "Ridge": ridge_cv_pipe,
    "Lasso": lasso_cv_pipe,
    "Elastic Net": en_grid,
}

for name, model in models.items():
    y_pred = model.predict(X_test)
    mse = mean_squared_error(y_test, y_pred)
    r2 = r2_score(y_test, y_pred)

```

```

fitted_pipe = (
    model.best_estimator_
    if name in ["Ridge", "Lasso", "Elastic Net"]
    else model
)
model_step = "elasticnet" if name == "Elastic Net" else "model"
fitted_model = fitted_pipe.named_steps[model_step]
n_nonzero = np.sum(fitted_model.coef_ != 0)

if name == "Lasso":
    alpha = model.best_params_["model__alpha"]
    l1_ratio = 1.0
elif name == "Elastic Net":
    alpha = model.best_params_["elasticnet__alpha"]
    l1_ratio = model.best_params_["elasticnet__l1_ratio"]
elif name == "Ridge":
    alpha = model.best_params_["model__alpha"]
    l1_ratio = 0.0
else:
    alpha = None
    l1_ratio = None

results.append({
    'Model': name,
    'Best alpha': alpha,
    'L1 ratio': l1_ratio,
    'Test MSE': mse,
    'Test R^2': r2,
    'Nonzero coefs': n_nonzero
})

pd.DataFrame(results)

```

	Model	Best alpha	L1 ratio	Test MSE	Test R ²	Nonzero coefs
0	OLS	NaN	NaN	1212.515486	0.491217	100
1	Ridge	12.328467	0.0	280.805623	0.882171	100
2	Lasso	0.497702	1.0	221.636260	0.906999	27
3	Elastic Net	0.497702	1.0	221.636260	0.906999	27

This example shows the main practical differences among ridge, lasso, and elastic net:

- Ridge shrinks coefficients continuously and is especially useful when predictors are strongly correlated.
- Lasso also shrinks coefficients, but can force some of them to be exactly zero, producing a sparse model.
- Elastic net combines both penalties, balancing sparsity with greater stability among correlated predictors. In this example, however, cross-validation selects `l1_ratio=1`, so the fitted elastic net model reduces to lasso.

In high-dimensional settings with collinearity and only a subset of truly relevant predictors, lasso can provide a more interpretable sparse model, ridge can produce more stable coefficient estimates, and elastic net offers a data-driven compromise between the two.

i Note

In this example, the best `l1_ratio` is 1. This is likely because the true model is sparse: only a small number of predictors contribute directly to the response, making variable selection especially useful. Thus, although the predictors are strongly correlated, the additional ridge penalty does not improve cross-validation performance for this particular dataset.

5 Dimension reduction

When the number of predictors is large or when predictors are highly correlated, directly fitting a regression model using the original predictors may lead to unstable estimates and poor predictive performance.

One approach to address this problem is dimension reduction. Instead of using the original predictors directly, we first construct a smaller number of derived variables (called components) and then perform regression using these components.

Two commonly used methods are

- Principal Component Regression (PCR)
- Partial Least Squares (PLS)

Both methods construct new variables that are linear combinations of the original predictors, but they differ in how those combinations are chosen.

5.1 Principal Component Regression (PCR)

PCR first applies principal component analysis (PCA) to the predictor matrix X , and then uses the leading principal components as predictors in a regression model.

The key idea is that most of the variation in the predictors often lies in a lower-dimensional subspace.

Definition 5.1 (Principal Component Regression). Let X be the $n \times p$ predictor matrix.

The principal components are defined as

$$Z_k = Xv_k,$$

where v_k is the k -th eigenvector of the Gram matrix $X^\top X$.

Principal component regression fits a linear model using the first m components:

$$y = \theta_0 + \theta_1 Z_1 + \theta_2 Z_2 + \cdots + \theta_m Z_m + \varepsilon.$$

Here

- $Z_1, \dots, Z_m$ are the first m principal components
- $m < p$ is chosen by cross-validation

Thus PCR replaces the original predictors with a smaller set of orthogonal components.

i Key feature of PCR

PCR constructs components using only the predictor matrix X , without considering the response y .

As a result, the directions that explain the most variation in X are not necessarily the directions that are most predictive of y .

5.1.1 Data matrix and its geometric views

Let $X \in \mathbb{R}^{n \times p}$ be a data matrix with n observations and p predictors. We assume columns are centered ($\sum_{i=1}^n x_{ij} = 0$) so that the origin represents the sample mean.

Column (variable) view: observation space $\mathbb{R}^n$

Each column of X_j is a vector in n -dimensional space, which represents one predictor across all observations.

- This space is often called the observation space.
- Geometry in this space describes relationships between predictors:
 - Inner products encode covariance.
 - After standardization, angles encode correlation.
 - Orthogonality corresponds to zero correlation.

Row (observation) view: feature space $\mathbb{R}^p$

Each row of $x_i^\top$ is a vector in p -dimensional space, which represents one observation across all predictors.

- This space is called the feature space.
- Geometry in this space describes relationships between observations:
 - Euclidean distance measures dissimilarity.
 - Clustering and nearest-neighbor methods operate in this space.

Example 5.1.

Click to expand.

To demonstrate the idea, we generate the following dataset. It contains 2 variables.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(0)

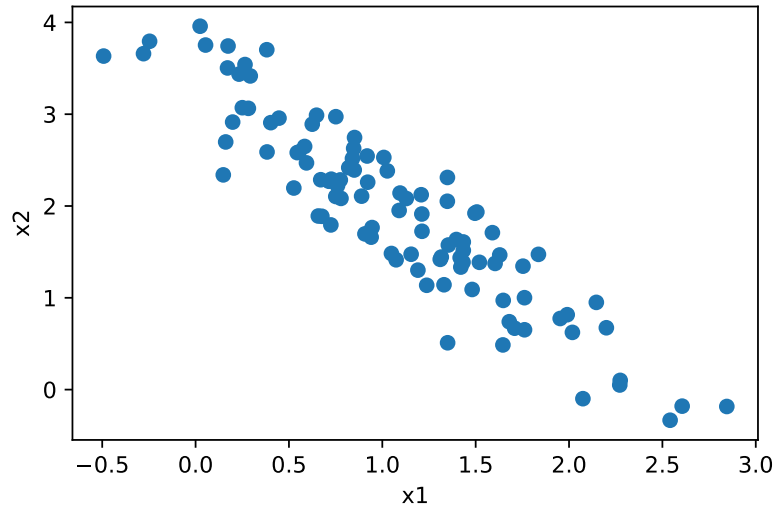
n = 100

Sigma = np.array([[0.5, -0.65], [-0.65, 1.0]])
mu = np.array([1, 2])

X = rng.multivariate_normal(mean=mu, cov=Sigma, size=n)

plt.scatter(X[:, 0], X[:, 1])

plt.xlabel("x1")
plt.ylabel("x2");
```



Each point is an observation and is represented by a point in the feature space.

5.1.2 Singular Value Decomposition (SVD)

The SVD provides the most direct link between the two geometric perspectives.

For a matrix with $\text{rank}(X) = r$

$$X = UDV^\top,$$

where

- $U \in \mathbb{R}^{n \times r}$, $U^\top U = I$
 - $D = \text{diag}(d_1, \dots, d_r)$, $d_1 \geq \dots \geq d_r > 0$
 - $V \in \mathbb{R}^{p \times r}$, $V^\top V = I$
-

5.1.3 Principal components (PC)

Principal component analysis (PCA) constructs new variables as linear combinations of the original predictors. It builds upon SVD.

Let $Z = XV = UD = [Z_1, \dots, Z_r] = [z_{ij}]$ denote the principal components. Each principal component has the form

$$Z_j = Xv_j = v_{1j}X_1 + v_{2j}X_2 + \dots + v_{pj}X_p$$

where $v_j = [v_{1j} \ \dots \ v_{pj}]^\top \in \mathbb{R}^p$ is a unit vector.

Loadings (directions)

- Each PC is a linear combination of the columns of X .
- **Loadings** (v_{ij}): The coefficients used to form PC Z_j are called loadings.
- Loadings describe directions in feature space.

Scores (coordinates)

- As a vector, each PC lies in $\mathbb{R}^n$ (observation space).
- **Scores** (z_{ij}): The coordinate of observation i along direction v_j is called the score.

Thus:

- rows of Z = transformed observations,
- columns of Z = principal components.

Example 5.2 (Centering and Scaling).

Click to expand.

We use the same previous dataset $\mathbf{X}$, centering it as $\mathbf{Xc}$ and standarizing it as $\mathbf{Xs}$.

```
Xc = X - X.mean(axis=0)
Xs = Xc / Xc.std(axis=0)
```

Now we plot them together with the princial compoenent. `pca()` from `sklearn` will automatically centering columns. In order to show the sceanario, we use `svd()` from `numpy.linalg` to do the computation.

```

import matplotlib.pyplot as plt

def plot_svd(ax, X, V, title, scale=3, length=5):
    ax.scatter(X[:, 0], X[:, 1])
    t = np.linspace(-length, length, 100)
    line = np.outer(t, V[:, 0])
    ax.plot(line[:, 0], line[:, 1], linewidth=2, color="red")

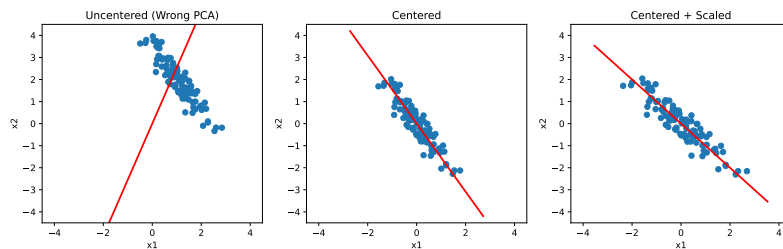
    ax.set_title(title)
    ax.set_xlabel("x1")
    ax.set_ylabel("x2")
    ax.set_xlim(-4.5, 4.5)
    ax.set_ylim(-4.5, 4.5)

V_raw = np.linalg.svd(X)[2].T
V_c = np.linalg.svd(Xc)[2].T
V_s = np.linalg.svd(Xs)[2].T

fig, axs = plt.subplots(1, 3, figsize=(15, 4))

plot_svd(axs[0], X, V_raw, "Uncentered (Wrong PCA)")
plot_svd(axs[1], Xc, V_c, "Centered")
plot_svd(axs[2], Xs, V_s, "Centered + Scaled")

```



The principal component finds the direction that maximizes the variance of the projected data.

If the data are not centered, the optimization is no longer based purely on variance. In this case, the objective involves both the covariance structure and the mean of the data.

As a result, when the mean is large, the first principal direction may be strongly influenced by the location of the data, and can

point roughly toward the data cloud rather than reflecting its intrinsic variation.

5.1.4 Covariance and Optimality

The sample covariance matrix is $S = \frac{1}{n-1}X^\top X$. By SVD,

$$S = \frac{1}{n-1}X^\top X = V \frac{D^2}{n-1} V^\top$$

- V : eigenvectors of covariance matrix
- $d_j^2/(n-1)$: eigenvalues of S
- $\text{Var}(Z_j) = d_j^2/(n-1)$

Therefore

- First PC: Direction v_1 that maximizes $\text{Var}(Xv_1)$ subject to $\|v_1\| = 1$.
- Subsequent PCs: Maximize remaining variance subject to orthogonality (that they are orthogonal to all previous PCs $v_k^\top v_j = 0$ for $j < k$).

5.1.5 Summary

Component	Matrix	Space	Interpretation
Loadings	V	Feature ($\mathbb{R}^p$)	Principal directions (rotation of axes)
Scores	$Z = UD$	Observa- tion ($\mathbb{R}^n$)	Transformed coordinates of observations
Normalized scores	U	Observa- tion ($\mathbb{R}^n$)	Directions of score vectors (unit length)
Singular values	D	—	Scaling of each component (spread)

Component	Matrix	Space	Interpretation
Variance	$D^2/(n-1)$	—	Importance of each principal component

- Columns of X : variables in $\mathbb{R}^n$ (relationships via angles)
- Rows of X : observations in $\mathbb{R}^p$ (relationships via distance)
- PCA finds orthogonal directions in feature space
- V gives directions, Z gives coordinates
- SVD unifies everything:

$$X = UDV^\top, \quad Z = XV = UD$$

5.1.6 Choosing the number of components (m)

Let $\text{rank}(X) = r$. PCA produces r components, but in practice we keep only the first $m \leq r$.

Define the truncated SVD:

$$X \approx X_m = U_m D_m V_m^\top,$$

where

- $U_m \in \mathbb{R}^{n \times m}$
- $D_m \in \mathbb{R}^{m \times m}$
- $V_m \in \mathbb{R}^{p \times m}$

Correspondingly,

$$Z_m = XV_m = U_m D_m$$

contains the first m principal components.

Keeping the m largest singular values gives the best rank- m approximation of X in least squares sense.

5.1.6.1 Variance explained

The total variance is

$$\sum_{j=1}^r \frac{d_j^2}{n-1},$$

and the proportion explained by the first m components is

$$\frac{\sum_{j=1}^m d_j^2}{\sum_{j=1}^r d_j^2}.$$

5.1.6.2 Choosing m for prediction (PCR)

In predictive settings, m is typically selected via cross-validation:

1. Compute principal components.
2. Fit regression models using $m = 1, 2, \dots, r$ components.
3. Evaluate validation error.
4. Choose m with the lowest validation error.

i Scree plot (Elbow method)

The scree plot (elbow method) plots d_j^2 against j and selects m at the point where the marginal gain in variance explained drops sharply (the “elbow”). This identifies the number of components that captures most of the variation in X .

However, PCA is unsupervised: directions that explain high variance in X do not necessarily have strong predictive power for y . Therefore, in predictive settings such as PCR, m is typically chosen via cross-validation rather than the elbow method.

For unsupervised tasks (e.g., representation or clustering), the scree plot remains a reasonable and commonly used choice.

5.2 PCR in Python

5.2.1 PCA

scikit-learn provides PCA for computing principal components. To demonstrate the code we use the same dataset from lasso.

```
import numpy as np
from sklearn.model_selection import train_test_split

rng = np.random.default_rng(0)

n = 150
p = 100
k = 12

Z = rng.normal(size=(n, k))
A = rng.normal(size=(k, p))
X = Z @ A + 0.8 * rng.normal(size=(n, p))

beta = np.zeros(p)
beta[:10] = [8, 7, 6, 5, 4, 3, 2, 1.5, 1, 0.5]

y = X @ beta + 12 * rng.normal(size=n)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=0
)
```

pca is an unsupervised learning method. Therefore to perform it we only need X.

```
from sklearn.decomposition import PCA

pca = PCA().fit(X_train)
```

All important information can be accessed from the returned object.

```

V = pca.components_.T          # loadings (V)
Z = pca.transform(X_train)     # scores (Z)
D = pca.singular_values_       # singular values (D)
var = pca.explained_variance_   # variance that are explained
ratio = pca.explained_variance_ratio_ # the ratio of variance that are explained

```

With `var` and `ratio`, we may plot the scree plot (although we won't really use it to select the number of components).

```

k = np.arange(1, len(var) + 1)

plt.figure()
plt.plot(k, ratio, marker="o")
plt.xlabel("Component index")
plt.ylabel("Proportion of variance explained")
plt.title("Scree Plot (Variance Ratio)")

cum = np.cumsum(ratio)

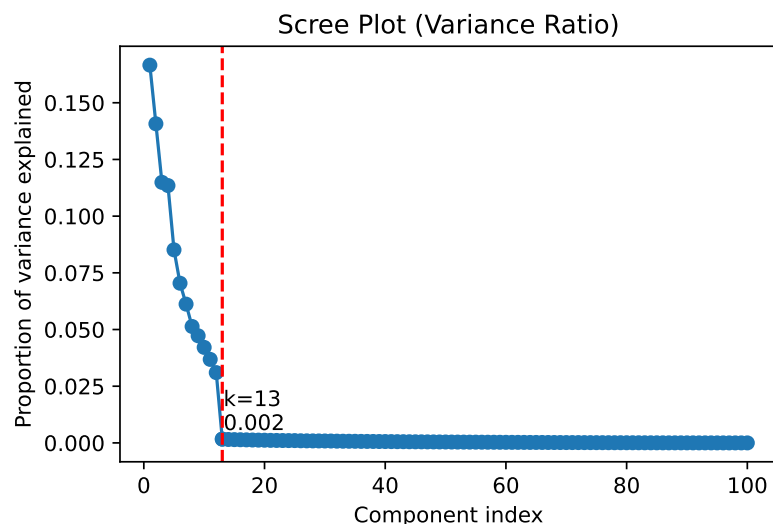
diff2 = np.diff(cum, 2)
elbow_idx = np.argmin(diff2) + 2
elbow_k = k[elbow_idx]
elbow_val = ratio[elbow_idx]
plt.axvline(elbow_k, linestyle="--", color="red") ①

plt.scatter(elbow_k, elbow_val)
plt.text(elbow_k + 0.2, elbow_val + 0.004, f"k={elbow_k}\n{elbow_val:.3f}")

```

① Use 2nd order difference to find the elbow.

```
Text(13.2, 0.005666697882320329, 'k=13\n0.002')
```



5.2.2 PCR

Using PCA to do regression, we have to standardize the data. Therefore we set up the pipeline.

We first use 13 components (which comes from the scree plot) as our starting point. We will tune it later using cross-validation.

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

pcr_pipe = Pipeline(
    [
        ("scaler", StandardScaler()),
        ("pca", PCA(n_components=13)),
        ("model", LinearRegression()),
    ]
)

pcr_pipe.fit(X_train, y_train)
y_pred_pcr = pcr_pipe.predict(X_test)
```

```
print(f"Test RMSE: {mean_squared_error(y_test, y_pred_pcr)}")
print(f"Test R^2 : {r2_score(y_test, y_pred_pcr)}")
```

```
Test RMSE: 348.7698605548008
Test R^2 : 0.8536528530700276
```

You may compare it with the results we get from `lasso` and `ridge`.

5.2.3 Choosing number of components by cross-validation

We use K -fold cross validation to find the best number of components. Our criterion is negative MSE.

```
from sklearn.model_selection import GridSearchCV

n_splits = 5
n_train_fold = int(X_train.shape[0] * (n_splits - 1) / n_splits)
max_components = min(n_train_fold, X_train.shape[1]) ①

param_grid = {"pca__n_components": list(range(1, max_components + 1))}

gs_pcr = GridSearchCV(
    pcr_pipe, param_grid, cv=n_splits, scoring="neg_mean_squared_error"
)
gs_pcr.fit(X_train, y_train)
gs_pcr.best_params_["pca__n_components"]
```

- ① The maximal possible number of components depends on the shape of X . However for cross-validation, the number of rows of X is reduced due to the split. Therefore we have to estimate this number in order to avoid “`n_components` out of bounds” warning.

It shows that 19 is the best number. Then we check the test result.

```
y_pred_pcr = gs_pcr.predict(X_test)

print(f"Test RMSE: {mean_squared_error(y_test, y_pred_pcr)}")
print(f"Test R^2 : {r2_score(y_test, y_pred_pcr)}")
```

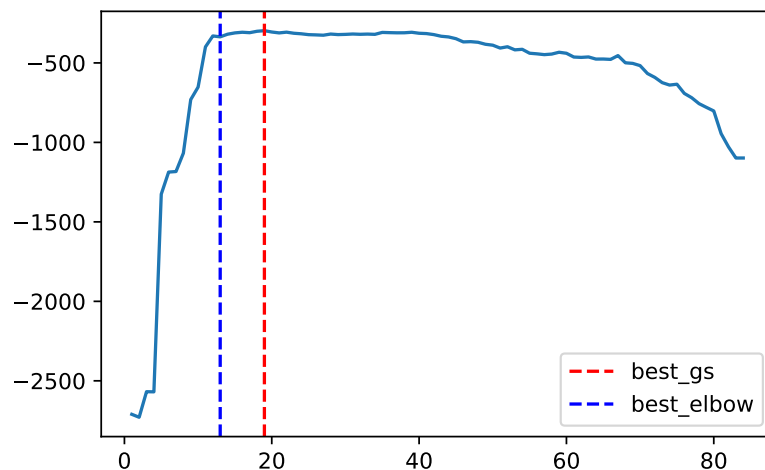
```
Test RMSE: 321.0762521471908
Test R^2 : 0.8652733542572647
```

It is slightly better than 13, but still worse than tuned `ridge` and `lasso`.

We could also look at the validation curve on the mean test score.

```
import matplotlib.pyplot as plt

plt.plot(param_grid["pca_n_components"], gs_pcr.cv_results_["mean_test_score"])
plt.axvline(19, linestyle="--", color="red", label="best_gs")
plt.axvline(13, linestyle="--", color="blue", label="best_elbow")
plt.legend()
```



5.3 Partial Least Squares (PLS)

One limitation of Principal Component Regression (PCR) is that it is unsupervised: the principal components depend only on X and ignore the response y . As a result, directions with large variance in X may not be relevant for predicting y .

Partial Least Squares (PLS) addresses this by constructing components that are guided by the response. It seeks directions in the predictor space that are both:

- informative about X , and
- strongly related to y

Definition 5.2 (Partial Least Squares). Partial least squares constructs a sequence of components

$$Z_k = Xw_k$$

where the weight vector w_k is chosen to maximize the covariance between the component and the response:

$$w_k = \arg \max_{\|w\|=1} \text{Cov}(Xw, y).$$

After constructing the component Z_k , both the predictor matrix X and the response y are deflated, and the procedure is repeated to construct additional components.

5.3.1 Algorithm intuition

PLS constructs components sequentially, with each step explicitly targeting predictive power for y , rather than variance in X .

Step 1: Find the direction. Choose a direction $w_k \in \mathbb{R}^p$ such that the component $Z_k = X_k w_k$ maximizes covariance with y_k .

$$w_k = \arg \max_{\|w\|=1} \text{Cov}(X_k w, y_k)$$

Step 2: Define the k -th component (also called the score vector):

$$Z_k = X_k w_k \in \mathbb{R}^n$$

- This is a linear combination of predictors
- Represents the direction in X most aligned with y_k

Step 3: Fit a simple regression of y_k on Z_k :

$$y_k = c_k Z_k + \text{residual}$$

This captures how strongly the component explains y_k .

Step 4: Deflation: remove the variation explained by Z_k from both X_k and y_k .

- Update response: $y_{k+1} = y_k - c_k Z_k$
- Update predictors: $X_{k+1} = X_k - Z_k p_k^\top$, where $p_k = \frac{X_k^\top Z_k}{Z_k^\top Z_k}$.
 - p_k are the loadings for reconstructing X
 - Deflation ensures the next components capture on new information

Step 5: Repeat the process for $k = 1, 2, \dots, m$.

5.3.2 Key properties

PLS has several important structural properties:

- The components $\{Z_1, Z_2, \dots\}$ are constructed sequentially.
- Each component explains the remaining variation in y .
- After deflation, the components are orthogonal.
- The ordering reflects predictive importance.

At each step, the main quantities have distinct roles:

Quantity	Interpretation
w_k	Direction in feature space used to combine predictors
$Z_k = X_k w_k$	Score (component), representing projected data

Quantity	Interpretation
p_k	Loading, describing how the component explains X

Together, these quantities describe how information is extracted from X and transferred to explain y .

5.3.3 Comparison with PCA

The key difference between PCA and PLS lies in how the directions are chosen:

Method	Criterion	Interpretation
PCA	maximize $\text{Var}(Xw)$	captures structure of X
PLS	maximize $\text{Cov}(Xw, y)$	targets prediction of y

PCA captures the internal structure of X , while PLS focuses on directions that are useful for predicting y .

5.3.4 Geometric meaning (Krylov Subspace)

The matrix X defines a cloud of points in $\mathbb{R}^p$ and the response y defines a direction in $\mathbb{R}^n$.

PLS finds directions w_k such that the projections $Z_k = Xw_k$ are maximally aligned with y in the observation space. In this sense, PLS can be understood geometrically as searching for directions in predictor space that align with the response.

To be more precise, it can be explained using Krylov subspaces.

Click to expand.

The Krylov subspace provides a structured way to generate directions for solving regression problems. Instead of searching the entire feature space $\mathbb{R}^p$, it builds a sequence of low-dimensional subspaces that are tailored to the problem.

Given a matrix $A \in \mathbb{R}^{p \times p}$ and a vector $b \in \mathbb{R}^p$, the Krylov subspace of order k is defined as

$$\mathcal{K}_k(A, b) = \text{span}\{b, Ab, A^2b, \dots, A^{k-1}b\}.$$

The subspaces are nested, meaning $\mathcal{K}_1 \subseteq \mathcal{K}_2 \subseteq \dots$, and the dimension cannot exceed $\text{rank}(A)$. Therefore, the sequence stabilizes after at most $\text{rank}(A)$ steps.

5.3.4.1 Krylov Subspace in Regression

In linear regression, we take

$$A = X^\top X, \quad b = X^\top y.$$

Then the Krylov subspace becomes

$$\mathcal{K}_k(X^\top X, X^\top y) = \text{span}\{X^\top y, (X^\top X)X^\top y, (X^\top X)^2 X^\top y, \dots\}.$$

The vector $X^\top y$ represents directions in the predictor space that are most correlated with the response, while $X^\top X$ encodes the covariance structure of the predictors. As a result, the Krylov subspace captures directions that are both aligned with y and consistent with the geometry of X .

5.3.4.2 Connection to PLS

Partial Least Squares constructs its directions sequentially within the Krylov subspace:

$$w_k \in \mathcal{K}_k(X^\top X, X^\top y).$$

This means that PLS does not search over all possible directions in $\mathbb{R}^p$. Instead, it expands a problem-dependent subspace step by step, starting from $X^\top y$ and incorporating higher-order interactions through $X^\top X$.

The number of components is therefore bounded by

$$m \leq \dim(\mathcal{K}_k) \leq \text{rank}(X).$$

i Note

The Krylov subspace can be viewed as a guided search path inside $\text{span}(X)$. The first direction is determined by $X^\top y$, and subsequent directions refine this information by incorporating the covariance structure of the predictors. In this sense, the Krylov subspace represents the portion of the feature space that is relevant for predicting y . Methods such as PLS operate within this space, focusing on directions that carry predictive signal rather than exploring the entire feature space.

5.4 PLS in Python

Although the algorithm is different, the code is very similar to PCR. PLS is implemented by `PLSRegression` from `sklearn.cross_decomposition`. The argument `scale=True` is by default so we don't need to attach standardizer to it.

We choose the default number of components (which is 2) here. The specific hyperparameter will be tuned later by cross-validation.

```
from sklearn.cross_decomposition import PLSRegression

pls = PLSRegression()

pls.fit(X_train, y_train)
```

We could use the following code to get access to the parameters.

```
W = pls.x_weights_    # weights, directions (W)
Z = pls.x_scores_     # scores, components (Z)
P = pls.x_loadings_   # loadings (P)
```

5.4.1 Example

We use the same simulated dataset as before.

```
import numpy as np
from sklearn.model_selection import train_test_split

rng = np.random.default_rng(0)

n = 150
p = 100
k = 12

Z = rng.normal(size=(n, k))
A = rng.normal(size=(k, p))
X = Z @ A + 0.8 * rng.normal(size=(n, p))

beta = np.zeros(p)
beta[:10] = [8, 7, 6, 5, 4, 3, 2, 1.5, 1, 0.5]

y = X @ beta + 12 * rng.normal(size=n)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=0
)
```

Similar to PCR, we need to choose the number of components. Since PLS is fit inside cross-validation, the maximum number of components should be no larger than the number of observations in each training fold.

```
from sklearn.model_selection import GridSearchCV
from sklearn.cross_decomposition import PLSRegression

n_splits = 5
n_train_fold = int(X_train.shape[0] * (n_splits - 1) / n_splits)
max_components = min(n_train_fold, X_train.shape[1])

param_grid = {"n_components": list(range(1, max_components + 1))}
```

```
gs_pls = GridSearchCV(
    PLSRegression(), param_grid, cv=n_splits, scoring="neg_mean_squared_error"
)
gs_pls.fit(X_train, y_train)
gs_pls.best_params_["n_components"]
```

4

After selecting the number of components by cross-validation, we evaluate the fitted model on the test set.

```
y_pred_pls = gs_pls.predict(X_test)

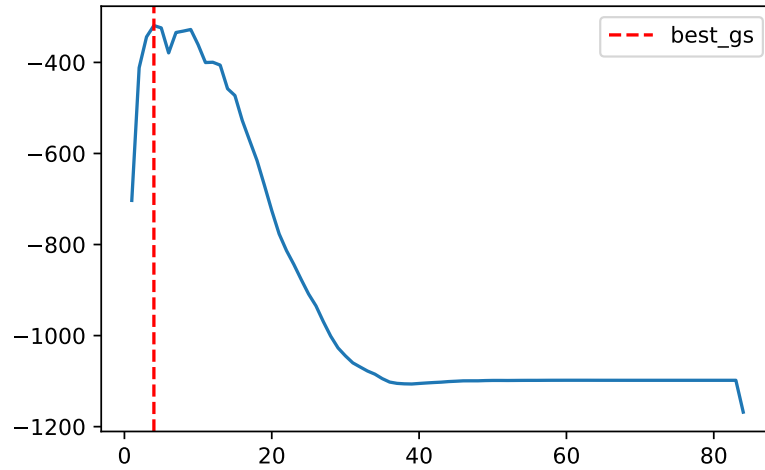
print(f"Test RMSE: {mean_squared_error(y_test, y_pred_pls)}")
print(f"Test R^2 : {r2_score(y_test, y_pred_pls)}")
```

```
Test RMSE: 333.9484272722302
Test R^2 : 0.8598720672844287
```

The validation curve shows how the cross-validation error changes as the number of PLS components increases.

```
import matplotlib.pyplot as plt

plt.plot(param_grid["n_components"], gs_pls.cv_results_["mean_test_score"])
plt.axvline(
    gs_pls.best_params_["n_components"],
    linestyle="--",
    color="red",
    label="best_gs",
)
plt.legend()
```



Important Notes

- `PLSRegression` automatically centers and scales the predictors by default.
- Unlike PCR, PLS uses both X and y when constructing components.
- The number of components should be chosen by cross-validation.
- Very large numbers of components are usually unnecessary and may lead to numerical warnings.

The results for this dataset are summarized in the following table.

	Test RMSE	Test R^2	Best param
Model			
OLS	1212.5155	0.491217	NaN
RidgeCV	274.2522	0.884921	alpha=10.0
LassoCV	221.7648	0.906945	alpha=0.4608014833934263
PCR	321.0763	0.865273	n_components=19
PLS	333.9484	0.859872	n_components=4

Note that PCR and PLS do not perform as well as ridge and lasso in this example. A key reason lies in how the dataset is generated.

- The coefficient vector β is sparse: only a small number of predictors (10 out of 100) are truly relevant.
- The noise level is relatively high, making signal recovery more difficult.
- More importantly, the variation in X is driven by a low-dimensional latent structure (ZA), while the true signal depends on a sparse set of original predictors. As a result, the directions of largest variance in X are not necessarily aligned with the directions that predict y .

Since PCR relies only on variance in X , and PLS is still influenced by this structure, both methods may fail to recover the true signal efficiently. In contrast, ridge and lasso operate directly on the original predictors, with lasso particularly well-suited for identifying sparse signals.

5.4.2 Another example

Now let us look at another example. In this setting, we construct X so that most of its variance comes from irrelevant noise, while the true signal has low variance but determines y . Specifically,

$$y = Z_1 b + \varepsilon_Y, \quad X = Z_1 A_1 + Z_2 A_2 + \varepsilon_X.$$

This setup has the following properties:

- Z_1 (signal) has low variance, while Z_2 (noise) has high variance.
- y depends only on Z_1 .
- Therefore, the predictive signal lies in low-variance but structured (low-rank) directions of X .
- For simplicity, we take Z_1 to be one-dimensional in the following example.

```
import numpy as np

rng = np.random.default_rng(42)

n = 120
p = 300
```



```

Z_1 = rng.normal(size=(n, 1))
A_1 = rng.normal(size=(1, p))
X_signal = Z_1 @ A_1 * 0.15

k_2 = 10
Z_2 = rng.normal(size=(n, k_2))
A_2 = rng.normal(size=(k_2, p))
X_noise = Z_2 @ A_2 * 15

X = X_noise + X_signal + 0.5 * rng.normal(size=(n, p))

y = 12 * Z_1.ravel() + 0.5 * rng.normal(size=n)

```

	Test RMSE	Test R ²	Best param
Model			
RidgeCV	34.7296	0.682587	alpha=0.1
LassoCV	22.8245	0.791395	alpha=0.0022916910232695206
PCR	17.2454	0.842385	n_components=51
PLS	16.2460	0.851518	n_components=12

In this example, PCR and especially PLS tend to outperform ridge and lasso.

- PCR can recover the signal if enough components are included, even though the signal lies in low-variance directions.
- PLS performs particularly well because it uses y when constructing components, allowing it to directly identify directions associated with the response.
- Ridge and lasso shrink coefficients based on the observed predictors. Since the signal is weak (low variance) and spread across many features through A_1 , it is harder for them to isolate and recover it.

Caution

Important predictive directions do not have to correspond to directions of large variance.

6 Logistic Regression

Logistic regression is a classification method for binary outcomes. It estimates the conditional probability of the positive class and then converts that probability into a class decision only after a threshold is chosen. Many of the materials were already covered in Chapter 2. We provide more details in this section.

6.1 Binary Response

6.1.1 Probability

Suppose $Y \in \{0, 1\}$. We usually call $Y = 1$ the positive class, success class, or class 1, and $Y = 0$ the negative class, failure class, or class 0.

For a given predictor vector x , define

$$p(x) = \Pr(Y = 1 \mid X = x).$$

Then

$$\Pr(Y = 0 \mid X = x) = 1 - p(x).$$

Note

Any probability is within $[0, 1]$. Therefore it is bounded.

6.1.2 Odds

If an event occurs with probability p , its odds are $\frac{p}{1-p}$. Odds compare the probability that the event occurs with the probability that it does not occur.

For binary classification,

$$\text{odds of class 1} = \frac{P(Y = 1 \mid X = x)}{P(Y = 0 \mid X = x)} = \frac{p(x)}{1 - p(x)}.$$

i Note

Odds take values in $(0, \infty)$. Comparing to probability, odds remove the upper bound of probability, but they are still restricted to be positive.

6.1.3 Logit and Log-Odds

To remove the positivity restriction of odds, we take a logarithm.

Definition 6.1 (Logit). The **logit** of a probability p is

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right).$$

It is also called the **log-odds**.

The inverse of the logit function is the logistic (sigmoid) function. It will be discussed next.

i Note

The logit transformation maps probabilities from $(0, 1)$ to the entire real line $(-\infty, \infty)$.

Example 6.1.

p	Odds	Logit	Interpretation
0.10	1/9	$\log(1/9) \approx -2.197$	Failure is much more likely than success
0.50	1	0	Success and failure are equally likely
0.75	3	$\log(3) \approx 1.099$	Success is three times as likely as failure

p	Odds	Logit	Interpretation
0.90	9	$\log(9) \approx 2.197$	Success is nine times as likely as failure

6.2 The Logistic Regression Model

Logistic regression assumes that the log-odds are linear in the predictors:

$$\text{logit}(p(x)) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

Equivalently,

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

This is the main equation of logistic regression.

6.2.1 From Log-Odds Back to Probability

Starting from

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = x^\top \beta,$$

exponentiating both sides gives

$$\frac{p(x)}{1 - p(x)} = \exp(x^\top \beta).$$

Solving for $p(x)$ gives

$$p(x) = \frac{\exp(x^\top \beta)}{1 + \exp(x^\top \beta)} = \frac{1}{1 + \exp(-x^\top \beta)}.$$

Definition 6.2 (The Sigmoid function). Define

$$\sigma(z) = \frac{1}{1 + \exp(-z)}.$$

This is the Sigmoid function, also called the logistic function.

With this definition, the logistic regression can be written as

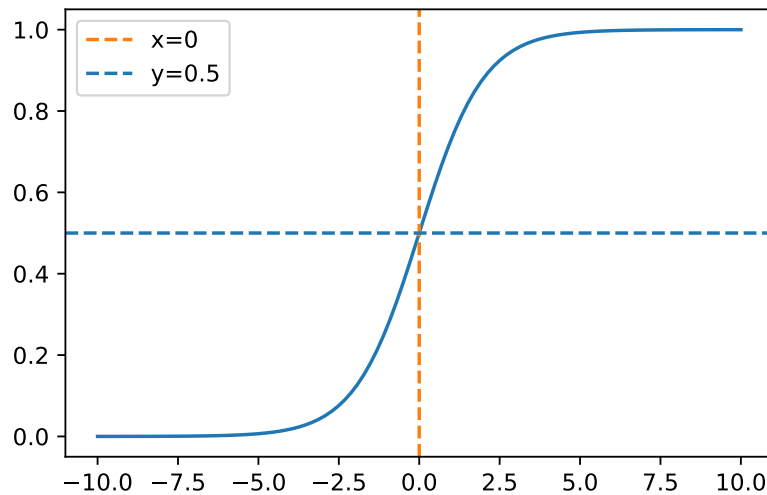
$$p(x) = \sigma(x^\top \beta) = \sigma(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = x^\top \beta).$$

The logistic function has several important properties.

1. $0 < \sigma(z) < 1$. So it always produces a valid probability for every real number z .
2. It is increasing: $z_1 < z_2 \Rightarrow \sigma(z_1) < \sigma(z_2)$.
3. $p(x)$ is

$$\begin{cases} \text{close to 0,} & \text{if } x^\top \beta \ll 0, \\ 0.5, & \text{if } x^\top \beta = 0, \\ \text{close to 1,} & \text{if } x^\top \beta \gg 0. \end{cases}$$

The curve is displayed as follows.



Historical Note

The logistic curve was introduced by Pierre-François Verhulst in 1838 [5] while studying population growth.

The term **logit** was introduced by Joseph Berkson in 1944 [6]. Berkson used the word logit for the logarithm of the odds and advocated the logistic function in bioassay problems.

Logistic regression later became a central example of generalized linear models. In GLM terminology, logistic regression uses:

- Bernoulli response distribution;
- logit link function;
- linear predictor $x^\top \beta$.

6.2.2 Logistic Regression as a Linear Classifier

Logistic regression produces probabilities, therefore it can also be used for classification.

Using the common threshold $t = 0.5$, we predict

$$\hat{y} = \begin{cases} 1, & p(x) \geq 0.5, \\ 0, & p(x) < 0.5. \end{cases}$$

Since

$$p(x) \geq 0.5 \iff x^\top \beta \geq 0,$$

the classification rule becomes

$$\hat{y} = \begin{cases} 1, & x^\top \beta \geq 0, \\ 0, & x^\top \beta < 0. \end{cases}$$

Therefore, the decision boundary is

$$x^\top \beta = 0.$$

i Geometric Interpretation

The decision boundary $x^\top \beta = 0$ is a line in two dimensions, a plane in three dimensions, and a hyperplane in higher dimensions.

Thus logistic regression is a linear classifier: Even though the probability curve is nonlinear, the decision boundary is linear in the original predictor space.

6.2.3 Interpreting the Coefficients

Consider the model

$$p(x) = \sigma(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p),$$

where $\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p = x^\top \beta$ is the linear predictor.

Suppose all other predictors are held fixed. Increasing x_j by one unit changes the linear predictor by β_j . Because the linear predictor represents the log-odds, increasing x_j by one unit changes the log-odds by β_j .

Thus:

- if $\beta_j > 0$, increasing x_j increases the log-odds of class 1;
- if $\beta_j < 0$, increasing x_j decreases the log-odds of class 1;
- if $\beta_j = 0$, increasing x_j does not change the log-odds.

For a one-unit increase in x_j , holding all other predictors fixed, the linear predictor increases by β_j . Since β_j is a change in log-odds, exponentiating gives a multiplicative change in odds, then the odds are expected to be multiplied by $\exp(\beta_j)$:

Consider

$$\log(\text{odds}_1) - \log(\text{odds}_2) = \beta_j.$$

Then

$$\frac{\text{odds}_1}{\text{odds}_2} = e^{\beta_j}.$$

Definition 6.3 (Odds Ratio). The quantity $\exp(\beta_j)$ is called the **odds ratio**.

Example 6.2 (Example). Suppose $\beta_j = 0.08$. Since

$$\exp(0.08) \approx 1.083,$$

Then a one-unit increase in x_j multiplies the odds by approximately 1.083. Equivalently, the odds increase by approximately 8.3%.

i Note

The coefficient β_j is not a direct change in probability since the relation between the logit and the probability is a sigmoid function (which is non-linear). So the probability change depends on the current value of $x^\top \beta$. For the same value of β_j , the probability change can be small when $p(x)$ is already close to 0 or close to 1, and larger when $p(x)$ is near 0.5.

6.3 Maximum Likelihood Estimation

Logistic regression is usually fit using maximum likelihood estimation.

Assume we are given a dataset $\{(x_i, y_i)\}_{i=1}^n$, where $y_i \in \{0, 1\}$. In logistic regression, we model the conditional probability that the response belongs to class 1:

$$p_i = \Pr(Y_i = 1 \mid X_i = x_i) = \sigma(x_i^\top \beta).$$

Since Y_i only takes the values 0 and 1, it is natural to model its conditional distribution using a Bernoulli distribution.

$$Y_i \mid X_i = x_i \sim \text{Bernoulli}(p_i).$$

The probability mass function of a Bernoulli random variable is

$$\Pr(Y_i = y_i) = p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Assuming the observations are independent and conditional on the predictors, the likelihood function is

$$L(\beta) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Taking the logarithm, the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)].$$

The maximum likelihood estimate is

$$\hat{\beta} = \arg \max_{\beta} \ell(\beta).$$

i Gradient descent

Unlike ordinary least squares, logistic regression does not have a simple closed-form solution for $\hat{\beta}$. In practice, numerical optimization methods are used.

Gradient descent is one of the most widely used optimization methods in machine learning. Although linear regression and logistic regression use different models and different loss functions, their gradient expressions have a similar matrix form. As a result, their gradient descent update formulas also look very similar.

Gradient descent is beyond the scope of this course, although it plays an important role in modern machine learning.

6.3.1 Binary Cross-Entropy and a Single Neuron

The negative log-likelihood is

$$-\ell(\beta) = -\sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)].$$

In machine learning, this is called binary cross-entropy loss or log loss. It is simply the negative of the log-likelihood mentioned in the previous section. The reason of this negative is that in machine learning it is usually preferred to lower the loss function.

Therefore logistic regression is not just a classification model. It is also a probabilistic model trained by minimizing a loss function that rewards accurate probability estimates.

What is more, a logistic regression model can be viewed as a single sigmoid neuron, since the model has the form

$$x \mapsto x^\top \beta \mapsto \sigma(x^\top \beta) \mapsto p(x).$$

This is essentially a neural network with no hidden layers and one output unit, with sigmoid activation and binary cross-entropy loss.

This connects logistic regression to modern machine learning methods, including neural networks, which often use cross-entropy loss for classification.

6.4 Regularized Logistic Regression

When there are many predictors, logistic regression can overfit. A common solution is to add regularization, like ridge and lasso for linear regression.

1. Ridge logistic regression penalizes large coefficients using an L^2 penalty:

$$-\ell(\beta) + \lambda \sum_{j=1}^p \beta_j^2.$$

This shrinks coefficients toward zero but usually does not set them exactly to zero.

2. Lasso logistic regression uses an L^1 penalty:

$$-\ell(\beta) + \lambda \sum_{j=1}^p |\beta_j|.$$

This can shrink some coefficients exactly to zero, producing a simpler model.

6.5 Logistic Regression in sklearn

We usually use `LogisticRegression` from `sklearn.linear_model` to implement logistic regression. The most commonly used solver in modern versions of scikit-learn is `lbfgs`, which is the default choice for `LogisticRegression()`. It is a quasi-Newton optimization method that usually converges much faster than simple gradient descent. In addition it is automatically extended to multiclass cases.

The basic code is as usual:

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

logit = make_pipeline(
    StandardScaler(),
    LogisticRegression(max_iter=5000),
)

logit.fit(X_train, y_train)

test_prob = logit.predict_proba(X_test)
test_pred = logit.predict(X_test)
```

Logistic regression is very similar to linear regression, that the scales of features matters. Therefore we need to set up a pipeline to standarize al features first.

The model actually has two types of outputs:

- `predict_proba()` returns estimated class probabilities. The second column is the estimated probability of the positive class.
- `predict()` returns class labels using the default threshold 0.5.

6.5.1 Regularization

The general idea of regularization in `LogisticRegression()` is based on the elastic-net style penalty, which is controlled by two arguments `l1_ratio` and `C`. Conceptually, the penalty behaves like

$$-\ell + \frac{1}{C}(\text{l1_ratio}\|\beta\|_1 + (1 - \text{l1_ratio})\|\beta\|_2^2).$$

- `l1_ratio` is to control the type of regularization:
 - `l1_ratio = 0`: pure L2 regularization;
 - `l1_ratio = 1`: pure L1 regularization;
 - `0 < l1_ratio < 1`: elastic-net regularization.
- `C`, which is $1/\lambda$, is to control the regularization strength:
 - `C=np.inf` means no penalty.

The exact formula used inside `sklearn` is more complicated which incorporates other aspects such as number of observations.

Note that some combinations of regularization and solvers are not compatible. For example,

- elastic-net regularization requires the `saga` solver;
- the default solve `lbfgs` requires `l2` so `l1_ratio` has to be 0.

The default setting for `LogisticRegression()` is that `l1_ratio=0`, `C=1` and `solver="lbfgs"`.

6.6 Example 1: Binary classification

We first consider a binary classification problem.

The Titanic dataset records information about passengers aboard the Titanic. Our goal is to predict whether a passenger survived. We load the dataset from the database from `seaborn`.

```
import seaborn as sns
from sklearn.model_selection import train_test_split

titanic = sns.load_dataset("titanic")
data = titanic[["survived", "age", "fare", "pclass"]].dropna()
X = data[["age", "fare", "pclass"]]
y = data["survived"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, stratify=y, random_state=1
)
```

Here we only use 3 features `age`, `fare` and `pclass` to predict `survived`.

We quickly fit the model with default setting.

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

logit = make_pipeline(StandardScaler(), LogisticRegression(max_iter=5000))
logit.fit(X_train, y_train)
test_prob = logit.predict_proba(X_test)[:, 1] ①
test_pred = logit.predict(X_test)
```

- ① For binary classification, the output of `.predict_proba()` contains two columns, each for a class. We want to test 1, so we choose column 1.

We focus on tuning `C` and the threshold. Note that the two hyperparameters take effects in different stage. `C` is about the training of the model, while the threshold is about prediction.

Therefore for logistic regression we usually tune `C` first and then find the best threshold afterwards.

6.6.1 Tuning `C`

We use cross-validation with the F1 score as the evaluation metric. Since the F1 score combines precision and recall, it is useful when we want to balance false positives and false negatives. In this example, we use it to select a model that balances the two types of classification errors when predicting passenger survival.

```
import numpy as np
from sklearn.model_selection import GridSearchCV

param_grid = {"logisticregression__C": np.logspace(-3, 3, 20)}
grid = GridSearchCV(logit, param_grid=param_grid, cv=5, scoring="f1")
grid.fit(X_train, y_train)
grid.best_params_
```

```
{'logisticregression__C': np.float64(2.976351441631316)}
```

6.6.2 Tuning the threshold

To tune the threshold, we use the helper function defined in Chapter 2.

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.metrics import confusion_matrix

def classification_summary(y_true, prob, threshold=0.5):
    pred = (prob >= threshold).astype(int)
    TN, FP, FN, TP = confusion_matrix(y_true, pred).ravel()

    return {
        "threshold": threshold,
        "TP": TP,
        "FP": FP,
        "FN": FN,
```

```

    "TN": TN,
    "accuracy": accuracy_score(y_true, pred),
    "precision": precision_score(y_true, pred, zero_division=0),
    "recall": recall_score(y_true, pred, zero_division=0),
    "f1": f1_score(y_true, pred, zero_division=0),
}

```

We may use the regular split-validation method. The model use the best C from the cross-validation from the previous section.

```

from sklearn.model_selection import train_test_split
import pandas as pd

X_model, X_valid, y_model, y_valid = train_test_split(
    X_train, y_train, test_size=0.25, stratify=y_train, random_state=1
)

best_C = grid.best_params_['logisticregression__C']

candidate_thresholds = np.linspace(0.05, 0.95, 91)

results = []
for t in candidate_thresholds:
    best_logit = make_pipeline(
        StandardScaler(),
        LogisticRegression(C=best_C, max_iter=5000),
    )
    best_logit.fit(X_model, y_model)
    valid_prob = best_logit.predict_proba(X_valid)[:, 1]
    results.append(classification_summary(y_valid, valid_prob, threshold=t))

threshold_df = pd.DataFrame(results)

best_row = threshold_df.loc[threshold_df["f1"].idxmax()] ①
best_row

```

① `best_row` is a Series object. Therefore although it is a row, it is displayed as a column.

```
threshold    0.360000
```

```
TP          39.000000
FP          17.000000
FN          12.000000
TN          57.000000
accuracy    0.768000
precision   0.696429
recall      0.764706
f1          0.728972
Name: 31, dtype: float64
```

We can fix the threshold and treat it as part of the prediction rule.

```
from sklearn.model_selection import FixedThresholdClassifier

final_logit = make_pipeline(
    StandardScaler(), LogisticRegression(C=best_C, max_iter=5000)
)
model_with_threshold = FixedThresholdClassifier(
    final_logit, threshold=best_row["threshold"]
)
model_with_threshold.fit(X_train, y_train)
model_with_threshold.score(X_test, y_test)
```

```
0.6465116279069767
```

i Note

Notice that changing the threshold does not change the fitted model. The estimated probabilities remain the same. The threshold only changes how those probabilities are converted into class labels.

6.7 Multiclass Extension

The basic logistic regression model handles binary outcomes. For problems with more than two classes, a common extension is multinomial logistic regression.

Suppose

$$Y \in 1, 2, \dots, K.$$

Instead of modeling the probability of a single class, multinomial logistic regression models the probability of every class simultaneously.

For each class k , define a linear predictor

$$\eta_k(x) = x^\top \beta_k = \beta_{0,k} + \beta_{1,k}x_1 + \dots + \beta_{p,k}x_p.$$

The predicted probability of class k is then given by the softmax function:

$$\Pr(Y = k \mid X = x) = \frac{\exp(\eta_k(x))}{\sum_{\ell=1}^K \exp(\eta_\ell(x))}.$$

The softmax function converts the collection of scores

$$\eta_1(x), \eta_2(x), \dots, \eta_K(x)$$

into probabilities that satisfy

$$0 \leq \Pr(Y = k \mid X = x) \leq 1$$

and

$$\sum_{k=1}^K \Pr(Y = k \mid X = x) = 1.$$

The softmax function plays the same role in multiclass classification that the sigmoid function plays in binary classification.

i Note

For $K=2$ classes, softmax reduces to the sigmoid model.

For prediction, the observation is assigned to the class with the largest predicted probability:

$$\hat{y} = \arg \max_k P(Y = k \mid X = x).$$

6.7.1 Cross-Entropy Loss

Suppose the true class labels are represented using one-hot encoding:

$$y_{ik} = \begin{cases} 1, & \text{if observation } i \text{ belongs to class } k, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$p_{ik} = P(Y_i = k \mid X_i = x_i)$$

denote the predicted probability from the softmax model.

The multiclass cross-entropy loss is

$$L = - \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log(p_{ik}).$$

Because exactly one value of y_{ik} equals 1 for each observation, this loss simply penalizes the model when it assigns a low probability to the true class.

Minimizing the multiclass cross-entropy loss is equivalent to maximizing the likelihood of the multinomial logistic regression model, just as binary cross-entropy corresponds to maximum likelihood estimation in ordinary logistic regression.

One-vs-Rest (OvR)

Another approach to multiclass classification is One-vs-Rest (OvR), also called One-vs-All.

Instead of fitting a single multinomial model, OvR trains one binary classifier for each class. For example, with three

classes A, B, and C, three separate models are fit:

- A versus not A
- B versus not B
- C versus not C

The final prediction is usually made using the classifier with the largest predicted probability.

Although OvR is simple and often works well in practice, multinomial logistic regression provides a more natural probabilistic model because all class probabilities are estimated simultaneously and automatically sum to one.

solver

There are many algorithms for fitting logistic regression models. In `scikit-learn`, these optimization algorithms are specified through the `solver` argument. Examples include `"lbfgs"`, `"newton-cg"`, `"newton-cholesky"`, `"sag"`, `"saga"`, and `"liblinear"`.

Different solvers use different optimization techniques and may have different computational properties. Some are better suited for large datasets, while others are designed for smaller problems or support additional features such as regularization.

Historically, some solvers handled multiclass classification using a One-vs-Rest (OvR) strategy, while others directly fit a multinomial logistic regression model using the soft-max formulation.

In this course, we focus on the multinomial (softmax) approach because it provides a natural extension of binary logistic regression and is widely used in modern machine learning. The details of optimization algorithms are beyond the scope of this course.

6.8 Example 2: Multinomial Logistic Regression

The Titanic example involved two classes. We now consider a multiclass classification problem using the wine dataset.

```
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split

wine = load_wine(as_frame=True)

X = wine.data[["alcohol", "flavanoids", "color_intensity"]]
y = wine.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, stratify=y, random_state=1
)
```

We fit a multinomial logistic regression model.

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

wine_logit = make_pipeline(
    StandardScaler(),
    LogisticRegression(max_iter=5000),
)

wine_logit.fit(X_train, y_train)

test_prob = wine_logit.predict_proba(X_test)
test_pred = wine_logit.predict(X_test)
```

6.8.1 Probabilities and Softmax

For multiclass problems, logistic regression uses the softmax formulation discussed earlier.

```
test_prob[:5]
```

```
array([[3.32849861e-02, 4.58338942e-02, 9.20881120e-01],
       [7.81045388e-02, 3.03513972e-02, 8.91544064e-01],
       [1.29661355e-02, 5.23981366e-05, 9.86981466e-01],
       [7.03100543e-03, 9.90470483e-01, 2.49851168e-03],
       [3.96520174e-02, 9.59921408e-01, 4.26574383e-04]])
```

Each row contains the estimated probabilities for all classes. The probabilities always sum to one.

```
test_prob[:5].sum(axis=1)
```

```
array([1., 1., 1., 1., 1.])
```

The predicted class is the class with the largest probability.

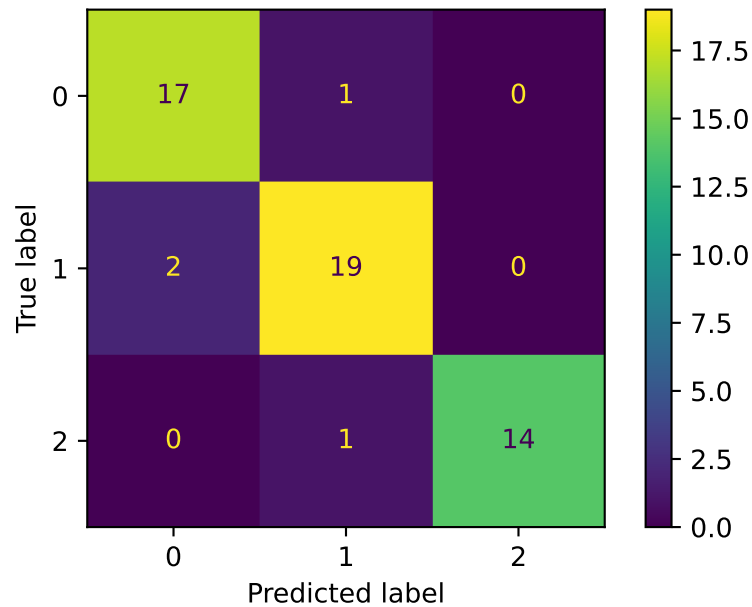
```
test_pred[:5]
```

```
array([2, 2, 2, 1, 1])
```

Unlike binary logistic regression, there is usually no threshold parameter to tune. The model simply predicts the class with the highest probability.

6.8.2 Confusion Matrix

```
from sklearn.metrics import ConfusionMatrixDisplay
ConfusionMatrixDisplay.from_predictions(y_test, test_pred)
```



We could also directly generate the classification report.

```
from sklearn.metrics import classification_report
print(classification_report(y_test, test_pred))
```

	precision	recall	f1-score	support
0	0.89	0.94	0.92	18
1	0.90	0.90	0.90	21
2	1.00	0.93	0.97	15
accuracy			0.93	54
macro avg	0.93	0.93	0.93	54
weighted avg	0.93	0.93	0.93	54

The report provides precision, recall, and F1 scores for each class separately.

i Note

For multiclass logistic regression, the regularization parameter C can be tuned in the same way as in the binary classification example.

However, there is usually no single threshold parameter to tune. The standard prediction rule is to choose the class with the largest predicted probability.

In this example, we skip the tuning step for C because the procedure is the same as before.

Part III

Nonparametric Supervised Learning

7 K-Nearest Neighbors

K-nearest neighbors (KNN) is a simple nonparametric prediction method. It does not estimate a fixed formula like linear regression. Instead, it predicts from nearby training observations. The method is local: points close to the new observation have influence, and distant points usually have none.

7.1 The KNN Model

KNN can be used for both regression and classification. The ideas are the same: use nearby observations to predict.

7.1.1 KNN Regression

Suppose the training data are

$$\{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^p, \quad y_i \in \mathbb{R}.$$

We want to estimate the regression function

$$f(x) = \mathbb{E}[Y \mid X = x].$$

Definition 7.1 (KNN regressor). For a new point x_0 , let $N_K(x_0)$ be the set of indices corresponding to the K training points closest to x_0 . The KNN regression estimate is

$$\hat{f}(x_0) = \frac{1}{K} \sum_{i \in N_K(x_0)} y_i.$$

That is, the predicted value is the average response among the K nearest neighbors of x_0 .

KNN regression estimates the conditional mean locally by averaging the responses of nearby observations.

Sometimes neighbors are assigned different weights. In that case, a weighted KNN regressor has the form

$$\hat{f}(x_0) = \sum_{i=1}^n w_i(x_0) y_i,$$

where

$$w_i(x_0) \geq 0, \quad \sum_{i=1}^n w_i(x_0) = 1.$$

Ordinary KNN regression uses

$$w_i(x_0) = \begin{cases} 1/K, & i \in N_K(x_0), \\ 0, & i \notin N_K(x_0). \end{cases}$$

7.1.2 KNN Classification

For classification, the response is a class label rather than a numerical response. Given a new point x_0 , KNN classification finds the K nearest training observations and predicts by majority vote.

Definition 7.2 (KNN classifier). For a new point x_0 , let $N_K(x_0)$ be the set of indices of the K nearest training observations. The KNN classifier predicts

$$\hat{G}(x_0) = \arg \max_g \sum_{i \in N_K(x_0)} I(y_i = g).$$

In words, KNN predicts the class receiving the largest number of votes among the K nearest neighbors.

For binary classification, KNN naturally provides a local estimate of the probability that an observation belongs to class 1:

$$\hat{p}(x_0) = \frac{1}{K} \sum_{i \in N_K(x_0)} I(y_i = 1).$$

Then predict class 1 if $\hat{p}(x_0)$ is larger than a chosen threshold, usually 0.5.

This estimate can be interpreted as the proportion of class-1 observations among the nearest neighbors. The threshold part is related to logistic regression.

i Note

Weighted KNN classification assigns larger weights to some neighbors, typically those closer to the query point. The predicted class is then determined by a weighted majority vote.

i Lazy learner

Based on the definition, KNN does not estimate model parameters during training. It mainly stores the training observations and waits until prediction time to use them. For this reason, KNN is often called a lazy learner.

The training phase is extremely simple because KNN does not estimate any model parameters. The algorithm mainly stores the training observations for later use. If the training data have n observations and p predictors, storing the data costs about $O(np)$.

The prediction cost can be much higher. For a single new observation, a brute-force KNN method computes its distance to all n training observations. This costs about $O(np)$. After computing the distances, the algorithm must identify the K nearest observations. A full ranking of all distances requires $O(n \log n)$ operations, although more efficient algorithms may reduce this cost. Taking all these into consideration, if we predict for m new observations,

the brute-force distance computation costs about $O(mnp)$. Thus, KNN is simple to train but can be relatively slow at prediction time, especially when the training set is large or the number of predictors is large.

i Comparison with Linear Models

Linear models are global models. A single fitted equation is used throughout the entire predictor space.

KNN is local. The prediction at x_0 is determined by nearby training observations.

Because of this difference:

- Linear models are simple and interpretable but may fail to capture complex nonlinear relationships.
- KNN can adapt to nonlinear patterns without specifying a functional form.

However, KNN may become unstable when nearby data are sparse or noisy, while linear models often provide more stable predictions.

7.1.3 Distance

The most common distance is Euclidean distance:

$$d(x, x') = \sqrt{\sum_{j=1}^p (x_j - x'_j)^2}.$$

KNN depends heavily on this distance. If one feature has a much larger scale than the others, it can dominate the distance calculation. This is why scaling is usually important.

Although Euclidean distance is the most common choice, other distances such as Manhattan distance can also be used.

i Scaling Matters

KNN relies entirely on distances between observations. If one predictor is measured on a much larger scale than another, it can dominate the distance calculation and effectively determine which observations are considered neighbors. For example, a variable measured in thousands may overwhelm another variable measured between 0 and 1. Therefore, KNN is usually applied after feature scaling, such as standardization. In practice, KNN is often combined with `StandardScaler()` in a pipeline.

7.1.4 The Role of K

The tuning parameter K controls the smoothness of the fitted model.

- Small K : uses very local information and can be sensitive to noise.
- Large K : averages over more observations and is more stable, but can be biased.

This is the usual bias-variance tradeoff. Smaller K produces a more flexible fit. Larger K produces a smoother fit.

In practice, K is typically chosen using cross-validation.

i Curse of Dimensionality

KNN can work well in low-dimensional problems, but it often struggles when the number of features is large. In high dimensions, data become sparse. As the number of features increases, the volume of the feature space grows rapidly. As a result, observations tend to be far apart from one another. Therefore local averaging becomes less meaningful. This is called the curse of dimensionality.

7.2 KNN in sklearn

`sklearn.neighbors` provides:

- `KNeighborsRegressor` for regression,
- `KNeighborsClassifier` for classification.

The estimator interface is the same as other `sklearn` models:

1. Create the model.
2. Fit the model using `.fit()`.
3. Predict using `.predict()`.
4. Usually coupled with `StandardScaler()` to create a pipeline.

The arguments here are straightforward. In most cases the default choice (other than `n_neighbors` of course) is good enough. For details please check the official documents.

7.3 Example 1: One-Dimensional Regression

We first simulate a one-dimensional regression problem based on a noisy sine curve.

```
import numpy as np
from sklearn.model_selection import train_test_split

rng = np.random.default_rng(1)

N = 100
X = rng.uniform(0, 2 * np.pi, size=N).reshape(-1, 1)
y = 2 * np.sin(X[:, 0]) + rng.normal(size=N)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=1)
```

We then generate a grid of values and apply the fitted model to the grid in order to estimate the underlying regression function.

```
X_grid = np.linspace(0, 2 * np.pi, 1001).reshape(-1, 1)
```

Fit a KNN regressor with $K = 3$.

```
from sklearn.neighbors import KNeighborsRegressor

knn = KNeighborsRegressor(n_neighbors=3)
knn.fit(X_train, y_train)

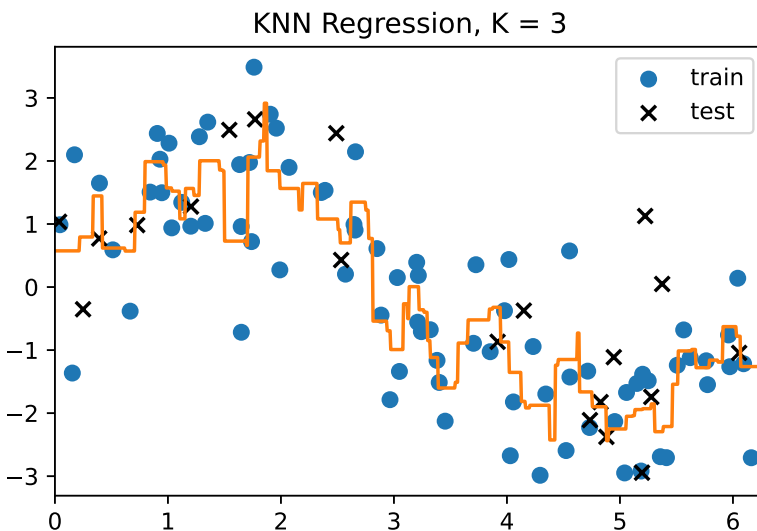
y_grid = knn.predict(X_grid)
```

Plot the fitted curve.

```
import matplotlib.pyplot as plt

plt.scatter(X_train.ravel(), y_train, s=40, label="train")
plt.scatter(X_test.ravel(), y_test, s=40, marker="x", color="black", label="test")
plt.plot(X_grid.ravel(), y_grid, color="C1")

plt.title("KNN Regression, K = 3")
plt.xlim(0, 2 * np.pi)
plt.legend();
```



i The dimension of X and X_{grid}

Please pay attention to the dimension of X and X_{grid} . We generate them as 2D arrays by `.reshape(-1, 1)`, and change them back to 1D array in plotting by `.ravel()`. Note that `.ravel()` is the same as `.reshape(-1)`. We choose it because it is shorter.

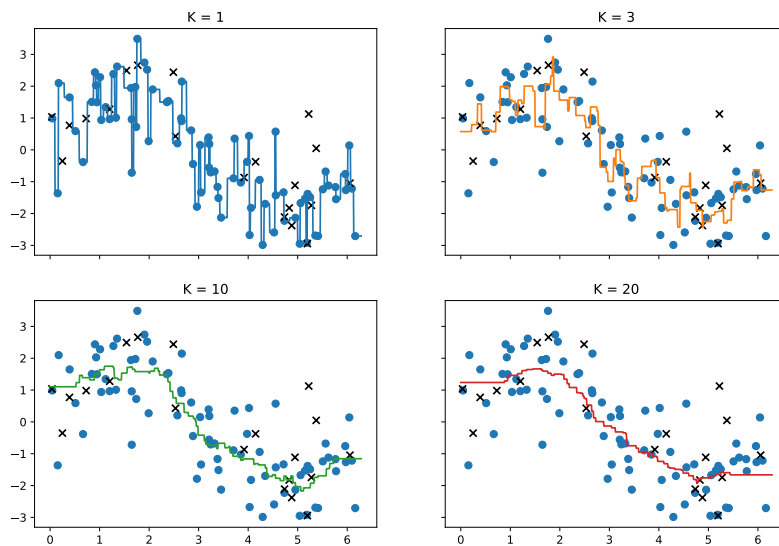
7.3.1 Comparing Different Values of K

The following plot compares several choices of K .

```
fig, axs = plt.subplots(2, 2, figsize=(12, 8), sharex=True, sharey=True)

for ax, k, color in zip(
    axs.ravel(),
    [1, 3, 10, 20],
    ["C0", "C1", "C2", "C3"],
):
    model = KNeighborsRegressor(n_neighbors=k)
    model.fit(X_train, y_train)
    pred = model.predict(X_grid)

    ax.scatter(X_train.ravel(), y_train, s=40, label="train")
    ax.scatter(X_test.ravel(), y_test, s=40, marker="x", color="black", label="test")
    ax.plot(X_grid.ravel(), pred, color=color)
    ax.set_title(f"K = {k}");
```

The fitted curve becomes smoother as K increases.

7.3.2 Choosing K by Cross-Validation

We now use cross-validation on the training set to select the value of K .

Since the training set contains 80 observations and we use $cv=5$, each training fold contains only 64 observations. Therefore we restrict the search range to values of K that are safely below this size.

```
from sklearn.model_selection import GridSearchCV

k_values = list(range(1, 61))

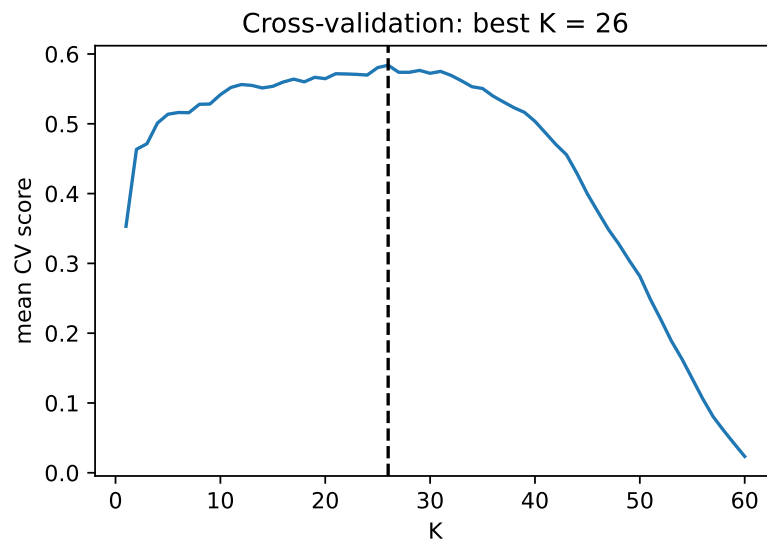
grid = GridSearchCV(KNeighborsRegressor(), param_grid={"n_neighbors": k_values}, cv=5)
grid.fit(X_train, y_train)

grid.best_params_
```

```
{'n_neighbors': 26}
```

Plot the cross-validation score.

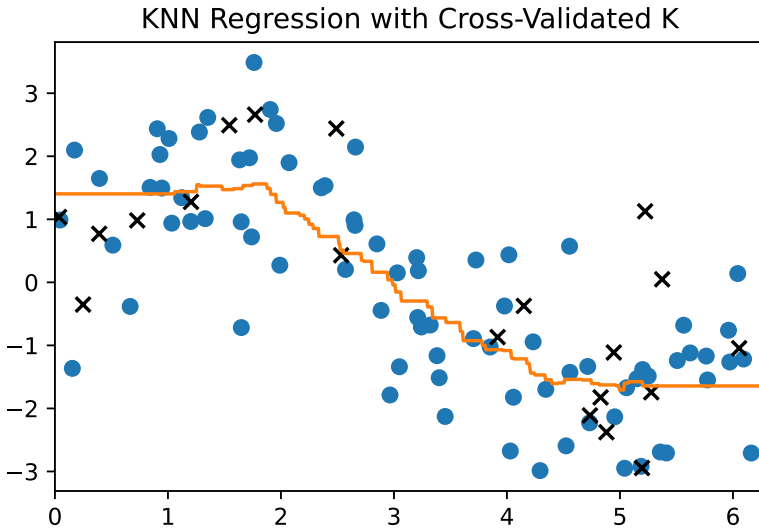
```
plt.plot(k_values, grid.cv_results_["mean_test_score"])
plt.axvline(
    x=grid.best_params_["n_neighbors"],
    color="black",
    linestyle="--",
)
plt.xlabel("K")
plt.ylabel("mean CV score")
plt.title(f"Cross-validation: best K = {grid.best_params_['n_neighbors']}");
```



Now plot the fitted curve using the selected value of K .

```
y_grid_best = grid.predict(X_grid)

plt.scatter(X_train.ravel(), y_train, s=40, label="train")
plt.scatter(X_test.ravel(), y_test, s=40, marker="x", color="black", label="test")
plt.plot(X_grid.ravel(), y_grid_best, color="C1")
plt.title("KNN Regression with Cross-Validated K")
plt.xlim(0, 2 * np.pi);
```



Finally, evaluate the selected model on the test set.

```
grid.score(X_test, y_test)
```

```
0.5552931849752354
```

7.4 Example: Iris Classification

We use the iris data from `sklearn.datasets`. To make visualization easy, we use only the first two features: sepal length and sepal width.

```
from sklearn import datasets
from sklearn.model_selection import train_test_split

iris = datasets.load_iris()

X = iris.data[:, :2]
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=1, stratify=y
)
```

Plot the training data.

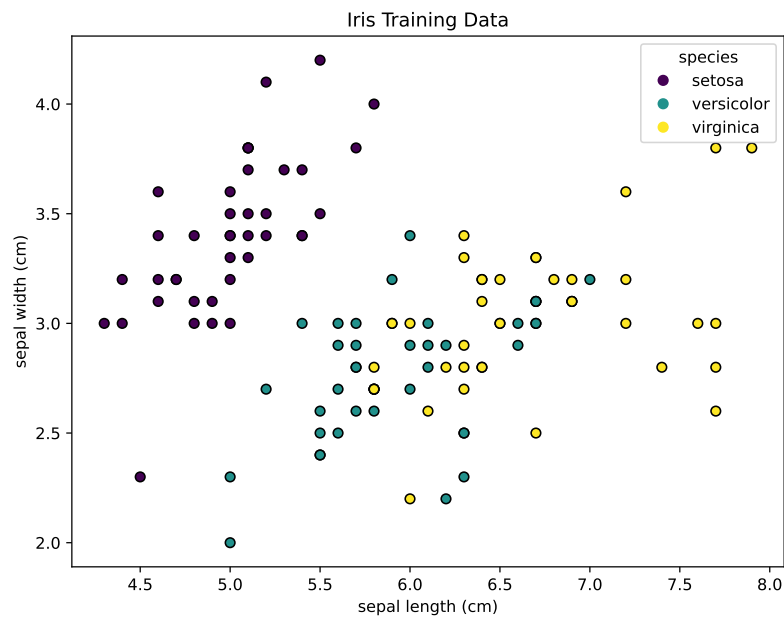
```
fig, ax = plt.subplots(figsize=(8, 6))

scatter = ax.scatter(
    X_train[:, 0],
    X_train[:, 1],
    c=y_train,
    edgecolor="k",
)

labels = ["setosa", "versicolor", "virginica"]
ax.legend(
    handles=scatter.legend_elements()[0],
    labels=labels,
    title="species",
)

ax.set_xlabel(iris.feature_names[0])
ax.set_ylabel(iris.feature_names[1])
ax.set_title("Iris Training Data")
```

```
Text(0.5, 1.0, 'Iris Training Data')
```



7.4.1 Pipeline

The usual KNN workflow includes scaling followed by the KNN model. A Pipeline keeps these steps together.

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier

pipe = make_pipeline(StandardScaler(), KNeighborsClassifier(n_neighbors=10))

pipe.fit(X_train, y_train)
y_pred = pipe.predict(X_test)
pipe.score(X_test, y_test)
```

0.7

7.4.2 Choosing K for Classification

To choose K , we tune `kneighborsclassifier__n_neighbors` inside the pipeline.

```

from sklearn.model_selection import GridSearchCV

n_values = list(range(1, 91))

param_grid = {"kneighborsclassifier__n_neighbors": n_values}

grid = GridSearchCV(pipe, param_grid=param_grid, cv=5, scoring="accuracy")

grid.fit(X_train, y_train)
grid.best_params_

```

```
{'kneighborsclassifier__n_neighbors': 5}
```

The best cross-validation score is:

```
grid.best_score_
```

```
np.float64(0.8583333333333332)
```

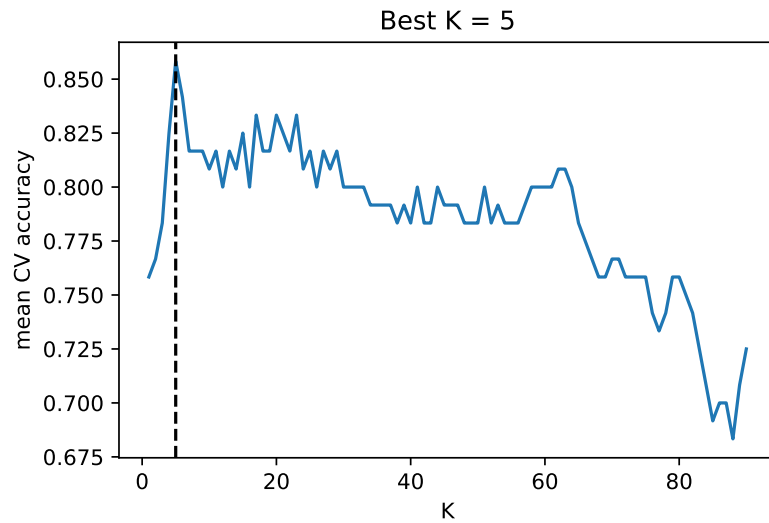
Plot the validation curve over different values of K .

```

plt.plot(n_values, grid.cv_results_["mean_test_score"])
plt.axvline(
    x=grid.best_params_["kneighborsclassifier__n_neighbors"],
    color="black",
    linestyle="--",
)
plt.xlabel("K")
plt.ylabel("mean CV accuracy")
plt.title(f"Best K = {grid.best_params_['kneighborsclassifier__n_neighbors']}")

```

```
Text(0.5, 1.0, 'Best K = 5')
```



7.4.3 Decision Boundary

We can visualize the decision boundary using `DecisionBoundaryDisplay`.

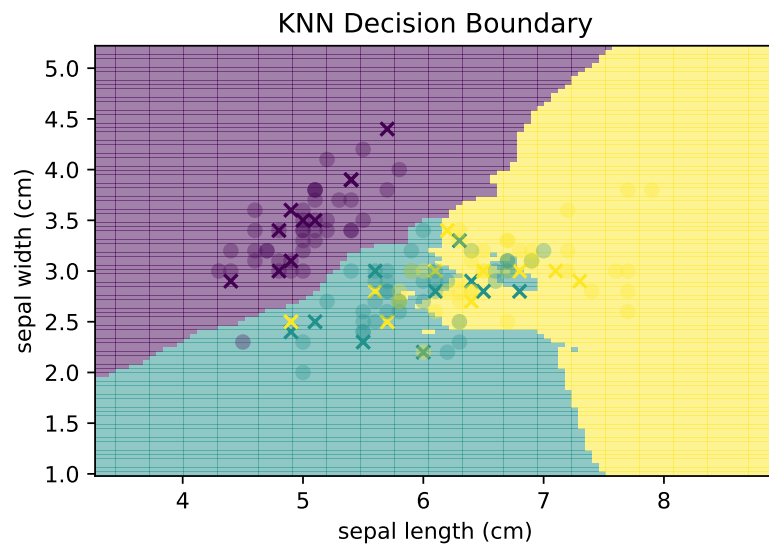
```
from sklearn.inspection import DecisionBoundaryDisplay

pipe.set_params(
    kneighborsclassifier__n_neighbors=grid.best_params_[
        "kneighborsclassifier__n_neighbors"
    ]
)
pipe.fit(X_train, y_train)

disp = DecisionBoundaryDisplay.from_estimator(
    pipe,
    X_train,
    response_method="predict",
    plot_method="pcolormesh",
    xlabel=iris.feature_names[0],
    ylabel=iris.feature_names[1],
    alpha=0.5,
)

disp.ax_.scatter(X_test[:, 0], X_test[:, 1], c=y_test, marker="x", label="test")
```

```
disp.ax_.scatter(X_train[:, 0], X_train[:, 1], c=y_train, alpha=0.3, label="train")  
plt.title("KNN Decision Boundary");
```



8 Kernel Smoothing

Kernel smoothing is a family of nonparametric methods for estimating smooth functions from data. Like KNN, it is based on local information: nearby observations should have more influence than distant observations.

The difference is that KNN uses a hard neighbor set, while kernel methods usually use smoothly decaying weights.

Note

This chapter focuses on **kernel smoothing**, including kernel density estimation and kernel regression. It does not cover kernel machines such as support vector machines or kernel ridge regression. These methods also use kernel functions, but their modeling goals are different.

Unlike many methods discussed earlier in this book, classical kernel smoothing is not fully integrated into the standard **sklearn** estimator framework. Although **sklearn** provides some related tools, such as **KernelDensity**, it does not provide a standard estimator for classical kernel regression.

In this chapter, we will use **KernelReg** from **statsmodels** to fit kernel regression models. For readers interested in the implementation details, we also include an optional section on building a custom regressor using the **sklearn** API and implementing the estimator from scratch. The goal is not only to understand how kernel regression works, but also to become familiar with practical tools from **statsmodels** and the basic structure of custom **sklearn** estimators.

8.1 From KNN to Kernel Weights

KNN regression predicts by averaging the K nearest responses. This gives equal weight to all K nearest neighbors and zero weight to every other observation. In other words, KNN is a weighted average with hard weights:

$$\hat{f}_{\text{KNN}}(x_0) = \sum_{i=1}^n w_i(x_0) y_i,$$

where

$$w_i(x_0) = \begin{cases} 1/K, & i \in N_K(x_0), \\ 0, & i \notin N_K(x_0). \end{cases}$$

Kernel smoothing keeps the same local-averaging idea but changes the weights. Instead of deciding whether a point is inside or outside the neighbor set, it assigns a smooth weight based on distance:

$$w_i(x_0) \propto K\left(\frac{x_0 - x_i}{h}\right).$$

Here $K(\cdot)$ is the kernel function and $h > 0$ is the bandwidth.

A kernel function converts distance into weight. It gives larger values to points near zero and smaller values to points far from zero. For density estimation, we usually require

$$K(u) \geq 0, \quad \int_{-\infty}^{\infty} K(u) du = 1.$$

The scaled kernel is

$$K_h(x, x_i) = \frac{1}{h} K\left(\frac{x - x_i}{h}\right).$$

The bandwidth h controls the distance scale. A small h makes the kernel narrow and local; a large h makes it wide and smooth.

Common choices include:

- Gaussian kernel: $K(u) = \frac{1}{\sqrt{2\pi}} \exp(-u^2/2)$.
- Uniform kernel: $K(u) = I(|u| \leq 1/2)$.
- Tricube kernel: $K(u) = (1 - |u|^3)^3 I(|u| \leq 1)$.

KNN fits are often discontinuous because the neighbor set can change abruptly as x_0 moves. Kernel smoothers, especially with smooth kernels such as the Gaussian kernel, usually produce smooth fitted curves.

In practice, the exact kernel shape is usually less important than the bandwidth. The bandwidth controls the main bias–variance tradeoff.

8.2 Kernel Density Estimation

Suppose a random variable has an unknown density function $f(x)$. Given a sample $\{x_1, \dots, x_n\}$, we would like to estimate this density from the observed data.

Kernel density estimation (KDE) is a nonparametric method for doing so. The kernel density estimator is

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^n K_h(x, x_i).$$

Equivalently,

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right).$$

The interpretation is simple: place a small smooth bump at every observation, add the bumps together, and divide by n .

The kernel function determines the shape of each bump, while the bandwidth h controls its width. As we will see, the bandwidth is usually the most important tuning parameter in kernel density estimation.

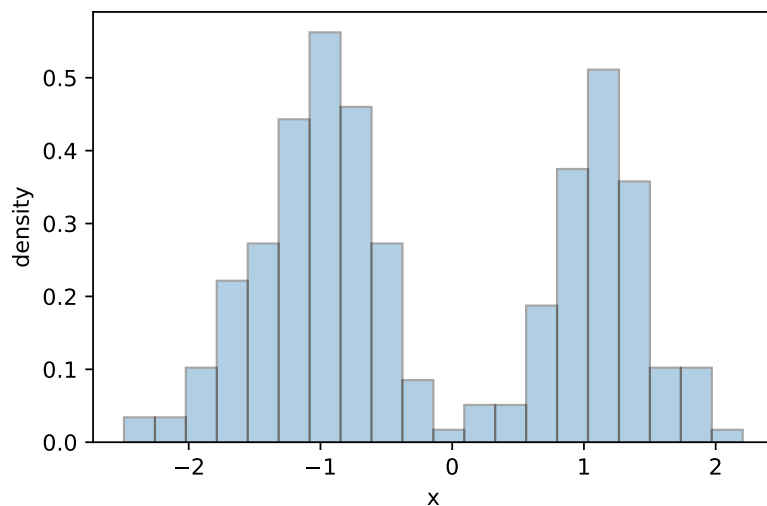
We demonstrate the idea with the following example. We use two normal distributions to generate a simple dataset. The dataset is `x`, and its histogram is displayed below.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1)

n1 = rng.normal(loc=-1.0, scale=0.55, size=150)
n2 = rng.normal(loc=1.2, scale=0.35, size=100)
x = np.concatenate([n1, n2])

plt.hist(x, bins=20, density=True, alpha=0.35, edgecolor="black")
plt.xlabel("x")
plt.ylabel("density");
```



8.2.1 Uniform Kernel and Histograms

A histogram estimates density by counting observations inside fixed bins. Kernel density estimation generalizes this idea by replacing fixed bins with a moving window.

The uniform kernel estimator is itself a valid kernel density estimator. Histograms and uniform KDE are closely related: histograms use fixed bins, while uniform KDE centers a window at each evaluation point and moves that window continuously across the domain.

Using the uniform kernel

$$K(u) = \begin{cases} 1, & |u| \leq 1/2, \\ 0, & \text{otherwise,} \end{cases}$$

the kernel density estimator becomes

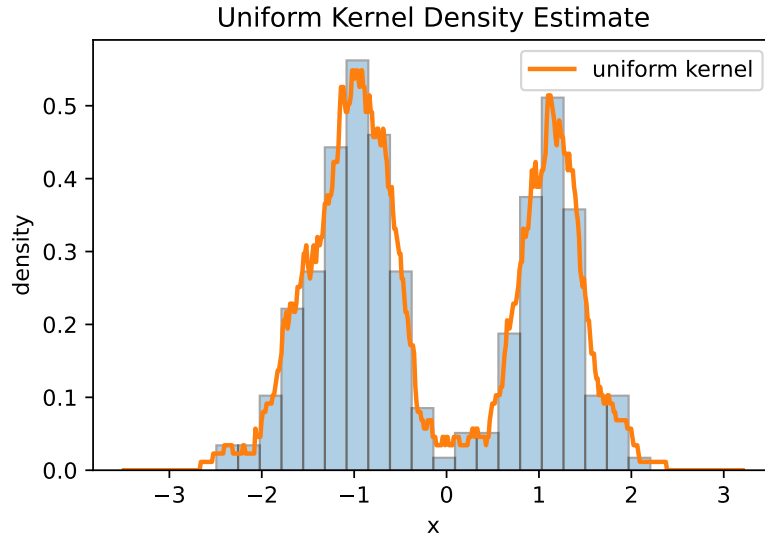
$$\hat{f}(x) = \frac{\#\{x_i : x_i \in [x - h/2, x + h/2]\}}{nh}.$$

This can be viewed as a moving histogram. Instead of counting observations in fixed bins, we center a window of width h at the evaluation point x and count the observations that fall inside the window. The value $h=0.35$ is chosen solely for illustration. In practice, selecting an appropriate bandwidth is an important problem and often requires careful study.

```
x_grid = np.linspace(x.min() - 1, x.max() + 1, 500)
h = 0.35

counts = [np.mean((x >= x0 - h / 2) & (x <= x0 + h / 2)) / h for x0 in x_grid]

plt.hist(x, bins=20, density=True, alpha=0.35, edgecolor="black")
plt.plot(x_grid, counts, color="C1", linewidth=2, label="uniform kernel")
plt.xlabel("x")
plt.ylabel("density")
plt.title("Uniform Kernel Density Estimate")
plt.legend();
```



8.2.2 Gaussian KDE

The uniform kernel gives equal weight to all observations inside the window and zero weight to observations outside. A Gaussian kernel uses a smoother weighting scheme in which nearby observations receive more weight than distant observations.

The Gaussian kernel is

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}.$$

Substituting this kernel into the kernel density estimator gives

$$\hat{f}(x) = \frac{1}{nh\sqrt{2\pi}} \sum_{i=1}^n \exp\left(-\frac{1}{2}\left(\frac{x-x_i}{h}\right)^2\right).$$

This estimator can be interpreted as placing a small Gaussian density centered at each observation and then summing the contributions from all observations. The bandwidth h controls the width of these Gaussian curves.

In practice, kernel density estimation is usually performed using software rather than by evaluating the formula directly.

`KernelDensity` from `sklearn.neighbors` provides an implementation of KDE with several kernel choices, including the Gaussian kernel.

```
from sklearn.neighbors import KernelDensity

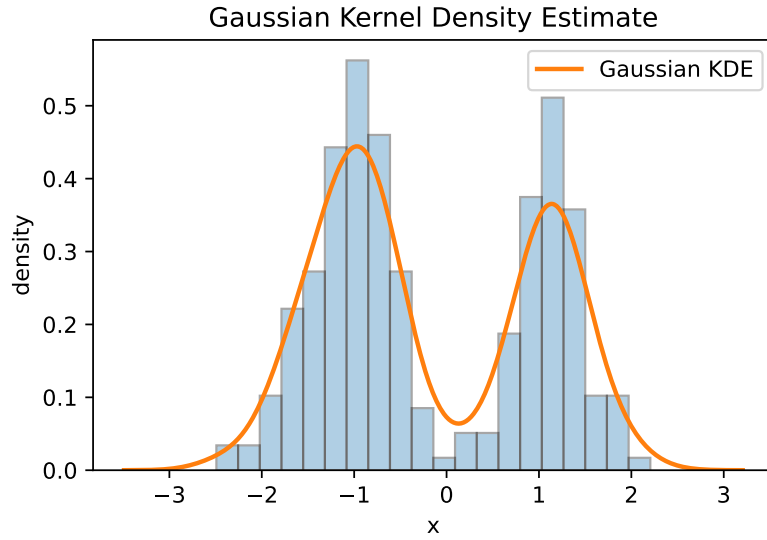
kde = KernelDensity(kernel="gaussian", bandwidth=0.25 * x.std())
kde.fit(x.reshape(-1, 1)) ①

log_density = kde.score_samples(x_grid.reshape(-1, 1))
density = np.exp(log_density)
```

- ① As with most `sklearn` estimators, `KernelDensity` expects a 2D predictor matrix. Therefore the 1D array `x` is reshaped to an $n \times 1$ matrix before fitting the model.

The method `.score_samples()` returns the logarithm of the estimated density rather than the density itself. This is done for numerical stability. Applying `np.exp()` converts the values back to the density scale.

```
plt.hist(x, bins=20, density=True, alpha=0.35, edgecolor="black")
plt.plot(x_grid, density, color="C1", linewidth=2, label="Gaussian KDE")
plt.xlabel("x")
plt.ylabel("density")
plt.title("Gaussian Kernel Density Estimate")
plt.legend();
```



8.2.3 Bias and Variance of KDE

To better understand why kernel density estimation works, let us examine the bias and variance of the estimator.

Assume $x_1, \dots, x_n$ are independent samples from a density f . The estimator is

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - x_i}{h}\right).$$

Theorem 8.1 (Bias of KDE). *Assume K is a symmetric kernel with finite second moment and f is twice continuously differentiable.*

$$\text{Bias} [\hat{f}(x)] = O(h^2).$$

Click for proof.

Since the observations are i.i.d., each term in the sum has the same expectation. Therefore

$$\begin{aligned}
E[\hat{f}(x)] &= \frac{1}{n} \sum_{i=1}^n E\left[\frac{1}{h} K\left(\frac{x-x_i}{h}\right)\right] \\
&= E\left[\frac{1}{h} K\left(\frac{x-X}{h}\right)\right] \\
&= \int \frac{1}{h} K\left(\frac{x-t}{h}\right) f(t) dt.
\end{aligned}$$

Let

$$u = \frac{x-t}{h}, \quad t = x - hu.$$

Then $dt = -hdu$. Therefore

$$\begin{aligned}
E[\hat{f}(x)] &= \int \frac{1}{h} K\left(\frac{x-t}{h}\right) f(t) dt \\
&= \int \frac{1}{h} K(u) f(x-hu) (-hdu) \\
&= \int K(u) f(x-hu) du.
\end{aligned}$$

Consider the Taylor series expansion

$$f(x-hu) = f(x) - huf'(x) + \frac{1}{2}h^2u^2f''(x) + O(h^3u^3).$$

We have

$$\begin{aligned}
E[\hat{f}(x)] &= \int K(u) \left[f(x) - huf'(x) + \frac{1}{2}h^2u^2f''(x) + O(h^3u^3) \right] du \\
&= \int K(u) f(x) du - \int K(u) huf'(x) du + \int \frac{1}{2}K(u) h^2u^2f''(x) du + \int K(u) O(h^3|u|^3) du \\
&= f(x) \int K(u) du - f'(x)h \int uK(u) du + \frac{1}{2}h^2f''(x) \int u^2K(u) du + O(h^3) \int |u|^3K(u) du.
\end{aligned}$$

Because a kernel function satisfies

$$\int K(u) du = 1,$$

and a symmetric kernel satisfies

$$\int uK(u)du = 0,$$

and

$$\int |u|^3 K(u)du < \infty,$$

we have

$$\begin{aligned} E[\hat{f}(x)] - f(x) &= \frac{1}{2}h^2 f''(x) \int u^2 K(u)du + O(h^3) \\ &= O(h^2). \end{aligned}$$

Then after plugged in, we have

$$\text{Bias} = E[\hat{f}(x)] - f(x) = O(h^2).$$

Note

The expectation of the KDE is a weighted average of nearby values of the true density. Therefore KDE estimates a smoothed version of the underlying density. In fact, the expectation can be viewed as a convolution of the density and a rescaled kernel.

Theorem 8.2 (Variance of KDE). *Assume f is continuous at x , and K satisfies $\int K^2(u) du < \infty$.*

$$\text{Var}[\hat{f}(x)] = O\left(\frac{1}{nh}\right).$$

Click for proof.

Since the observations are independent,

$$\text{Var}[\hat{f}(x)] = \frac{1}{n^2} \sum_{i=1}^n \text{Var} \left[\frac{1}{h} K \left(\frac{x - x_i}{h} \right) \right].$$

All terms have the same variance, so

$$\text{Var}[\hat{f}(x)] = \frac{1}{n} \text{Var} \left[\frac{1}{h} K \left(\frac{x-X}{h} \right) \right].$$

Using the definition of variance,

$$\text{Var} \left[\frac{1}{h} K \left(\frac{x-X}{h} \right) \right] = \text{E} \left[\frac{1}{h^2} K^2 \left(\frac{x-X}{h} \right) \right] - \left\{ \text{E} \left[\frac{1}{h} K \left(\frac{x-X}{h} \right) \right] \right\}^2.$$

For the first term,

$$\begin{aligned} \text{E} \left[\frac{1}{h^2} K^2 \left(\frac{x-X}{h} \right) \right] &= \int \frac{1}{h^2} K^2 \left(\frac{x-t}{h} \right) f(t) dt \\ &= \frac{1}{h} \int K^2(u) f(x-hu) du. \end{aligned}$$

When h is small and f is assumed smooth near x ,

$$f(x-hu) = f(x) + O(h|u|).$$

Therefore

$$\text{E} \left[\frac{1}{h^2} K^2 \left(\frac{x-X}{h} \right) \right] = \frac{f(x)}{h} \int K^2(u) du + \frac{O(h)}{h} \int |u| K^2(u) du.$$

Since $f(x)$ and $\int K^2(u) du$ are constants, then

$$\text{E} \left[\frac{1}{h^2} K^2 \left(\frac{x-X}{h} \right) \right] = O \left(\frac{1}{h} \right).$$

The second term is

$$\left\{ \text{E} \left[\frac{1}{h} K \left(\frac{x-X}{h} \right) \right] \right\}^2 = \left\{ \text{E} [\hat{f}(x)] \right\}^2.$$

From the bias calculation,

$$\text{E}[\hat{f}(x)] = f(x) + O(h^2),$$

so this squared term is

$$\begin{aligned}(f(x) + O(h^2))^2 &= f(x)^2 + 2f(x)O(h^2) + O(h^4) \\ &= f(x)^2 + O(h^2) \\ &= O(1).\end{aligned}$$

Hence

$$\text{Var} \left[\frac{1}{h} K \left(\frac{x - X}{h} \right) \right] = O\left(\frac{1}{h}\right) - O(1) = O\left(\frac{1}{h}\right).$$

Finally,

$$\text{Var}[\hat{f}(x)] = \frac{1}{n} O\left(\frac{1}{h}\right) = O\left(\frac{1}{nh}\right).$$

Therefore the pointwise mean squared error of KDE estimator satisfies

$$\text{MSE} [\hat{f}(x)] = \left(\text{Bias} [\hat{f}(x)] \right)^2 + \text{Var} [\hat{f}(x)] = O(h^4) + O\left(\frac{1}{nh}\right).$$

Reducing the bandwidth decreases the magnitude of the bias but increases the variance. Bandwidth selection therefore involves a bias-variance tradeoff.

i Note

Balancing the two dominant terms, $h^4 \asymp \frac{1}{nh}$, which implies $h \asymp n^{-1/5}$. This explains why many rule-of-thumb bandwidth formulas involve the factor $n^{\{-1/5\}}$.

i The Practical Impact of Bandwidth

The bandwidth controls the smoothness of the density estimate and has a much larger effect on the result than the choice of kernel.

- **Small Bandwidth (Low Bias / High Variance):** The estimate follows the observed data very closely and can capture fine-scale features. However, it is often sensitive to random sampling variation and may produce spurious peaks that do not correspond to genuine structure in the underlying distribution.
- **Large Bandwidth (High Bias / Low Variance):** The estimate is smoother and less sensitive to random fluctuations. However, excessive smoothing can obscure important characteristics of the distribution, such as skewness, heavy tails, or multiple modes.

As a result, the same dataset can produce substantially different density estimates when different bandwidth values are used.

For most practical applications, selecting an appropriate bandwidth is far more important than selecting a particular kernel function. Consequently, bandwidth selection is the central tuning problem in kernel density estimation.

8.2.4 Rule-of-Thumb Bandwidth Selection for KDE

Numerous bandwidth selection methods have been proposed.

The asymptotic bias-variance analysis above implies that an optimal bandwidth should decrease roughly at the rate $h = O(n^{-1/5})$.

For a Gaussian kernel, a widely used rule-of-thumb bandwidth is

$$h = 1.06 \hat{\sigma} n^{-1/5}.$$

This formula is derived under the assumption that the underlying density is approximately Gaussian and is commonly known as Silverman's rule of thumb.

The following Python code implements this rule.

```
h = 1.06 * np.std(x, ddof=1) * len(x) ** (-1 / 5)
```

Rule-of-thumb formulas are computationally inexpensive and often provide reasonable starting values. However, they rely on simplifying assumptions and may perform poorly when the underlying density differs substantially from those assumptions.

8.2.5 Bandwidth Selection by Visual Inspection

Bandwidth selection for KDE can be performed using cross-validation, plug-in methods, or other data-driven procedures.

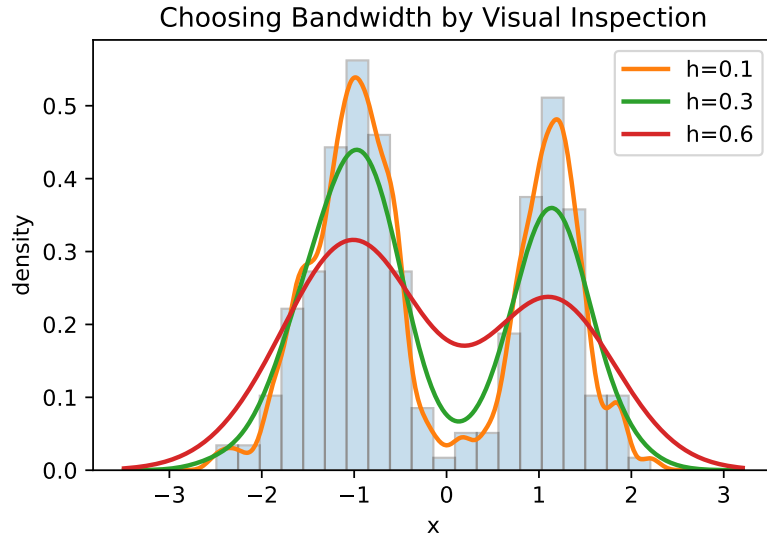
However, KDE is often used for exploratory visualization. In such settings, practitioners frequently inspect several nearby bandwidth values visually to assess the stability of the estimated density.

```
bandwidths = [0.1, 0.3, 0.6]

plt.hist(x, bins=20, density=True, alpha=0.25, edgecolor="black")

for bw in bandwidths:
    kde = KernelDensity(kernel="gaussian", bandwidth=bw)
    kde.fit(x.reshape(-1, 1))
    density = np.exp(kde.score_samples(x_grid.reshape(-1, 1)))
    plt.plot(x_grid, density, linewidth=2, label=f"h={bw}")

plt.xlabel("x")
plt.ylabel("density")
plt.title("Choosing Bandwidth by Visual Inspection")
plt.legend();
```



The figure compares KDE estimates with several bandwidth values.

- With $h=0.1$, the density estimate contains many small peaks and valleys. These fluctuations are likely caused by sampling noise rather than genuine features of the underlying distribution.
- With $h=0.6$, the estimate is much smoother, but some local features have disappeared.
- With $h=0.3$, the estimate preserves the main shape of the distribution while avoiding excessive noise.

A common visual strategy is to choose the smallest bandwidth that removes obvious noise while preserving important features such as peaks, valleys, skewness, or multiple modes.

8.3 Kernel Regression

Kernel regression estimates

$$f(x) = E[Y \mid X = x].$$

The most common kernel regression estimator is the Nadaraya-Watson estimator.

Definition 8.1 (Nadaraya-Watson estimator). Given training data (x_i, y_i) , the Nadaraya-Watson estimator is

$$\hat{f}(x) = \frac{\sum_{i=1}^n K_h(x, x_i) y_i}{\sum_{i=1}^n K_h(x, x_i)}.$$

This is a weighted average of the responses. The weights are

$$w_i(x) = \frac{K_h(x, x_i)}{\sum_{j=1}^n K_h(x, x_j)}.$$

Thus

$$\hat{f}(x) = \sum_{i=1}^n w_i(x) y_i, \quad \sum_{i=1}^n w_i(x) = 1.$$

For the Gaussian kernel,

$$K_h(x, x_i) = \frac{1}{h\sqrt{2\pi}} \exp\left\{-\frac{(x - x_i)^2}{2h^2}\right\}.$$

8.3.1 Comparing KNN and Kernel Regression

The following example compares a KNN regression curve with a Gaussian kernel regression curve on the same data.

We first generate a one-feature dataset and a grid for plotting predictions. The data are generated from

$$y = 2 \sin(x) + \varepsilon,$$

so the true regression function is $f(x) = 2 \sin(x)$. We will use this function as a reference when evaluating the fitted models.


```
rng = np.random.default_rng(1)

X = rng.uniform(0, 2 * np.pi, size=40)
y = 2 * np.sin(X) + rng.normal(size=len(X))
X_grid = np.linspace(0, 2 * np.pi, 500)
```

For comparison we fit a KNN regressor with `n_neighbors=10`. Since the purpose is to show the idea, for simplicity we do not split train-test or tune K .

```
from sklearn.neighbors import KNeighborsRegressor

knn = KNeighborsRegressor(n_neighbors=10)
knn.fit(X.reshape(-1, 1), y)
y_knn = knn.predict(X_grid.reshape(-1, 1))
```

Now we fit a Nadaraya-Watson estimator using `KernelReg` from `statsmodels`.

```
from statsmodels.nonparametric.kernel_regression import KernelReg

h = 0.25
kr = KernelReg(endog=y, exog=X, reg_type="lc", bw=[h], var_type="c")
y_kernel, _ = kr.fit(X_grid)
```

The bandwidth is chosen to be 0.25. This is only a simple starting value; bandwidth tuning is discussed later.

```
fig, axs = plt.subplots(1, 2, figsize=(12, 4), sharey=True)

axs[0].scatter(X, y, s=35, facecolors="none", edgecolors="black")
axs[0].plot(X_grid, 2 * np.sin(X_grid), color="C0", label="true function")
axs[0].plot(X_grid, y_knn, color="C1", label="KNN")
axs[0].set_title("KNN Regression")

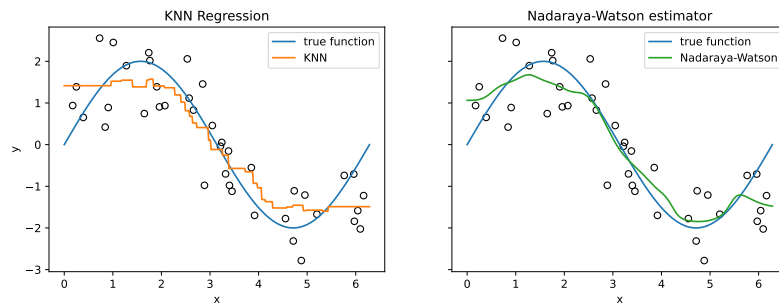
axs[1].scatter(X, y, s=35, facecolors="none", edgecolors="black")
axs[1].plot(X_grid, 2 * np.sin(X_grid), color="C0", label="true function")
axs[1].plot(X_grid, y_kernel, color="C2", label="Nadaraya-Watson")
axs[1].set_title("Nadaraya-Watson estimator")
```

```

for ax in axs:
    ax.set_xlabel("x")
    ax.legend()

axs[0].set_ylabel("y");

```



The KNN fit is piecewise constant because the prediction is the average of a fixed set of nearest neighbors. As the target point moves, the neighbor set may suddenly change, causing abrupt jumps in the fitted curve.

The Nadaraya-Watson estimator fit changes smoothly because every observation contributes to the prediction. The weights vary continuously with distance, producing a smooth estimate of the regression function.

8.3.2 Understanding KernelReg

The API of `KernelReg` from `statsmodels` differs substantially from the APIs used by most `sklearn` estimators, so we briefly explain the most important arguments and methods. More details can be found in the [official document](#).

8.3.2.1 Different fit

Unlike most `sklearn` estimators, `KernelReg` does not have a separate training step. The training data are supplied when the object is created, and the `fit()` method is used to evaluate the fitted regression function at new points. In other words, `y_hat, _ = kr.fit(X)` performs prediction at the input locations `X`.

- The first returned value is the estimated conditional mean.
- The second returned value contains estimated marginal effects.
- In this chapter, we only use the fitted values, so the marginal effects are discarded by assigning them to `_`.

8.3.2.2 endog and exog

The model is initialized using `endog=y` and `exog=X`.

- `endog`: endogenous variable
- `exog`: exogenous variable
- These terms come from econometrics. In a statistical learning context, you can simply think of them as `endog=y` and `exog=X`.

8.3.2.3 Regressor type

`reg_type="lc"` specifies the regression estimator.

- "lc" stands for local constant regression, which corresponds to the Nadaraya-Watson estimator introduced earlier.
- "ll" stands for local linear regression.
- The default is "ll".

8.3.2.4 Bandwidth

`bw` specifies the bandwidth.

- It can be supplied directly as a numeric value, as in this example.
- Explicit bandwidth values should be provided as a list. This design allows a separate bandwidth for each predictor. For one-dimensional regression, the list contains only one element.
- Alternatively, setting `bw="cv_ls"` instructs `KernelReg` to select the bandwidth automatically using least-squares cross-validation. This will be discussed later.

8.3.2.5 Variable type

`var_type` specifies the type of each predictor variable:

- "c": continuous
- "u": unordered discrete
- "o": ordered discrete

`var_type="c"` indicates that the predictor is continuous. For multiple predictors, `var_type` should contain one character for each predictor. For example, "ccc" indicates three continuous predictors.

8.3.3 Local Averaging at One Target Point

The key idea of kernel regression is local weighted averaging. In this section we use the previous example to visualize this behavior: at a target point x_0 , observations closer to x_0 receive larger kernel weights.

If we just want to predict the value at x_0 we can directly apply `KernelReg`. However it doesn't return all the weights. Therefore here we manually implement the algorithm in this section. Note that `y_kernel` is the one fitted with `KernelReg` from previous codes.

Recall that the Nadaraya-Watson estimate at x_0 is the weighted average

$$\hat{f}(x_0) = \sum_{i=1}^n w_i(x_0) y_i, \quad \text{where} \quad w_i(x_0) = \frac{K_h(x_0, x_i)}{\sum_{j=1}^n K_h(x_0, x_j)}.$$

with a Gaussian kernel,

$$K_h(x_0, x_i) = \frac{1}{h\sqrt{2\pi}} \exp \left\{ -\frac{(x_0 - x_i)^2}{2h^2} \right\}.$$

Because the Nadaraya-Watson estimator normalizes the weights, the constant $1/(h\sqrt{2\pi})$ cancels out. Therefore, for regression weights, it is enough to use

$$\text{raw_weights} = \exp \left\{ -\frac{(x_0 - x_i)^2}{2h^2} \right\}.$$

We first compute the kernel weights for the target point $x_0 = 3$ with $h = 0.25$.

```
x0 = 3.0
h = 0.25

raw_weights = np.exp(-0.5 * ((x0 - X) / h) ** 2)
fhat_x0 = np.sum(raw_weights * y) / np.sum(raw_weights)
```

We then use the weights to determine the size of each plotted point, so that larger circles correspond to larger kernel weights. The factor 1200 is chosen only for visualization.

```
point_sizes = 1200 * raw_weights / raw_weights.sum()
```

We can now plot the data together with the fitted curve, using the assigned point sizes to represent the kernel weights.

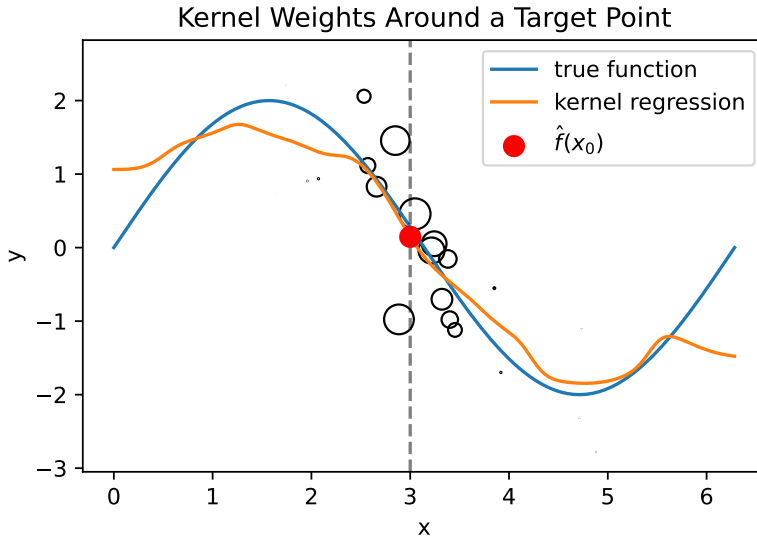
```
import matplotlib.pyplot as plt

plt.scatter(X, y, s=point_sizes, facecolors="none", edgecolors="black")
plt.plot(X_grid, 2 * np.sin(X_grid), color="C0", label="true function")
plt.plot(X_grid, y_kernel, color="C1", label="kernel regression")

plt.scatter([x0], [fhat_x0], color="red", s=80, zorder=3, label=r"$\hat{f}(x_0)$") ①
plt.axvline(x0, color="black", linestyle="--", alpha=0.5)

plt.xlabel("x")
plt.ylabel("y")
plt.title("Kernel Weights Around a Target Point")
plt.legend();
```

① We specifically plot the estimated point.



The circle sizes are proportional to the kernel weights. Points near x_0 receive larger weights and therefore have a greater influence on the local average. Points farther away receive smaller weights and contribute less to the estimate.

We may apply the same code to different target points to show the effects of different weights.

```
import matplotlib.pyplot as plt

x0_values = [1.0, 2.5, 4.0, 5.5]
h = 0.25

fig, axes = plt.subplots(2, 2, figsize=(10, 8))
axes = axes.ravel() ①

for ax, x0 in zip(axes, x0_values):
    raw_weights = np.exp(-0.5 * ((x0 - X) / h) ** 2)
    point_sizes = 1200 * raw_weights / raw_weights.sum()
    fhat_x0 = np.sum(raw_weights * y) / np.sum(raw_weights)

    ax.scatter(X, y, s=point_sizes, facecolors="none", edgecolors="black")

    ax.plot(X_grid, 2 * np.sin(X_grid), color="C0", label="true function")
    ax.plot(X_grid, y_kernel, color="C1", label="kernel regression")
```

```

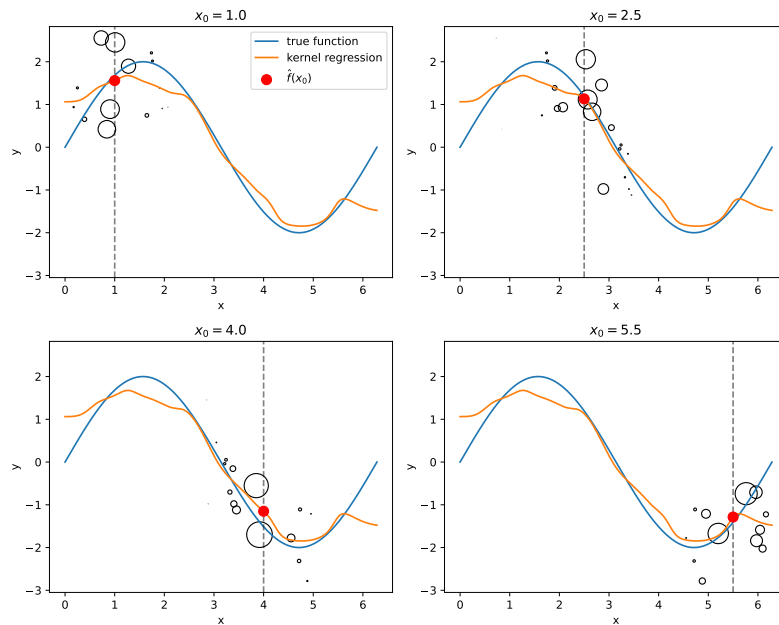
ax.scatter([x0], [fhat_x0], color="red", s=80, zorder=3, label=r"$\hat{f}(x_0)$")
ax.axvline(x0, color="black", linestyle="--", alpha=0.5)

ax.set_title(f"$x_0={x0}$")
ax.set_xlabel("x")
ax.set_ylabel("y")

axes[0].legend()
plt.tight_layout();

```

- ① The original `axes` object is a 2-by-2 array, so we would need to use `axes[0, 0]` to access the upper-left plot. After flattening it with `ravel()`, we can use `axes[0]`, `axes[1]`, and so on. This makes the loop simpler because we do not need to track the row and column indices explicitly.



8.3.4 Effect of Bandwidth in Kernel Regression

Bandwidth plays the same role in kernel regression as it does in kernel density estimation. A small bandwidth gives more weight to nearby observations and produces a more flexible fit,

while a large bandwidth averages over a wider neighborhood and produces a smoother fit.

Under suitable regularity conditions, it can be shown that kernel regression exhibits the same bias-variance tradeoff as KDE: decreasing the bandwidth reduces bias but increases variance, while increasing the bandwidth has the opposite effect. The exact formulas are more complicated than in KDE, so we focus on the visual behavior here.

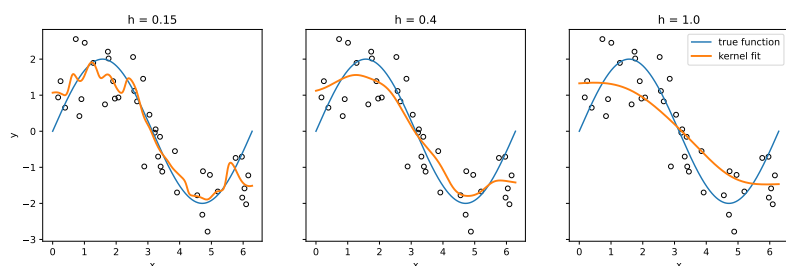
```
X_grid = np.linspace(0, 2 * np.pi, 500)

fig, axs = plt.subplots(1, 3, figsize=(14, 4), sharey=True)

for ax, bw in zip(axs, [0.15, 0.4, 1.0]):
    kr = KernelReg(endog=y, exog=X, bw=[bw], reg_type='lc', var_type='c')
    y_hat, _ = kr.fit(X_grid)

    ax.scatter(X, y, s=25, facecolors="none", edgecolors="black")
    ax.plot(X_grid, 2 * np.sin(X_grid), color="C0", label="true function")
    ax.plot(X_grid, y_hat, color="C1", linewidth=2, label="kernel fit")
    ax.set_title(f"h = {bw}")
    ax.set_xlabel("x")

axs[0].set_ylabel("y")
axs[-1].legend();
```



The figure illustrates the effect of bandwidth on the fitted curve.

- When the bandwidth is small ($h = 0.15$), the fit follows local fluctuations closely. This reduces smoothing bias but increases variability.

- With a moderate bandwidth ($h = 0.4$), the fit captures the overall trend while remaining reasonably smooth.
- When the bandwidth is large ($h = 1.0$), the fit becomes overly smooth and may miss important features of the underlying function.

As the bandwidth increases, variance decreases but bias increases. Choosing an appropriate bandwidth therefore requires balancing these two sources of error.

i Kernel Regression MSE

For one-dimensional kernel regression, the asymptotic bias is typically of order $O(h^2)$, while the variance is of order $O(\frac{1}{nh})$. Consequently, the mean squared error behaves roughly as $O(h^4) + O(\frac{1}{nh})$, which mirrors the KDE case. Therefore, although the exact formulas differ from those in KDE, the same bias-variance tradeoff leads to bandwidth selection rules of a similar form.

8.3.5 Bandwidth Selection by Cross-Validation

Rule-of-thumb methods are fast and easy to use, but they rely on assumptions that may not hold in practice. Since kernel regression is typically used for prediction, cross-validation is often preferred as a more data-driven approach to bandwidth selection.

`KernelReg` from `statsmodels` can automatically select the bandwidth by cross-validation when `bw="cv_ls"` is specified. In this case, leave-one-out cross-validation is used to search for the bandwidth that minimizes the prediction error.

The cross-validation procedure is performed automatically when the model is initialized. The selected bandwidth can then be inspected through the `bw` attribute.

```
from statsmodels.nonparametric.kernel_regression import KernelReg

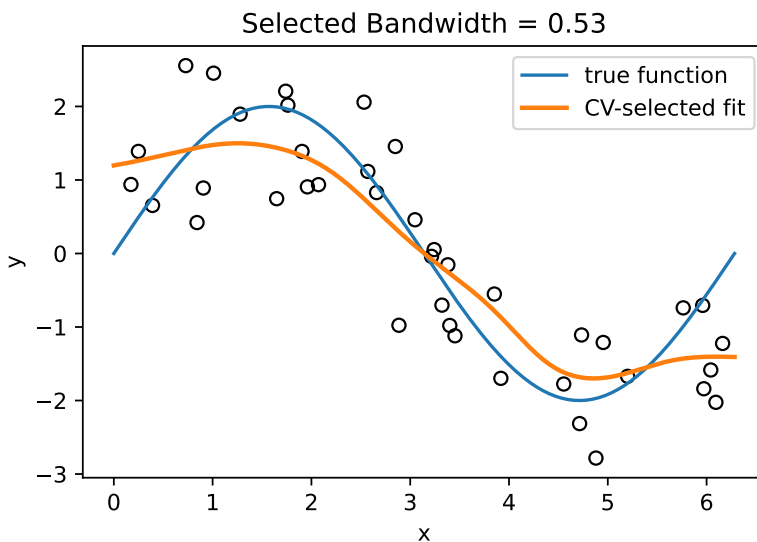
kr = KernelReg(endog=y, exog=X, var_type="c", reg_type="lc", bw="cv_ls")
kr.bw
```

```
array([0.53263357])
```

With the selected bandwidth, we can plot the fitted curve.

```
y_best, _ = kr.fit(X_grid)

plt.scatter(X, y, s=35, facecolors="none", edgecolors="black")
plt.plot(X_grid, 2 * np.sin(X_grid), color="C0", label="true function")
plt.plot(X_grid, y_best, color="C1", linewidth=2, label="CV-selected fit")
plt.xlabel("x")
plt.ylabel("y")
plt.title(f"Selected Bandwidth = {kr.bw.item():.2f}")
plt.legend();
```



8.4 Extensions

The basic Nadaraya-Watson estimator is a local constant estimator. It can be extended in several directions. Two common extensions are local linear/polynomial regression and LOWESS.

8.4.1 Local Linear Regression

Around each target point x_0 , local constant regression fits a weighted constant. Local linear regression instead fits a weighted line:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n K_h(x_0, x_i) [y_i - \beta_0 - \beta_1(x_i - x_0)]^2.$$

The predictors are centered at the target point x_0 . As a result, the fitted intercept directly estimates the regression function at that location.

$$\hat{f}(x_0) = \hat{\beta}_0.$$

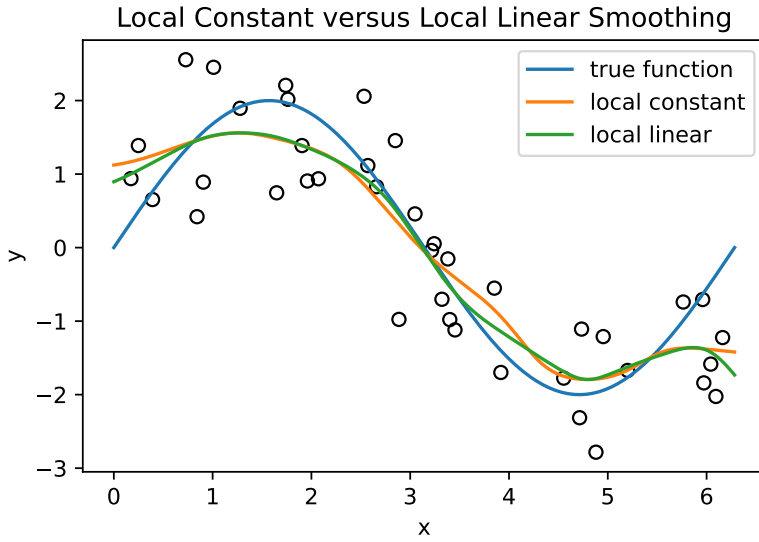
The `statsmodels` implementation `KernelReg` supports both local constant (Nadaraya-Watson) and local linear kernel regression through the `reg_type` parameter. Use `reg_type="lc"` for local constant regression and `reg_type="ll"` for local linear regression.

```
from statsmodels.nonparametric.kernel_regression import KernelReg

# Local constant regression
kr_lc = KernelReg(endog=y, exog=X, var_type="c", reg_type="lc", bw=[0.4])
y_lc, _ = kr_lc.fit(X_grid)

# Local linear regression
kr_ll = KernelReg(endog=y, exog=X, var_type="c", reg_type="ll", bw=[0.4])
y_ll, _ = kr_ll.fit(X_grid)

plt.scatter(X, y, s=35, facecolors="none", edgecolors="black")
plt.plot(X_grid, 2 * np.sin(X_grid), color="C0", label="true function")
plt.plot(X_grid, y_lc, color="C1", label="local constant")
plt.plot(X_grid, y_ll, color="C2", label="local linear")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Local Constant versus Local Linear Smoothing")
plt.legend();
```



Local linear regression often improves boundary behavior. Near the boundary of the data, a local average can be biased because observations are available on only one side of the target point. A local line can extrapolate slightly toward the boundary, reducing this additional bias.

8.4.2 Local Polynomial Regression

Local linear regression is a special case of local polynomial regression. At each target point x_0 , local polynomial regression fits

$$\min_{\beta_0, \dots, \beta_d} \sum_{i=1}^n K_h(x_0, x_i) \left[y_i - \beta_0 - \sum_{r=1}^d \beta_r (x_i - x_0)^r \right]^2.$$

As in local linear regression, centering the predictors at x_0 ensures that the fitted intercept corresponds to the estimated function value at the target point.

$$\hat{f}(x_0) = \hat{\beta}_0.$$

The degree d controls the local shape:

- $d = 0$: local constant regression (Nadaraya-Watson),
- $d = 1$: local linear regression,
- $d = 2$: local quadratic regression.

Higher-degree local polynomials can reduce bias because they capture more local curvature. However, they also have higher variance and may become unstable when the bandwidth is small or when only a few observations receive substantial kernel weight.

Although local polynomial regression can be formulated for any degree d , the `KernelReg` implementation in `statsmodels` provides only local constant and local linear smoothing. The `scikit-learn` library does not currently include local polynomial regression, although custom implementations can be built using the standard estimator API. We provide an example of such a custom implementation at the end of this chapter.

8.4.3 LOWESS

LOWESS, short for locally weighted scatterplot smoothing, is a local linear regression method that combines nearest-neighbor neighborhoods with distance-based weights.

For a target point x_0 , LOWESS first selects a neighborhood around x_0 . The neighborhood size is controlled by a span parameter, usually denoted by α . If the dataset contains n observations, the closest $m = \lceil \alpha n \rceil$ points are used in the local fit and form the local set $N(x_0)$.

Let

$$d(x_0) = \max_{i \in N(x_0)} |x_i - x_0|$$

be the distance from x_0 to the farthest point in the neighborhood. LOWESS assigns distance weights using the tricube kernel:

$$w_i = \begin{cases} \left(1 - \left|\frac{x_i - x_0}{d(x_0)}\right|^3\right)^3, & |x_i - x_0| < d(x_0), \\ 0, & \text{otherwise.} \end{cases}$$

At x_0 , LOWESS fits a weighted local linear model:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n w_i [y_i - \beta_0 - \beta_1(x_i - x_0)]^2.$$

The fitted value is the fitted intercept:

$$\hat{f}(x_0) = \hat{\beta}_0.$$

Thus, LOWESS can be viewed as a local linear regression method that uses a nearest-neighbor span instead of a fixed bandwidth. The LOWESS span can therefore be interpreted as an adaptive bandwidth: it fixes the number of nearby observations, so the corresponding distance scale changes with the local data density.

i LOWESS vs. kernel regression

LOWESS can be viewed as a variant of local linear kernel smoothing. Both methods fit a weighted local line around each target point.

The main difference is how the neighborhood is defined: local linear kernel regression uses a fixed bandwidth, whereas LOWESS uses a fixed fraction of nearby observations, leading to an adaptive bandwidth. Therefore, LOWESS can be viewed as a local linear smoother with an adaptive neighborhood size.

8.4.3.1 LOWESS in Python

In Python, LOWESS is available from `statsmodels.nonparametric.smoothers_lowess`.

```
from statsmodels.nonparametric.smoothers_lowess import lowess

lowess_result = lowess(endog=y, exog=X, frac=0.25, it=0, return_sorted=True)
```

Similar to `KernelReg`, unlike an estimator object, `lowess` directly computes and returns the smoothed values in a single

function call. The argument `return_sorted` controls the format of the output.

- If `return_sorted=True`, the output is a two-column NumPy array. The first column contains the sorted `X` values, and the second column contains the corresponding fitted values.
- If `return_sorted=False`, the output is a one-dimensional NumPy array of fitted values in the original order of `X`.
- If `xvals` is provided, LOWESS evaluates the fitted smoother at those values. In this case, `return_sorted` is ignored, and the returned array is always one-dimensional.
- `frac` is the span parameter α . This is the most important hyperparameter.

i LOWESS is often used for visualization

LOWESS was originally developed as a smoothing method for exploratory data analysis and visualization. Consequently, many software implementations emphasize plotting and curve estimation rather than predictive modeling. This is why the default output uses `return_sorted=True`. When `return_sorted=True`, the returned x -values are sorted from low to high, so the fitted curve can be plotted directly.

Like kernel regression, LOWESS can also be used for prediction by evaluating the fitted smoother at new values through the `xvals` argument. However, in practice LOWESS is most commonly used for visualizing trends and exploring nonlinear relationships rather than for building predictive models.

8.4.3.2 Robust LOWESS

LOWESS can also perform robustification iterations. After an initial fit, observations with large residuals receive smaller robustness weights. The final fit uses both distance-based weights and robustness weights, so outliers have less influence on the fitted curve.

In `statsmodels`, the argument `it` specifies the number of robustification iterations.

Although `it` can take any nonnegative integer value, most applications use either `it=0` for ordinary LOWESS or `it=3` for robust LOWESS. Note that the original LOWESS paper recommended three robustness iterations, which is why `it=3` became the common default in many software packages.

So usually we start from setting `it=0`, and then try `it=3` to see whether it is better.

We fit both ordinary LOWESS (`it=0`) and robust LOWESS (`it=3`) using the same value of `frac`. Because `return_sorted=True`, each result contains the sorted predictor values in the first column and the corresponding fitted values in the second column.

```
from statsmodels.nonparametric.smoothers_lowess import lowess

lowess_standard = lowess(
    endog=y, exog=X, frac=0.25, it=0, return_sorted=True
)

lowess_robust = lowess(
    endog=y, exog=X, frac=0.25, it=3, return_sorted=True
)
```

We then plot the two fitted curves together for comparison.

```
import matplotlib.pyplot as plt

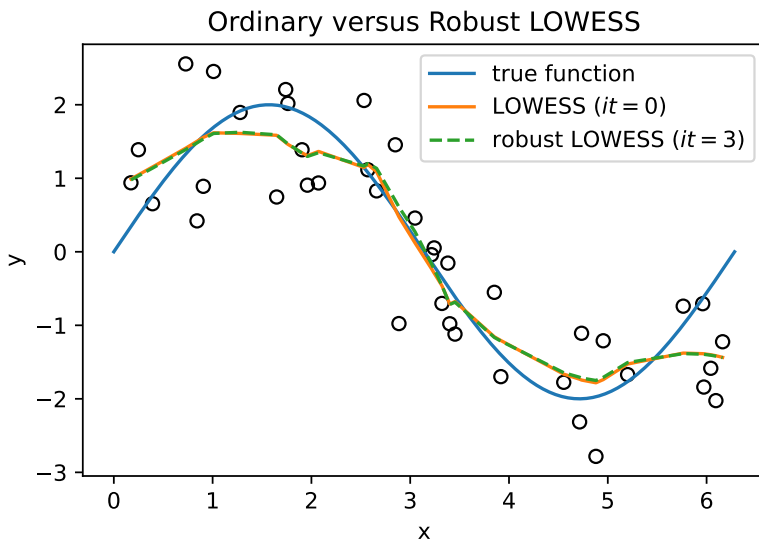
plt.scatter(X, y, s=35, facecolors="none", edgecolors="black")
plt.plot(X_grid, 2 * np.sin(X_grid), label="true function")
plt.plot(
    lowess_standard[:, 0],
    lowess_standard[:, 1],
    label="LOWESS ($it=0$)",
)
plt.plot(
    lowess_robust[:, 0],
    lowess_robust[:, 1],
```



```

        linestyle="--",
        label="robust LOWESS ($it=3$)",
    )
plt.xlabel("x")
plt.ylabel("y")
plt.title("Ordinary versus Robust LOWESS")
plt.legend()

```



The two fitted curves are quite similar because this dataset does not contain any strong outliers. To make the effect of robustification easier to see, we manually add a few outliers to a copy of the response values.

```

y_outlier = y.copy()
outlier_indices = np.argsort(X)[[8, 20, 32]]
y_outlier[outlier_indices] += [5, -6, 5]

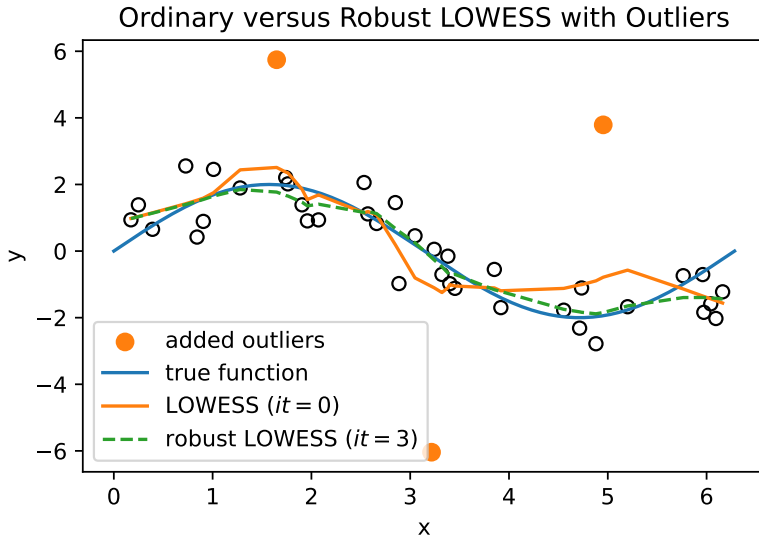
lowess_standard_outlier = lowess(
    endog=y_outlier, exog=X, frac=0.25, it=0, return_sorted=True
)

lowess_robust_outlier = lowess(
    endog=y_outlier, exog=X, frac=0.25, it=3, return_sorted=True
)

```

We now compare ordinary and robust LOWESS on the data with outliers.

```
plt.scatter(X, y_outlier, s=35, facecolors="none", edgecolors="black")
plt.scatter(
    X[outlier_indices],
    y_outlier[outlier_indices],
    s=55,
    color="C1",
    label="added outliers",
    zorder=3,
)
plt.plot(X_grid, 2 * np.sin(X_grid), label="true function")
plt.plot(
    lowess_standard_outlier[:, 0],
    lowess_standard_outlier[:, 1],
    label="LOWESS ($it=0$)",
)
plt.plot(
    lowess_robust_outlier[:, 0],
    lowess_robust_outlier[:, 1],
    linestyle="--",
    label="robust LOWESS ($it=3$)",
)
plt.xlabel("x")
plt.ylabel("y")
plt.title("Ordinary versus Robust LOWESS with Outliers")
plt.legend();
```



8.4.3.3 Choosing frac

The parameter `frac` controls the amount of smoothing.

- Smaller `frac` uses fewer nearby observations and produces a more flexible curve.
- Larger `frac` uses more nearby observations and produces a smoother curve.

The span parameter `frac` can be selected by cross-validation when prediction accuracy is the main goal. However, LOWESS is often used as an exploratory visualization tool, so in practice `frac` is frequently chosen by comparing several values visually.

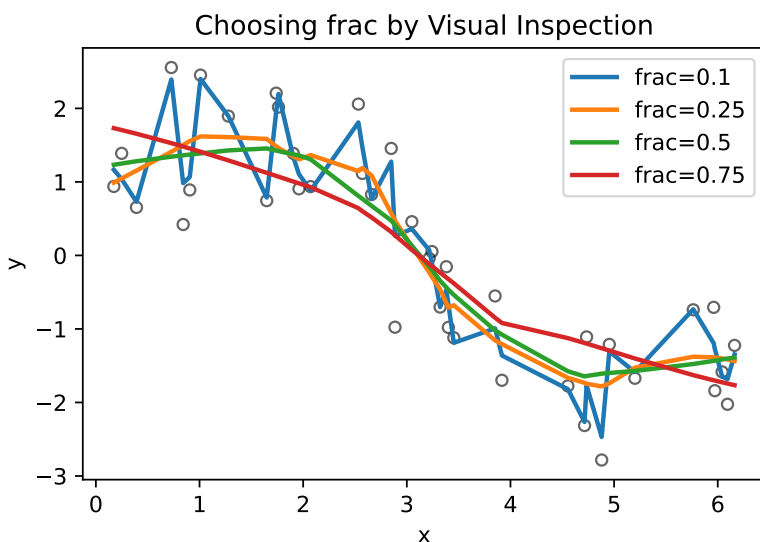
```
from statsmodels.nonparametric.smoothers_lowess import lowess

fracs = [0.1, 0.25, 0.5, 0.75]

plt.scatter(X, y, s=25, facecolors="none", edgecolors="black", alpha=0.6)

for frac in fracs:
    result = lowess(endog=y, exog=X, frac=frac, it=0, return_sorted=True)
    plt.plot(result[:, 0], result[:, 1], linewidth=2, label=f"frac={frac}")
```

```
plt.xlabel("x")
plt.ylabel("y")
plt.title("Choosing frac by Visual Inspection")
plt.legend()
```



Smaller values of `frac` produce more flexible curves that may follow noise, while larger values produce smoother curves that may miss local structure. The goal is to find a value that captures the overall pattern without introducing excessive variability.

The curve with `frac=0.1` follows many small fluctuations and appears to be fitting noise. The curve with `frac=0.75` is heavily smoothed and begins to miss local features of the data. Values between 0.25 and 0.5 provide a reasonable compromise, preserving the overall trend while avoiding excessive variability.

i Visual bandwidth selection

Kernel smoothing methods involve a smoothing parameter:

- bandwidth h in kernel density estimation and kernel regression;
- span `frac` in LOWESS.

Smaller values produce more flexible estimates with lower

bias but higher variance. Larger values produce smoother estimates with higher bias but lower variance.

When visualization is the primary goal, a common strategy is to compare several values and choose the smallest amount of smoothing that removes obvious noise while preserving the main structure of the data.

8.4.4 Multiple Dimensions

For $x_i \in \mathbb{R}^p$, a common Gaussian kernel is

$$K_h(x, x_i) \propto \exp\left(-\frac{\|x - x_i\|^2}{2h^2}\right).$$

This is similar to KNN in that it relies on distances in feature space. Therefore, scaling matters.

In multiple dimensions, kernel smoothing faces the curse of dimensionality. As p increases, local neighborhoods become sparse. To get enough nearby points, the bandwidth must grow, which increases bias.

For p dimensions, bandwidth rates are often of order

$$n^{-1/(p+4)}.$$

This gets slower as p increases, which is another way to see the curse of dimensionality.

Bandwidth conventions

Different software packages may define bandwidth differently. Some use a raw scale h inside the kernel. Others use a span, a nearest-neighbor fraction, or an internally rescaled bandwidth. Always check the documentation before comparing bandwidth values across implementations.

8.5 Optional: Building a Custom sklearn Regressor

Click to expand.

8.5.1 Nadaraya-Watson estimator

Recall the formula for the Nadaraya-Watson estimator:

$$\hat{f}(x) = \frac{\sum_{i=1}^n K_h(x, x_i) y_i}{\sum_{i=1}^n K_h(x, x_i)}.$$

We can translate this formula directly into a Python function.

```
def kernel_regression_predict(X, y, X_new, bandwidth):
    diff = X_new[:, None] - X[None, :]
    weights = np.exp(-0.5 * (diff / bandwidth) ** 2)
    weights = weights / weights.sum(axis=1, keepdims=True)
    return weights @ y

h = 0.25
y_kernel = kernel_regression_predict(X, y, X_grid, bandwidth=h)
```

Here `diff = X_new[:, None] - X[None, :]` uses numpy broadcasting to create a matrix of pairwise differences. It was briefly discussed in Section C.3.3. If `X_new` contains m points and `X` contains n points, then `diff` has shape (m, n) , where the (i, j) entry is the difference between the i th prediction point and the j th training point.

This simple function is useful for understanding the algorithm. We now turn it into a custom `sklearn` regressor so that it can be used with tools such as `GridSearchCV`.

8.5.2 Guidelines for Developing a Custom Scikit-learn Regressor

According to the [official sklearn developer guide](#), a custom regression estimator should inherit from both `RegressorMixin`

and `BaseEstimator` to integrate smoothly with the `sklearn` ecosystem. A minimal regressor typically implements three methods: `__init__`, `fit`, and `predict`.

1. The Role of `BaseEstimator`

`BaseEstimator` provides the infrastructure that allows an estimator to work with `sklearn` utilities such as pipelines, model selection tools, and cloning.

- It automatically implements `.get_params()` and `.set_params()`. These methods are required by tools such as `GridSearchCV`, `RandomizedSearchCV`, and `Pipeline`.
- The `__init__` method should only store the supplied parameters. Every constructor argument should be assigned directly to an attribute with the same name, such as `self.bandwidth = bandwidth`.
- The constructor should not
 - modify parameter values,
 - perform input validation,
 - perform data-dependent computations,
 - store training data.
- Data validation and data-dependent computations should be performed inside `fit()` or `predict()`.

By following these rules, the estimator automatically supports:

- cloning via `sklearn.base.clone`,
- parameter inspection,
- compatibility with pipelines and model-selection tools.

2. The Role of `RegressorMixin`

- `RegressorMixin` identifies the estimator as a regression model and provides a default scoring method.
- The mixin automatically implements the `.score()` method, which returns the coefficient of determination, or R^2 , by default.

- A regressor should implement `.fit(X, y)` and `.predict(X)`, where `y` is usually a numerical target variable.
- Attributes estimated from the training data should end with a trailing underscore, such as `self.coef_`, `self.intercept_`, or `self.X_train_`. This convention distinguishes learned quantities from hyperparameters.
- The `fit()` method should always return the estimator itself. This allows method chaining such as `model.fit(X, y).predict(X_new)`.

3. Recommended Practices

`sklearn` provides `validate_data` and `check_is_fitted` from `sklearn.utils.validation` to perform input validation and fitted-state checks consistently with built-in estimators.

Although these utilities are not required for simple educational examples, they are recommended for production-quality estimators.

In addition, it is recommended to validate all arguments besides the input dataset. For example, the bandwidth in our case should be positive.

8.5.3 `sklearn` regressor

```
from sklearn.base import BaseEstimator, RegressorMixin
from sklearn.utils.validation import validate_data, check_is_fitted

class NadarayaWatsonRegressor(RegressorMixin, BaseEstimator):
    def __init__(self, bandwidth=1.0):
        self.bandwidth = bandwidth

    def fit(self, X, y):
        if self.bandwidth <= 0:
            raise ValueError("bandwidth must be positive.")
        X, y = validate_data(self, X, y, y_numeric=True)
        self.X_train_ = X
        self.y_train_ = y
```



```

    return self

def predict(self, X_new):
    check_is_fitted(self, ["X_train_", "y_train_"])
    X_new = validate_data(self, X_new, reset=False)

    diff = X_new[:, None, :] - self.X_train_[None, :, :]
    squared_distances = np.sum(diff**2, axis=2)

    weights = np.exp(-0.5 * squared_distances / self.bandwidth**2)
    weights = weights / weights.sum(axis=1, keepdims=True)

    return weights @ self.y_train_

```

This `predict` method is closely related to the `kernel_regression_predict` function defined earlier. The main difference, apart from the validation checks, is that this version supports multidimensional input. That is, `X` may contain more than one feature column.

8.5.4 Grid search

With the custom regressor, we can seamlessly connect it to the `sklearn` ecosystem. For example, we can start to perform grid search.

```

from sklearn.model_selection import GridSearchCV

bandwidth_grid = {"bandwidth": np.linspace(0.1, 1.2, 23)}

grid = GridSearchCV(NadarayaWatsonRegressor(), param_grid=bandwidth_grid, cv=5)

grid.fit(X.reshape(-1, 1), y)
grid.best_params_

```

```
{'bandwidth': np.float64(0.5499999999999999)}
```

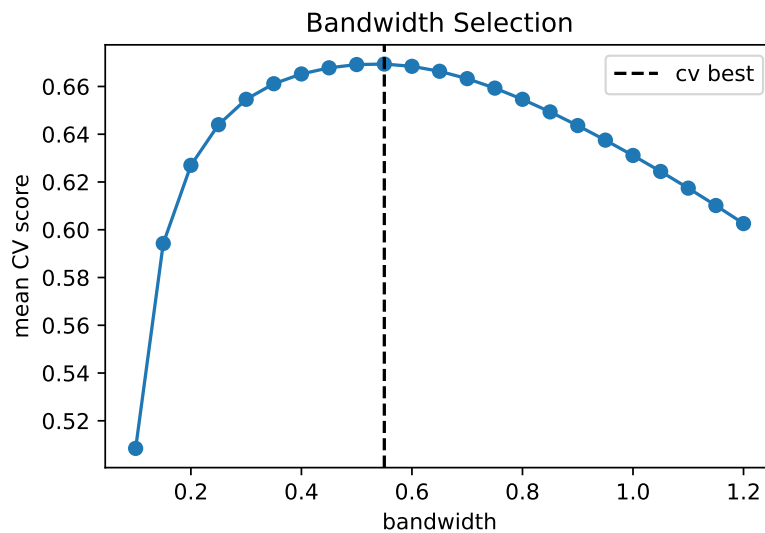
Plot the cross-validation curve.

```

cv_score = grid.cv_results_["mean_test_score"]
bandwidth_values = bandwidth_grid["bandwidth"]

plt.plot(bandwidth_values, cv_score, marker="o")
plt.axvline(grid.best_params_["bandwidth"], color="black", linestyle="--", label='cv best')
plt.xlabel("bandwidth")
plt.ylabel("mean CV score")
plt.title("Bandwidth Selection")
plt.legend();

```



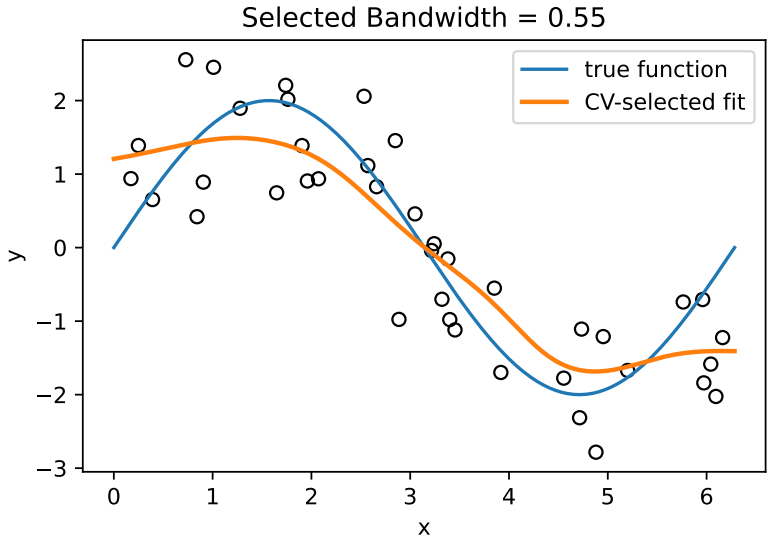
Fit the selected model.

```

best_kernel = grid.best_estimator_
y_best = best_kernel.predict(X_grid.reshape(-1, 1))

plt.scatter(X, y, s=35, facecolors="none", edgecolors="black")
plt.plot(X_grid, 2 * np.sin(X_grid), color="C0", label="true function")
plt.plot(X_grid, y_best, color="C1", linewidth=2, label="CV-selected fit")
plt.xlabel("x")
plt.ylabel("y")
plt.title(f"Selected Bandwidth = {grid.best_params_['bandwidth']:.2f}")
plt.legend();

```



9 Tree-Based Methods

A decision tree is another nonparametric method, which learns a collection of simple if-then rules directly from the data.

The basic idea is:

1. Split the feature space into regions.
2. Continue splitting each region into smaller regions.
3. Use the observations inside each final region to make predictions.

9.1 Basic Tree Terminology

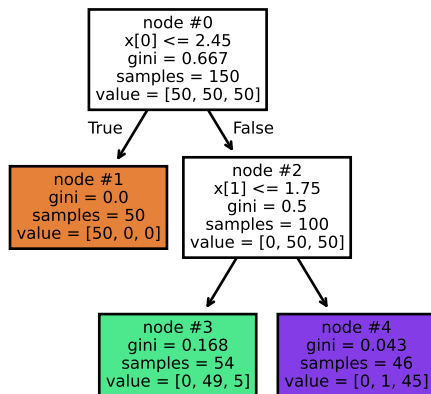
A decision tree recursively divides the training observations into smaller groups. Each node is a position in the tree together with the observations assigned to it.

- A node is a position in the tree. We use G to denote the set of training observations in a node, and $|G|$ denotes the number of observations in that set. Equivalently, G identifies a subset of the rows of X .
- The root node is the node at the top of the tree. It contains all training observations.
- A split applies a rule such as $x_j \leq c$ to a node and divides it into two child nodes. The original node is their parent node.
- An internal node, also called a decision node, is a node that has been split.
- A leaf node, also called a terminal node, is a node that is not split further. Each leaf contains a final group of observations used to make a prediction.
- The depth of a node is the number of parent-child connections traversed when moving from the root to that node. The root has depth 0.

- The depth of a tree is the largest depth among all nodes in the tree. Equivalently, it is the largest number of edges on any path from the root to a leaf.
- A subtree consists of a node and all of its descendants. Pruning replaces a larger subtree with a leaf.
- A binary tree is a tree in which each internal node has two children. CART constructs binary trees. Note that there are other algorithms that can build trees, and they may not create binary trees.

For classification, a leaf usually predicts the most common class or the class proportions among its observations. For regression, a leaf usually predicts the mean response among its observations. A classification node is pure when all its observations belong to the same class.

Example 9.1 (Example: A Classification Tree).

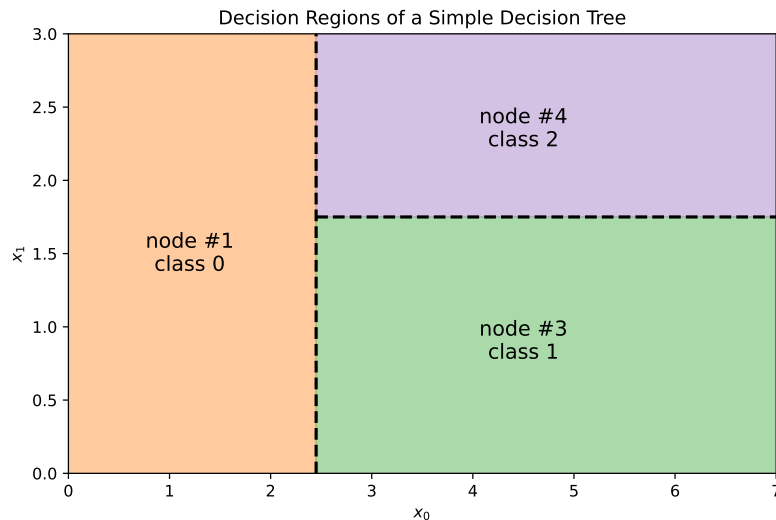


This is a classification tree. The tree repeatedly asks simple questions such as

- Is $x[0] \leq 2.45$?
- Is $x[1] \leq 1.75$?

Each split divides the feature space into smaller regions.

This tree actually create the following splits in the feature space.



We could also go over all basic terminologies in this example.

- Node #0 is the root node. It contains all 150 observations in the dataset and has depth 0.
- Node #0 is an internal node. Its split sends observations to node #1 or node #2, which are its child nodes.
- Node #2 is also an internal node. It is a child of node #0 and the parent of nodes #3 and #4.
- Nodes #1, #3, and #4 are leaf nodes because they are not split further. Their depths are 1, 2, and 2, respectively.
- The depth of this tree is 2, which is the largest depth among its nodes.
- This is a binary tree because every internal node has exactly two children.
- One example of splits: node #0 uses the split $x_0 \leq 2.45$ to send observations to node #1 or node #2.
- One example of subtree: the subtree rooted at node #2 consists of nodes #2, #3, and #4.
- Each leaf makes a prediction using the observations it contains. For example, node #1 predicts class 0, the most common class among its observations.
- Node #1 is pure because all observations in it belong to class 0. Its Gini impurity is therefore 0.

Based on the idea of a decision tree, a tree makes decisions by

repeatedly asking questions of the form $x_j \leq c$. Therefore, a tree has two important properties:

- It depends directly on the original features.
- The scale of a feature is usually not important.

As a result, we should expect that a tree is:

- sensitive to feature construction or feature combinations:
A tree can only split on the features that are provided. If the useful signal is hidden in a combination such as $x_1 + x_2$ or $x_1 x_2$, a tree may need many splits to approximate that relationship.
- not sensitive to monotone transformations of a feature:
For example, replacing x by $\log(x)$ or $2x + 5$ does not change the ordering of the observations. Since tree splits mainly depend on ordering, the tree can usually produce an equivalent split after the transformation.

9.2 CART: Classification and Regression Trees

There are many algorithms for constructing decision trees. The most common one is called CART (Classification and Regression Trees) [7].

CART has the following properties:

- It creates only binary splits;
- It can perform both classification and regression tasks:
 - For classification, common splitting criteria include Gini impurity and entropy;
 - For regression, common splitting criteria include residual sum of squares (RSS) and mean squared error (MSE).

9.2.1 Splitting a Dataset

At each step, CART searches over possible splits and chooses the split that produces the largest reduction in impurity (or prediction error).

i Algorithm: Split the Dataset

Inputs: A dataset $S = [\text{features}, \text{target}]$

Outputs: The best split (k, t)

1. For each feature x_k :
 1. For each candidate split value t :
 1. Split the dataset into two subsets: $G_l = \{\text{data with } x_k \leq t\}$, $G_r = \{\text{data with } x_k > t\}$.
 2. Compute the split cost function $J(k, t)$
 3. Keep the best split found so far
 2. Return the split (k, t) with the smallest cost

The splitting algorithm is applied recursively.

i Algorithm: CART

Inputs: A labeled dataset S and a maximal tree depth `max_depth`

Outputs: A decision tree

1. Start with the original dataset S
2. If the node is not pure and stopping conditions are not met:
 1. Find the best split
 2. Split the node into: G_l and G_r
 3. Apply the same procedure recursively to both child nodes
3. Stop when:
 - the maximal tree depth is reached, or
 - the node becomes pure, or

- further splitting is impossible

When making predictions, a classification tree uses the most common class label in a region as the predicted value, while a regression tree uses the mean of the response values in the region as the predicted value.

9.2.2 Classification Trees: Gini Impurity

Classification trees are used when the response variable is categorical. A classification tree partitions the feature space into piecewise constant regions, where each region predicts either a class label or class probabilities.

In these notes, we use Gini impurity to measure the quality of a split.

Assume that a node contains n observations divided into K classes. Let $p_i = \frac{n_i}{n}$ be the proportion of class i . If we classify observations according to the class proportions:

- the probability an observation belongs to class i is p_i
- the probability we incorrectly classify it is $1 - p_i$

Therefore the probability of misclassifying an observation from class i is

$$p_i(1 - p_i).$$

Summing over all classes gives

$$\sum_{i=1}^K p_i(1 - p_i).$$

Since

$$\sum_{i=1}^K p_i = 1,$$

we obtain

$$\text{Gini} = \sum_{i=1}^K p_i(1 - p_i) = 1 - \sum_{i=1}^K p_i^2.$$

Definition 9.1 (Gini Impurity). The **Gini impurity** of a node is

$$\text{Gini} = 1 - \sum_{i=1}^K p_i^2,$$

where p_i is the proportion of class i inside the node.

Example 9.2 (Pure node). Suppose a node contains observations from only one class.

Then one of the probabilities equals 1 and all others equal 0.

Therefore

$$\text{Gini} = 1 - 1^2 = 0.$$

This is the minimum possible impurity.

A node with Gini impurity 0 is called a pure node.

Example 9.3 (Two-Class Example). Suppose we have two classes.

Let

$$p_2 = 1 - p_1.$$

Then

$$\text{Gini} = 1 - p_1^2 - (1 - p_1)^2 = 2p_1 - 2p_1^2.$$

When:

- $p_1 = 0$ or 1 , the impurity is 0;
- $p_1 = 0.5$, the impurity is maximized.

Therefore, impurity is largest when the classes are perfectly balanced.

The impurity of a node alone is not enough. CART evaluates a split by examining the impurity of the child nodes after splitting.

Suppose a node G is split into G_l and G_r . The split cost is

$$J = \frac{|G_l|}{|G|} \text{Gini}(G_l) + \frac{|G_r|}{|G|} \text{Gini}(G_r).$$

A good split produces child nodes with low impurity, ideally creating nodes that are close to pure.

The cost of a tree is obtained by applying this weighting recursively down to the leaves. It is therefore a weighted sum of the Gini impurity of each leaf, with weights given by the proportion of samples falling into that leaf.

Entropy

Entropy is another popular impurity measure. Its formula is

$$\text{Entropy} = - \sum_{i=1}^K p_i \log(p_i).$$

Entropy and Gini impurity usually produce very similar trees.

Because Gini impurity is slightly simpler computationally and conceptually, we mainly focus on it.

9.2.3 Regression Trees: RSS

Regression trees are used when the response variable is quantitative. Instead of predicting a class label, they predict numerical values.

Suppose a leaf node contains responses $y_1, y_2, \dots, y_n$. The prediction for that region is usually the sample mean:

$$\hat{y} = \frac{1}{n} \sum_{i=1}^n y_i.$$

Therefore, regression trees approximate the regression function using piecewise constant regions.

For regression trees, CART typically minimizes the residual sum of squares (RSS). Sometimes the mean squared error (MSE) is used instead. Since MSE differs from RSS only by a constant scaling factor, minimizing one is equivalent to minimizing the other.

Suppose a split divides the data into two regions G_l and G_r . The split cost is

$$\text{RSS} = \sum_{i \in G_l} (y_i - \bar{y}_l)^2 + \sum_{i \in G_r} (y_i - \bar{y}_r)^2,$$

where $\bar{y}_l$ and $\bar{y}_r$ are the mean responses within the two regions. The best split is the one that minimizes this quantity.

9.2.4 Trees are local

A tree makes predictions using only observations inside a small region of the feature space. Suppose a tree partitions the feature space into regions. The prediction at a point x depends only on the region containing x . As a result, tree predictions are local:

- nearby regions may have very different predictions
- only observations in the same local region affect the prediction
- changing distant observations usually does not affect the prediction

This is similar in spirit to KNN, where prediction also depends primarily on nearby observations.

9.2.5 Trees are nonlinear

A tree creates many if-then rules. Because of these recursive splits:

- the prediction function can change abruptly
- variable effects depend on the region
- interactions are created automatically

The prediction surface is therefore discontinuous and piecewise constant. Therefore, trees are nonlinear models.

9.2.6 Automatic interaction effects

One important property of trees is that they automatically create interaction effects.

For example, consider a tree with depth 3. One path through the tree may look like:

- if $x_1 \leq 1$
- and $x_2 \leq 2$
- and $x_3 \leq 3$
- then predict $y = 1.1$

Notice that the prediction depends on the combination of all three conditions simultaneously. Therefore, the effect of one variable may depend on the values of earlier splitting variables. In other words, the effect of x_3 is not independent of x_1 and x_2 . The split on x_3 is only relevant inside the region defined by the earlier splits.

Therefore, trees naturally model interactions between variables.

This is very different from a standard additive linear model such as

$$f(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3,$$

where each variable contributes independently unless explicit interaction terms like $x_1 x_2$ are manually added.

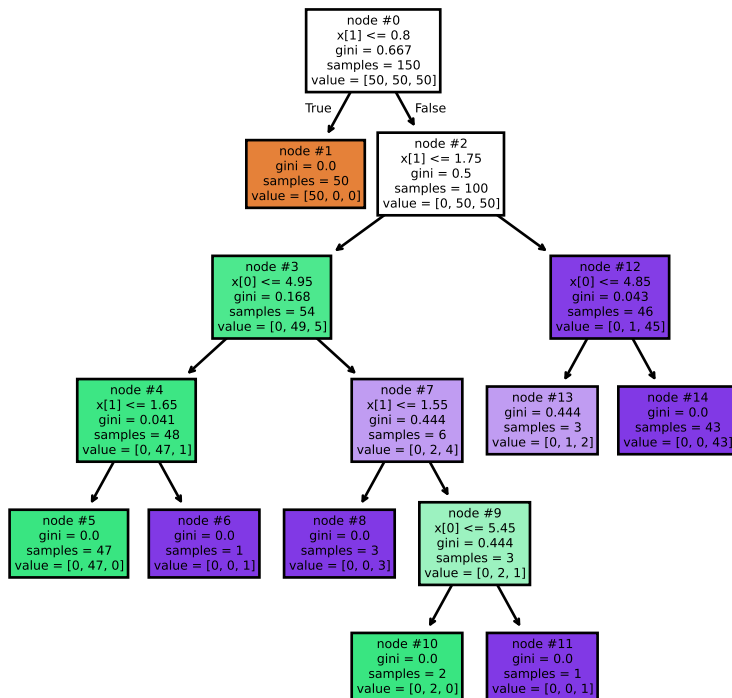
9.3 Tuning hyperparameters

Building a decision tree is the same as recursively splitting the training dataset. If we are allowed to keep splitting it, almost every leaf may become pure: its Gini impurity is 0 for classification, or all observations in the leaf have the same response value for regression.

Example 9.4 (ALMOST all leaves are pure).

Click to expand.

In this example, we let the tree grow as far as possible. It only stops when (almost) all leaves are pure, even if a leaf contains only ONE observation (like #6 and #11).



In this example there is one exception. #13 node is not pure. We could list all data in node #13.

	x[0]	x[1]	y
0	4.8	1.8	1
1	4.8	1.8	2
2	4.8	1.8	2

All the data points in this node has the same feature while their labels are different. They cannot be split further purely based on features.

As the tree continues splitting the feature space into smaller regions, it learns increasingly fine details of the training set. Although this increases flexibility, it can also make the model overly sensitive to noise. Therefore, we usually do not want the tree to grow indefinitely.

There are two common ways to control a single tree growth:

1. Pruning: first grow a large tree, then remove unnecessary branches afterward.
2. Pre-pruning: stop the tree early by imposing restrictions during training.

9.3.1 Pruning

Pruning removes a subtree by replacing its root with a leaf node. The pruning method introduced in CART is called Minimal Cost-Complexity Pruning (MCCP).

More specifically, the cost function is modified by adding a complexity penalty controlled by the parameter `ccp_alpha`. The parameter `ccp_alpha` is called the complexity parameter and is always ≥ 0 . Let T be a subtree, and let $|L(T)|$ denote the number of leaves in T . The cost-complexity criterion is

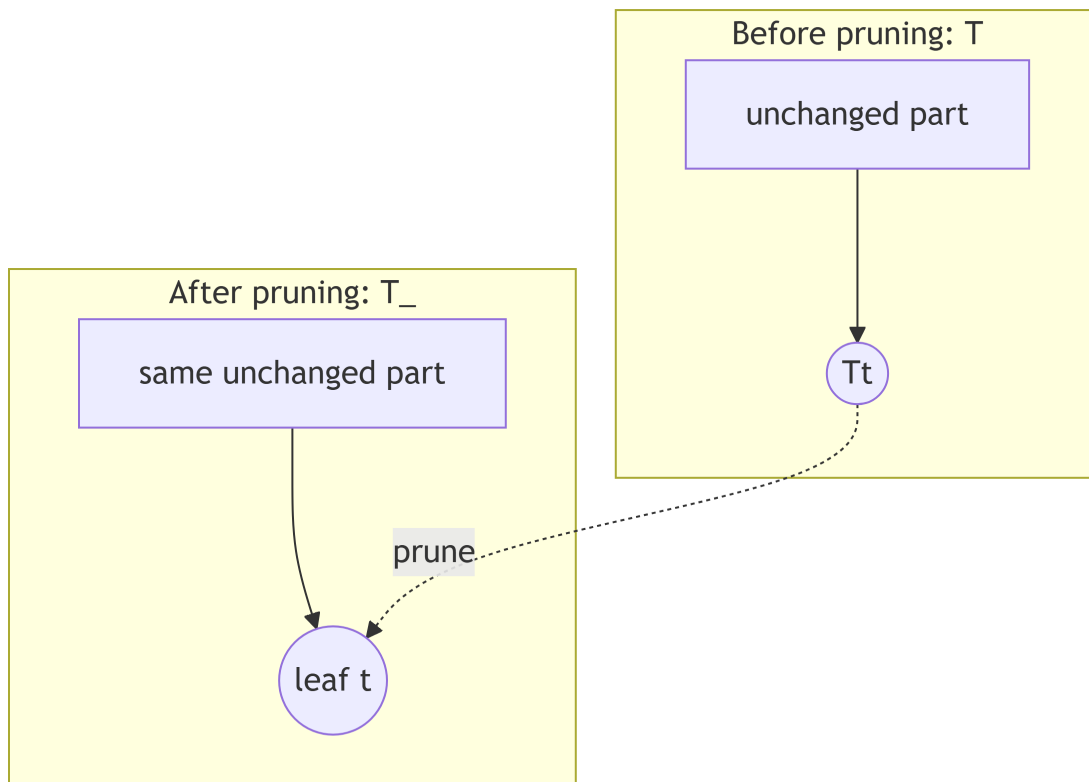
$$C_\alpha(T) = J(T) + \alpha|L(T)|,$$

where:

- $J(T)$ is the total training cost of T , based on weighted Gini impurity for classification and RSS for regression,
- α corresponds to `ccp_alpha` in the code.

The idea is to find a tree who has the smallest cost-complexity criterion for a certain α .

Example 9.5 (The split cost). The cost comparison can be understood by considering whether a subtree should be retained or pruned. Let T_t be the subtree of T rooted at an internal node t , and let T_- be the tree obtained by replacing T_t with a single leaf t .



The leaf t is already counted in T_- . Expanding it into T_t therefore adds $|L(T_t)| - 1$ leaves, so

$$|L(T)| - |L(T_-)| = |L(T_t)| - 1.$$

The reduction in cost from retaining T_t is

$$\begin{aligned} C_\alpha(T_-) - C_\alpha(T) &= (J(T_-) + \alpha|L(T_-)|) - (J(T) + \alpha|L(T)|) \\ &= (J(T_-) - J(T)) - \alpha(|L(T_t)| - 1). \end{aligned}$$

The subtree is worth retaining only when the cost is reduced $C_\alpha(T) < C_\alpha(T_-)$, which means that

$$J(T_-) - J(T) > \alpha(|L(T_t)| - 1).$$

Thus, the reduction in training cost must exceed the complexity penalty for the additional leaves. When $\alpha = 0$, there is no complexity penalty, so the procedure behaves like an ordinary decision tree and larger trees are generally preferred.

If the optimization problem were solved independently for each value of α , there would be no obvious reason to expect the resulting optimal subtrees to be related. However [7] proved that if $\alpha_1 > \alpha_2$, then the optimal subtree for α_1 can always be obtained by pruning subtrees from the optimal subtree for α_2 . Therefore, the MCCP algorithm views pruning as gradually increasing `ccp_alpha` from 0 to larger values, generating a nested sequence of subtrees from the fully grown tree. The goal is to find a tree that balances predictive performance and model complexity.

In practice, we first compute the `ccp_alpha` path, which contains the values of `ccp_alpha` at which the optimal subtree changes. We then use a validation set or cross-validation to evaluate the subtrees along this path and select an appropriate value of `ccp_alpha`. If several candidate trees achieve similar predictive performance, we usually prefer the simpler tree, corresponding to a larger `ccp_alpha`, because it is often more stable and easier to interpret.

9.3.2 Pre-pruning

We often impose some early stopping conditions to limit the growth of a tree. This strategy is called pre-pruning. Some common early stopping conditions are:

- `max_depth`: maximum tree depth
- `min_samples_split`: minimum number of observations required to split a node
- `min_samples_leaf`: minimum number of observations in a leaf node

- `max_leaf_nodes`: maximum number of leaves

Smaller trees are usually more stable but may underfit. Larger trees are more flexible but may overfit. A larger tree is not automatically better, even if it achieves lower training error.

However, pre-pruning can be greedy. A split that appears only mildly useful at an early stage may later enable highly informative splits deeper in the tree. If we stop the tree too early, we may miss important structure in the data.

Therefore, in modern practice, pre-pruning is usually not the primary method for selecting the final model complexity. Instead, we mainly rely on pruning by choosing the best `ccp_alpha` through validation or cross-validation. Pre-pruning is often used only as a guardrail to prevent the initial tree from becoming excessively large. In practice, we usually set relatively mild early stopping limits to stabilize the tree-growing process and then rely on pruning to determine the final tree size.

i Tuning in practice

In classical CART, `ccp_alpha` is the primary complexity parameter and is usually selected by cross-validation after growing a large tree.

In modern practice, pre-pruning parameters and `ccp_alpha` are often tuned together, especially for computational efficiency and stability. In other words, methods such as `GridSearchCV` can be used to tune all hyperparameters simultaneously.

However, when the search space becomes too large, this approach can become computationally expensive. In that case, we often return to the more traditional CART-style strategy: we first coarsely identify a reasonable range for the early stopping hyperparameters and then fine-tune `ccp_alpha`.

9.4 Trees in Python

We use `DecisionTreeClassifier` and `DecisionTreeRegressor` from `sklearn.tree` to implement decision tree models. The API is almost identical to other `sklearn` models: `.fit()`, `.predict()`, `.score()`, and other familiar methods are used in the same way.

1. We could set `max_depth`, `min_samples_leaf`, `ccp_alpha`, etc. for the tree model where their meaning is straightforward.
2. The method `.cost_complexity_pruning_path()` can be used to find the critical `ccp_alpha` values for pruning.
3. We could also set `random_state`. The randomness in `sklearn` decision trees mainly comes from the internal feature ordering used during the split search. Even when `splitter="best"` is used, `sklearn` randomly permutes the feature order before evaluating splits. Therefore, when multiple splits produce the same impurity reduction, different trees may be generated unless `random_state` is fixed.

Note

Decision trees are often described as methods that naturally handle categorical predictors. This is true conceptually, but `sklearn` decision trees still require numeric input. Therefore categorical predictors must be encoded before fitting the model. For nominal categorical variables, one-hot encoding is usually a safe choice.

However, one-hot encoding for the target variable is not necessary because `sklearn` classifiers can directly work with class labels.

9.4.1 Example 1: Classify iris dataset

The `iris` dataset is a classic classification dataset. The goal is to use four numeric flower measurements to classify the species of iris. The dataset contains 150 observations and 3 classes.

More information can be found in the `sklearn` documentation for datasets.

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=1
)
```

The following code shows the basic `sklearn` API.

```
from sklearn.tree import DecisionTreeClassifier

clf = DecisionTreeClassifier(random_state=2)
clf.fit(X_train, y_train)
clf.score(X_test, y_test)
```

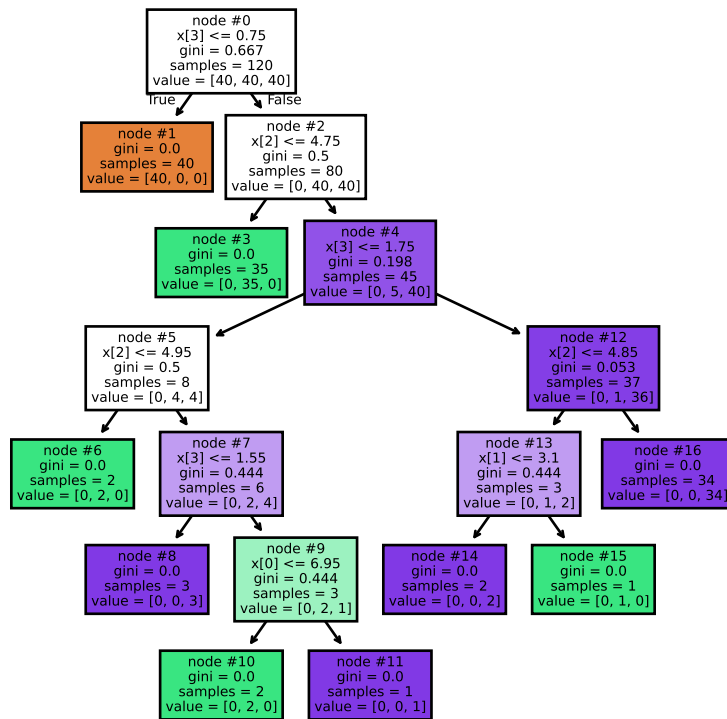
0.9666666666666667

After fitting the model, we can display the tree using `plot_tree()`. In many real examples, the tree may be too large to read clearly, so we use `matplotlib` to control the size and resolution of the plot.

```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(5, 5), dpi=300)
plot_tree(clf, filled=True, impurity=True, node_ids=True); ①
```

- ① `filled=True` fills the nodes with colors. `impurity=True` displays the Gini impurity. `node_ids=True` displays the node ID.



Now we prune the tree. We first find all critical `ccp_alpha` values.

```
ccp_path = clf.cost_complexity_pruning_path(X_train, y_train)
ccp_alphas = ccp_path.ccp_alphas[:-1]
```

The last `ccp_alpha` is usually removed because it is large enough to prune the tree down to a single root node. In most applications, that model is too simple to be useful.

We then train one model for each effective `ccp_alpha`. After fitting the trees, we record both predictive performance and model complexity. Here we use `accuracy` as our metric, which comes automatically from `clf.score()`.

```
from sklearn.metrics import accuracy_score

clfs = []
```

```

for ccp_alpha in ccp_alphas:
    clf = DecisionTreeClassifier(random_state=42, ccp_alpha=ccp_alpha)
    clf.fit(X_train, y_train)
    clfs.append(clf)

node_counts = [clf.tree_.node_count for clf in clfs]
train_acc = [clf.score(X_train, y_train) for clf in clfs]
test_acc = [clf.score(X_test, y_test) for clf in clfs]

```

Here we use a single train/test split for simplicity (by abusing notations since we actually use test set as validation set). In addition to accuracy, we also record the number of nodes because it is a useful measure of tree complexity.

A larger tree usually has lower bias but higher variance. A smaller tree is often more interpretable and more stable.

The results are easier to compare visually.

```

import matplotlib.pyplot as plt

best_alpha = 0.02

fig, ax = plt.subplots(2, 1, figsize=(7, 7), sharex=True)

ax[0].plot(ccp_alphas, node_counts, marker="o", drawstyle="steps-post") ①
ax[1].plot(ccp_alphas, train_acc, marker="o", drawstyle="steps-post", label="train")
ax[1].plot(ccp_alphas, test_acc, marker="o", drawstyle="steps-post", label="test")

ax[0].axvline(best_alpha, color="black", linestyle="--", label="best ccp_alpha")
ax[1].axvline(best_alpha, color="black", linestyle="--", label="best ccp_alpha")

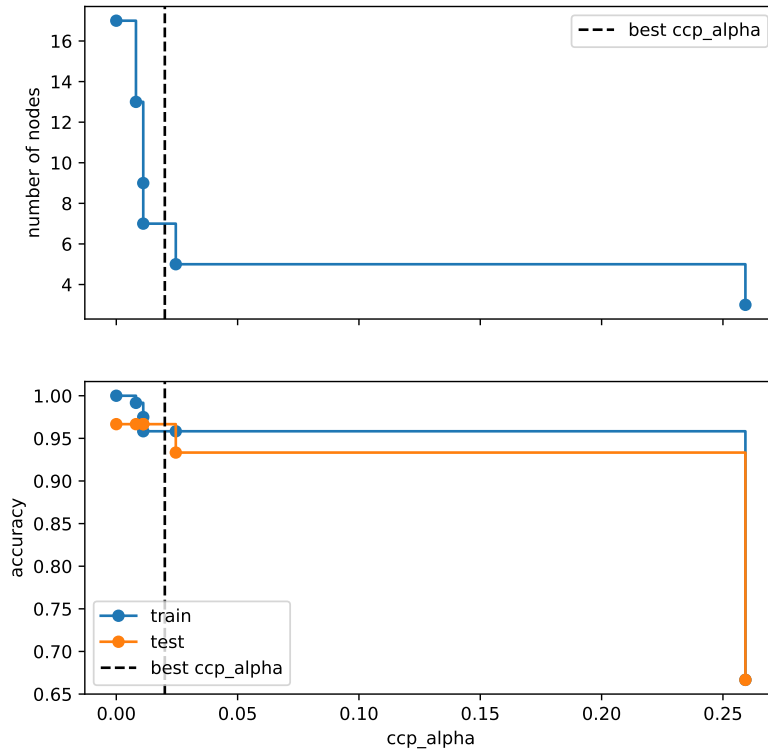
ax[1].set_xlabel("ccp_alpha")
ax[0].set_ylabel("number of nodes")
ax[1].set_ylabel("accuracy")

ax[0].legend()
ax[1].legend(loc="lower left")

```

- ① For pruning-related plots, `drawstyle="steps-post"` is important because the tree structure stays the same within

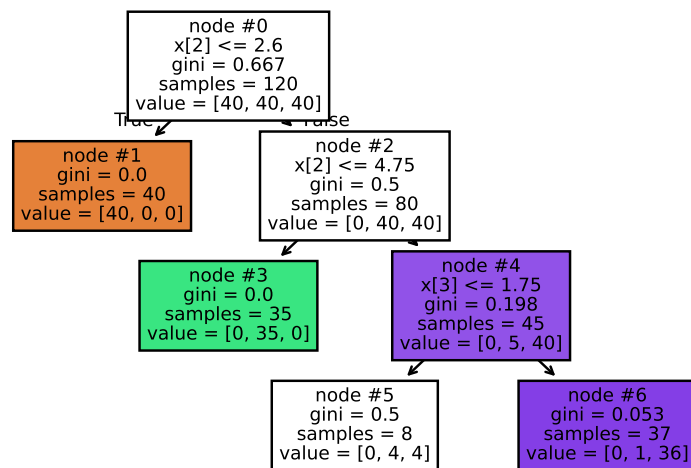
each interval of `ccp_alpha` values. The tree only changes after reaching the next pruning threshold. Each critical `ccp_alpha` corresponds to pruning one or more weak internal nodes, so the pruning path is piecewise constant.



Based on the plot, from $0.0 \leq \text{ccp_alpha} < 0.02452452452452452$, the test accuracy stay the same. Therefore we would like to choose the one with the least number of nodes, which is 0.01111111111111113. Since all α values in the range give the same tree, we hand pick `ccp_alpha=0.02` for simplicity.

We can now build the final tree and display it.

```
clf = DecisionTreeClassifier(random_state=42, ccp_alpha=best_alpha)
clf.fit(X_train, y_train)
plot_tree(clf, filled=True, impurity=True, node_ids=True);
```



i Note

Although the hyperparameters with the highest validation score are often preferred, model complexity should also be taken into consideration. This is why the number of nodes is displayed.

In this `iris` example, the dataset is very small and the differences between validation scores are negligible. In such situations, a slightly simpler tree is often preferred because it is more interpretable and potentially more stable while achieving similar predictive performance.

In larger datasets, small differences in validation scores are often more meaningful, so predictive performance may be prioritized more heavily.

9.4.2 Example 2: Regression with the diabetes dataset

We now look at a regression example. The `diabetes` dataset uses 10 baseline variables to predict a quantitative measure of disease progression one year after baseline. More information can be found in the `sklearn` documentation and the original diabetes dataset [description](#).


```

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split

X, y = load_diabetes(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=1)

```

The overall workflow is similar to the `iris` example. In this example, we choose to use cross-validation instead of a single validation split in order to obtain a more stable estimate of model performance. This is not specific to regression trees; either approach can be used for both classification and regression problems.

We first create a temporary tree to compute the critical `ccp_alpha` values. The method `cost_complexity_pruning_path()` internally grows a tree in order to compute the effective pruning path, although the estimator itself is not stored as a fitted model.

```

from sklearn.tree import DecisionTreeRegressor

base_tree = DecisionTreeRegressor(random_state=1)
path = base_tree.cost_complexity_pruning_path(X_train, y_train)
ccp_alphas = path.ccp_alphas[:-1]

```

This follows the classical CART pruning strategy:

1. Grow a large tree.
2. Compute the pruning path.
3. Use validation or cross-validation to select the final sub-tree.

Now we run `GridSearchCV` over these `ccp_alpha` values. We use R^2 as the model metric in this example.

```

from sklearn.model_selection import GridSearchCV
from sklearn.metrics import r2_score

params = param_grid = {"ccp_alpha": ccp_alphas}
grid = GridSearchCV(

```

```

    DecisionTreeRegressor(random_state=1), param_grid=params, scoring="r2"
)

grid.fit(X_train, y_train)

best_tree = grid.best_estimator_

y_pred_train = best_tree.predict(X_train)
y_pred_test = best_tree.predict(X_test)

print(f"Best ccp_alpha: {grid.best_params_['ccp_alpha']}")
print(f"Training R^2: {r2_score(y_train, y_pred_train)}")
print(f"Testing R^2: {r2_score(y_test, y_pred_test)}")

```

```

Best ccp_alpha: 105.18472145590351
Training R^2: 0.5347863093947497
Testing R^2: 0.1922738375305083

```

We can visualize the cross-validation curve. `GridSearchCV` records validation scores, but it does not automatically record tree complexity measures such as the number of nodes. Therefore, if we want to display the complexity curve, we need to refit the models for the same `ccp_alpha` values and record their node counts.

```

cv_score = grid.cv_results_["mean_test_score"]

node_counts = []
for ccp_alpha in ccp_alphas:
    tree = DecisionTreeRegressor(random_state=1, ccp_alpha=ccp_alpha)
    tree.fit(X_train, y_train)
    node_counts.append(tree.tree_.node_count)

best_alpha = grid.best_params_["ccp_alpha"]

fig, ax = plt.subplots(2, 1, figsize=(7, 7), sharex=True)

ax[0].plot(ccp_alphas, node_counts, marker="o", drawstyle="steps-post")
ax[1].plot(ccp_alphas, cv_score, marker="o", drawstyle="steps-post", label="CV score")

```

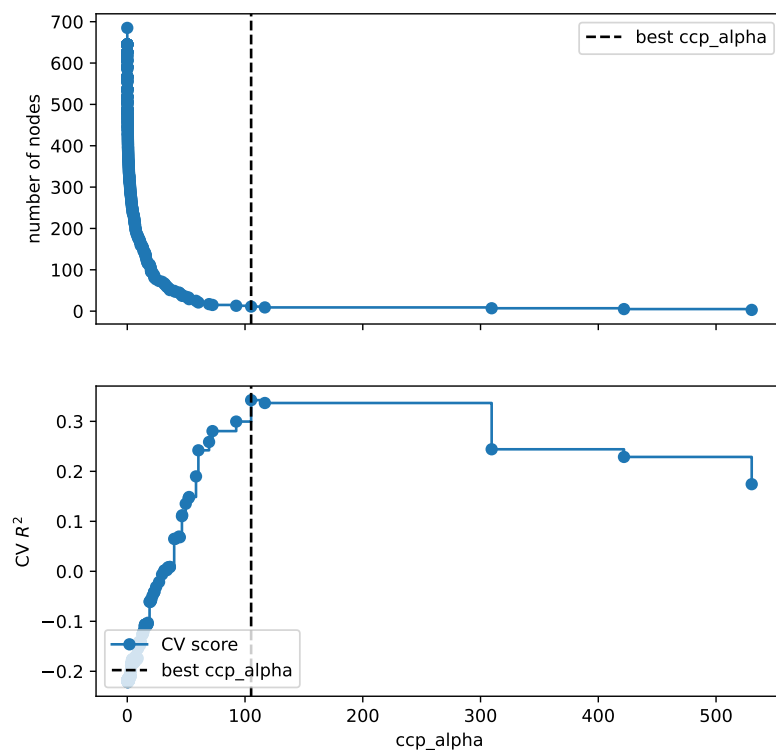
```

ax[0].axvline(best_alpha, color="black", linestyle="--", label="best ccp_alpha")
ax[1].axvline(best_alpha, color="black", linestyle="--", label="best ccp_alpha")

ax[1].set_xlabel("ccp_alpha")
ax[0].set_ylabel("number of nodes")
ax[1].set_ylabel("CV  $R^2$ ")

ax[0].legend()
ax[1].legend(loc="lower left")

```



From the plot, our selected `ccp_alpha=np.float64(105.18472145590351)` is reasonable since it gives a relative strong cross-validation score without producing an unnecessarily large tree.

 Warning

A single regression tree often has unstable predictive performance on moderate-sized noisy datasets. Therefore, the test R^2 may not look high in this example. This is one reason why ensemble methods such as random forests and boosting are often preferred in practice.

10 Bagging and Random Forests

In earlier chapters we studied single decision trees. A tree is flexible because it can capture nonlinear relationships and interactions without specifying them in advance. However, a single tree is also unstable. A small change in the training data may produce a very different tree, especially when the tree is grown deeply.

Bagging and random forests are ensemble methods designed to reduce this instability.

The central idea is:

Instead of trusting one unstable tree, build many unstable trees and average them.

This chapter first discusses bagging. Random forests will then be introduced as an improvement of bagging. Since bagging and random forests use almost the same aggregation idea for regression and classification, we will mainly use regression notation. For classification, averages are usually replaced by majority vote or averaged class probabilities.

10.1 Ensemble Averaging

Suppose we observe training data

$$\{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^p, \quad y_i \in \mathbb{R}.$$

An ensemble combines many fitted models

$$\hat{f}_1, \hat{f}_2, \dots, \hat{f}_B$$

into one final predictor. For regression, the simplest ensemble average is

$$\hat{f}_{\text{ens}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x).$$

Each individual model may be noisy. However, the average can be much more stable.

To see why, suppose each $\hat{f}_b(x)$ has variance σ^2 , and suppose for a moment that the fitted models are independent. Then

$$\text{Var} \left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x) \right) = \frac{\sigma^2}{B}.$$

Thus averaging can greatly reduce variance.

In practice, the fitted models are not independent. If the pairwise correlation between two fitted models is approximately ρ , then

Click to expand.

$$\text{Corr}(\hat{f}_i(x), \hat{f}_j(x)) \approx \rho, \quad i \neq j.$$

Then

$$\text{Cov}(\hat{f}_i(x), \hat{f}_j(x)) = \text{Corr}(\hat{f}_i(x), \hat{f}_j(x)) \sqrt{\text{Var}(\hat{f}_i(x)) \text{Var}(\hat{f}_j(x))} \approx \rho \sigma^2.$$

Therefore,

$$\begin{aligned}
\text{Var} \left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x) \right) &= \frac{1}{B^2} \left[\sum_{b=1}^B \text{Var}(\hat{f}_b(x)) + \sum_{i \neq j} \text{Cov}(\hat{f}_i(x), \hat{f}_j(x)) \right] \\
&\approx \frac{1}{B^2} [B\sigma^2 + B(B-1)\rho\sigma^2] \\
&= \sigma^2 \left[\frac{1}{B} + \frac{B-1}{B}\rho \right] \\
&= \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.
\end{aligned}$$

Then

$$\text{Var} \left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x) \right) \approx \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

This formula is very important. Increasing B reduces the second term, but the first term remains. Therefore, if the individual trees are highly correlated, averaging has limited benefit.

10.2 Bagging

Bagging is short for bootstrap aggregation.

Assume the original training set has n data. A bootstrap sample is a sample drawn from the original training data with replacement. Each bootstrap sample has size n , but because sampling is done with replacement, some observations appear more than once and some observations do not appear at all.

Let $Z = \{(x_1, y_1), \dots, (x_n, y_n)\}$ be the original training set. Bagging constructs bootstrap samples $Z^{*1}, Z^{*2}, \dots, Z^{*B}$. For each bootstrap sample Z^{*b} , we fit a model $\hat{f}_b(x)$. The bagged regression estimate is then

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x).$$

For classification, the final prediction is usually based on majority vote:

$$\hat{G}_{\text{bag}}(x) = \text{mode}\{\hat{G}_1(x), \dots, \hat{G}_B(x)\}.$$

Equivalently, we may average estimated class probabilities and choose the class with the largest average probability.

Bias-variance view

Bagging is useful when the base learner is unstable. A decision tree is a typical unstable learner. Different bootstrap samples lead to different trees, because each tree sees a slightly different version of the data.

For a decision tree, bagging usually uses large trees rather than heavily pruned trees. The reason is that a large tree has low bias but high variance. Bagging is designed to reduce variance, so it is natural to start with a flexible base learner.

A single deep tree usually has low bias and high variance. Bagging keeps the low bias and reduces the variance by averaging many deep trees.

Bagging estimators

Although bagging is most commonly applied to decision trees, it can also be used with other types of models.

10.2.1 Out-of-Bag Error

Although a bootstrap sample has size n , it does not contain all n observations.

Definition 10.1 (Out-of-bag observations). Each bootstrap sample contains about 63.2% of the original observations as distinct observations. The remaining observations are called **out-of-bag** (OOB) observations for that bootstrap sample.

Click to expand.

For a fixed observation i , the probability that it is not selected in one draw is

$$1 - \frac{1}{n}.$$

The probability that it is never selected in n draws is

$$\left(1 - \frac{1}{n}\right)^n.$$

When n is large,

$$\left(1 - \frac{1}{n}\right)^n \rightarrow e^{-1} \approx 0.368.$$

Therefore, the probability that observation i appears at least once is approximately

$$1 - e^{-1} \approx 0.632,$$

and it is the expected proportion of distinct observations in a sample.

The OOB observations give a natural way to estimate prediction error.

For observation i , consider only the trees for which observation i was out-of-bag. Average those trees to get an OOB prediction:

$$\hat{f}_{\text{OOB},i} = \frac{1}{|B_i|} \sum_{b \in B_i} \hat{f}_b(x_i),$$

where B_i is the set of trees that did not use observation i in their bootstrap sample.

The OOB mean squared error is

$$\text{MSE}_{\text{OOB}} = \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{OOB},i}\right)^2.$$

All other metrics have OOB versions in the straightforward way.

OOB error is useful because it behaves like an internal validation estimate. We do not need to set aside a separate validation set just to estimate the error of the bagged model. Therefore, for bagging, in addition to single-split validation and cross-validation, we can also use OOB validation to tune hyperparameters.

10.3 Random Forests

Random forests are a modification of bagging for decision trees. In addition to the diversity by sampling observations from Bagging, Random forests create additional diversity by sampling features during tree construction.

The random forest algorithm for regression is:

1. Draw a bootstrap sample from the training data.
2. Grow a large regression tree on this bootstrap sample.
3. **At each split, randomly choose m predictors from the full set of p predictors.**
4. **Find the best split only among those m predictors.**
5. Repeat this process for B trees.
6. Average the predictions.

The final random forest regressor is

$$\hat{f}_{\text{RF}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x).$$

The formula looks the same as bagging. The difference is how the trees are grown.

The key difference from Bagging(Trees) is the Step 3 and 4. In ordinary bagging, all predictors are available at every split. If there is one very strong predictor, many trees may split on that same predictor near the top of the tree. As a result, the trees become similar.

Similar trees are highly correlated. From the variance formula,

$$\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2,$$

we know that high correlation limits the benefit of averaging.

Now random forests reduce this issue by forcing each split to consider only a random subset of predictors. This prevents the strongest predictors from dominating every tree. The trees become less correlated, and averaging becomes more effective.

10.3.1 max_features

How many predictors are considered at each split? In `sklearn`, this is controlled by `max_features`. This is a very important hyperparameter because it affects the correlations between trees.

Large `max_features` means each tree can choose from many predictors. Individual trees may become stronger, but the trees may also become more similar to each other.

Small `max_features` forces trees to explore different predictors. This usually decreases the correlations between trees, but each individual tree may become weaker.

Thus `max_features` controls an important tradeoff between tree strength and tree correlation.

Common traditional choices are:

- for classification, `max_features`= $\sqrt{p}$
- for regression, `max_features`= $p/3$

These recommendations originated from the original Random Forest literature [8] and later became popular defaults in many software implementations [3]. They are mainly based on empirical experience and the general intuition that:

- classification forests often benefit more from tree diversity
- regression forests often benefit more from stronger individual trees

In modern practice, however, `max_features` is usually treated as a tuning hyperparameter and is commonly selected using validation or cross-validation.

10.3.2 Random Forest as Adaptive Local Averaging

A random forest can also be viewed as a data-adaptive local averaging method.

For one tree, two points are close if they fall in the same terminal node. Let

$$I_b(x, x_i) = \begin{cases} 1, & \text{if } x \text{ and } x_i \text{ are in the same leaf of tree } b, \\ 0, & \text{otherwise.} \end{cases}$$

Across the whole forest, define the similarity

$$K_{\text{RF}}(x, x_i) = \frac{1}{B} \sum_{b=1}^B I_b(x, x_i).$$

This measures the proportion of trees in which x and x_i fall into the same terminal node. The quantity $K_{\text{RF}}(x, x_i)$ can be interpreted as a data-adaptive similarity measure between x and x_i .

The random forest prediction can be written as a weighted average

$$\hat{f}_{\text{RF}}(x) = \sum_{i=1}^n w_i(x) y_i,$$

where observations that frequently share terminal nodes with x receive larger weights.

More details about the adaptive nearest-neighbor interpretation of random forests can be found in [9] and [3].

Note

The adaptive-neighbor interpretation helps explain why random forests often perform much better than classical distance-based methods such as KNN in complex datasets. In KNN, neighborhoods are determined by a fixed geometric distance. In contrast, random forests learn the

neighborhood structure directly from the data through recursive splitting. Two observations are considered close if the forest repeatedly places them into the same terminal nodes, so the similarity automatically adapts to the importance of variables and their relationships with the response variable.

As a result, the neighborhood automatically adapts to:

- important variables
- interaction effects
- nonlinear structures
- local heterogeneity of the feature space

This viewpoint also provides intuition for why random forests are often robust in high-dimensional settings. Instead of relying on a global distance metric, the forest constructs many local data-dependent neighborhoods and averages over them.

From a theoretical perspective, this interpretation connects random forests to classical nonparametric smoothing methods such as KNN and kernel regression, helping explain random forests using the broader language of statistical learning theory.

10.3.3 Feature Importance

There are two commonly used measures of feature importance. We briefly discuss them here. More detailed examples and implementations can be found in the [scikit-learn documentation](#).

10.3.3.1 Mean Decrease in Impurity

One common measure is impurity-based importance, also called Mean Decrease in Impurity (MDI). This importance measure is specific to tree-based methods because it depends directly on the tree splitting process.

Recall that when building a tree, each split is selected to reduce a splitting impurity (or loss), such as:

- squared error for regression
- Gini impurity for classification

Impurity-based importance measures how much a variable decreases the splitting impurity across all splits in all trees. Variables that frequently produce large impurity reductions receive larger importance values.

For a single tree, the feature importance is simply the total impurity reduction contributed by that variable across all splits in the tree. However, because a single decision tree is often unstable and has high variance, these importance values may vary substantially from one tree to another. In bagging and Random Forests, the importance values are averaged across many trees, producing a much more stable and reliable measure of variable importance.

10.3.3.2 Permutation importance

Another common measure is permutation importance. Unlike impurity-based importance, permutation importance is a general model-agnostic method and is not specific to trees or Random Forests.

The basic idea is:

1. Measure prediction error on validation data.
2. Randomly permute one predictor.
3. Measure prediction error again.

Permuting a predictor destroys the relationship between that predictor and the response while keeping the marginal distribution unchanged. If permuting a predictor substantially worsens predictive performance, then the model strongly relies on that predictor, indicating high importance.

Permutation importance is often easier to interpret because it is directly tied to predictive performance rather than the internal structure of the model.

MDI vs Permutation Importance

The two methods answer different questions:

- MDI explains how much the model used a variable during training.
- Permutation importance measures how much the model depends on that variable for prediction.

In general, if the main goal is to understand predictive performance and generalization, permutation importance is usually more relevant. However, MDI is much faster to compute because it is obtained directly during training. This advantage becomes especially important when the dataset contains many features or when the model is very large.

In many practical situations, MDI and permutation importance may produce similar rankings of variables. In addition, MDI can also help us investigate the internal structure of the model by showing how frequently and how effectively variables are used for splitting.

Limitations of MDI

Continuous predictors, predictors with many possible split points, or high-cardinality categorical variables may have more opportunities to create impurity reductions during splitting. Therefore, MDI may favor these variables.

Limitations of Permutation Importance

When there are redundant features or highly correlated predictors, permuting one variable may not substantially reduce predictive performance because other correlated variables can still provide similar information. As a result, permutation importance may underestimate the importance of correlated predictors.

10.4 Different Types of Randomization in Tree Ensembles

One of the main ideas behind ensemble tree methods is to introduce randomness in order to reduce variance. However, there are many different ways to introduce randomness. Different methods randomize different parts of the tree-building process.

The general tradeoff is:

- more randomness usually decreases variance
- but too much randomness may increase bias

The goal is therefore to make trees different while still keeping each tree reasonably strong.

10.4.1 (Almost) No randomness: Single Decision Tree

A standard CART tree contains almost no randomness.

At each node:

1. all predictors are considered
2. all candidate split points are searched
3. the best split is selected greedily

As a result:

- trees are highly adaptive
- trees usually have low bias
- but trees often have very high variance

Small changes in the training data may completely change the structure of the tree.

In reality, there is still a small amount of randomness in CART. For example, when multiple features produce identical impurity reductions, the algorithm may randomly choose among them.

In practice, in `sklearn`, this is implemented partly by randomizing the order of features. The `random_state` parameter controls this randomness.

10.4.2 Bootstrap Randomness: Bagging

Bagging introduces randomness through bootstrap resampling.

For each tree:

1. sample observations with replacement
2. grow a large tree using the bootstrap sample
3. aggregate all trees together

The splitting procedure itself is unchanged:

- all predictors are still considered
- the best split is still searched greedily

Therefore, bagging mainly creates diversity by changing the training data seen by each tree.

The individual trees are still strong learners, but averaging reduces variance.

10.4.3 Feature Randomness: Random Forest

Random Forest adds another layer of randomness. At each split:

1. randomly select a subset of predictors
2. search for the best split only among those predictors

This has several important effects:

- strong predictors cannot dominate every split
- trees become less correlated
- averaging becomes more effective

However, the split threshold is still optimized greedily. For a chosen predictor, Random Forest still searches all candidate split points.

A natural question is why not randomly choose a subset of predictors once for the entire tree. The reason is that this would usually create trees that are too weak.

Suppose an important predictor is excluded at the beginning. Then the entire tree loses access to that predictor forever.

Random Forest instead performs randomization locally:

- each node receives a new random subset of predictors
- important predictors still have many opportunities to appear
- trees remain reasonably strong

This creates a balance between tree strength and tree diversity.

10.4.4 Threshold Randomness: Extra Trees

Extra Trees (Extremely Randomized Trees) introduces even more randomness. At each split:

1. randomly choose a subset of predictors (similar to Random Forest)
2. randomly generate candidate split thresholds, typically one threshold for each selected predictor
3. select the best split among those randomly generated thresholds

Unlike Random Forest:

- the algorithm does not exhaustively search all candidate split points
- the split thresholds themselves are randomized

This additional randomness usually:

- further decreases correlation between trees
- further reduces variance
- slightly increases bias

10.5 Bagging in Python

We now discuss implementation details in `sklearn`.

There are two main approaches.

1. Use `BaggingRegressor` or `BaggingClassifier` with a chosen base estimator.
2. Use `RandomForestRegressor` or `RandomForestClassifier` when the base estimator is a decision tree and we also want random feature selection.

In the initial demo, we will only talk about Classifier since it is very similar to Regressor. We will use the following generated dataset as the demo example.

```
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split

X, y = make_classification(
    n_samples=1200,
    n_features=10,
    n_informative=3,
    n_redundant=2,
    n_repeated=0,
    n_clusters_per_class=3,
    flip_y=0.03,
    random_state=1,
)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1, stratify=y
)
```

10.5.1 Basic Bagging classifier

The following code creates a bagged ensemble of regression trees.

```

from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier

bag = BaggingClassifier(
    estimator=DecisionTreeClassifier(random_state=0),
    n_estimators=300,
    max_samples=1.0,
    bootstrap=True,
    oob_score=True,
    random_state=0,
)

```

The important arguments are:

- **estimator**: the base model. For classical tree bagging, this is a decision tree.
- **n_estimators**: the number of bootstrap models.
- **max_samples**: the sample size used for each bootstrap sample.
- **bootstrap**: whether sampling is done with replacement. **True** gives bagging; **False** gives pasting.
- **oob_score**: whether to compute out-of-bag performance.

After creating the model, we use the same `.fit()` and `.predict()` and `.score()` workflow as before.

```

bag.fit(X_train, y_train)
bag.score(X_test, y_test)

```

```
0.9138888888888889
```

10.5.2 Random Forest Classifier

A random forest classifier is similar, but the random feature selection is built in.

```

from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=300,
    max_features=1.0,
    bootstrap=True,
    oob_score=True,
    random_state=0
)

rf.fit(X_train, y_train)
rf.score(X_test, y_test)

```

0.9138888888888889

The most important additional argument is `max_features`.

For example:

```

RandomForestClassifier(max_features=1.0)
RandomForestClassifier(max_features="sqrt")
RandomForestClassifier(max_features=0.5)

```

- 1.0: consider all predictors at each split. In this case, it is reduced to a `BaggingRegressor`.
- "sqrt": consider about $\sqrt{p}$ predictors at each split.
- 0.5: consider half of the predictors at each split.

When `max_features=1.0`, a random forest regressor is very close to bagging with decision trees. When `max_features` is smaller, the trees become more decorrelated.

10.5.3 Useful Attributes

After fitting a bagging or random forest model, several attributes are useful. This example shows random forest, but bagging also supports these attributes.

- `.estimators_` stores the fitted trees, and we could indicate which tree we want to use by calling the index.

```
rf.estimators_[0]
```

```
criterion criterion: 'gini'  
{ "gini", "entropy",  
  "log_loss"},  
default="gini"
```

The function to
measure the
quality of a split.
Supported criteria
are

"gini" for the Gini
impurity and
"log_loss" and
"entropy" both for
the
Shannon
information gain,
see :ref:'tree_math-
ematical_formula-
tion'.

```
splitter splitter: 'best'  
{ "best",  
  "random"},  
default="best"
```

The strategy used
to choose the split
at each node.
Supported
strategies are
"best" to choose
the best split and
"random" to choose
the best random
split.

<code>max_depth</code>	None
<code>max_depth: int,</code>	
<code>default=None</code>	

The maximum depth of the tree. If None, then nodes are expanded until all leaves are pure or until all leaves contain less than `min_samples_split` samples.

`min_samples_split` 2

`min_samples_split`: int or float, default=2

The minimum number of samples required to split an internal node:

- If int, then consider ‘`min_samples_split`’ as the minimum number.
- If float, then ‘`min_samples_split`’ is a fraction and ‘`ceil(min_samples_split * n_samples)`’ are the minimum number of samples for each split.

.. versionchanged::

0.18

Added float values for fractions.

`min_samples_leaf` 1

`min_samples_leaf`:
int or float,
default=1

The minimum
number of samples
required to be at a
leaf node.

A split point at
any depth will only
be considered if it
leaves at
least “`min_sam-
ples_leaf`” training
samples in each of
the left and
right branches.

This may have the
effect of smoothing
the model,
especially in
regression.

- If int, then
consider
“`min_samples_leaf`”
as the minimum
number.

- If float, then
“`min_samples_leaf`”
is a fraction and
“`ceil(min_sam-
ples_leaf *
n_samples)`” are
the minimum
number of samples
for each node.

.. versionchanged::
0.18

Added float values
for fractions.

<code>min_weight_fraction_leaf</code>	<code>0.0</code>
---------------------------------------	------------------

`min_weight_fraction_leaf: float, default=0.0`

The minimum weighted fraction of the sum total of weights (of all the input samples) required to be at a leaf node. Samples have equal weight when `sample_weight` is not provided.

`max_features` 1.0
`max_features: int,`
`float or {"sqrt",`
`"log2"},`
`default=None`

The number of features to consider when looking for the best split:

- If int, then consider `'max_features'` features at each split.
- If float, then `'max_features'` is a fraction and `'max(1, int(max_features * n_features_in_))'` features are considered at each split.
- If "sqrt", then `'max_features=sqrt(n_features)'`.
- If "log2", then `'max_features=log2(n_features)'`.
- If None, then `'max_features=n_features'`.

.. note::

The search for a split does not stop until at least one valid partition of the node samples is found, even if it requires to effectively inspect more than `"max_features"` features.

`random_state` 209652396
`random_state: int,`
`RandomState`
instance or `None`,
`default=None`

Controls the randomness of the estimator. The features are always randomly permuted at each split, even if “`splitter`” is set to “`best`”. When “`max_features < n_features`”, the algorithm will select “`max_features`” at random at each split before finding the best split among them. But the best found split may vary across different runs, even if “`max_features=n_features`”. That is the case, if the improvement of the criterion is identical for several splits and one split has to be selected at random. To obtain a deterministic behaviour during fitting, “`random_state`” has to be fixed to an integer. See :term:‘Glossary’ for details.

<code>max_leaf_nodes</code>	None
<code>max_leaf_nodes:</code>	
int, default=None	

Grow a tree with
“max_leaf_nodes”
in best-first
fashion.
Best nodes are
defined as relative
reduction in
impurity.
If None then
unlimited number
of leaf nodes.

`min_impurity_decrease` 0.0
`min_impurity_decrease`: float,
default=0.0

A node will be split if this split induces a decrease of the impurity greater than or equal to this value.

The weighted impurity decrease equation is the following::

$$\begin{aligned} & N_t / N * \\ & (\text{impurity} - N_t_R / N_t * \\ & \text{right_impurity} \\ & - N_t_L / N_t * \\ & \text{left_impurity}) \end{aligned}$$

where “N” is the total number of samples, “N_t” is the number of samples at the current node, “N_t_L” is the number of samples in the left child, and “N_t_R” is the number of samples in the right child.

“N”, “N_t”, “N_t_R” and “N_t_L” all refer to the weighted sum, 318 if “sample_weight” is passed.

.. versionadded::
0.19

<code>class_weight</code>	None
<code>class_weight: dict,</code>	
<code>list of dict or</code>	
<code>"balanced",</code>	
<code>default=None</code>	

Weights associated with classes in the form “{class_label: weight}”.

If None, all classes are supposed to have weight one.

For multi-output problems, a list of dicts can be provided in the same order as the columns of y.

Note that for multioutput (including multilabel) weights should be defined for each class of every column in its own dict. For example, for four-class multilabel classification weights should be [{0: 1, 1: 1}, {0: 1, 1: 5}, {0: 1, 1: 1}, {0: 1, 1: 1}] instead of [{1:1}, {2:5}, {3:1}, {4:1}].

The “balanced” mode uses the values of y to automatically adjust weights inversely proportional to class frequencies in the input data

ccp_alpha 0.0

ccp_alpha:
non-negative float,
default=0.0

Complexity
parameter used for
Minimal
Cost-Complexity
Pruning. The
subtree with the
largest cost
complexity that is
smaller than
“ccp_alpha“ will
be chosen. By
default, no pruning
is performed. See
:ref:‘mini-
mal_cost_com-
plexity_pruning‘
for details. See
:ref:‘sphx_glr_auto_ex-
am-
ples_tree_plot_cost_com-
plexity_prun-
ing.py‘
for an example of
such pruning.

.. versionadded::
0.22

`monotonic_cst` None

`monotonic_cst`:
array-like of int of
shape (n_features),
default=None

Indicates the
monotonicity
constraint to
enforce on each
feature.

- 1: monotonic
increase
- 0: no constraint
- -1: monotonic
decrease

If `monotonic_cst` is
None, no
constraints are
applied.

Monotonicity
constraints are not
supported for:

- multiclass
classifications (i.e.
when 'n_classes >
2'),
- multioutput
classifications (i.e.
when 'n_outputs_> 1'),
- classifications
trained on data
with missing
values.

The constraints
hold over the
probability of the
positive class.

Read ³²¹more in the
:ref:'User Guide '.

.. versionadded::
1.4

- `.oob_score_` gives the OOB score if `oob_score=True`.
For classification, `bag.oob_score_` is the OOB accuracy.
For regression, it is the OOB R^2 score.

```
rf.oob_score_
```

```
0.888095238095238
```

10.5.3.1 Feature importance

MDI can be directly computed using `.feature_importances_` after the model is fitted.

```
rf.feature_importances_
```

```
array([0.02767912, 0.05947287, 0.11152095, 0.02443874, 0.02509798,
       0.25352125, 0.02526593, 0.03385812, 0.26526717, 0.17387786])
```

For permutation importance, we use `permutation_importance` from `sklearn.inspection`.

```
from sklearn.inspection import permutation_importance

pi = permutation_importance(rf, X_test, y_test, n_repeats=30, random_state=1)

pi.importances_mean
```

```
array([-5.27777778e-03,  1.12037037e-02,  8.38888889e-
02,  9.25925926e-05,
        4.62962963e-04,  1.86111111e-01, -1.20370370e-
03,  2.96296296e-03,
        2.05000000e-01,  9.92592593e-02])
```

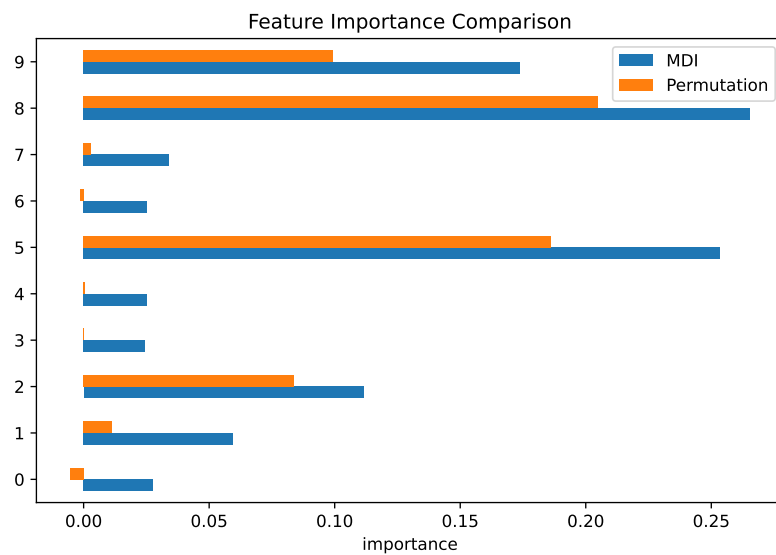
In this example you may see that the two are very similar to each other.

```
import pandas as pd
import matplotlib.pyplot as plt

importance_df = pd.DataFrame(
    {"MDI": rf.feature_importances_, "Permutation": pi.importances_mean}
)
importance_df.plot.barh(figsize=(8, 5))

plt.xlabel("importance")
plt.title("Feature Importance Comparison")
```

```
Text(0.5, 1.0, 'Feature Importance Comparison')
```



10.6 Tuning Hyperparameters

Bagging and random forests are usually easier to tune than boosting. Still, several parameters matter.

Parameter	Main effect
<code>n_estimators</code>	More trees reduce Monte Carlo variation, but increase computation.
<code>max_features</code>	Controls tree correlation. Smaller values create more randomness.
<code>max_samples</code>	Controls bootstrap sample size. Smaller values create more diversity.
<code>min_samples_leaf</code>	Controls smoothness of each tree. Larger values reduce overfitting.
<code>max_depth</code>	Limits tree depth. Often left as <code>None</code> , but can help noisy data.

A practical tuning strategy is:

1. Choose a reasonably large `n_estimators` (so it is highly possible to be stable).
2. Tune `max_features`, `min_samples_leaf`, `max_samples` or `max_depth` with cross-validation or OOB error.
3. Make `n_estimators` smaller to find the smallest stable number.

`n_estimators`

For bagging / random forests, `n_estimators` is usually not a regularization parameter in the usual sense. Increasing the number of trees usually stabilizes the model rather than causing overfitting. The main cost is computation. Therefore when reading the score vs number of trees plot, we are not looking at the highest score, we are looking at when the score is stabilizing.

⚠ 1 and 1.0

These two represent different things.

- `max_samples=1` means that the max number of samples we choose is 1.
- `max_samples=1.0` means that we want to choose all samples that the percent is $1.0=100\%$.

i Note

Comparing to single tree/boosting, we tend to use larger trees in bagging. Therefore we don't need `ccp_alpha`, and we could use larger `max_depth` and smaller `min_samples_leaf`.

i OOB validation

We could use oob score to tune hyperparameters. However we have to manually write the tuning code.

```
from sklearn.ensemble import RandomForestClassifier

max_features_list = [0.3, 0.5, 0.8, 1.0]
best_score = -1
best_param = None

for mf in max_features_list:
    rf = RandomForestClassifier(
        n_estimators=300, max_features=mf, oob_score=True, random_state=0, n_jobs=-1
    )
    rf.fit(X_train, y_train)
    if rf.oob_score_ > best_score:
        best_score = rf.oob_score_
        best_param = mf

print(best_param)
```

0.3

i Cross-validation

We could still use cross-validation to tune a random forest classifier.

```
from sklearn.model_selection import GridSearchCV
from sklearn.ensemble import RandomForestClassifier

param_grid = {
    "max_features": [0.3, 0.6, 1.0],
    "min_samples_leaf": [1, 3, 5],
    "max_samples": [0.6, 0.8, 1.0],
}

rf = RandomForestClassifier(n_estimators=300, random_state=0)

grid = GridSearchCV(rf, param_grid=param_grid, cv=5, n_jobs=-1)

grid.fit(X_train, y_train)

best_rf = grid.best_estimator_
grid.best_params_

{'max_features': 0.3, 'max_samples': 0.8, 'min_samples_leaf': 1}
```

i Randomized Search

If the dataset is large, a full grid can be slow. In that case, `RandomizedSearchCV` is often a better choice. In this case, only a few random combinations will be searched. In the following example, we only search 30 combinations.

```

from sklearn.model_selection import RandomizedSearchCV
from scipy.stats import randint, uniform

param_dist = {
    "max_features": uniform(0.2, 0.8),
    "min_samples_leaf": randint(1, 20),
    "max_samples": uniform(0.5, 0.5),
}

rf = RandomForestClassifier(n_estimators=300, random_state=0)

search = RandomizedSearchCV(
    rf, param_distributions=param_dist, n_iter=30, cv=5, random_state=0, n_jobs=-1
)

search.fit(X_train, y_train)
search.best_params_

{'max_features': np.float64(0.9220787804235238),
 'max_samples': np.float64(0.7249749949556138),
 'min_samples_leaf': 2}

```

10.6.1 n_jobs

When we reach this stage, you may notice that training can become quite slow. This is expected because, in `GridSearchCV`, we need to evaluate many combinations of hyperparameters. For each combination, we may need to train many trees (due to bagging or Random Forests), and the entire process must be repeated multiple times because of cross-validation (for example, `cv=5`).

Although many models need to be trained, an important advantage is that these training tasks are largely independent. Therefore, they can often be trained simultaneously using multiple CPU cores.

In contrast, boosting methods that we will discuss later are fundamentally sequential. Each tree depends on the results of the previous trees, so the trees must be trained one after another.

As a result, boosting methods are generally less parallelizable at the tree level.

`sklearn` supports multi-core CPUs through the `n_jobs` argument.

- `n_jobs=1` means single-core execution
- `n_jobs=k` uses exactly `k` CPU cores
- `n_jobs=-1` uses all available CPU cores
- `n_jobs=None` uses the library default, which is currently equivalent to 1 in most `sklearn` functions

Many functions in `sklearn` support this argument. For details, consult the official documentation for the specific function being used.

One important practical issue is nested parallelism. When multiple nested functions both support `n_jobs`, it is usually best to parallelize only the outermost function.

For example, `GridSearchCV(RandomForestClassifier(n_jobs=1), n_jobs=-1)` is generally preferred over `GridSearchCV(RandomForestClassifier(n_jobs=-1), n_jobs=-1)` because allowing both levels to parallelize simultaneously may create too many worker processes or threads, leading to excessive overhead and sometimes even slower performance.

Example 10.1 (time it). We first write a helper function to compute the runtime.

```
from time import perf_counter
from contextlib import contextmanager

@contextmanager
def timer(name="runtime"):
    start = perf_counter()
    yield
    end = perf_counter()

    print(f"{name}: {end - start:.4f} seconds")
```

Then use it to test the time for different `n_jobs`.


```

param_grid = {
    "max_features": [0.3, 0.6, 1.0],
    "min_samples_leaf": [1, 3, 5],
    "max_samples": [0.6, 0.8, 1.0],
}

rf = RandomForestClassifier(n_estimators=300, random_state=0)

grid = GridSearchCV(rf, param_grid=param_grid, cv=5, n_jobs=-1)
with timer("n_jobs=-1"):
    grid.fit(X_train, y_train)

grid = GridSearchCV(rf, param_grid=param_grid, cv=5, n_jobs=1)
with timer("n_jobs=1"):
    grid.fit(X_train, y_train)

```

n_jobs=-1: 20.4251 seconds
n_jobs=1: 76.9425 seconds

10.7 Example: Diabetes dataset

We now use the diabetes regression dataset from `sklearn`. In order to compare results using trees, we use the same split with `random_state=1`.

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

X, y = load_diabetes(return_X_y=True)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=1)

```

We compare three models:

1. a single regression tree,
2. bagging with regression trees,
3. a random forest.

```
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import BaggingRegressor, RandomForestRegressor

tree = DecisionTreeRegressor(random_state=1, ccp_alpha=105.185)

bag = BaggingRegressor(
    estimator=DecisionTreeRegressor(random_state=1),
    n_estimators=300,
    oob_score=True,
    random_state=1,
)

rf = RandomForestRegressor(
    n_estimators=300,
    max_features="sqrt",
    oob_score=True,
    random_state=1,
)
```

```
tree.fit(X_train, y_train)
tree.score(X_test, y_test)
```

0.1922738375305083

```
bag.fit(X_train, y_train)
bag.score(X_test, y_test)
```

0.2951908085625511

```
rf.fit(X_train, y_train)
rf.score(X_test, y_test)
```

0.3555401846697034

The single tree is expected to be unstable. Bagging and random forests usually improve test performance by averaging many trees.

For the bagging and random forest models, we can also check OOB performance.

```
bag.oob_score_
```

```
0.4567923788468702
```

```
rf.oob_score_
```

```
0.47245609860472904
```

OOB R^2 gives an internal estimate of predictive performance based only on the training data.

10.7.1 Tuning the Random Forest

We showed Bagging in the previous section. Since it overlaps with RandomForest, we will only focus on the Random Forest model in this section. We first tune `max_features`, `min_samples_leaf`, and `max_samples` by cross-validation, with a relative small number of estimators.

```
from sklearn.model_selection import GridSearchCV

rf_base = RandomForestRegressor(n_estimators=300, random_state=1)

param_grid = {
    "max_features": ["sqrt", 0.5, 1.0],
    "min_samples_leaf": [1, 3, 5, 10],
    "max_samples": [0.6, 0.8, 1.0],
}

grid = GridSearchCV(rf_base, param_grid=param_grid, cv=5, n_jobs=-1)

grid.fit(X_train, y_train)
```

```
print(f"Best parameters: {grid.best_params_}")
print(f"Best CV R^2: {grid.best_score_}")
```

```
Best parameters: {'max_features': 'sqrt', 'max_samples': 1.0, 'min_samples_leaf': 5}
Best CV R^2: 0.45220837115313606
```

Now evaluate the tuned model on the test set.

```
grid.best_estimator_.score(X_test, y_test)
```

```
0.3777186499485139
```

10.7.2 Find the Number of Trees

The number of trees controls how stable the ensemble average is. The following code fits forests with different numbers of trees and records OOB performance.

```
tree_counts = [10, 25, 50, 100, 200, 300, 400, 800]
oob_scores = []
test_r2 = []

for n_trees in tree_counts:
    model = RandomForestRegressor(
        n_estimators=n_trees,
        max_features="sqrt",
        max_samples=1.0,
        min_samples_leaf=5,
        oob_score=True,
        random_state=1,
        n_jobs=-1,
    )
    model.fit(X_train, y_train)
    oob_scores.append(model.oob_score_)
    test_r2.append(model.score(X_test, y_test))

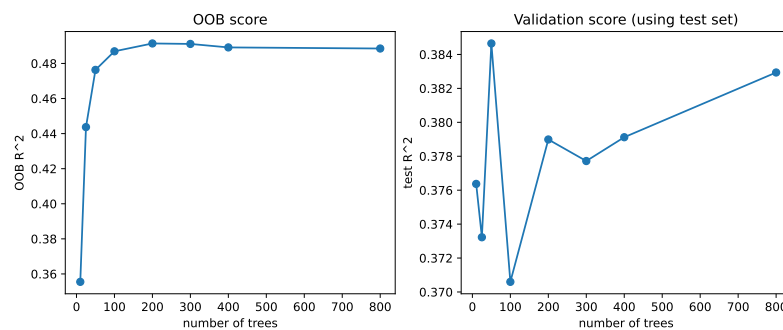
fig, ax = plt.subplots(1, 2, figsize=(11, 4))
```

```

ax[0].plot(tree_counts, oob_scores, marker="o")
ax[0].set_xlabel("number of trees")
ax[0].set_ylabel("OOB R^2")
ax[0].set_title("OOB score")

ax[1].plot(tree_counts, test_r2, marker="o")
ax[1].set_xlabel("number of trees")
ax[1].set_ylabel("test R^2")
ax[1].set_title("Validation score (using test set)");

```



As mentioned previously, when reading these two plots, it is important not to focus on the single highest point. Instead, we look for the point where the curve becomes stable and nearly flat. In this example, around 200 trees appears to be a reasonable choice.

Therefore our best model so far is

```

best_rf = RandomForestRegressor(
    n_estimators=200,
    max_features="sqrt",
    max_samples=1.0,
    min_samples_leaf=5,
    oob_score=True,
    random_state=1,
    n_jobs=-1
)
best_rf.fit(X_train, y_train)

```

n_estimators	200
n_estimators: int, default=100	

The number of
trees in the forest.

.. versionchanged::
0.22
The default value
of “n_estimators“
changed from 10 to
100
in 0.22.

```

criterion criterion: 'squared_error'
{"squared_error",
 "absolute_error",
 "friedman_mse",
 "poisson"}, default="squared_error"

```

The function to measure the quality of a split. Supported criteria are

"squared_error" for the mean squared error, which is equal to variance reduction as feature selection criterion and minimizes the L2 loss using the mean of each terminal node, "friedman_mse", which uses mean squared error with Friedman's improvement score for potential splits, "absolute_error" for the mean absolute error, which minimizes the L1 loss using the median of each terminal node, and "poisson" which uses reduction in Poisson deviance to find splits. Training using "absolute_error" is significantly slower than when using "squared_error".

.. versionadded:: 0.18

Mean Absolute

<code>max_depth</code>	None
<code>max_depth: int,</code>	
<code>default=None</code>	

The maximum depth of the tree. If None, then nodes are expanded until all leaves are pure or until all leaves contain less than `min_samples_split` samples.

`min_samples_split` 2

`min_samples_split`: int or float, default=2

The minimum number of samples required to split an internal node:

- If int, then consider ‘`min_samples_split`’ as the minimum number.
- If float, then ‘`min_samples_split`’ is a fraction and ‘`ceil(min_samples_split * n_samples)`’ are the minimum number of samples for each split.

.. versionchanged::

0.18

Added float values for fractions.

`min_samples_leaf` 5

`min_samples_leaf`:
int or float,
default=1

The minimum
number of samples
required to be at a
leaf node.

A split point at
any depth will only
be considered if it
leaves at
least “`min_sam-
ples_leaf`” training
samples in each of
the left and
right branches.

This may have the
effect of smoothing
the model,
especially in
regression.

- If int, then
consider
“`min_samples_leaf`”
as the minimum
number.

- If float, then
“`min_samples_leaf`”
is a fraction and
“`ceil(min_sam-
ples_leaf *
n_samples)`” are
the minimum
number of samples
for each node.

.. versionchanged::
0.18

Added float values
for fractions.

<code>min_weight_fraction_leaf</code>	<code>0.0</code>
---------------------------------------	------------------

`min_weight_fraction_leaf: float, default=0.0`

The minimum weighted fraction of the sum total of weights (of all the input samples) required to be at a leaf node. Samples have equal weight when `sample_weight` is not provided.

```
max_features      'sqrt'
max_features:
{"sqrt", "log2",
None}, int or float,
default=1.0
```

The number of features to consider when looking for the best split:

- If int, then consider 'max_features' features at each split.
- If float, then 'max_features' is a fraction and 'max(1, int(max_features * n_features_in_))' features are considered at each split.
- If "sqrt", then 'max_features=sqrt(n_features)'.
- If "log2", then 'max_features=log2(n_features)'.
- If None or 1.0, then 'max_features=n_features'.

.. note::
The default of 1.0 is equivalent to bagged trees and more randomness can be achieved by setting smaller values, e.g. 0.3.

.. versionchanged:: 1.1
The default of 'max_features'

<code>max_leaf_nodes</code>	None
<code>max_leaf_nodes:</code>	
int, default=None	

Grow trees with
“max_leaf_nodes”
in best-first
fashion.
Best nodes are
defined as relative
reduction in
impurity.
If None then
unlimited number
of leaf nodes.

`min_impurity_decrease` 0.0
`min_impurity_decrease`: float,
default=0.0

A node will be split if this split induces a decrease of the impurity greater than or equal to this value.

The weighted impurity decrease equation is the following::

$$\begin{aligned} & N_t / N * \\ & (\text{impurity} - N_t_R / N_t * \\ & \text{right_impurity} \\ & - N_t_L / N_t * \\ & \text{left_impurity}) \end{aligned}$$

where “N” is the total number of samples, “N_t” is the number of samples at the current node, “N_t_L” is the number of samples in the left child, and “N_t_R” is the number of samples in the right child.

“N”, “N_t”, “N_t_R” and “N_t_L” all refer to the weighted sum, 342 if “sample_weight” is passed.

.. versionadded::
0.19

<code>bootstrap</code>	<code>True</code>
<code>bootstrap: bool,</code>	
<code>default=True</code>	

Whether bootstrap samples are used when building trees. If False, the whole dataset is used to build each tree.

<code>oob_score</code>	True
------------------------	------

`oob_score`: bool or callable,
default=False

Whether to use out-of-bag samples to estimate the generalization score.

By default, :func:`~sklearn.metrics.r2\_score` is used.

Provide a callable with signature `'metric(y_true, y_pred)'` to use a custom metric.

Only available if `'bootstrap=True'`.

For an illustration of out-of-bag (OOB) error estimation, see the example

:ref:`sphx\_glr\_auto\_examples\_ensemble\_plot\_ensemble\_oob.py`.

`n_jobs n_jobs: int, -1`
`default=None`

The number of jobs
to run in parallel.

`:meth:‘fit‘`,
`:meth:‘predict‘`,
`:meth:‘decision_path‘` and
`:meth:‘apply‘` are
all parallelized over
the

trees. “None”
means 1 unless in a
`:obj:‘joblib.parallel_backend‘`
context. “-1”
means using all
processors. See
`:term:‘Glossary`
‘ for more details.

`random_state` 1
`random_state: int,`
`RandomState`
instance or `None`,
`default=None`

Controls both the randomness of the bootstrapping of the samples used when building trees (if “bootstrap=True”) and the sampling of the features to consider when looking for the best split at each node (if “max_features < n_features”). See :term:‘Glossary’ for details.

`verbose verbose: 0`
`int, default=0`

Controls the verbosity when fitting and predicting.

<code>warm_start</code>	False
<code>warm_start: bool,</code>	
<code>default=False</code>	

When set to
“True“, reuse the
solution of the
previous call to fit
and add more
estimators to the
ensemble,
otherwise, just fit a
whole
new forest. See
`:term:‘Glossary ‘`
and
`:ref:‘tree_ensem-
ble_warm_start‘`
for details.

ccp_alpha 0.0

ccp_alpha:
non-negative float,
default=0.0

Complexity
parameter used for
Minimal
Cost-Complexity
Pruning. The
subtree with the
largest cost
complexity that is
smaller than
“ccp_alpha“ will
be chosen. By
default, no pruning
is performed. See
:ref:‘mini-
mal_cost_com-
plexity_pruning‘
for details. See
:ref:‘sphx_glr_auto_ex-
am-
ples_tree_plot_cost_com-
plexity_prun-
ing.py‘
for an example of
such pruning.

.. versionadded::
0.22

`max_samples` 1.0

`max_samples`: int
or float,
default=None

If `bootstrap` is
True, the number
of samples to draw
from `X`
to train each base
estimator.

- If None (default),
then draw
`'X.shape[0]'`
samples.
- If int, then draw
`'max_samples'`
samples.
- If float, then draw
`'max(round(n_sam-
ples *
max_samples), 1)'`
samples. Thus,
`'max_samples'`
should be in the
interval `'(0.0, 1.0]'`.

.. versionadded::
0.22

monotonic_cst monotonic_cst: array-like of int of shape (n_features), default=None	None
Indicates the monotonicity constraint to enforce on each feature. - 1: monotonically increasing - 0: no constraint - -1: monotonically decreasing	
If monotonic_cst is None, no constraints are applied.	
Monotonicity constraints are not supported for: - multioutput regressions (i.e. when 'n_outputs_ > 1'), - regressions trained on data with missing values.	
Read more in the :ref:'User Guide '	
.. versionadded:: 1.4	

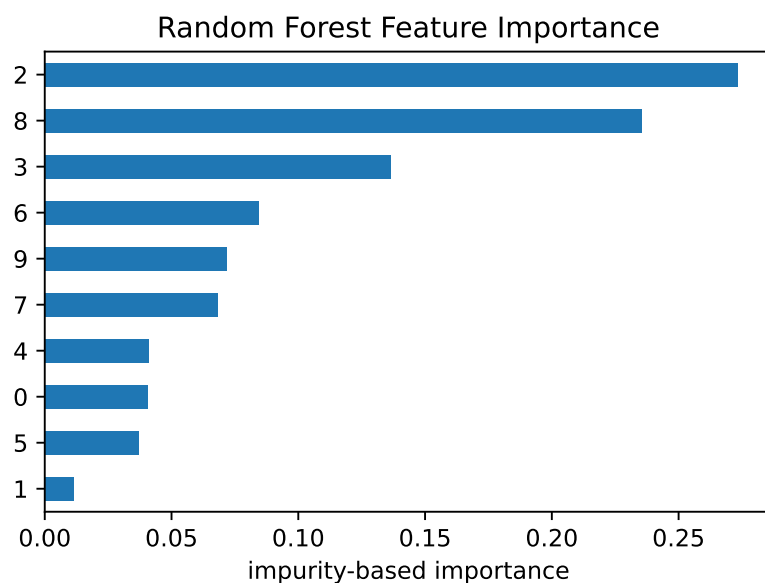
10.7.2.1 Variable Importance

We first examine impurity-based feature importance.

```
importance = pd.Series(best_rf.feature_importances_).sort_values(ascending=True)

importance.plot(kind="barh")
plt.xlabel("impurity-based importance")
plt.title("Random Forest Feature Importance")
```

```
Text(0.5, 1.0, 'Random Forest Feature Importance')
```



These values measure how much each variable reduces prediction error inside the trees. They are useful for a quick summary, but they should not be treated as causal effects.

Permutation importance gives a more direct performance-based measure.

```
from sklearn.inspection import permutation_importance

perm = permutation_importance(
    best_rf, X_test, y_test, n_repeats=30, random_state=1, n_jobs=-1
```

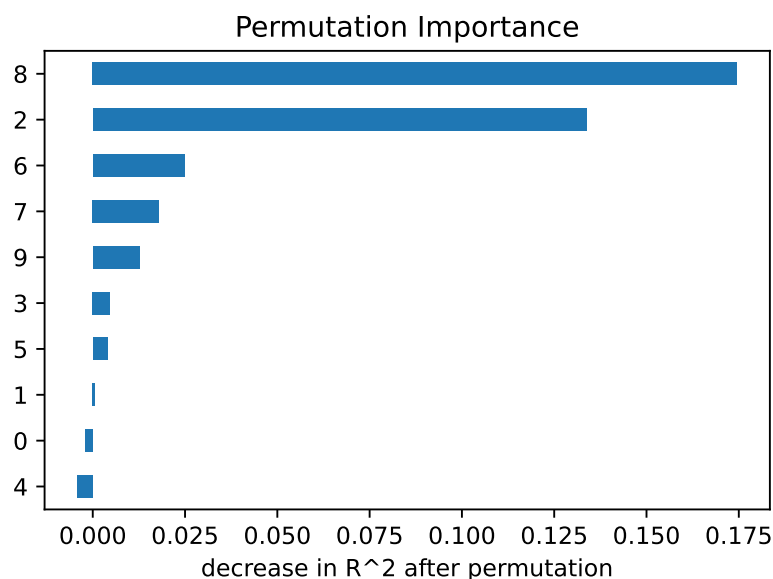
```

)
perm_importance = pd.Series(perm.importances_mean).sort_values(ascending=True)

perm_importance.plot(kind="barh")
plt.xlabel("decrease in R^2 after permutation")
plt.title("Permutation Importance")

```

```
Text(0.5, 1.0, 'Permutation Importance')
```



If permuting a predictor substantially worsens predictions, then the fitted random forest relies on that predictor. If the importance is near zero, the predictor may not add much predictive information beyond the others.

From the two plots, feature 2 and 8 are the two most important features. If we go back to check the description of the dataset, they are `bmi`: body mass index and `s5`: `ltg`, possibly log of serum triglycerides level.

11 Boosting

Boosting is another ensemble method. Like bagging and random forests, it combines many trees into one model. The difference is how the trees are built.

Bagging and random forests build trees independently and then average them.

Boosting builds trees sequentially. Each new tree is trained to improve the current ensemble.

The central idea is:

Build many weak learners one at a time, and let each new learner focus on what the current model still gets wrong.

This chapter first discusses AdaBoost. AdaBoost is historically important and is the cleanest way to see the boosting idea. We then discuss the margin effect of AdaBoost and how to tune its hyperparameters. Finally, we discuss gradient boosting and use XGBoost as the main implementation example.

11.1 AdaBoost

AdaBoost (Adaptive Boosting) was introduced by Yoav Freund and Robert E. Schapire in 1995 [10] as one of the first highly successful boosting algorithms. Their original work showed that many weak learners, each performing only slightly better than random guessing, could be combined into a highly accurate classifier. AdaBoost became a foundational method in statistical learning, and it strongly influenced the later development of modern boosting methods such as Gradient Boosting and XGBoost.

The original AdaBoost algorithm was proposed as a classification method. Regression extensions were developed later after the success of AdaBoost classifiers. In this course, we mainly study AdaBoost classification because it introduces the core ideas of boosting in a simple and historically important setting.

For regression problems, we will move on to gradient boosting methods, which have become the dominant boosting framework in modern machine learning.

To keep the notation simple, assume the response is binary:

$$y_i \in \{-1, +1\}.$$

AdaBoost constructs a classifier of the form

$$F_M(x) = \sum_{m=1}^M \alpha_m G_m(x),$$

where:

- $G_m(x)$ is the m -th weak classifier,
- α_m is the weight assigned to that classifier,
- M is the number of boosting rounds.

The final prediction is

$$\hat{G}(x) = \text{sign}\{F_M(x)\}.$$

Thus AdaBoost is a weighted voting method. Classifiers with larger α_m have more influence in the final vote.

11.1.1 Weak Learners

AdaBoost is built from weak learners.

A weak learner is a classifier that is only slightly better than random guessing. In tree-based AdaBoost, the most common weak learner is a decision stump, which is a tree with only one split.

In `sklearn`, a decision stump is created by

```
from sklearn.tree import DecisionTreeClassifier

DecisionTreeClassifier(max_depth=1)
```

A decision stump is extremely simple and usually underfits when used alone. One of the most surprising and important ideas behind AdaBoost is that by combining many such weak learners sequentially, the resulting ensemble can become a highly accurate and powerful model.

11.1.2 Weights of observations

In AdaBoost, there are two different types of weights.

- Estimator weights (α_m) determine how much influence each weak learner has in the final ensemble.
- Observation weights ($w_i^{(m)}$) determine which training observations receive more attention when fitting the next learner.

During training, AdaBoost repeatedly updates the weights of the training observations. Observations that are difficult to classify receive larger weights, so future learners focus more on them.

To understand this process, we need to discuss how a decision tree is trained on a weighted dataset.

The main idea is very simple. In an ordinary dataset, quantities such as class frequencies and node sample counts are computed using the number of observations. For a weighted dataset, these counts are replaced by weighted counts, that is, by the sum of the observation weights.

For example, suppose a node contains observations with weights $w_1, w_2, \dots, w_n$. Then the effective sample size of the node becomes $\sum_{i=1}^n w_i$. Similarly, class proportions are computed using weighted proportions rather than ordinary proportions.

Therefore, training a decision tree on weighted data is conceptually very similar to ordinary tree training. We simply replace

ordinary counts by weighted counts throughout the splitting calculations.

Example 11.1. Consider the following data:

Table 11.1

x0	x1	y	Weight
1.0	2.1	+	0.5
1.0	1.1	+	0.125
1.3	1.0	-	0.125
1.0	1.0	-	0.125
2.0	1.0	+	0.125

The weighted Gini impurity is

$$\text{WeightedGini} = 1 - (0.5 + 0.125 + 0.125)^2 - (0.125 + 0.125)^2 = 0.375.$$

11.1.3 The AdaBoost Algorithm

Suppose we have training data

$$(x_1, y_1), \dots, (x_n, y_n), \quad y_i \in \{-1, +1\}.$$

Initialize observation weights:

$$w_i^{(1)} = \frac{1}{n}.$$

Then AdaBoost proceeds as follows.

1. For boosting round $m = 1, \dots, M$, fit a weak classifier G_m using the current weights.
2. Compute the weighted classification error by the m -th classifier:

$$\text{err}_m = \frac{\text{the total weights of misclassified observations}}{\text{the total weights}} = \frac{\sum_{\text{misclassified}} w_i^{(m)}}{\sum_{\text{all}} w_i^{(m)}}.$$

3. Assign a weight to the weak classifier:

$$\alpha_m = \frac{1}{2} \log \left(\frac{1 - \text{err}_m}{\text{err}_m} \right).$$

4. Update observation weights and then normalize the weights so they sum to one:

$$w_i^{(m+1)} \xleftarrow{\text{normalize}} \hat{w}_i^{(m+1)} \leftarrow \begin{cases} w_i^{(m)} \exp(-\alpha_m) & \text{if observation } i \text{ is correctly classified by the } m\text{th weak learner} \\ w_i^{(m)} \exp(\alpha_m) & \text{if observation } i \text{ is misclassified by the } m\text{th weak learner} \end{cases}$$

that

$$w_i^{(m+1)} = \frac{\hat{w}_i^{(m+1)}}{\sum_j \hat{w}_j^{(m+1)}}.$$

It is easy to see from the updating rule that AdaBoost puts more weight on observations that were misclassified.

5. Repeat the above process until the stop conditions are met.

The final model is then

$$F_M(x) = \sum_{m=1}^M \alpha_m G_m(x),$$

and the final prediction is

$$\hat{G}(x) = \text{sign} \{F_M(x)\}.$$

i Learning Rate and the $\frac{1}{2}$ Factor

In the classical AdaBoost derivation, the classifier weight is

$$\alpha_m = \frac{1}{2} \log \left(\frac{1 - \text{err}_m}{\text{err}_m} \right).$$

In practice, many implementations additionally introduce a learning rate $\eta > 0$ and use

$$\alpha_m = \eta \cdot \frac{1}{2} \log \left(\frac{1 - \text{err}_m}{\text{err}_m} \right).$$

Some textbooks instead write

$$\alpha_m = \eta \cdot \log \left(\frac{1 - \text{err}_m}{\text{err}_m} \right),$$

that is, without the factor $\frac{1}{2}$.

This does not fundamentally change the algorithm. The factor $\frac{1}{2}$ can simply be absorbed into the learning rate by replacing η with $\eta/2$. Therefore the two formulas differ only by a rescaling of the step size.

We mainly use the classical form with $\frac{1}{2}$ since it naturally appears in the standard derivation from exponential loss minimization and margin theory.

The learning rate controls how much each weak learner contributes to the final ensemble.

- A larger learning rate makes the model change more aggressively at each iteration.
- A smaller learning rate produces more gradual updates.

In practice, smaller learning rates are often combined with a larger number of estimators.

i Margin

$yF(x)$ is called the margin of the model F on the observation (x, y) . Since AdaBoost uses $\text{sign}\{F(x)\}$ as the final prediction rule, the margin can be interpreted as a confidence score for the prediction.

- If the margin is positive, the prediction is correct.
- If the margin is negative, the prediction is incorrect.
- A larger positive margin indicates a more confident prediction.
- A margin close to zero indicates uncertainty.

Notice that the observation weight updating formula can be rewritten as

$$w_i^{(m+1)} = w_i^{(m)} \exp(-\alpha_m y_i G_m(x_i)),$$

where $y_i G_m(x_i)$ is exactly the margin of the weak learner G_m on observation (x_i, y_i) .

Therefore, AdaBoost increases the weights of observations with small or negative margins and decreases the weights of observations with large positive margins. In this sense, AdaBoost can be viewed as an algorithm that attempts to improve the margin distribution by pushing margins toward larger positive values.

Margin theory is one of the central ideas in boosting and helps explain why boosting methods often continue to improve test performance even after the training error has already reached zero. We will discuss this topic in more detail later.

i Why the Learner Weight Makes Sense

The classifier weight is

$$\alpha_m = \eta \cdot \log \left(\frac{1 - \text{err}_m}{\text{err}_m} \right).$$

- If err_m is small, then the classifier is useful and α_m is large.
- If err_m is close to 0.5, then the classifier is close to random guessing and α_m is close to zero.
- If $\text{err}_m > 0.5$, then the classifier is worse than random guessing. In practice, this means the weak learner is not doing its job.

11.1.4 Why Does AdaBoost Work?

One of the most surprising aspects of AdaBoost is that it often performs extremely well even when each individual learner is very weak.

A useful way to understand boosting is through the bias-variance tradeoff.

- Bagging and Random Forests mainly reduce variance by averaging many independent trees.

- Boosting mainly reduces bias by sequentially correcting the errors made by the current model.

At each boosting round, the new learner focuses more on the observations that are currently difficult to classify. As more learners are added, many simple decision rules combine into a much more flexible decision boundary. This sequential correction process allows AdaBoost to gradually improve the model while still using very simple base learners such as decision stumps.

In the early boosting rounds, the model often becomes substantially more accurate. However, if boosting continues too aggressively, variance may eventually increase and overfitting may occur. Several techniques are commonly used to control this effect, like shallow trees, smaller learning rates, subsampling and early stopping. Thus boosting can often achieve highly accurate models while still maintaining good generalization performance.

From a mathematical viewpoint, understanding *why* AdaBoost works so well mathematically became one of the major research topics in statistical learning during the late 1990s and early 2000s. Many theoretical studies showed that AdaBoost is closely connected to several important topics, including additive modeling, exponential loss minimization and margin theory.

One particularly influential result showed that AdaBoost can be interpreted as a stagewise additive optimization procedure. Important theoretical works include [11], [12], [13], and [14]. These studies greatly influenced the later development of modern boosting methods such as Gradient Boosting and XGBoost.

11.2 AdaBoost in Python

The basic APIs are as usual. We use `AdaBoostClassifier` from `sklearn.ensemble` to implement AdaBoost. All the regular APIs are the same to other models.

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
```



```

ada = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1),
    n_estimators=800,
    learning_rate=0.1,
    random_state=1,
)

ada.fit(X_train, y_train)

```

i SAMME and SAMME.R

The original AdaBoost algorithm was designed for binary classification. Later extensions such as SAMME and SAMME.R generalized AdaBoost to multiclass classification problems. In **sklearn**, multiclass classification for **AdaBoostClassifier** is implemented using the SAMME algorithm by default.

However, these variants are less important in modern statistical learning practice and will not be studied in detail in this course. One reason is that these extensions substantially increase the algorithmic complexity while introducing relatively limited new conceptual ideas beyond the original AdaBoost framework. In particular, SAMME.R relies heavily on predicted class probabilities, making the method more sensitive to the quality of probability estimation and numerical stability.

More importantly, many of the key ideas behind SAMME.R already overlap with the additive optimization viewpoint used in modern gradient boosting methods. As a result, modern frameworks such as Gradient Boosting and XGBoost provide a more general, flexible, and unified approach for both regression and classification problems. Consequently, we will focus primarily on the original AdaBoost classifier as an accessible introduction to the core ideas of boosting before moving directly to Gradient Boosting. Reflecting a similar shift in modern machine learning practice, **sklearn** has also deprecated SAMME.R in recent versions due to its reliance on probability estimation and its conceptual overlap with more general gradient boosting frameworks.

11.2.1 The Margin Effect of AdaBoost

AdaBoost is often explained as an algorithm that reduces training error. However, its behavior is more subtle than that.

For a binary classifier with labels $y_i \in \{-1, +1\}$, define the margin of observation i as

$$\text{margin}_i = y_i F_M(x_i).$$

The sign of the margin determines whether the observation is classified correctly:

- if $\text{margin}_i > 0$, the observation is correctly classified;
- if $\text{margin}_i < 0$, the observation is misclassified.

The magnitude of the margin measures confidence. A large positive margin means the ensemble strongly supports the correct class.

11.2.2 Why Margins Matter

AdaBoost often continues to improve test performance even after the training error has reached zero. This can seem surprising. If the training error is already zero, what is left to improve?

The answer is margins.

After all training observations are classified correctly, AdaBoost may still increase the margins. Larger margins mean the classifier is more confident and more stable. This can improve generalization.

Margin view

Training error only asks whether the margin is positive.
The margin view also asks how positive it is.

This is one reason AdaBoost can work well in practice. It does not only try to classify points correctly. It also tends to push correctly classified points farther away from the decision boundary.

11.2.3 Margin study in Python

The following example uses a manually crafted dataset. We track training error, test error, and margins as the number of boosting rounds increases.

```
import numpy as np
from sklearn.model_selection import train_test_split

rng = np.random.default_rng(1)

n = 3000
p = 10
X = rng.normal(size=(n, p))

score = (
    1.2 * (X[:, 0] > -0.6)
    + 1.0 * (X[:, 0] > 0.4)
    - 1.1 * (X[:, 1] > 0.0)
    + 0.9 * (X[:, 2] > 0.5)
    - 0.8 * (X[:, 3] > -0.2)
    + 0.7 * ((X[:, 4] > 0) & (X[:, 5] > 0))
    - 0.7 * ((X[:, 6] < 0) & (X[:, 7] > 0.3))
    + 0.5 * (X[:, 8] > 0.1)
    - 0.5 * (X[:, 9] > -0.4)
)
y = 2 * (score > np.quantile(score, 0.52)).astype(int) - 1

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.5, stratify=y, random_state=1
)
```

We first train the model.

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier

ada = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=2, random_state=1),
    n_estimators=2000,
```

```

    learning_rate=0.2,
    random_state=1,
)

ada.fit(X_train, y_train)

```

estimator	DecisionTreeC...an-
estimator: object,	dom_state=1)
default=None	

The base estimator from which the boosted ensemble is built. Support for sample weighting is required, as well as proper “classes_” and “n_classes_” attributes. If “None”, then the base estimator is

:class:~sklearn.tree.DecisionTreeClassifier

initialized with ‘max_depth=1’.

.. versionadded:: 1.2

‘base_estimator’ was renamed to ‘estimator’.

n_estimators	2000
n_estimators: int, default=50	

The maximum number of estimators at which boosting is terminated.

In case of perfect fit, the learning procedure is stopped early.

Values must be in the range $[1, \infty)$.

learning_rate	0.2
learning_rate: float, default=1.0	

Weight applied to each classifier at each boosting iteration. A higher learning rate increases the contribution of each classifier.

There is a trade-off between the 'learning_rate' and 'n_estimators' parameters.

Values must be in the range $(0.0, \infty)$.

`random_state` 1
`random_state: int,`
`RandomState`
instance or `None`,
default=`None`

Controls the
random seed given
at each ‘estimator’
at each
boosting iteration.
Thus, it is only
used when
‘estimator’ exposes
a ‘`random_state`’.
Pass an int for
reproducible
output across
multiple function
calls.
See :term:‘Glossary
‘.

criterion criterion: 'gini'
{ "gini", "entropy",
"log_loss"},
default="gini"

The function to
measure the
quality of a split.
Supported criteria
are
"gini" for the Gini
impurity and
"log_loss" and
"entropy" both for
the
Shannon
information gain,
see :ref:'tree_math-
ematical_formula-
tion'.

splitter splitter: 'best'
{ "best",
"random"},
default="best"

The strategy used
to choose the split
at each node.
Supported
strategies are
"best" to choose
the best split and
"random" to choose
the best random
split.

<code>max_depth</code>	2
<code>max_depth: int,</code>	
<code>default=None</code>	

The maximum depth of the tree. If None, then nodes are expanded until all leaves are pure or until all leaves contain less than `min_samples_split` samples.

`min_samples_split` 2

`min_samples_split`: int or float, default=2

The minimum number of samples required to split an internal node:

- If int, then consider ‘`min_samples_split`’ as the minimum number.
- If float, then ‘`min_samples_split`’ is a fraction and ‘`ceil(min_samples_split * n_samples)`’ are the minimum number of samples for each split.

.. versionchanged::

0.18

Added float values for fractions.

`min_samples_leaf` 1
`min_samples_leaf`:
int or float,
default=1

The minimum
number of samples
required to be at a
leaf node.

A split point at
any depth will only
be considered if it
leaves at
least “`min_sam-
ples_leaf`” training
samples in each of
the left and
right branches.

This may have the
effect of smoothing
the model,
especially in
regression.

- If int, then
consider
“`min_samples_leaf`”
as the minimum
number.
- If float, then
“`min_samples_leaf`”
is a fraction and
“`ceil(min_sam-
ples_leaf *
n_samples)`” are
the minimum
number of samples
for each node.

.. versionchanged::
0.18
Added float values
for fractions.

<code>min_weight_fraction_leaf</code>	<code>0.0</code>
---------------------------------------	------------------

`min_weight_fraction_leaf: float, default=0.0`

The minimum weighted fraction of the sum total of weights (of all the input samples) required to be at a leaf node. Samples have equal weight when `sample_weight` is not provided.

max_features	None
max_features: int, float or {"sqrt", "log2"}, default=None	

The number of features to consider when looking for the best split:

- If int, then consider 'max_features' features at each split.
- If float, then 'max_features' is a fraction and 'max(1, int(max_features * n_features_in_))' features are considered at each split.
- If "sqrt", then 'max_features=sqrt(n_features)'.
- If "log2", then 'max_features=log2(n_features)'.
- If None, then 'max_features=n_features'.

.. note::

The search for a split does not stop until at least one valid partition of the node's samples is found, even if it requires to effectively inspect more than "max_features" features.

`random_state` 1
`random_state: int,`
`RandomState`
instance or `None`,
default=`None`

Controls the randomness of the estimator. The features are always randomly permuted at each split, even if “`splitter`” is set to “`best`”. When “`max_features < n_features`”, the algorithm will select “`max_features`” at random at each split before finding the best split among them. But the best found split may vary across different runs, even if “`max_features=n_features`”. That is the case, if the improvement of the criterion is identical for several splits and one split has to be selected at random. To obtain a deterministic behaviour during fitting, “`random_state`” has to be fixed to an integer. See :term: ‘Glossary’ for details.

<code>max_leaf_nodes</code>	None
<code>max_leaf_nodes:</code>	
int, default=None	

Grow a tree with
“max_leaf_nodes”
in best-first
fashion.
Best nodes are
defined as relative
reduction in
impurity.
If None then
unlimited number
of leaf nodes.

`min_impurity_decrease` 0.0
`min_impurity_decrease`: float,
default=0.0

A node will be split if this split induces a decrease of the impurity greater than or equal to this value.

The weighted impurity decrease equation is the following::

$$\begin{aligned} & N_t / N * \\ & (\text{impurity} - N_t_R / N_t * \\ & \text{right_impurity} \\ & - N_t_L / N_t * \\ & \text{left_impurity}) \end{aligned}$$

where “N” is the total number of samples, “N_t” is the number of samples at the current node, “N_t_L” is the number of samples in the left child, and “N_t_R” is the number of samples in the right child.

“N”, “N_t”, “N_t_R” and “N_t_L” all refer to the weighted sum, ³⁷⁵ if “sample_weight” is passed.

.. versionadded::
0.19

<code>class_weight</code>	None
<code>class_weight: dict,</code>	
<code>list of dict or</code>	
<code>"balanced",</code>	
<code>default=None</code>	

Weights associated with classes in the form “{class_label: weight}”.

If None, all classes are supposed to have weight one.

For multi-output problems, a list of dicts can be provided in the same order as the columns of y.

Note that for multioutput (including multilabel) weights should be defined for each class of every column in its own dict. For example, for four-class multilabel classification weights should be [{0: 1, 1: 1}, {0: 1, 1: 5}, {0: 1, 1: 1}, {0: 1, 1: 1}] instead of [{1:1}, {2:5}, {3:1}, {4:1}].

The “balanced” mode uses the values of y to automatically adjust weights inversely proportional to class frequencies in the input data

ccp_alpha 0.0

ccp_alpha:
non-negative float,
default=0.0

Complexity
parameter used for
Minimal
Cost-Complexity
Pruning. The
subtree with the
largest cost
complexity that is
smaller than
“ccp_alpha“ will
be chosen. By
default, no pruning
is performed. See
[:ref:‘mini-
mal_cost_com-
plexity_pruning‘](#)
for details. See
[:ref:‘sphx_glr_auto_ex-
am-
ples_tree_plot_cost_com-
plexity_prun-
ing.py‘](#)
for an example of
such pruning.

.. versionadded::
0.22

`monotonic_cst` None

`monotonic_cst`:
array-like of int of
shape (n_features),
default=None

Indicates the
monotonicity
constraint to
enforce on each
feature.

- 1: monotonic
increase
- 0: no constraint
- -1: monotonic
decrease

If `monotonic_cst` is
None, no
constraints are
applied.

Monotonicity
constraints are not
supported for:

- multiclass
classifications (i.e.
when ‘n_classes >
2’),
- multioutput
classifications (i.e.
when ‘n_outputs_> 1’),
- classifications
trained on data
with missing
values.

The constraints
hold over the
probability of the
positive class.

Read ³⁷⁸more in the
:ref:‘User Guide ‘.

.. versionadded::
1.4

Then since the model contains the information of all trees, we could directly read them and use them about how the model works if we only use the first several trees. Here we use `.staged_predict()` and `.staged_decision_function()` methods.

- `.staged_predict()`: returns the predictions after each boosting round.
- `.staged_decision_function()`: returns the decision scores after each boosting round.

We would like to record all the training and test scores.

```
from sklearn.metrics import accuracy_score

train_acc = []
test_acc = []

for pred_train, pred_test in zip(
    ada.staged_predict(X_train), ada.staged_predict(X_test)
):
    train_acc.append(accuracy_score(y_train, pred_train))
    test_acc.append(accuracy_score(y_test, pred_test))
```

In addition, the 10th-percentile margin is of our interesting, since we treat $\text{margin} < 10\%$ as no confidence.

```
train_margin_10 = []
test_margin_10 = []

for score_train, score_test in zip(
    ada.staged_decision_function(X_train),
    ada.staged_decision_function(X_test),
):
    train_margins = y_train * score_train
    test_margins = y_test * score_test

    train_margin_10.append(np.quantile(train_margins, 0.1))
    test_margin_10.append(np.quantile(test_margins, 0.1))
```

We also need to test when the training score becomes 1.0, and when the test score is the highest. Then we have the interesting output.

```
first_full_train = next((i + 1 for i, s in enumerate(train_acc) if s == 1.0), None)
best_test_round = np.argmax(test_acc) + 1

print("First round with training accuracy 1:", first_full_train)
print("Test accuracy at that round:", test_acc[first_full_train - 1])
print("Best test accuracy:", np.max(test_acc))
print("Best test round:", best_test_round)
print("Final training accuracy:", train_acc[-1])
print("Final test accuracy:", test_acc[-1])
```

```
First round with training accuracy 1: 426
Test accuracy at that round: 0.9746666666666667
Best test accuracy: 0.9806666666666667
Best test round: 1121
Final training accuracy: 1.0
Final test accuracy: 0.98
```

We could plot the curves. From the plot, we can see that after the training score becomes 1.0, the model is still improving and the test score reaches the highest long after.

```
import matplotlib.pyplot as plt

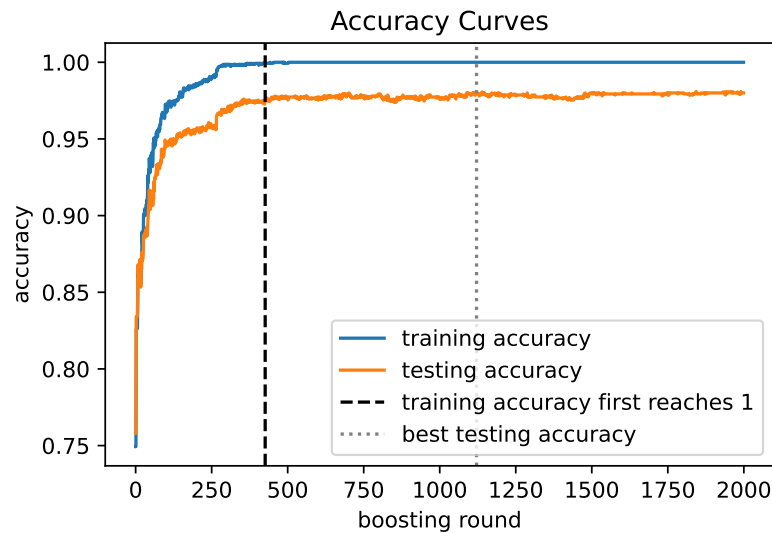
plt.plot(train_acc, label="training accuracy")
plt.plot(test_acc, label="testing accuracy")

plt.axvline(
    first_full_train,
    color="black",
    linestyle="--",
    label="training accuracy first reaches 1",
)

plt.axvline(best_test_round, color="gray", linestyle=":", label="best testing accuracy")

plt.xlabel("boosting round")
```

```
plt.ylabel("accuracy")
plt.title("Accuracy Curves")
plt.legend();
```



We may also plot the 10th-percentile margin curve. The 10th-percentile margin represents the lower tail of the margin distribution, that is, the examples on which the classifier is least confident.

From the plot, we can see that even after the training accuracy reaches 1.0, the margins may still continue to change. This shows that **AdaBoost** is not merely optimizing training accuracy; it continues to adjust the additive model in order to improve the margin distribution and increase confidence.

- If the lower-percentile margin continues to increase while test performance remains stable or improves, this suggests that the model is still improving its confidence on difficult cases and is not yet clearly overfitting.
- If the lower-percentile margin decreases together with decreasing validation or test performance, this suggests that the model may be becoming less reliable on difficult cases and may be starting to overfit.

```

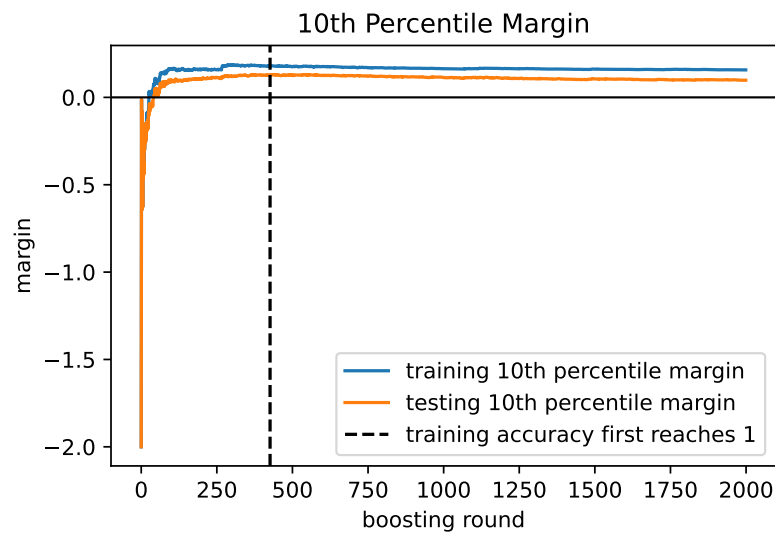
plt.plot(train_margin_10, label="training 10th percentile margin")
plt.plot(test_margin_10, label="testing 10th percentile margin")

plt.axvline(
    first_full_train,
    color="black",
    linestyle="--",
    label="training accuracy first reaches 1",
)

plt.axhline(0, color="black", linewidth=1)

plt.xlabel("boosting round")
plt.ylabel("margin")
plt.title("10th Percentile Margin")
plt.legend();

```



11.3 Tuning AdaBoost Hyperparameters

The most important AdaBoost hyperparameters are:

Parameter	Role
<code>n_estimators</code>	Number of weak learners.
<code>learning_rate</code>	Shrinks the contribution of each learner.
<code>estimator__max_depth</code>	Complexity of the base tree.
<code>estimator__min_samples_leaf</code>	Smoothness of each base tree.

The two most important parameters are `n_estimators` and `learning_rate`.

11.3.1 Learning Rate and Number of Trees

There is a strong tradeoff:

- smaller `learning_rate` usually needs larger `n_estimators`;
- larger `learning_rate` learns faster but can overfit or become unstable;
- smaller `learning_rate` often gives smoother and more stable models.

11.3.2 Base Tree Depth

Decision stumps use `max_depth=1`. They are common because AdaBoost was originally designed around weak learners. Using deeper trees allows each boosting round to model interactions. In most cases, AdaBoost use depths from 1 to 3. Deeper trees are stronger learners, but they also increase the risk of overfitting.

11.3.3 Grid Search for AdaBoost

The following code tunes AdaBoost by grid search cross-validation.

```
from sklearn.model_selection import GridSearchCV
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier

ada = AdaBoostClassifier(
```

```

    estimator=DecisionTreeClassifier(random_state=0),
    random_state=0,
)

param_grid = {
    "n_estimators": [100, 300, 600],
    "learning_rate": [0.01, 0.05, 0.1, 0.5],
    "estimator__max_depth": [1, 2, 3],
    "estimator__min_samples_leaf": [1, 5, 10],
}

grid = GridSearchCV(ada, param_grid=param_grid, cv=5, scoring="accuracy", n_jobs=-1)

grid.fit(X_train, y_train)
grid.best_params_

```

Tuning should be done using validation data or cross-validation.
The test set should be saved for the final evaluation.

Noisy data

AdaBoost can be sensitive to mislabeled observations and outliers.

Because it repeatedly increases the weights of difficult observations, noisy points may receive too much attention.

11.3.4 Example: AdaBoost on a Nonlinear Classification Problem

We first use a synthetic nonlinear classification problem. This example shows that many weak learners can create a flexible decision boundary.

```

from sklearn.datasets import make_moons
from sklearn.model_selection import train_test_split

X, y = make_moons(n_samples=1500, noise=0.25, random_state=1)

X_train, X_test, y_train, y_test = train_test_split(

```



```
X, y, test_size=0.3, stratify=y, random_state=1
)
```

Fit a baseline stump and an AdaBoost model.

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier

stump = DecisionTreeClassifier(max_depth=1, random_state=1)

ada = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1, random_state=1),
    n_estimators=300,
    learning_rate=0.05,
    random_state=1,
)

stump.fit(X_train, y_train)
ada.fit(X_train, y_train)

print("Stump accuracy:", stump.score(X_test, y_test))
print("AdaBoost accuracy:", ada.score(X_test, y_test))
```

```
Stump accuracy: 0.8488888888888889
AdaBoost accuracy: 0.9222222222222223
```

Now tune the AdaBoost model.

```
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score

ada_base = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(random_state=1), random_state=1
)

param_grid = {
    "n_estimators": [100, 300, 600],
    "learning_rate": [0.03, 0.05, 0.1],
    "estimator__max_depth": [1, 2],
```

```

    "estimator__min_samples_leaf": [1, 5],
}

grid = GridSearchCV(
    ada_base, param_grid=param_grid, cv=5, scoring="accuracy", n_jobs=-1
)

grid.fit(X_train, y_train)
best_ada = grid.best_estimator_

print("Best parameters:", grid.best_params_)
print("Best CV accuracy:", grid.best_score_)
print("Test accuracy:", best_ada.score(X_test, y_test))

```

```

Best parameters: {'estimator__max_depth': 2, 'estimator__min_samples_leaf': 5, 'learning_rate'
Best CV accuracy: 0.9466666666666667
Test accuracy: 0.94

```

11.4 XGBoost and Gradient Boosting

The core idea of boosting is to sequentially train models in order to correct the errors made by earlier models. A natural idea is therefore:

1. fit a model
2. compute the residuals
3. fit a new model to the residuals
4. repeat

We mainly consider regression problems at the current stage.

11.4.1 Additive model

The above idea leads directly to **Gradient Boosting**. When the models used are trees, the model is also called Gradient Boosted Decision Trees (GBDT).

Gradient Boosting builds an additive model of the form

$$F_M(x) = \sum_{m=1}^M \beta_m h_m(x)$$

where

- $h_m(x)$ is usually a small regression tree
- β_m is a coefficient
- the model is built sequentially

Unlike bagging, where trees are trained independently, boosting trains each new tree based on the current errors of the model.

11.4.2 Functional gradient viewpoint

In ordinary optimization problems, we optimize finitely many parameters. For example, in linear regression we optimize the coefficient vector

$$\beta = (\beta_1, \dots, \beta_p).$$

Gradient Boosting takes a different viewpoint. Instead of directly optimizing finitely many parameters, it attempts to optimize the prediction function itself.

The model is built additively:

$$F_{m+1}(x) = F_m(x) + f_m(x),$$

where each new function $f_m(x)$ is usually a small regression tree.

Thus each boosting step adds a new basis function to the model. Since the space of possible functions is extremely large, Gradient Boosting is often interpreted as a form of gradient descent in function space, also called functional gradient descent.

Conceptually:

- ordinary gradient descent updates parameters
- Gradient Boosting updates functions

11.4.3 Regression tree fitting

In order to find the best model $F(x)$, we would like to minimize the empirical loss function

$$L(F) = \sum_{i=1}^n l(y_i, F(x_i)),$$

where $\{(x_i, y_i)\}$ is the training dataset. One of the most common choices for the loss function is the squared-error loss

$$L(F) = \sum_{i=1}^n (y_i - F(x_i))^2.$$

Assume the current model is $F_m(x)$. After adding a new tree $f_m(x)$, the next model becomes

$$F_{m+1}(x) = F_m(x) + f_m(x).$$

The new tree $f_m(x)$ is chosen to reduce the loss function as much as possible.

Now consider the first-order Taylor expansion of the loss function:

$$l(y_i, F_m(x_i) + f_m(x_i)) \approx l(y_i, F_m(x_i)) + g_i f_m(x_i),$$

where

$$g_i = \left. \frac{\partial l(y_i, F_m(x_i))}{\partial F_m(x_i)} \right|_{F=F_m}.$$

This gives a linear approximation of the loss function in terms of the update $f_m(x_i)$.

Since the first-order Taylor approximation is linear, the negative gradient gives the direction of steepest descent. Therefore we define the pseudo-residual

$$r_i = -g_i = - \left. \frac{\partial l(y_i, F(x_i))}{\partial F(x_i)} \right|_{F=F_m}.$$

The pseudo-residual plays the role of the error signal that the next tree should fit.

Assume that the new tree splits the dataset into regions $R_1, \dots, R_s$, and for each region R_j a value s_j is assigned. We would like the tree to approximate the pseudo-residuals r_i . Therefore we choose the tree to minimize

$$\sum_{j=1}^s \sum_{x_i \in R_j} (r_i - s_j)^2.$$

For a fixed region R_j , the optimal value of s_j is the mean of the observations inside that region:

$$s_j = \frac{1}{|R_j|} \sum_{x_i \in R_j} r_i.$$

Therefore, the new tree $f_m(x)$ is simply a regression tree (trained using RSS) with target values r_i .

i Pseudo-residual

r_i is called a *pseudo-residual* because it is technically a negative gradient rather than an ordinary residual. However, it plays the role of the residual in Gradient Boosting. For squared-error loss

$$l(y, F(x)) = (y - F(x))^2,$$

we have

$$r_i = 2(y_i - F_m(x_i)),$$

which is proportional to the ordinary residual. Therefore, under L^2 loss, Gradient Boosting can literally be interpreted as fitting residuals.

11.4.4 XGBoost: Initial idea

XGBoost stands for **Extreme Gradient Boosting**. Its mathematical foundation can be viewed as a modified and improved version of GBDT.

XGBoost became one of the most successful shallow machine learning models in modern practice. In addition to its mathematical improvements, it also introduced many engineering innovations that significantly improved efficiency and scalability. After the success of XGBoost, several related models such as LightGBM and CatBoost were also developed and became widely used.

We mainly focus on the mathematical ideas behind XGBoost. The motivation is very similar to GBDT. We consider the additive model

$$F_{m+1}(x) = F_m(x) + f_m(x),$$

and would like to minimize the empirical loss

$$L(F) = \sum_{i=1}^n l(y_i, F(x_i)).$$

Assume the current model is $F_m(x)$. After adding a new tree $f_m(x)$, we obtain

$$F_{m+1}(x) = F_m(x) + f_m(x).$$

Instead of using only first-order information as in GBDT, XGBoost uses a local second-order Taylor approximation of the loss function at each boosting iteration:

$$l(y_i, F_m(x_i) + f_m(x_i)) \approx l(y_i, F_m(x_i)) + g_i f_m(x_i) + \frac{1}{2} h_i f_m(x_i)^2,$$

where

$$g_i = \left. \frac{\partial l(y_i, F(x_i))}{\partial F(x_i)} \right|_{F=F_m},$$

and

$$h_i = \left. \frac{\partial^2 l(y_i, F(x_i))}{\partial F(x_i)^2} \right|_{F=F_m}.$$

Here

- g_i measures the gradient
- h_i measures the curvature of the loss function

Thus, XGBoost uses both gradient and curvature information.

11.4.5 Intuition of the quadratic approximation

The first-order Taylor expansion used in GBDT gives only a linear approximation of the loss function. In contrast, the second-order Taylor expansion gives a quadratic approximation:

$$g_i f + \frac{1}{2} h_i f^2.$$

As a function of f , its derivative is

$$g_i + h_i f.$$

Setting this equal to zero gives the critical point

$$f = -\frac{g_i}{h_i}.$$

This can be viewed as the approximately optimal update suggested by the local quadratic approximation of the loss function.

11.4.6 Tree structure in XGBoost

Assume that the new tree splits the dataset into regions

$$R_1, \dots, R_s,$$

and assigns a prediction value s_j to region R_j . Then for every observation inside region R_j , we have

$$f_m(x_i) = s_j.$$

Substituting this into the quadratic approximation gives the approximate objective

$$\sum_{j=1}^s \sum_{x_i \in R_j} \left(g_i s_j + \frac{1}{2} h_i s_j^2 \right).$$

For a fixed tree structure, we minimize this expression with respect to s_j .

Differentiating with respect to s_j gives

$$\sum_{x_i \in R_j} (g_i + h_i s_j) = 0.$$

Therefore,

$$s_j = -\frac{\sum_{x_i \in R_j} g_i}{\sum_{x_i \in R_j} h_i}.$$

If we define

$$G_j = \sum_{x_i \in R_j} g_i, \quad H_j = \sum_{x_i \in R_j} h_i,$$

then the optimal leaf value becomes

$$s_j = -\frac{G_j}{H_j}.$$

 Important difference from ordinary CART

This tree is NOT an ordinary CART regression tree trained with RSS.

- In GBDT, we explicitly train a CART regression tree to fit the pseudo-residuals.
- In XGBoost, the leaf values are derived directly from the quadratic approximation of the objective function itself.

Therefore, although XGBoost trees may look structurally similar to CART trees, mathematically they are solving a different optimization problem.

11.4.7 Regularization

So far we derived the basic XGBoost objective using the second-order Taylor approximation.

However, minimizing only the training loss may produce overly complex trees and lead to overfitting. Therefore XGBoost introduces an explicit regularization term to control tree complexity.

Suppose the new tree $f_m(x)$ has

- T leaves
- leaf values $s_1, \dots, s_T$

XGBoost defines the regularization term

$$\Omega(f_m) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T s_j^2.$$

Therefore the objective becomes

$$L(F_m + f_m) + \Omega(f_m).$$

Here

- γT penalizes the number of leaves
- $\frac{1}{2}\lambda \sum s_j^2$ penalizes large leaf values

Using the second-order Taylor approximation and grouping terms by regions gives

$$\sum_{j=1}^T \left[G_j s_j + \frac{1}{2} (H_j + \lambda) s_j^2 \right] + \gamma T.$$

For a fixed tree structure, we minimize the objective with respect to s_j . Differentiating gives

$$G_j + (H_j + \lambda) s_j = 0.$$

Therefore the optimal leaf value becomes

$$s_j = -\frac{G_j}{H_j + \lambda}.$$

Compared with the non-regularized version

$$s_j = -\frac{G_j}{H_j},$$

the regularization term enlarges the denominator and shrinks the leaf predictions toward zero. This effect is similar to ridge regularization in linear regression.

Substituting the optimal leaf values back into the objective gives

$$-\frac{1}{2} \sum_{j=1}^T \frac{G_j^2}{H_j + \lambda} + \gamma T.$$

This quantity is used by XGBoost to evaluate the quality of candidate tree splits.

Therefore, unlike ordinary CART trees which use RSS or Gini impurity, XGBoost derives its splitting criterion directly from the regularized optimization objective.

11.4.8 Learning rate

In practice, although XGBoost computes the best update tree f_m , it usually does not add the full tree update directly to the model. Instead, the update is scaled by a learning rate (also called the shrinkage parameter):

$$F_{m+1}(x) = F_m(x) + \eta f_m(x),$$

where

$$0 < \eta \leq 1.$$

Thus every new tree contributes only partially to the final model.

Equivalently, after computing the optimal leaf value

$$s_j = -\frac{G_j}{H_j + \lambda},$$

XGBoost actually uses the scaled leaf prediction

$$\eta s_j.$$

Therefore shrinkage reduces the influence of each individual tree.

The parameter η is called the **learning rate** or shrinkage parameter.

i XGBoost in practice

The mathematical foundation of XGBoost is built on three main ideas:

- second-order Taylor approximation of the loss function
- explicit regularization on tree complexity
- shrinkage through the learning rate

In addition to these mathematical ideas, XGBoost also introduced many important engineering optimizations such as:

- parallel tree construction
- efficient handling of sparse data
- cache-aware memory access
- approximate split finding algorithms

These engineering improvements are one of the main reasons why XGBoost became highly successful in practical machine learning applications.

11.4.9 GBDT and XGBoost for classification

Both GBDT and XGBoost can naturally be extended from regression to classification problems. The overall framework remains the same:

- additive tree model
- boosting optimization
- sequentially adding trees
- minimizing a loss function

The main difference is the choice of the loss function.

For binary classification, boosting methods commonly use logistic loss

$$l(y, p) = -[y \log p + (1 - y) \log(1 - p)].$$

The model output $F(x)$ is converted into probabilities using the logistic function

$$p(x) = \frac{1}{1 + e^{-F(x)}}.$$

For multiclass classification, softmax-based loss functions are commonly used.

The difference between GBDT and XGBoost remains the same as in regression:

- GBDT uses first-order gradients (pseudo-residuals)
- XGBoost uses second-order Taylor approximation together with regularization

Thus both methods provide a unified framework for regression and classification problems.

i AdaBoost as Gradient Boosting with exponential loss

AdaBoost can be interpreted as a special case of Gradient Boosting using the exponential loss function

$$l(y, F(x)) = e^{-yF(x)},$$

where

$$y \in \{-1, 1\}.$$

The empirical loss becomes

$$L(F) = \sum_{i=1}^n e^{-y_i F(x_i)}.$$

This loss heavily penalizes observations with negative margins

$$y_i F(x_i).$$

In particular:

- correctly classified points with large positive margins receive very small loss
- misclassified points receive exponentially large loss

Therefore minimizing exponential loss naturally increases the weights of difficult observations, which explains the adaptive reweighting behavior of AdaBoost.

Taking derivative gives

$$\frac{\partial l(y_i, F(x_i))}{\partial F(x_i)} = -y_i e^{-y_i F(x_i)}.$$

Thus the pseudo-residual becomes

$$r_i = y_i e^{-y_i F(x_i)}.$$

Observations with small or negative margins receive much larger pseudo-residuals and therefore larger influence in the next boosting step.

From this viewpoint, AdaBoost can be interpreted as functional gradient descent under exponential loss.

Gradient Boosting generalizes this idea by allowing arbitrary differentiable loss functions rather than only exponential loss.

11.5 XGBoost in Python

XGBoost is implemented using the library **XGBoost**. You may go to the [official website](#) for more information. We will cover the basic usage in this lecture.

A typical XGBoost regression model looks like this.

```
from xgboost import XGBRegressor

xgb = XGBRegressor(
    objective="reg:squarederror",
    n_estimators=500,
    learning_rate=0.05,
    max_depth=3,
    subsample=0.8,
    colsample_bytree=0.8,
    reg_lambda=1.0,
    random_state=0,
    n_jobs=-1,
)

xgb.fit(X_train, y_train)
```

The `objective="reg:squarederror"` is to indicate the loss function. The default is `"reg:squarederror"` that is regression with squared loss. More loss function can be found in the [documnet](#).

The most important hyperparameters are:

Parameter	Meaning
<code>n_estimators</code>	Number of boosting rounds.
<code>learning_rate</code>	Learning rate.
<code>max_depth</code>	Depth of each tree.
<code>subsample</code>	Fraction of observations used for each tree.
<code>colsample_bytree</code>	Fraction of predictors used for each tree.
<code>reg_lambda</code>	L2 regularization on leaf weights.
<code>reg_alpha</code>	L1 regularization on leaf weights.

Most of them are straightforward. I only discuss these since they are not from the theory.

i `subsample`

`subsample` controls the fraction of training observations randomly sampled for each boosting round. For example `subsample=0.8` means that each new tree is trained using only 80% of the observations.

This introduces additional randomness into boosting, similar to bagging in Random Forests. Smaller values of `subsample` and reduce variance and improve generalization. However, if the value is too small, the model may underfit.

i `colsample_bytree`

`colsample_bytree` controls the fraction of predictors randomly sampled for each tree. For example `colsample_bytree=0.7` means that each tree only considers 70% of the predictors.

This idea is similar to feature subsampling in Random Forests. It reduces correlation between trees and can improve robustness and generalization performance.

i `reg_alpha`

`reg_alpha` is the L1 regularization parameter on leaf weights. Previously we introduced the L2 regularization

term $\frac{1}{2}\lambda \sum s_j^2$, which corresponds to ridge-style regularization.

XGBoost can also include an L1 penalty

$$\alpha \sum |s_j|,$$

which is controlled by `reg_alpha`. L1 regularization tends to shrink some leaf values exactly to zero, producing sparser models and sometimes improving interpretability and robustness. This corresponds to lasso-style regularization. We don't introduce L1 regularization in the first place because it might make the math formula too complicated.

Example grid:

```
from sklearn.model_selection import GridSearchCV
from xgboost import XGBRegressor

xgb = XGBRegressor(random_state=1, n_jobs=1)

param_grid = {
    "n_estimators": [200, 500, 800],
    "learning_rate": [0.03, 0.05, 0.1],
    "max_depth": [2, 3, 4],
    "subsample": [0.7, 0.9, 1.0],
    "colsample_bytree": [0.7, 0.9, 1.0],
    "reg_lambda": [1.0, 5.0, 10.0],
}

grid = GridSearchCV(xgb, param_grid=param_grid, cv=5, n_jobs=-1)

grid.fit(X_train, y_train)
```

i n_jobs

Although for boosting parallel computing is in the nature as in bagging, XGBoost successfully support it and offers `n_jobs` to accelerate. The API is similar, so we could use `n_jobs=-1` to indicate using all CPU cores.

Similar to bagging, for a nested struture like `GridSearchCV`

with `XGBoost`, we only apply `n_jobs` to one of them. In general, if it is a small dataset, we let `GridSearchCV` use multiple cores. If it is a large dataset, we let `XGBoost` use multiple cores.

11.6 Tuning XGBoost

If there are enough computing resources, to tune `XGBoost` is no different from other models. However since the model becomes more complicated, it usually takes longer to go through the whole grid search. In this case, usually we make some modifications to the workflow.

11.6.1 RandomSearchCV

We already show the code of `RandomSearchCV` before. Here we give more explanations.

`RandomSearchCV` is to search random combinations of hyperparameters. Therefore it is used to do a rough search in a very large parameter space. Other than the `param_grid` like `GridSearchCV`, `RandomSearchCV` needs `n_iter` to indicate how many combinations one wants to test.

Since now we can search in a very large space, usually we make a really large space, by directly choosing hyperparameters based on distributions. We use distributions from the library `scipy.stats`. Here we give an examples.

```
from scipy.stats import randint, uniform

param_distributions = {
    "n_estimators": randint(100, 1000),
    "max_depth": randint(2, 8),
    "learning_rate": uniform(0.01, 0.19),
    "subsample": uniform(0.6, 0.4),
    "colsample_bytree": uniform(0.6, 0.4),
    "reg_lambda": uniform(0, 10),
}
```

```
random_search = RandomizedSearchCV(  
    xgb, param_distributions=param_distributions, n_iter=30, cv=5, random_state=0  
)
```

11.6.2 Workflow

A practical tuning workflow is often a two-stage process.

First, run `RandomizedSearchCV` over a relatively large parameter space. This gives a rough idea of promising regions in the hyperparameter space.

Then, based on the best combination found by random search, choose a smaller and more refined grid around those values and run `GridSearchCV`.

If the refined grid is still too slow, we can tune parameters in smaller batches. A common strategy is:

1. tune model complexity parameters first, such as `max_depth`
2. tune learning rate
3. tune sampling parameters, such as `subsample` and `colsample_bytree`
4. tune regularization parameters, such as `reg_lambda` and `reg_alpha`

This workflow is not guaranteed to find the global best combination, but it is usually much more practical than running one huge exhaustive grid search.

11.6.3 Early Stopping

Boosting may eventually overfit if too many trees are added. Early stopping helps prevent this by monitoring validation performance during training.

In principle, `n_estimators` can be tuned together with other hyperparameters using grid search or similar methods. In practice, however, it is often determined separately through early stopping.

Conceptually, the procedure is simple:

1. Split the training data into a training part and a validation part.
2. Fit trees sequentially.
3. Monitor the validation error after each boosting round.
4. Stop when the validation performance no longer improves.

In XGBoost, this is commonly implemented using an evaluation set together with `early_stopping_rounds`. For example, `early_stopping_rounds=20` means that training stops if the validation score does not improve for 20 consecutive boosting rounds.

```
xgb = XGBRegressor(early_stopping_rounds=20)
xgb.fit(X_train, y_train, eval_set=[(X_val, y_val)])
```

Exact early-stopping syntax can differ across XGBoost versions, so it is useful to check the installed version when writing production code. The current syntax comes from [XGBoost 3.2.0](#).

i Modern hyperparameter tuning

In modern machine learning practice, workflows often use tools such as **Optuna** and various **AutoML** frameworks for hyperparameter optimization.

Unlike **GridSearchCV**, which exhaustively searches a fixed parameter grid, these methods use adaptive search strategies to explore promising hyperparameter regions more efficiently. This is especially useful for large models such as **XGBoost**, **LightGBM**, and neural networks, where exhaustive grid search may become computationally expensive.

Many modern tuning frameworks are related to ideas from Bayesian optimization. In practice, these methods often attempt to use information from previous trials to guide future searches toward more promising regions of the parameter space.

We only briefly mention these topics here since they have become mainstream tools in modern machine learning workflows:

- Bayesian optimization
- TPE (Tree-structured Parzen Estimator)
- acquisition functions
- Gaussian process optimization
- AutoML systems

A full discussion of topics is beyond the scope of this course.

11.7 Example: XGBoost on the Diabetes Data

We now use the diabetes regression dataset from `sklearn` introduced earlier. The response is a quantitative measure of disease progression one year after baseline.

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split

X, y = load_diabetes(return_X_y=True)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=1)
```

In general you should not touch test sets. However in this example for simplicity we abuse it by using test sets as validation sets.

11.7.1 Baseline model

We first build a baseline model, with all default hyperparameters.

```
from xgboost import XGBRegressor

xgb_baseline = XGBRegressor(random_state=1, n_jobs=-1)
xgb_baseline.fit(X_train, y_train)
xgb_baseline.score(X_test, y_test)
```

0.12475668653462724

11.7.2 RandomSearchCV

The first round we use random search to try to find a good enough point. Note that we choose a relatively large `n_estimators` which we will tune in the next early stopping stage.

```
from scipy.stats import randint, uniform
from sklearn.model_selection import RandomizedSearchCV

param_distributions = {
    "max_depth": randint(2, 8),
    "learning_rate": uniform(0.01, 0.19),
    "subsample": uniform(0.6, 0.4),
    "colsample_bytree": uniform(0.6, 0.4),
    "reg_lambda": uniform(0, 10),
}

random_search = RandomizedSearchCV(
    XGBRegressor(n_estimators=2000, random_state=1, n_jobs=1),
    param_distributions=param_distributions,
    n_iter=20,
    cv=5,
    random_state=0,
    n_jobs=-1,
)

random_search.fit(X_train, y_train)
random_search.best_params_
```

```
{'colsample_bytree': np.float64(0.8650107467800177),
 'learning_rate': np.float64(0.012578610766300869),
 'max_depth': 3,
 'reg_lambda': np.float64(8.379449074988038),
 'subsample': np.float64(0.6384393631575852)}
```

The purpose of this random search is to find a rough range of the best combination in order for the refined grid search.

According to the result, we may construct a smaller range for refined grid search.

```
param_grid_refined = {
    "learning_rate": [0.008, 0.0125, 0.018],
    "max_depth": [2, 3, 4],
    "subsample": [0.55, 0.64, 0.72],
    "colsample_bytree": [0.80, 0.87, 0.95],
    "reg_lambda": [5.0, 8.5, 12.0],
}
```

11.7.3 Early stopping

Before we run the refined grid search, we need to determine the `n_estimators` using early stopping.

```
X_tr, X_val, y_tr, y_val = train_test_split(
    X_train, y_train, test_size=0.2, random_state=0
)
xgb_es = XGBRegressor(
    n_estimators=3000,
    **random_search.best_params_,
    early_stopping_rounds=50,
    random_state=0,
    n_jobs=-1,
)
xgb_es.fit(X_tr, y_tr, eval_set=[(X_val, y_val)], verbose=False)
best_round = xgb_es.best_iteration + 1
best_round
```

281

11.7.4 Refined grid search

Now we can run refined grid search using `n_estimators=best_round` (which is 281 in this example).

```

from sklearn.model_selection import GridSearchCV

xgb_base = XGBRegressor(n_estimators=best_round, random_state=0, n_jobs=1)

param_grid_refined = {
    "learning_rate": [0.008, 0.0125, 0.018],
    "max_depth": [2, 3, 4],
    "subsample": [0.55, 0.64, 0.72],
    "colsample_bytree": [0.80, 0.87, 0.95],
    "reg_lambda": [5.0, 8.5, 12.0],
}

grid_search = GridSearchCV(xgb_base, param_grid=param_grid_refined, cv=5, n_jobs=-1)
grid_search.fit(X_train, y_train)

```

Sometimes a full refined grid search is still computationally expensive. In this case, a common practical strategy is to tune the hyperparameters in stages.

i Stage 1: tune tree complexity

```

xgb_base = XGBRegressor(n_estimators=best_round, random_state=0, n_jobs=1)

param_grid_1 = {"max_depth": [1, 2, 3]}

grid_1 = GridSearchCV(xgb_base, param_grid=param_grid_1, cv=5, n_jobs=-1)

grid_1.fit(X_train, y_train)
grid_1.best_params_

{'max_depth': 1}

```

i Stage 2: tune learning rate around the selected number of trees

```
xgb_stage2 = XGBRegressor(  
    n_estimators=best_round, random_state=0, n_jobs=1, **grid_1.best_params_  
)  
  
param_grid_2 = {"learning_rate": [0.035, 0.05, 0.07]}  
  
grid_2 = GridSearchCV(xgb_stage2, param_grid=param_grid_2, cv=5, n_jobs=-1)  
  
grid_2.fit(X_train, y_train)  
grid_2.best_params_  
  
{'learning_rate': 0.07}
```

i Stage 3: tune sampling parameters

```
xgb_stage3 = XGBRegressor(  
    n_estimators=best_round,  
    random_state=0,  
    n_jobs=1,  
    **grid_1.best_params_,  
    **grid_2.best_params_,  
)  
  
param_grid_3 = {"subsample": [0.60, 0.66, 0.75], "colsample_bytree": [0.85, 0.90, 0.95]}  
  
grid_3 = GridSearchCV(xgb_stage3, param_grid=param_grid_3, cv=5, n_jobs=-1)  
  
grid_3.fit(X_train, y_train)  
grid_3.best_params_  
  
{'colsample_bytree': 0.85, 'subsample': 0.75}
```


i Stage 4: tune regularization

```
xgb_stage4 = XGBRegressor(
    n_estimators=best_round,
    random_state=0,
    n_jobs=1,
    **grid_1.best_params_,
    **grid_2.best_params_,
    **grid_3.best_params_,
)

param_grid_4 = {"reg_lambda": [2.0, 3.25, 5.0]}

grid_4 = GridSearchCV(xgb_stage4, param_grid=param_grid_4, cv=5, n_jobs=-1)

grid_4.fit(X_train, y_train)
grid_4.best_params_

{'reg_lambda': 2.0}
```

Finally we put all hyperparameters together for a final training.

```
best_params = {
    **grid_1.best_params_,
    **grid_2.best_params_,
    **grid_3.best_params_,
    **grid_4.best_params_,
}

final_model = XGBRegressor(
    n_estimators=best_round, random_state=0, n_jobs=-1, **best_params
)

final_model.fit(X_train, y_train)
final_model.score(X_test, y_test)
```

0.415029538595333

It doesn't have to be exactly the same as the full grid search result. However the result will be very similar.

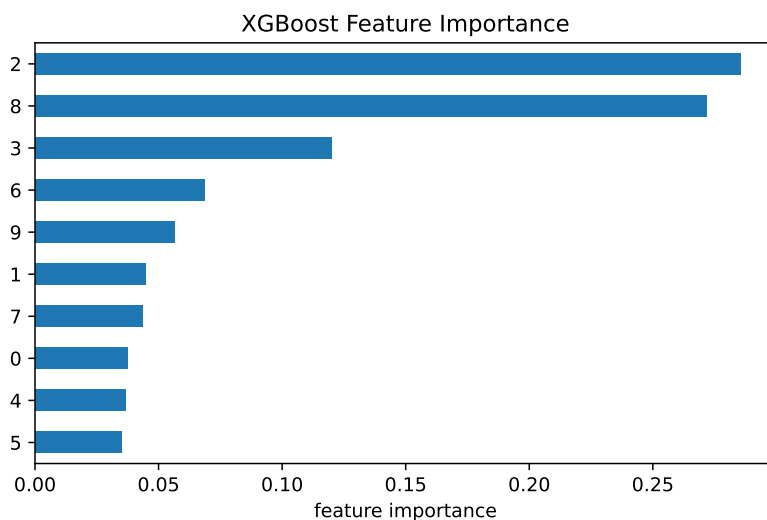
11.7.5 Feature Importance

XGBoost provides impurity-like feature importance through `.feature_importances_`. Conceptually, this is similar to MDI in Random Forests since both measure how much a feature improves the training objective during tree construction.

However, they are not exactly the same. In Random Forests, MDI is based on reductions in RSS or Gini impurity, while in XGBoost the gain is computed from reductions in the regularized boosting objective.

```
import matplotlib.pyplot as plt
importance = pd.Series(final_model.feature_importances_).sort_values(ascending=True)

plt.figure(figsize=(7, 4))
importance.plot(kind="barh")
plt.xlabel("feature importance")
plt.title("XGBoost Feature Importance");
```



Feature importance describes how much the fitted model uses each predictor for prediction. It should not be interpreted as a causal effect.

Part IV

Unsupervised Learning

12 Clustering

Clustering is an unsupervised learning problem. We observe only a feature matrix $X \in \mathbb{R}^{n \times p}$ and seek to partition the observations into groups whose members are similar. Unlike classification, there are no training labels that define the target categories.

Clustering is commonly used to explore latent subpopulations, summarize a large dataset with a small number of groups or prototypes, organize observations for visualization, and generate hypotheses for later scientific analysis.

 A cluster is not automatically a real category

A clustering algorithm produces cluster assignments according to its objective and tuning choices. The result may reflect meaningful structure, but it may also reflect feature scaling, outliers, an unsuitable dissimilarity measure, or a forced choice of the number of clusters. Interpretation is part of the analysis.

Clustering methods differ in their definition of a group. In the notes we focus on three major direct clustering paradigms, represented by one typical method from each:

Family	Typical method	What defines a cluster?
Centroid-based	K-means	observations close to the same center
Hierarchical	agglomerative hierarchical clustering	observations joined through a sequence of merges

Family	Typical method	What defines a cluster?
Density-based	DBSCAN	observations connected through dense regions

Many other clustering approaches exist, including graph-based and representation-based methods. The three paradigms here are emphasized because they give distinct direct definitions of what constitutes a cluster. We use one representative algorithm from each paradigm to illustrate the main ideas.

i Distance, Similarity, and Scaling

Most clustering methods require a notion of similarity or dissimilarity. For numerical data, a common choice is Euclidean distance: $d(x_i, x_j) = \|x_i - x_j\|_2 = \sqrt{\sum_{\ell=1}^p (x_{i\ell} - x_{j\ell})^2}$.

The distance defines what the algorithm means by *similar*. Changing the features, their scales, or the dissimilarity measure changes the clustering problem. Features with larger numerical ranges can dominate distance calculations, so standardization is often useful when feature scales are not directly comparable.

Euclidean distance is not the only option. Manhattan distance, cosine similarity, or a domain-specific measure may be more appropriate depending on the structure one hopes to discover.

i Scaling is a modeling decision

Standardize when numerical scales are arbitrary or incomparable. Do not standardize automatically when the original magnitudes are scientifically meaningful, such as comparable pixel intensities or measurements already expressed on a common scale.

12.1 Evaluating a Clustering

Evaluating clustering differs from evaluating supervised learning. In classification or regression, the response variable defines what a correct prediction means, and performance can be measured on held-out labeled data. In clustering, group membership is not observed during fitting. The algorithm produces cluster assignments, but several different assignments may be reasonable for the same dataset depending on the features, scaling, distance, and scientific question.

For this reason, clustering evaluation usually separates two questions:

- Are the fitted clusters geometrically coherent in the observed feature space?
- If reference labels are available, how well do the cluster assignments agree with those external labels?

Although some evaluation measures are commonly associated with particular clustering methods, the focus here is on the metrics themselves; the corresponding algorithms are discussed in later sections.

12.1.1 WCSS (or Inertia)

Internal clustering metrics use only the observations and their cluster assignments. They can compare values of K , initializations, or methods, but only when the observations, features, scaling, and distance are held fixed. They measure geometric properties of the fitted clustering, not whether the clusters correspond to meaningful external categories.

Two common internal metrics are WCSS and the silhouette value.

The examples below use a small hypothetical dataset to illustrate the mechanics of the evaluation. We first visualize one set of cluster assignments.

```

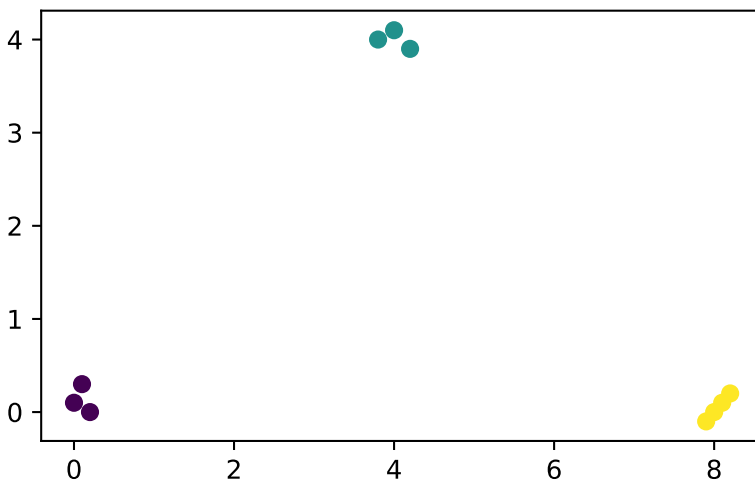
import numpy as np
import matplotlib.pyplot as plt

X_eval = np.array([
    [0.0, 0.1],
    [0.2, 0.0],
    [0.1, 0.3],
    [4.0, 4.1],
    [4.2, 3.9],
    [3.8, 4.0],
    [8.0, 0.0],
    [8.2, 0.2],
    [7.9, -0.1],
    [8.1, 0.1],
])

cluster_eval = np.array([0, 0, 0, 1, 1, 1, 2, 2, 2, 2])

plt.scatter(X_eval[:, 0], X_eval[:, 1], c=cluster_eval)
plt.gcf().set_size_inches(5, 3)

```



The within-cluster sum of squares (WCSS) measures the total squared distance from each observation to the center of its assigned cluster. If μ_k is the center of cluster C_k , then

$$\text{WCSS} = \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|_2^2.$$

WCSS can be computed once cluster assignments and centers are specified. For arbitrary cluster labels, a natural choice is to set each center to the mean of its assigned observations. If only centers are given, as in the K-means assignment step, each observation is first assigned to its nearest center and WCSS is then computed from those assignments.

For K-means, WCSS is both the fitting objective and an internal measure of cluster compactness. It plays a role similar to the residual sum of squares (RSS) in regression: smaller values indicate that observations are closer to their fitted centers.

WCSS can be computed for any assignment of observations to clusters using any methods. However, it is most naturally associated with K-means because K-means explicitly minimizes WCSS during fitting. For clustering methods designed to detect non-spherical or density-based structures, WCSS may not provide a meaningful assessment of clustering quality.

In `sklearn`, WCSS is also called inertia and is stored as `.inertia_` on a fitted `KMeans` object. This is the WCSS for the fitted centers, specifically the best solution among the initializations tried. There is no general `sklearn` API for computing WCSS from arbitrary centers or labels.

```
from sklearn.cluster import KMeans

kmeans_eval = KMeans(n_clusters=3, n_init=20, random_state=1)
kmeans_eval.fit(X_eval)
kmeans_eval.inertia_
```

0.2666666666666665

Click to expand.

For this example, the three fitted groups have centers

$$\mu_0 = \frac{(0.0, 0.1) + (0.2, 0.0) + (0.1, 0.3)}{3} = \left(0.1, \frac{2}{15}\right),$$

$$\mu_1 = \frac{(4.0, 4.1) + (4.2, 3.9) + (3.8, 4.0)}{3} = (4.0, 4.0),$$

$$\mu_2 = \frac{(8.0, 0.0) + (8.2, 0.2) + (7.9, -0.1) + (8.1, 0.1)}{4} = (8.05, 0.05).$$

Plugging these centers into the WCSS formula gives

$$\begin{aligned} \text{WCSS}_0 &= \left\| (0.0, 0.1) - \left(0.1, \frac{2}{15}\right) \right\|^2 + \left\| (0.2, 0.0) - \left(0.1, \frac{2}{15}\right) \right\|^2 + \left\| (0.1, 0.3) - \left(0.1, \frac{2}{15}\right) \right\|^2 \\ &= \frac{1}{15}, \end{aligned}$$

$$\begin{aligned} \text{WCSS}_1 &= \|(4.0, 4.1) - (4.0, 4.0)\|^2 + \|(4.2, 3.9) - (4.0, 4.0)\|^2 + \|(3.8, 4.0) - (4.0, 4.0)\|^2 \\ &= 0.10, \end{aligned}$$

$$\begin{aligned} \text{WCSS}_2 &= \|(8.0, 0.0) - (8.05, 0.05)\|^2 + \|(8.2, 0.2) - (8.05, 0.05)\|^2 + \|(7.9, -0.1) - (8.05, 0.05)\|^2 + \|(8.1, 0.1) - (8.05, 0.05)\|^2 \\ &= 0.10. \end{aligned}$$

Therefore,

$$\text{WCSS} = \frac{1}{15} + 0.10 + 0.10 \approx 0.267.$$

Smaller inertia means observations are closer to their assigned centers. However, inertia always decreases as K increases and becomes zero when every distinct observation is its own cluster. Therefore, its absolute minimum cannot be used to choose K .

12.1.2 Silhouette Value

The Silhouette value is another internal evaluation measure. It evaluates both within-cluster cohesion and separation from nearby clusters. For observation i , define:

- $a(i)$ as its average distance to other observations in its own cluster;
- $b(i)$ as the smallest average distance to observations in another cluster.

The silhouette value is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}.$$

It lies between -1 and 1 :

- values near 1 indicate a well-separated observation;
- values near 0 indicate an observation near a cluster boundary;
- negative values suggest that another cluster may be closer.

The overall silhouette score is the average of $s(i)$ over all observations. Larger values are preferred. It requires at least two nonempty clusters.

```
from sklearn.metrics import silhouette_score

silhouette_score(X_eval, cluster_eval)
```

0.952562475841761

Click to expand.

For example, consider the first observation $x_1 = (0.0, 0.1)$, which belongs to cluster 0. Its average distance to the other observations in its own cluster is

$$a(1) = \frac{\|(0.0, 0.1) - (0.2, 0.0)\| + \|(0.0, 0.1) - (0.1, 0.3)\|}{2} \approx 0.224.$$

Next compute its average distance to each other cluster:

$$d(1, C_1) = \frac{\|(0.0, 0.1) - (4.0, 4.1)\| + \|(0.0, 0.1) - (4.2, 3.9)\| + \|(0.0, 0.1) - (3.8, 4.0)\|}{3} \\ \approx 5.589,$$

and

$$d(1, C_2) = \frac{\|(0.0, 0.1) - (8.0, 0.0)\| + \|(0.0, 0.1) - (8.2, 0.2)\| + \|(0.0, 0.1) - (7.9, -0.1)\| + \|(0.0, 0.1) - (8.1, 0.1)\|}{4} \\ \approx 8.051.$$

Therefore,

$$b(1) = \min\{5.589, 8.051\} = 5.589,$$

and the silhouette value for this observation is

$$s(1) = \frac{b(1) - a(1)}{\max\{a(1), b(1)\}} = \frac{5.589 - 0.224}{5.589} \approx 0.960.$$

The Silhouette score for the whole dataset is the mean of the score for each individual observation.

This score is close to 1, indicating that most observations are much closer to observations in their assigned cluster than to observations in other clusters.

12.1.3 Stability

A useful clustering should not change drastically under small, reasonable changes to the analysis. Stability asks whether the same broad grouping appears when the data or fitting procedure is slightly perturbed.

Common checks include rerunning the method with different random seeds, changing tuning parameters slightly, refitting after small changes to preprocessing, or clustering bootstrap or subsampled versions of the data. The numeric cluster labels themselves do not need to match, because labels are arbitrary.

What matters is whether similar observations remain grouped together.

Stability is not the same as correctness. A clustering can be stable because a method has a strong geometric bias, even if the resulting groups are not scientifically meaningful. It should therefore be considered together with internal scores, visual inspection when possible, and interpretation in the original features.

12.1.4 Interpretation in the Original Features

Numerical scores do not explain what a cluster represents. After fitting a clustering, the groups should be described using the original variables. For tabular data, a simple interpretation step is to summarize each feature by cluster.

```
import pandas as pd

cluster_summary = (
    pd.DataFrame(X_eval, columns=["x1", "x2"])
    .assign(cluster=cluster_eval)
    .groupby("cluster")
    .agg(["mean", "std", "size"])
)

cluster_summary
```

	x1			x2		
	mean	std		mean	std	
cluster			size			
0	0.10	0.100000	3	0.133333	0.152753	3
1	4.00	0.200000	3	4.000000	0.100000	3
2	8.05	0.129099	4	0.050000	0.129099	4

 No single score validates a clustering

Internal metrics and Rand-index-based agreement measures answer different questions. A useful clustering should be geometrically coherent, reasonably stable, and interpretable for the problem at hand.

12.1.5 Optional: Rand Index

Click to expand.

When reference labels are available, they can be used after clustering for external validation. This setting is common in simulations, benchmark datasets, or applications where external categories are known but intentionally withheld from the clustering algorithm.

External validation does not turn clustering into supervised learning because the reference labels are not used during fitting. Instead, it asks how well the cluster assignments agree with an external reference labeling.

The Rand index compares cluster assignments with reference labels by considering all pairs of observations. A pair is counted as an agreement if the two observations are assigned to the same cluster and have the same reference label, or if they are assigned to different clusters and have different reference labels. The index lies between 0 and 1, with larger values indicating stronger agreement.

Let a be the number of observation pairs that are grouped together by both labelings. That is, the two observations have the same cluster assignment and also have the same reference label. Let b be the number of observation pairs that are separated by both labelings. That is, the two observations have different cluster assignments and also have different reference labels. Since there are $\binom{n}{2}$ total pairs, the Rand index is

$$\text{RI} = \frac{a + b}{\binom{n}{2}}.$$

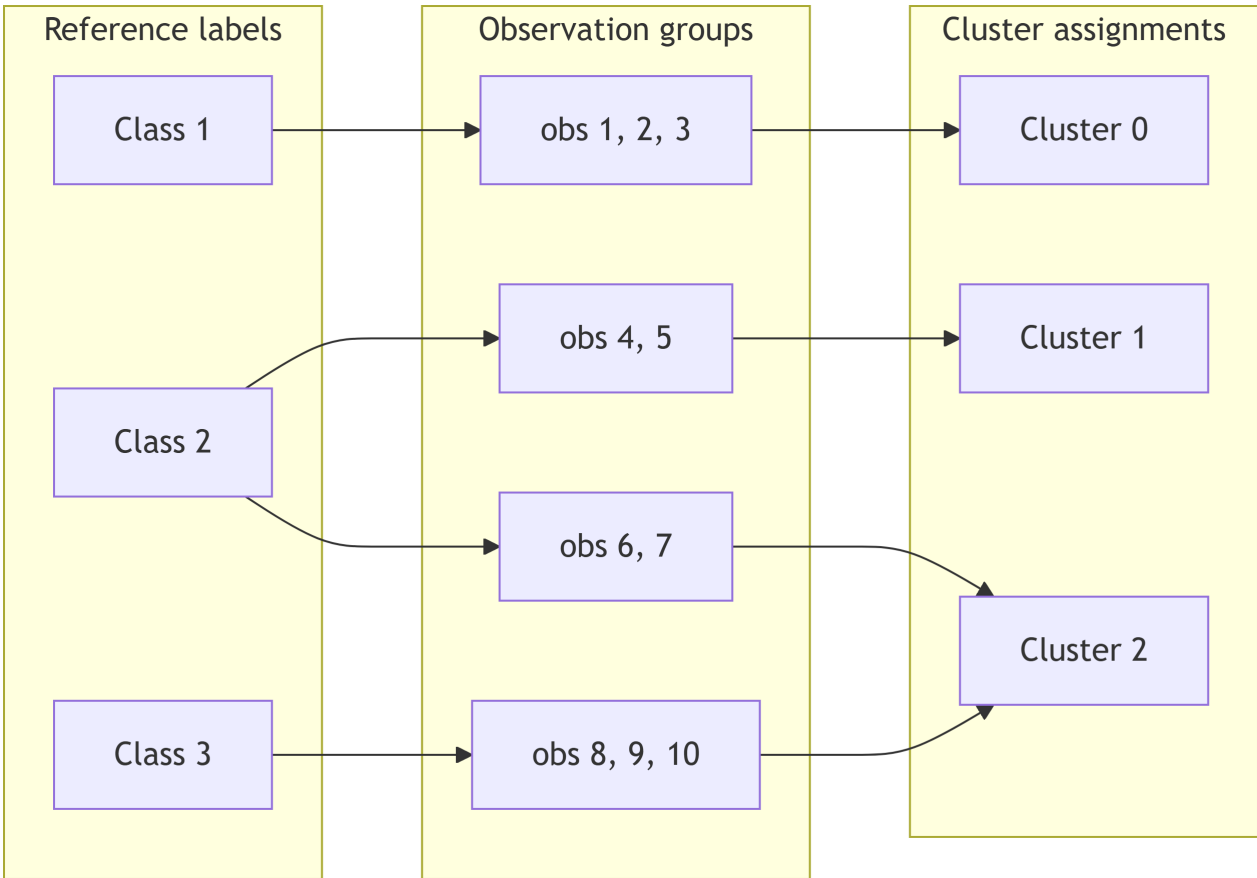
```
from sklearn.metrics import rand_score

reference_labels = np.array([1, 1, 1, 2, 2, 2, 2, 3, 3, 3])
cluster_assignments = np.array([0, 0, 0, 1, 1, 2, 2, 2, 2, 2])

rand_score(reference_labels, cluster_assignments)
```

0.7777777777777778

The diagram below shows where the two labelings agree and disagree. The reference labels define groups $\{1, 2, 3\}$, while the clustering algorithm returns cluster labels $\{0, 1, 2\}$. The mismatch occurs because reference class 2 is split across clusters 1 and 2, and cluster 2 also contains observations from reference class 3.



The diagram also gives a quick way to compute the four pair counts:

- a : pairs together in both labelings. Add $\binom{m}{2}$ within each middle observation group: $a = \binom{3}{2} + \binom{2}{2} + \binom{2}{2} + \binom{3}{2} = 8$.
- c : pairs together in the reference labels but separated by the cluster assignments. Reference class 2 is split into groups of sizes 2 and 2, so $c = 2 \times 2 = 4$.
- d : pairs separated by the reference labels but together in the cluster assignments. Cluster 2 contains 2 observations from reference class 2 and 3 observations from reference class 3, so $d = 2 \times 3 = 6$.
- b : pairs separated by both labelings. The fastest calculation is subtraction from all pairs:

$$b = \binom{10}{2} - a - c - d = 45 - 8 - 4 - 6 = 27.$$

There are $\binom{10}{2} = 45$ observation pairs. Among them, $a = 8$ pairs are grouped together by both labelings, and $b = 27$ pairs are separated by both labelings. Therefore,

$$\text{RI} = \frac{8 + 27}{45} \approx 0.778.$$

12.1.6 Optional: Adjusted Rand Index

Click to expand.

The Rand index can be high even when two labelings agree partly by chance, especially when there are many pairs of observations. The adjusted Rand index corrects for this chance agreement.

Example 12.1 (A relatively high Rand index can be misleading). Suppose the reference labels contain two groups of equal size, but the clustering algorithm assigns every observation to its own cluster:

Reference label	1	1	1	1	1	2	2	2	2	2
Cluster assignment	0	1	2	3	4	5	6	7	8	9

This clustering does not recover the two reference groups. However, the Rand index is still above 0.5 because many pairs from different reference classes are also placed in different clusters. In this example, $a = 0$ and $b = 25$, so

$$\text{RI} = \frac{0 + 25}{\binom{10}{2}} = \frac{25}{45} \approx 0.556.$$

```

from sklearn.metrics import rand_score

reference_labels = [1, 1, 1, 1, 1, 2, 2, 2, 2, 2]
cluster_assignments = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

rand_score(reference_labels, cluster_assignments)

```

0.5555555555555556

The adjusted Rand index compares the observed pairwise agreement with the agreement expected from random labelings with similar group sizes. Suppose n_{ij} is the number of observations that are in reference class i and cluster j . Let

$$a_i = \sum_j n_{ij}, \quad b_j = \sum_i n_{ij}.$$

Thus, a_i is the size of reference class i , and b_j is the size of cluster j . Then the adjusted Rand index is

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}.$$

The formula has three main parts:

- $\sum_{ij} \binom{n_{ij}}{2}$ is the observed number of pairs that are placed together in both the reference labels and the cluster assignments.
- $\frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}$ is the expected number of such agreeing pairs under random labelings with the same class sizes and cluster sizes.
- $\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right]$ is the average of the largest pair counts implied by the two marginal labelings.

Thus, the adjusted Rand index has the form

$$\text{ARI} = \frac{\text{observed agreement} - \text{expected agreement}}{\text{maximum agreement} - \text{expected agreement}}.$$

The key difference between the Rand index and the adjusted Rand index is that the Rand index measures raw pairwise agreement, while the adjusted Rand index measures pairwise agreement after correcting for chance. This correction matters because many pairs may agree simply because they are separated in both labelings, even when the clustering does not recover the reference groups.

Its interpretation is:

- 1 means the two labelings are identical, up to relabeling of the clusters;
- values near 0 mean the agreement is about what would be expected by chance;
- negative values are possible when the agreement is worse than expected by chance.

This correction is useful because cluster labels themselves are arbitrary. For example, cluster assignments [0, 0, 1, 1] and [2, 2, 5, 5] describe the same grouping even though the numeric labels are different.

Example 12.2 (A relatively high Rand index can be misleading (cont.)). Continue from the Example [12.1](#).

```
from sklearn.metrics import adjusted_rand_score

reference_labels = [1, 1, 1, 1, 1, 2, 2, 2, 2, 2]
cluster_assignments = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

adjusted_rand_score(reference_labels, cluster_assignments)
```

0.0

The adjusted Rand index is 0, correctly indicating that the observed agreement is no greater than what would be expected by chance.

The same score can compare two clustering runs on the same observations. This is useful for assessing whether the fitted clustering is stable under small changes in preprocessing, initialization, or resampling.

```
cluster_run_1 = np.array([0, 0, 0, 1, 1, 1, 2, 2, 2, 2])
cluster_run_2 = np.array([1, 1, 1, 0, 0, 0, 2, 2, 2, 2])

adjusted_rand_score(cluster_run_1, cluster_run_2)
```

1.0

This example gives high agreement even though the numeric cluster labels have changed. That is appropriate because the labels identify groups only up to relabeling.

12.2 Centroid-Based: K-Means Clustering

K-means represents each cluster by its center. Let $C_1, \dots, C_K$ be a partition of the observations, and let

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

be the center of cluster k .

Definition 12.1 (K-means objective). K-means chooses the partition that minimizes the within-cluster sum of squares (WCSS):

$$\text{WCSS} = \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|_2^2.$$

In **sklearn**, this objective is also called **inertia**.

The standard algorithm alternates between two steps:

1. assign each observation to its nearest center;
2. replace each center by the mean of its assigned observations.

Each step decreases, or leaves unchanged, the K-means objective. The algorithm eventually reaches a local optimum, but not necessarily the global optimum. For this reason, software usually tries several random initializations.

12.2.1 K-means in Python

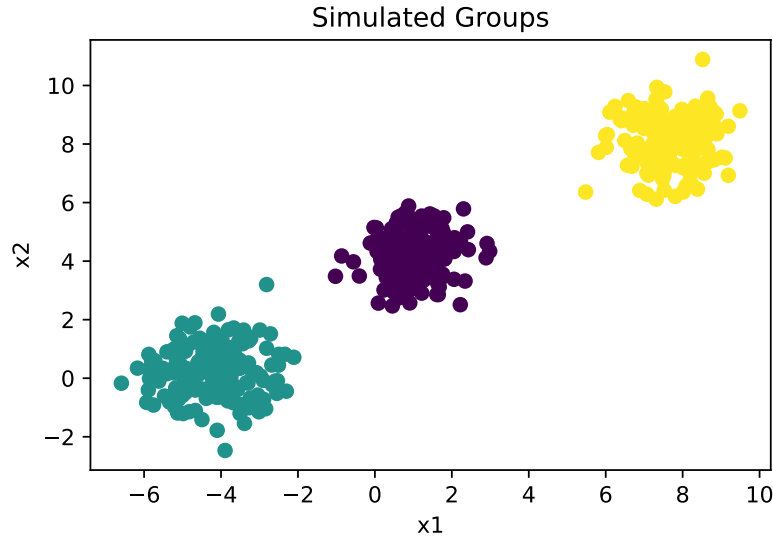
We use a simulated dataset to illustrate K-means in Python.

We use `make_blobs` from `sklearn.datasets` to generate 500 observations from three latent groups. The data have two features so that the result can be visualized directly. The colors below show the generating labels. They are not used when fitting K-means.

```
import numpy as np
from sklearn.datasets import make_blobs
import matplotlib.pyplot as plt

X, true_group = make_blobs(
    n_samples=500,
    n_features=2,
    centers=3,
    cluster_std=[0.7, 0.9, 0.8],
    random_state=3,
)

plt.scatter(X[:, 0], X[:, 1], c=true_group)
plt.xlabel("x1")
plt.ylabel("x2")
plt.title("Simulated Groups");
```



`KMeans` follows the usual `sklearn` interface. Since clustering is performed on the observed data themselves, we fit the model to the feature matrix and obtain cluster assignments for the observations.

For this first illustration, we set `n_clusters=3` so that the fitted result can be compared visually with the simulated groups. This does not represent how K is chosen in practice. In real applications, the number of clusters is usually unknown and must be treated as a tuning choice.

As mentioned previously, it is recommended to use multiple random initializations to reduce sensitivity to a poor starting configuration. The argument `n_init=20` runs K-means from 20 different initial center configurations and retains the solution with the smallest inertia among those runs.

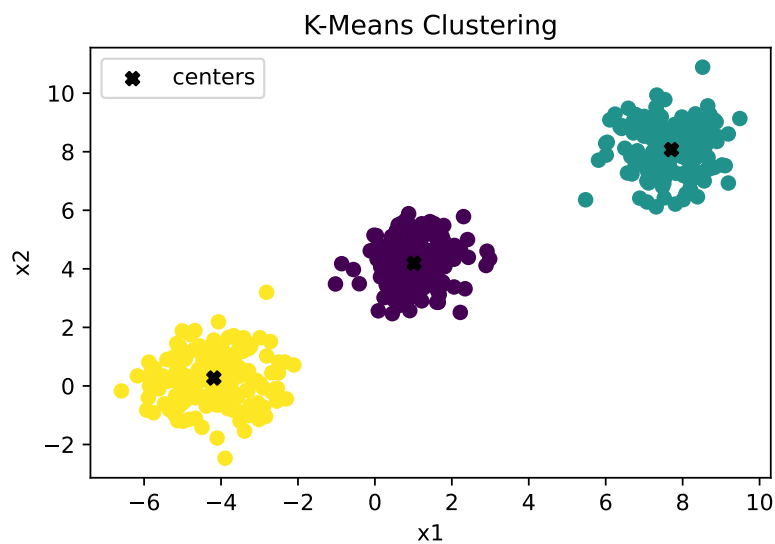
```
from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=3, n_init=20, random_state=1)
kmeans.fit(X)
cluster = kmeans.predict(X) ①
```

① You may also use `cluster = kmeans.fit_predict(X)` to combine the two commands together.

`cluster` is an array of cluster assignments, one for each observation. We can then plot the observations with colors indicating their assigned clusters.

```
plt.scatter(X[:, 0], X[:, 1], c=cluster)
plt.scatter(
    kmeans.cluster_centers_[0, 0],
    kmeans.cluster_centers_[0, 1],
    marker="X",
    color="black",
    label="centers",
)
plt.xlabel("x1")
plt.ylabel("x2")
plt.title("K-Means Clustering")
plt.legend();
```



The fitted centers are available from the model.

```
kmeans.cluster_centers_
```

```
array([[ 1.01494239,  4.19111967],
       [ 7.70936555,  8.07424709],
       [-4.1867326 ,  0.2695436 ]])
```

Label switching

Cluster labels are arbitrary. A cluster labeled 0 in one fit may be labeled 2 in another fit even when the two clusterings are equivalent. Interpret the groups through their members and feature summaries, not the numeric label.

12.2.2 Tuning Hyperparameters

The most important K-means hyperparameter is `n_clusters`, the number of centers to fit. A larger value gives a more detailed clustering and lowers the within-cluster sum of squares, but it may split groups that are better interpreted together. A smaller value gives a coarser summary but may merge distinct groups. There is no universally correct criterion because the useful level of grouping depends on the scientific question.

Two common diagnostics for choosing `n_clusters` are inertia and the silhouette score. The elbow plot examines inertia over a range that includes $K = 1$. The silhouette score compares within-cluster cohesion with separation from neighboring clusters and is computed only for $K \geq 2$.

```
from sklearn.metrics import silhouette_score

inertia_k = range(1, 11)
inertias = []
for k in inertia_k:
    model = KMeans(n_clusters=k, n_init=20, random_state=1)
    model.fit(X)
    inertias.append(model.inertia_)

silhouette_k = range(2, 11)
silhouettes = []
for k in silhouette_k:
    model = KMeans(n_clusters=k, n_init=20, random_state=1)
    labels = model.fit_predict(X)
    silhouettes.append(silhouette_score(X, labels))
```

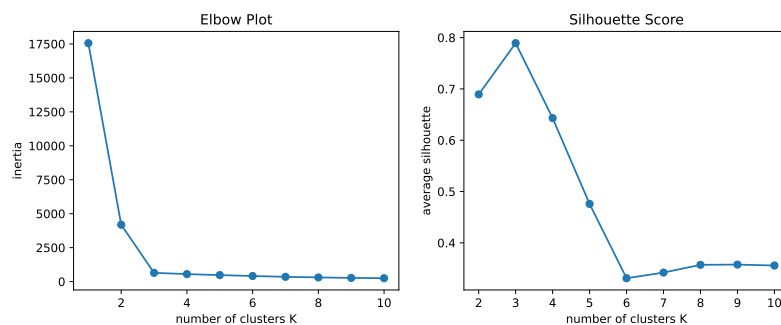

Plot the two diagnostics together.

```
import matplotlib.pyplot as plt

fig, axs = plt.subplots(1, 2, figsize=(11, 4))

axs[0].plot(inertia_k, inertias, marker="o")
axs[0].set_xlabel("number of clusters K")
axs[0].set_ylabel("inertia")
axs[0].set_title("Elbow Plot")

axs[1].plot(silhouette_k, silhouettes, marker="o")
axs[1].set_xlabel("number of clusters K")
axs[1].set_ylabel("average silhouette")
axs[1].set_title("Silhouette Score");
```



For inertia, the elbow heuristic looks for the point after which additional clusters produce only modest improvement. In this example, the largest improvements occur for small values of K , and $K = 3$ is a plausible elbow.

For the silhouette score, larger values are preferred. In this example, the average silhouette score is largest at $K = 3$ over the range considered.

The two diagnostics can disagree because they measure different aspects of the cluster assignments. The final choice of K should be guided by the analysis goal, stability, and interpretability, not by a single plot alone. In this simulated example, the diagnostics are consistent with the illustrative choice $K = 3$.

Other K-means hyperparameters control how the optimization is carried out:

- `init` controls how the initial centers are chosen. The default `"k-means++"` spreads initial centers out and is usually preferable to purely random initialization.
- `n_init` controls how many initial center configurations are tried. Since K-means can stop at a local optimum, using several starts makes the result less dependent on one unlucky initialization.
- `max_iter` is the maximum number of assignment-update steps for one initialization.
- `tol` controls the convergence tolerance. Smaller values ask the algorithm to keep iterating until the centers change very little.
- `random_state` makes randomized initialization reproducible.

For routine analyses, it is common to use `init="k-means++"` (which is the default value in the current version of `sklearn`) and tune `n_clusters`, while using a reasonably large `n_init`.



Diagnostics do not define the scientific answer

WCSS and the silhouette score evaluate geometric compactness and separation under the chosen features, scaling, and distance. They do not show that the clusters are stable, scientifically meaningful, or useful for a later decision.

12.2.3 When K-Means Works Well

Because K-means uses squared Euclidean distance to a center, it is most effective when clusters are:

- compact and approximately spherical;
- separated by their centers;
- reasonably similar in size and spread;
- not dominated by extreme outliers.

It can perform poorly on curved clusters, clusters with very different densities, or data where a mean is not an appropriate prototype.

```

from sklearn.datasets import make_circles
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

X_circle, true_circle_group = make_circles(
    n_samples=500, factor=0.45, noise=0.05, random_state=1
)

circle_kmeans = KMeans(n_clusters=2, n_init=20, random_state=1)
circle_cluster = circle_kmeans.fit_predict(X_circle)

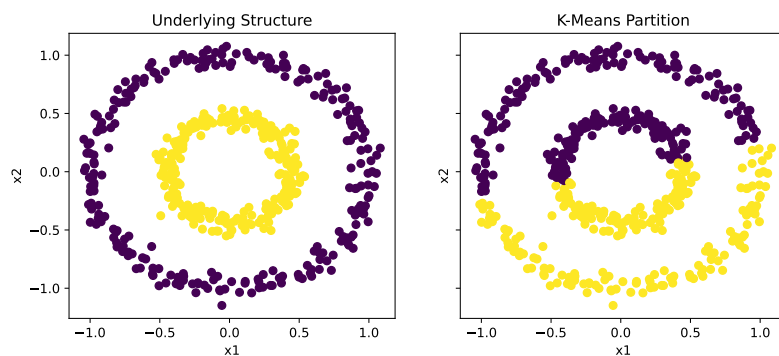
fig, axes = plt.subplots(1, 2, figsize=(10, 4), sharex=True, sharey=True)

axes[0].scatter(X_circle[:, 0], X_circle[:, 1], c=true_circle_group)
axes[0].set_title("Underlying Structure")

axes[1].scatter(X_circle[:, 0], X_circle[:, 1], c=circle_cluster)
axes[1].set_title("K-Means Partition")

for ax in axes:
    ax.set_xlabel("x1")
    ax.set_ylabel("x2")

```



K-means divides the feature space into convex regions, so it cannot recover the two rings.

12.3 Hierarchical: Agglomerative Clustering

Hierarchical clustering builds a nested sequence of clusterings. Agglomerative hierarchical clustering begins with each observation in its own cluster and repeatedly merges the two “closest” clusters until all observations belong to one cluster.

The result is a tree called a **dendrogram**. Unlike K-means, the hierarchy can be fit before choosing the final number of clusters.

12.3.1 Linkage

At each step of agglomerative hierarchical clustering, we merge the two closest clusters. To determine which clusters are closest, we first need to define the distance between clusters. A linkage rule specifies how this distance is computed.

For two clusters G and H , common linkage rules include:

- Single linkage uses the closest pair: $d_{\text{single}}(G, H) = \min_{i \in G, j \in H} d(x_i, x_j)$.
- Complete linkage uses the farthest pair: $d_{\text{complete}}(G, H) = \max_{i \in G, j \in H} d(x_i, x_j)$.
- Average linkage uses the average pairwise distance: $d_{\text{avg}}(G, H) = \frac{1}{|G||H|} \sum_{i \in G, j \in H} d(x_i, x_j)$.
- Ward linkage merges the pair of clusters that produces the smallest increase in the within-cluster sum of squares (WCSS). Unlike the previous linkage rules, Ward linkage is based on cluster variance rather than pairwise distances, and therefore tends to produce compact, approximately spherical clusters.

Different linkage rules represent different notions of what makes two clusters close. Single linkage can recover elongated or irregularly shaped clusters, but it is sensitive to the chaining effect, where intermediate points connect otherwise separate groups. Complete linkage and Ward linkage are less affected by chaining and tend to produce more compact clusters.

12.3.2 Dendrogram

A dendrogram is a visual record of the sequence of merges. At the beginning, each observation is its own cluster. At each step, the two closest current clusters are merged according to the chosen linkage rule. The height of the merge records the dissimilarity at which the two branches are joined.

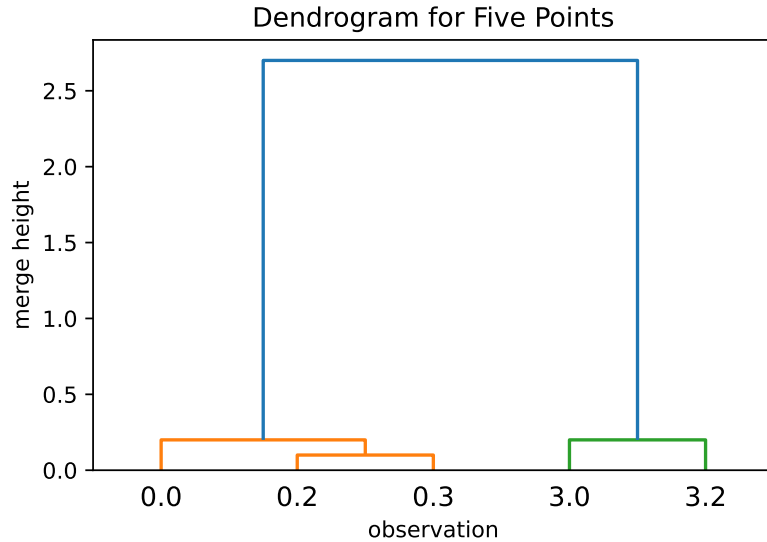
The merge height is not always the distance between two individual observations. It is the dissimilarity between the two clusters being merged, as defined by the linkage rule. For single linkage, the height is the smallest pairwise distance between the two clusters. For complete linkage, it is the largest pairwise distance; for average linkage, it is the average pairwise distance; and for Ward linkage, it is based on the increase in within-cluster variation caused by the merge.

For example, consider five one-dimensional observations:

0.0, 0.2, 0.3, 3.0, 3.2.

Using single linkage, the tree is built as follows:

1. Merge 0.2 and 0.3, the closest pair, at height 0.1.
2. Merge 0.0 with the cluster $\{0.2, 0.3\}$ at height 0.2.
3. Merge 3.0 and 3.2 at height 0.2.
4. Merge the two larger clusters $\{0.0, 0.2, 0.3\}$ and $\{3.0, 3.2\}$ at height 2.7.



The toy dendrogram shows the main interpretation: observations that merge at low heights are similar under the chosen linkage rule, while branches that merge only at high heights are more separated.

12.3.3 Hierarchical clustering in Python

In `sklearn`, agglomerative hierarchical clustering is fit using `AgglomerativeClustering`. Here we apply it to the simulated dataset from K-means. Like other `sklearn` estimators, it provides `fit_predict` for returning cluster assignments.

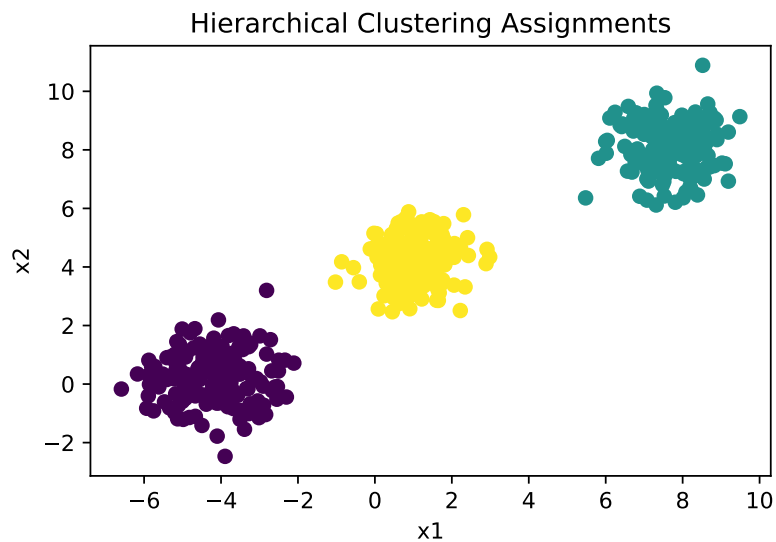
```
from sklearn.cluster import AgglomerativeClustering
hierarchical_model = AgglomerativeClustering(n_clusters=3, linkage="single")
hierarchical_cluster = hierarchical_model.fit_predict(X)
```

Here `n_clusters=3` is chosen for illustration. We can visualize the resulting cluster assignments in the original feature space.

```
import matplotlib.pyplot as plt

plt.scatter(X[:, 0], X[:, 1], c=hierarchical_cluster)
plt.xlabel("x1")
```

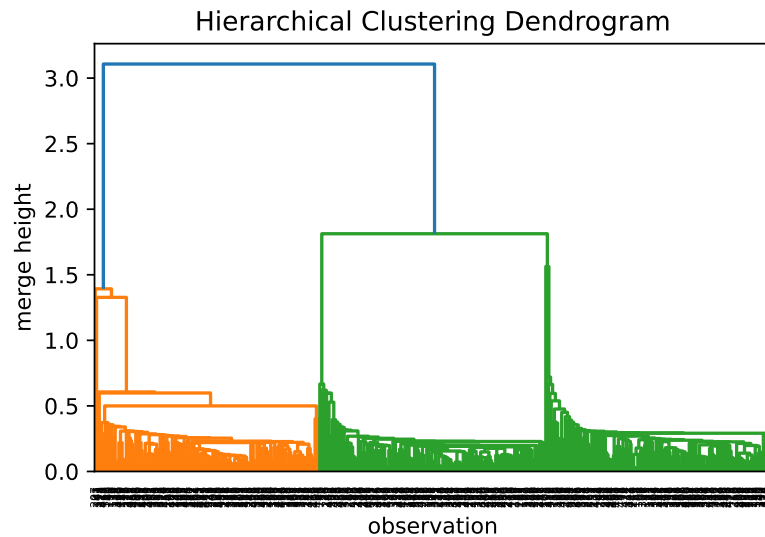
```
plt.ylabel("x2")
plt.title("Hierarchical Clustering Assignments");
```



`sklearn` can fit an agglomerative hierarchical clustering model and return the cluster assignments, but it does not provide a function to directly plot a dendrogram. For visualization, we use `linkage` from `scipy.cluster.hierarchy` to compute the hierarchy and `dendrogram` to display the resulting merge tree.

```
from scipy.cluster.hierarchy import linkage, dendrogram

Z = linkage(X, method="single")
dendrogram(Z)
plt.xlabel("observation")
plt.ylabel("merge height")
plt.title("Hierarchical Clustering Dendrogram");
```



The merge height records the dissimilarity at which two clusters are merged. The dendrogram summarizes the fitted hierarchy, not uncertainty about the hierarchy.

i Cutting the Dendrogram

Hierarchical clustering separates fitting the hierarchy from choosing the final flat clustering. One conceptual approach is to inspect the dendrogram and make a horizontal cut at a merge height. This keeps merges below the cut and separates branches that would only merge above the cut. The cut height is often chosen by looking for a large vertical gap in the dendrogram, but this is only a heuristic. It should be checked against the analysis goal, stability, validation scores, and interpretability.

i `n_clusters` versus a height cut

In `sklearn`, the usual `AgglomerativeClustering(n_clusters=...)` interface asks for an exact number of clusters. This is not the same as drawing a horizontal line at a fixed dendrogram height. The algorithm can return any requested value of `n_clusters`, even when several merges occur at the

same height. Operationally, this is equivalent to following the merge sequence determined by the implementation and stopping when exactly `n_clusters` groups remain, rather than choosing all merges below a fixed height. If several candidate merges have the same height, the order is implementation-determined, not random.

If the goal is a strict height-based cut, use `scipy.cluster.hierarchy.linkage` with `fcluster`, or use `AgglomerativeClustering` with `distance_threshold` and `n_clusters=None`. In this chapter, we focus mainly on the `sklearn` `n_clusters` interface because it is simple and matches the way the examples compare flat cluster assignments.

12.3.4 Comparing Linkages

To make the effect of linkage visible, we use a dataset whose geometry is not compact or spherical. The two-moons data have two curved groups, so different linkage rules can produce noticeably different flat clusterings.

Although the data are generated from two underlying groups, we set `n_clusters=4` here to make the behavior of the linkage rules easier to compare.

```
from sklearn.cluster import AgglomerativeClustering
from sklearn.datasets import make_moons
import matplotlib.pyplot as plt

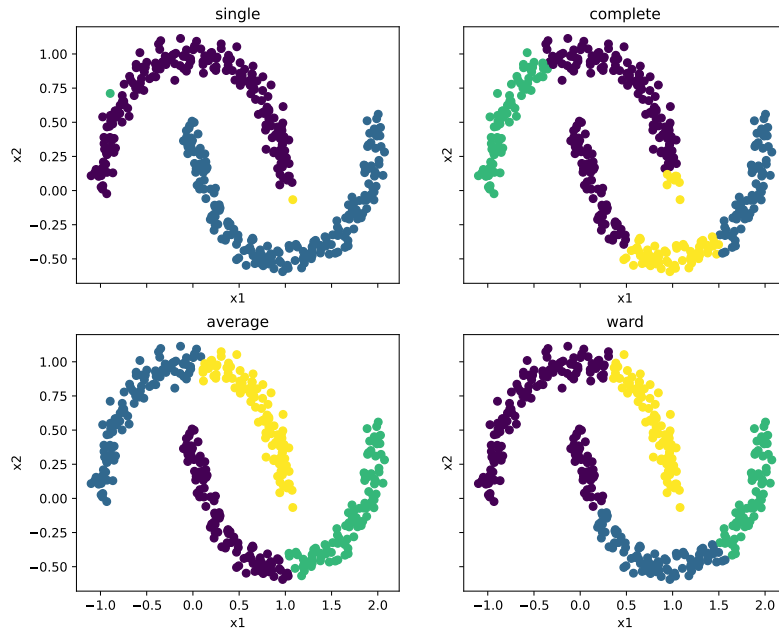
X_ex, _ = make_moons(n_samples=400, noise=0.06, random_state=6)

methods = ["single", "complete", "average", "ward"]
fig, axs = plt.subplots(2, 2, figsize=(10, 8), sharex=True, sharey=True)

for ax, method in zip(axs.ravel(), methods):
    model = AgglomerativeClustering(n_clusters=4, linkage=method)
    labels = model.fit_predict(X_ex)

    ax.scatter(X_ex[:, 0], X_ex[:, 1], c=labels)
```

```
ax.set_title(method)
ax.set_xlabel("x1")
ax.set_ylabel("x2");
```



The cluster labels are arbitrary, so the important feature is the shape of the split:

- Single linkage uses the closest pair of observations between two clusters. It favors connected shapes and can follow long curves. In the plot, it keeps most of each moon as a connected chain, and the extra clusters appear where a small gap breaks the chain.
- Complete linkage uses the farthest pair of observations between two clusters. It favors clusters with small diameter, so it tends to produce compact, ball-like groups rather than long or stretched groups. In the plot, it cuts the moons into shorter pieces instead of preserving the full curved shapes.
- Average linkage uses the average distance over all pairs of observations between two clusters. It is a compromise between connectivity and compactness: it is less sensitive to one nearby pair than single linkage, but less strict about

the farthest pair than complete linkage. In the plot, it still splits the moons where the average distance to the rest of the arc becomes large.

- Ward linkage merges clusters that cause the smallest increase in WCSS. It favors compact, variance-minimizing clusters, similar in spirit to K-means. In the plot, its cuts look like local spatial blocks rather than moon-shaped pieces, because the curved groups are not compact around a single center.

i Complete linkage and Ward linkage

Complete linkage and Ward linkage both tend to produce compact, ball-like clusters, but they measure compactness differently.

- Complete linkage is based on cluster diameter. It avoids merges that would make the farthest two observations in a cluster too far apart.
- Ward linkage is based on within-cluster variation. It avoids merges that would cause a large increase in WCSS, where each cluster is summarized by its mean.

Thus, complete linkage is diameter-based compactness, while Ward linkage is variance-based compactness.

They can differ when cluster sizes or shapes are uneven. Complete linkage reacts strongly to the most distant pair of observations, so one outlying point can prevent a merge by making the cluster diameter too large. Ward linkage reacts to the overall increase in WCSS, so it also depends on cluster sizes and centroids. If two large clusters have nearby centers, Ward linkage may still merge them even when a few observations are relatively far apart, because those observations may not dominate the total WCSS increase.

12.3.5 When Hierarchical Clustering Works Well

Hierarchical clustering is especially useful when the nested structure itself is meaningful. For example, it is often used to organize genes, documents, products, or other observations where relationships at multiple levels of granularity are informative.

It is most useful when:

- the hierarchy itself is part of the interpretation;
- one wants to inspect several possible numbers of clusters from the same fitted tree;
- the dataset is small or moderate in size;
- the chosen linkage rule matches the geometry of the groups.

However, agglomerative clustering can be computationally expensive for very large datasets, and early merges cannot be undone. A suboptimal early merge can affect the rest of the tree.

12.4 Density-Based: DBSCAN

Density-based spatial clustering of applications with noise (DBSCAN) defines clusters as connected regions of sufficiently high local density. Unlike K-means or hierarchical clustering, DBSCAN does not require the number of clusters to be specified in advance.

The main idea is that a cluster should be a dense region separated from other dense regions by lower-density space. This makes DBSCAN useful when clusters are curved, irregularly shaped, or when some observations should be treated as noise rather than forced into a cluster.

DBSCAN uses two main parameters:

- **eps**: the radius of a local neighborhood;
- **min_samples**: the number of observations required for a dense neighborhood, including the observation itself.

These parameters define three types of observations:

- A **core point** has at least `min_samples` observations in its `eps` neighborhood, including itself.
- A **border point** is not a core point, but lies within the `eps` neighborhood of a core point.
- A **noise point** is neither a core point nor a border point.

12.4.1 How DBSCAN Forms Clusters

DBSCAN begins by checking the `eps` neighborhood of each observation. If an observation has at least `min_samples` observations in that neighborhood, counting itself, it becomes a core point. A core point can start a new cluster if it has not already been reached from another core point.

Once a core point starts a cluster, DBSCAN adds all observations within its `eps` neighborhood. If any of those observations are also core points, their own `eps` neighborhoods are added as well. This expansion continues until no new core points can be reached. All core points connected in this way belong to the same cluster, and non-core points reached from those core points are included as border points.

Observations that are not reachable from any core point are labeled as noise. Thus, DBSCAN does not assign points by nearest center. It forms clusters by following chains of dense neighborhoods, and the number of clusters is whatever number of separated dense regions the algorithm finds.

12.4.2 DBSCAN in Python

We use DBSCAN from `sklearn.cluster` to implement the method.

```
from sklearn.cluster import DBSCAN
from sklearn.datasets import make_circles

X_circle, true_circle_group = make_circles(
    n_samples=500, factor=0.45, noise=0.05, random_state=1
)
```

```
circle_dbscan = DBSCAN(eps=0.13, min_samples=4)
circle_dbscan_group = circle_dbscan.fit_predict(X_circle)
```

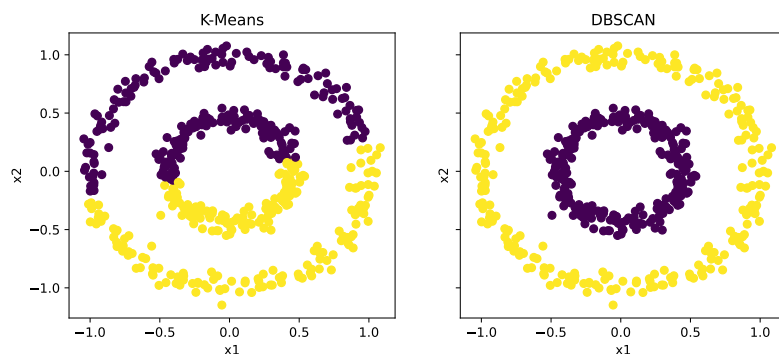
```
import matplotlib.pyplot as plt
```

```
fig, axes = plt.subplots(1, 2, figsize=(10, 4), sharex=True, sharey=True)
```

```
axes[0].scatter(X_circle[:, 0], X_circle[:, 1], c=circle_cluster)
axes[0].set_title("K-Means")
```

```
axes[1].scatter(X_circle[:, 0], X_circle[:, 1], c=circle_dbscan_group)
axes[1].set_title("DBSCAN")
```

```
for ax in axes:
    ax.set_xlabel("x1")
    ax.set_ylabel("x2")
```



In `sklearn`, noise observations receive the label `-1`. As with all clustering labels, the nonnegative cluster labels have no ordinal meaning.

12.4.3 Choosing `eps` and `min_samples`

The most important DBSCAN hyperparameter is `eps`, because it controls the scale at which density is measured. A smaller `eps` makes neighborhoods smaller and the density requirement stricter. This may split clusters or label more observations as

noise. A larger `eps` makes neighborhoods larger and can merge groups through low-density bridges.

The parameter `min_samples` controls how much local support is required for a point to be considered a core point. Larger values make the method more conservative and usually produce more noise points.

A practical workflow is to first set `min_samples`, then use the data to select `eps`. A common starting choice is to set `min_samples` to about twice the number of features. In this two-dimensional example, we use 4.

i Checking `min_samples`

There is no supervised-style cross-validation rule for choosing `min_samples`. A reasonable value should give a stable and interpretable density structure: nearby choices should not radically change the main clusters, and the fraction of noise points should be plausible for the problem. A lower noise fraction is not automatically better, because forcing sparse points into clusters may hide outliers or connect groups through weak bridges.

Several diagnostics can help. For a chosen `min_samples`, the k-distance plot should ideally show a reasonably clear elbow, suggesting a defensible density scale. As a starting point, two-dimensional data often use `min_samples=4` or 5; more generally, one may start around $p + 1$ or $2p$, where p is the number of features.

If external labels are available, the Rand index or adjusted Rand index can be used as additional diagnostics.

After choosing `min_samples`, compute the distance from each observation to the neighbor that determines whether it is a core point. Since `min_samples` includes the observation itself in `sklearn`, this is the distance to the `min_samples`-th nearest observation when the observation itself is included as the first neighbor. Points inside dense regions should have small such distances, while isolated or low-density points should have larger distances. Sorting these distances gives a diagnostic plot. A visible elbow in the plot suggests a candidate value of `eps`.

In Python we can use `NearestNeighbors` from `sklearn.neighbors` to compute these distances. The result is then sorted and plotted.

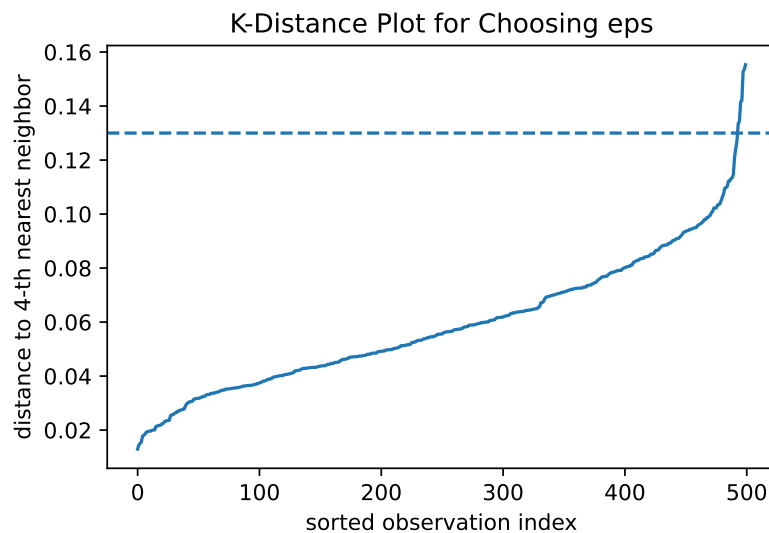
```
from sklearn.neighbors import NearestNeighbors

min_samples = 4

neighbors = NearestNeighbors(n_neighbors=min_samples)
neighbors.fit(X_circle)

distances, indices = neighbors.kneighbors(X_circle)
k_distances = np.sort(distances[:, -1])

plt.plot(k_distances)
plt.axhline(y=0.13, ls="--")
plt.xlabel("sorted observation index")
plt.ylabel(f"distance to {min_samples}-th nearest neighbor")
plt.title("K-Distance Plot for Choosing eps");
```



A candidate `eps` can be drawn as a horizontal line on this plot. Observations with `k`-distances below the line satisfy the core-point density requirement. Observations above the line are not core points; they may later become border points if they lie within `eps` of a core point, or they may be labeled as noise.

Thus, decreasing **eps** usually creates fewer core points, more noise, and a greater chance that a dense region is split into pieces. Increasing **eps** usually creates more core points and a greater chance that nearby dense regions are merged through a bridge of points.

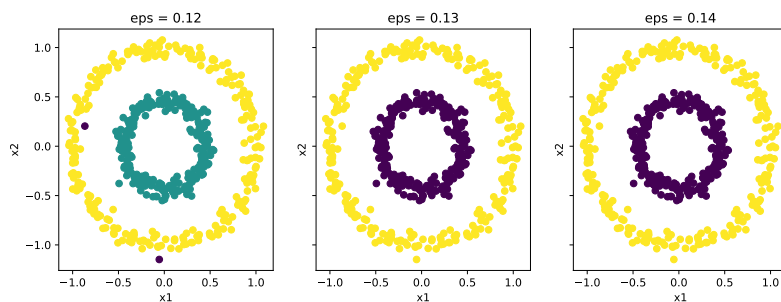
In this example, the curve remains relatively flat for most observations and then increases more sharply for observations whose local neighborhoods are sparser. An **eps** value near the bend separates the dense ring structure from the lower-density gaps. We next check nearby values.

```
eps_values = [0.12, 0.13, 0.14]

fig, axs = plt.subplots(1, 3, figsize=(12, 4), sharex=True, sharey=True)

for ax, eps in zip(axs, eps_values):
    labels = DBSCAN(eps=eps, min_samples=min_samples).fit_predict(X_circle)

    ax.scatter(X_circle[:, 0], X_circle[:, 1], c=labels)
    ax.set_title(f"eps = {eps}")
    ax.set_xlabel("x1")
    ax.set_ylabel("x2")
```



If small changes in **eps** lead to very different cluster assignments, the density structure may not be stable at a single scale. In that case, the fitted clusters should be interpreted cautiously, or a method designed for varying density levels may be more appropriate.

In this example, from 0.12 to 0.14 the clustering structure is relatively stable. The color labels may change because cluster

labels are arbitrary, but the two-ring grouping is essentially the same. Therefore, 0.13 is a reasonable choice within this stable range.

Silhouette scores are only a rough diagnostic for DBSCAN

One can compute silhouette scores over a grid of `eps` and `min_samples` values, but this is not ordinary cross-validation because clustering has no supervised prediction error. The score is also harder to interpret for DBSCAN: treating noise as an extra cluster can distort the score, while removing noise ignores one of DBSCAN's main outputs.

Silhouette scores also favor compact, well-separated groups, whereas DBSCAN is often used for curved or irregular clusters. Use them only as a secondary diagnostic alongside the k-distance plot, the number of clusters, the fraction of noise points, visual inspection when possible, and sensitivity to nearby parameter values.

DBSCAN does not tune `n_clusters`

DBSCAN determines the number of clusters from the density structure induced by `eps` and `min_samples`. This is useful when the number of groups is unknown, but it also means that the result can change substantially when the density scale changes.

12.4.4 When DBSCAN Works Well

DBSCAN is most useful when:

- clusters have irregular or curved shapes;
- clusters are separated by sparse regions;
- noise points should be explicitly identified;
- a single density scale is reasonable for the dataset.

It is less effective when cluster densities differ substantially. One global value of `eps` may be too small for a sparse cluster and

too large for a dense cluster.

12.5 Comparing the Three Methods

The three direct methods answer different versions of the clustering question. The key difference is their definitions of a cluster: what each method treats as evidence that observations belong together:

Method	Informal rule	Works especially well when
K-means	Points belong together if they are close to the same center.	clusters are roughly spherical, compact, and similar in size
Hierarchical clustering	Points belong together if they are joined through a sequence of merges.	one wants to explore grouping structure at several resolutions
DBSCAN	Points belong together if they lie in the same sufficiently dense region.	clusters are dense regions, noise is present, and the number of clusters is unknown

No method is uniformly best. The geometry of the data, the intended definition of a cluster, and the scientific question should determine which paradigm is appropriate. A useful comparison does not require every method to look successful; a poor-looking fit can be informative when it reveals the method's assumptions.

- K-means is fast, simple, and often the practical default when K is known and clusters are roughly compact and center-based. It can revise assignments during iteration, but it forces every observation into one of the K clusters and is not designed for curved shapes or explicit noise.

- Agglomerative hierarchical clustering is most useful when the merge sequence or dendrogram is part of the interpretation. It can show several possible resolutions from one fitted tree, but early merges cannot be undone and classical agglomerative methods scale poorly because they rely on many pairwise dissimilarities.
- DBSCAN is useful when clusters are dense regions separated by sparse space. It does not require `n_clusters` and can label observations as noise, but it is sensitive to the density scale set by `eps` and `min_samples` and does not show nested structure.

We show three examples below illustrating the strengths and limitations of each method. These examples are not a ranking. They show that each method is answering a different question: nearest center, merge tree, or density connectivity. In the examples, the hierarchical clustering result uses single linkage. When a method performs poorly on a particular geometry, the failure is still useful if it matches the method's definition of a cluster.

Since all datasets are generated, we know the true label for each observation. Therefore, we can use ARI as an external metric to compare the clustering results on each dataset. ARI is optional and was introduced earlier in the notes.

In the curves-with-noise example, the artificially added noise points are treated as a separate reference group when computing ARI. This is a teaching device for simulated data, not something usually available in real applications. It favors methods that can identify noise points, such as DBSCAN, and penalizes methods such as K-means and hierarchical clustering that must assign every observation to a cluster.

12.5.1 Example: Curves with Noise

```
from sklearn.datasets import make_blobs, make_moons
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.metrics import adjusted_rand_score
import numpy as np
```

```

import matplotlib.pyplot as plt

rng = np.random.default_rng(2)

X_moon, true_moon_group = make_moons(n_samples=450, noise=0.06, random_state=2)
moon_noise = rng.uniform(low=[-1.5, -0.8], high=[2.5, 1.3], size=(30, 2))
X_moon_noise = np.vstack([X_moon, moon_noise])
true_moon_noise_group = np.concatenate(
    [true_moon_group, np.repeat(-1, len(moon_noise))]
)

kmeans_labels = KMeans(n_clusters=2, n_init=20, random_state=1).fit_predict(
    X_moon_noise
)

single_labels = AgglomerativeClustering(n_clusters=2, linkage="single").fit_predict(
    X_moon_noise
)

dbscan_labels = DBSCAN(eps=0.16, min_samples=6).fit_predict(X_moon_noise)

kmeans_ari = adjusted_rand_score(true_moon_noise_group, kmeans_labels)
single_ari = adjusted_rand_score(true_moon_noise_group, single_labels)
dbscan_ari = adjusted_rand_score(true_moon_noise_group, dbscan_labels)

fig, axs = plt.subplots(1, 3, figsize=(12, 3.5), sharex=False, sharey=False)

axs[0].scatter(X_moon_noise[:, 0], X_moon_noise[:, 1], c=kmeans_labels, s=16)
axs[0].set_title(f"K-Means\nARI = {kmeans_ari:.2f}")

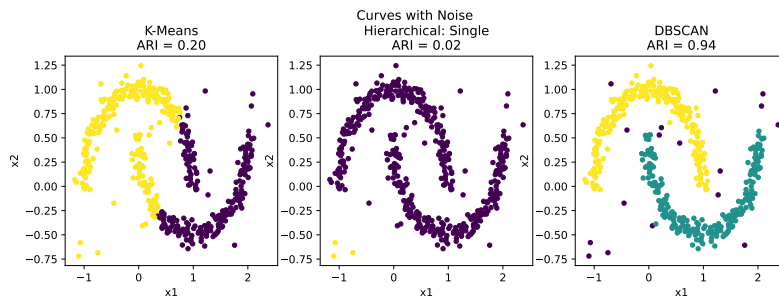
axs[1].scatter(X_moon_noise[:, 0], X_moon_noise[:, 1], c=single_labels, s=16)
axs[1].set_title(f"Hierarchical: Single\nARI = {single_ari:.2f}")

axs[2].scatter(X_moon_noise[:, 0], X_moon_noise[:, 1], c=dbscan_labels, s=16)
axs[2].set_title(f"DBSCAN\nARI = {dbscan_ari:.2f}")

for ax in axs:
    ax.set_xlabel("x1")
    ax.set_ylabel("x2")

```

```
fig.suptitle("Curves with Noise", y=1.05);
```



The moon dataset contains two curved dense regions plus scattered noise.

- DBSCAN works best here because it defines clusters as dense connected regions, so it can follow the two curved shapes and leave scattered observations as noise.
- K-means has trouble because each cluster is represented by a center, which breaks the curved geometry into center-based regions.
- Hierarchical clustering using single linkage can follow connected shapes, but scattered noise may create unwanted connections or unbalanced cuts.

12.5.2 Example: Unequal Densities

```
X_unequal_density, true_unequal_density_group = make_blobs(
    n_samples=[180, 220, 260],
    centers=[[-4.0, 0], [0.0, 0], [4.5, 0]],
    cluster_std=[0.18, 0.55, 1.15],
    random_state=8,
)

kmeans_labels = KMeans(n_clusters=3, n_init=20, random_state=1).fit_predict(
    X_unequal_density
)

single_labels = AgglomerativeClustering(n_clusters=3, linkage="single").fit_predict(
```

```

X_unequal_density
)

dbscan_labels = DBSCAN(eps=0.40, min_samples=6).fit_predict(X_unequal_density)

kmeans_ari = adjusted_rand_score(true_unequal_density_group, kmeans_labels)
single_ari = adjusted_rand_score(true_unequal_density_group, single_labels)
dbscan_ari = adjusted_rand_score(true_unequal_density_group, dbscan_labels)

fig, axs = plt.subplots(1, 3, figsize=(12, 3.5), sharex=False, sharey=False)

axs[0].scatter(X_unequal_density[:, 0], X_unequal_density[:, 1], c=kmeans_labels, s=16)
axs[0].set_title(f"K-Means\nARI = {kmeans_ari:.2f}")

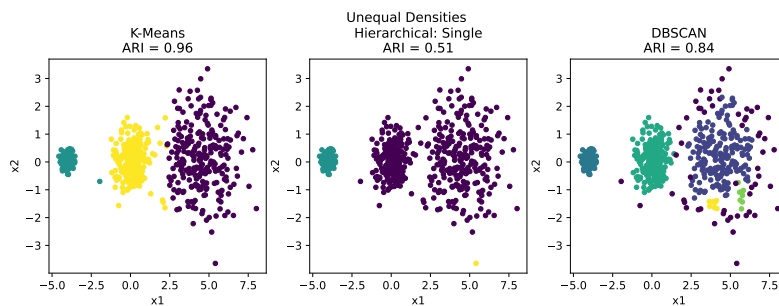
axs[1].scatter(X_unequal_density[:, 0], X_unequal_density[:, 1], c=single_labels, s=16)
axs[1].set_title(f"Hierarchical: Single\nARI = {single_ari:.2f}")

axs[2].scatter(X_unequal_density[:, 0], X_unequal_density[:, 1], c=dbscan_labels, s=16)
axs[2].set_title(f"DBSCAN\nARI = {dbscan_ari:.2f}")

for ax in axs:
    ax.set_xlabel("x1")
    ax.set_ylabel("x2")

fig.suptitle("Unequal Densities", y=1.05);

```



The unequal-density dataset illustrates a limitation of DBSCAN. The three groups have very different spreads.

- K-means is relatively effective here because the groups are still roughly center-based, even though their densities

differ.

- DBSCAN is less effective here than K-means and is more sensitive to the chosen `eps`, because one global `eps` may be appropriate for the dense group but too small for the sparse group, or appropriate for the sparse group but too large for the dense group.
- Hierarchical clustering using single linkage can also be unstable here because a few close points between groups can create early connections that affect the later merges.

12.5.3 Example: Uneven Chains

```
rng_chain = np.random.default_rng(9)
t_chain = np.concatenate([np.linspace(0, 1.1, 170), np.linspace(1.1, np.pi, 35)])
upper_chain = np.column_stack([np.cos(t_chain), np.sin(t_chain)])
lower_chain = np.column_stack([1 - np.cos(t_chain), 0.5 - np.sin(t_chain)])
X_chain = np.vstack([upper_chain, lower_chain]) + rng_chain.normal(
    scale=0.025, size=(2 * len(t_chain), 2)
)
true_chain_group = np.concatenate(
    [np.repeat(0, len(t_chain)), np.repeat(1, len(t_chain))]
)

kmeans_labels = KMeans(n_clusters=2, n_init=20, random_state=1).fit_predict(X_chain)

single_labels = AgglomerativeClustering(n_clusters=2, linkage="single").fit_predict(
    X_chain
)
dbscan_labels = DBSCAN(eps=0.08, min_samples=6).fit_predict(X_chain)

kmeans_ari = adjusted_rand_score(true_chain_group, kmeans_labels)
single_ari = adjusted_rand_score(true_chain_group, single_labels)
dbscan_ari = adjusted_rand_score(true_chain_group, dbscan_labels)

fig, axs = plt.subplots(1, 3, figsize=(12, 3.5), sharex=False, sharey=False)

axs[0].scatter(X_chain[:, 0], X_chain[:, 1], c=kmeans_labels, s=16)
axs[0].set_title(f"K-Means\nARI = {kmeans_ari:.2f}")
```



```

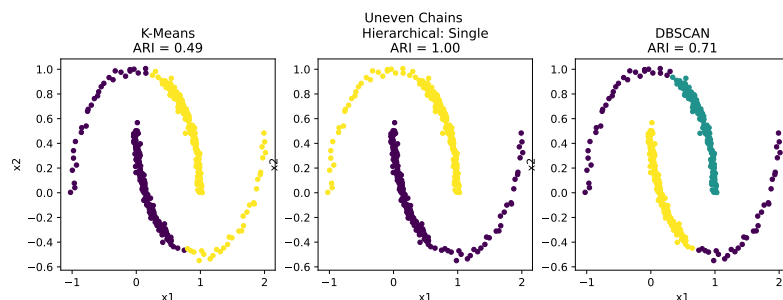
axs[1].scatter(X_chain[:, 0], X_chain[:, 1], c=single_labels, s=16)
axs[1].set_title(f"Hierarchical: Single\nARI = {single_ari:.2f}")

axs[2].scatter(X_chain[:, 0], X_chain[:, 1], c=dbscan_labels, s=16)
axs[2].set_title(f"DBSCAN\nARI = {dbscan_ari:.2f}")

for ax in axs:
    ax.set_xlabel("x1")
    ax.set_ylabel("x2")

fig.suptitle("Uneven Chains", y=1.05);

```



The uneven-chain dataset illustrates a case where hierarchical clustering using single linkage can be useful. The groups are best described as connected chains with uneven spacing.

- Single linkage can follow each chain through nearest-neighbor connections, so it is the most natural fit among the three methods.
- K-means instead cuts the plane into center-based regions, which mixes parts of the two chains.
- DBSCAN can label the thinner parts on both sides of the chains as noise when one global density scale is used. In this example, those side segments are not merged into one cluster; they are mostly excluded as noise because they are too sparse under the chosen `eps` and `min_samples`.

13 Unsupervised Dimension Reduction

In the supervised dimension-reduction chapter, PCA appeared inside principal component regression (PCR). There the workflow was

$$X \longrightarrow Z = XV_m \longrightarrow \hat{y}.$$

The principal components were useful only if they helped predict the response. The number of components was therefore chosen by cross-validation against prediction error.

This chapter uses the same algebra for a different purpose. There is no response y . The workflow is instead

$$X \longrightarrow Z \longrightarrow \hat{X}.$$

The question is no longer “does the reduced representation predict y ?” but “what structure of the data matrix does the reduced representation preserve?” This change makes reconstruction, compression, visualization, denoising, and distance preservation more central than prediction.

We will use PCA as the main example, revisiting PCA, SVD, and related concepts only where they help explain dimension reduction as a way of building useful low-dimensional representations of a data matrix.

13.1 How to Read the Data Matrix

Let $X \in \mathbb{R}^{n \times p}$ contain n observations and p features. We usually center the columns before applying PCA, so the origin represents the sample mean.

This X has two complementary geometric views. The coordinates of a vector determine the space it lives in. Therefore

- A row has one coordinate for each feature, so it is a vector in feature space $\mathbb{R}^p$.
- A column has one coordinate for each observation, so it is a vector in observation space $\mathbb{R}^n$.

This is not only notation. It determines which objects can be compared, combined, or projected. Row operations compare observations by their feature values. Column operations compare features by their behavior across observations.

13.1.1 Rows: observations in feature space

The i -th row

$$x_i^\top = [x_{i1}, \dots, x_{ip}]$$

is one observation represented as a point in feature space $\mathbb{R}^p$. The axes of this space are the features. For example, if the features are height, weight, and age, then each person is a point in a three-dimensional feature space.

In this view, we ask questions about observations:

- Which observations are close to each other?
- Which observations are outliers?
- Can the observations be represented well by a lower-dimensional plane?

Distances between rows $\|x_i - x_\ell\|$ compare observations. Clustering, nearest neighbors, and visualization usually start from this row-side view. Dimension reduction replaces each high-dimensional row $x_i^\top$ with a lower-dimensional row $z_i^\top$.

13.1.2 Columns: features in observation space

The j -th column

$$X_j = [x_{1j}, \dots, x_{nj}]^\top$$

is one feature measured across all observations. It is a vector in observation space $\mathbb{R}^n$. The axes of this space are the individual observations. For example, the height column records the height value for person 1, person 2, and so on. In that sense, height is a point in an n -dimensional space whose coordinates are people.

In this view, we ask questions about features:

- Which features vary together across observations?
- Which features are redundant?
- Which weighted combinations of features create useful new features?

Inner products between columns $X_j^\top X_k$ measure feature co-movement. After centering, $X_j^\top X_k / (n - 1)$ is the sample covariance between features j and k . After standardizing columns, the angle between X_j and X_k is related to their correlation.

13.1.3 A matrix is a bridge between two spaces

Let $e_i \in \mathbb{R}^n$ select observation i , and let $f_j \in \mathbb{R}^p$ select feature j . Then

$$e_i^\top X f_j = x_{ij}.$$

This single number connects the two spaces: it is the measurement of feature j on observation i . Reading across a row fixes an observation and varies the feature index. Reading down a column fixes a feature and varies the observation index.

More generally, X can combine many features or many observations at once. If $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^p$, then

$$Xv \in \mathbb{R}^n, \quad u^\top X \in \mathbb{R}^p, \quad u^\top Xv \in \mathbb{R}.$$

- The vector v gives weights for the features. Multiplying Xv makes one new feature column, with one value for each observation.
- The vector u gives weights for the observations. Multiplying $u^\top X$ makes one new row, with one value for each feature.

Example 13.1 (Constructing a feature with Xv). Suppose the rows of X are people and the columns are standardized features such as height, weight, and age. To construct a new feature, choose weights for the columns. For example, let

$$v = \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

where the first two coordinates correspond to standardized height and standardized weight. Then

$$Xv = 0.6X_1 + 0.8X_2.$$

This is a new feature column in observation space $\mathbb{R}^n$. Its i -th entry is

$$x_i^\top v = 0.6x_{i1} + 0.8x_{i2},$$

a size-like score for person i . The vector v is the recipe for the new feature, and Xv lists that feature's values for all people.

Example 13.2 (Aggregating observations with $u^\top X$). Now mix rows instead of columns. The row $x_i^\top$ contains all feature values for person i . If

$$u = \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

then left-multiplying by $u^\top$ gives

$$u^\top X = \frac{1}{2}x_1^\top + \frac{1}{2}x_2^\top,$$

the average values across all features for the first two people. This is still a vector in feature space $\mathbb{R}^p$: it has one coordinate for height, one for weight, one for age, and so on.

If instead

$$u = \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

then

$$u^\top X = \frac{1}{2}x_1^\top - \frac{1}{2}x_2^\top,$$

which is a contrast between the first two people across all features. Positive entries show features where person 1 is larger than person 2, and negative entries show features where person 2 is larger than person 1.

13.1.4 How to Interpret Principal Component Vectors

To interpret the principal component vectors, start from the singular value decomposition of the centered data matrix $X \in \mathbb{R}^{n \times p}$. If $\text{rank}(X) = r$, then

$$X = UDV^\top, \quad Z = XV = UD,$$

where

- $U \in \mathbb{R}^{n \times r}$ and $V \in \mathbb{R}^{p \times r}$ have orthonormal columns;
- $D = \text{diag}(d_1, \dots, d_r)$ contains the positive singular values in decreasing order.

The columns of V are the principal component directions in feature space. For component j ,

$$v_j \in \mathbb{R}^p, \quad Z_j = Xv_j \in \mathbb{R}^n.$$

This is exactly the operation Xv discussed above: PCA chooses a special feature-space direction v_j and uses Xv_j to combine the original feature columns into one new score column.

The vector v_j is the loading vector. It gives the weights used to combine the original feature columns:

$$Z_j = v_{1j}X_1 + \cdots + v_{pj}X_p.$$

The vector Z_j is the score vector. It records the value of this new component feature for every observation:

$$z_{ij} = x_i^\top v_j.$$

Equivalently, since $Z = UD$, we have $Z_j = d_j u_j$. The column u_j gives the normalized score direction in observation space, while the singular value d_j gives its scale.

Example 13.3 (Reading loading signs and magnitudes). Suppose one component uses two standardized features:

$$z_{i1} = 0.7x_{i1} + 0.6x_{i2}.$$

The two loadings have similar magnitude and the same sign. Both features therefore contribute strongly, and both push the score in the same direction: increasing either feature increases z_{i1} , holding the other feature fixed. Observations score high when both features are high, and low when both features are low.

Now compare this with

$$z_{i1} = 0.7x_{i1} - 0.6x_{i2}.$$

The magnitudes are still similar, so both features matter. But the signs are opposite: the first feature increases the score, while the second decreases it. Observations score high when x_{i1} is large relative to x_{i2} , and low when x_{i2} is large relative to x_{i1} . The component is a contrast between the two features.

A feature with a near-zero loading would contribute little to this component. The overall sign is arbitrary: multiplying both the loading vector and the score vector by -1 gives the same component. Therefore, signs should be read relative to each other within the same component.

Thus a principal component vector is not just a list of coefficients. It is a direction in feature space, and its entries tell us how to combine the original features to form a new score column. Entries with the same sign capture shared variation among features, while entries with opposite signs define a contrast between features.

Definition 13.1 (Explained variance). For centered data, the variance explained by component j is

$$\lambda_j = \text{Var}(Z_j) = \frac{d_j^2}{n-1}.$$

The explained variance ratio of component j is

$$\frac{\lambda_j}{\lambda_1 + \cdots + \lambda_r} = \frac{d_j^2}{d_1^2 + \cdots + d_r^2}.$$

The cumulative explained variance ratio for the first m components is

$$\frac{\sum_{j=1}^m d_j^2}{\sum_{j=1}^r d_j^2}.$$

13.2 PCA as Reconstruction

With loadings and scores in place, PCA can be read as a reconstruction method. Assume X is centered and has rank r . Its singular value decomposition is

$$X = UDV^\top = d_1 u_1 v_1^\top + \cdots + d_r u_r v_r^\top,$$

where $d_1 \geq \cdots \geq d_r > 0$. Keeping only the first m terms gives

$$\hat{X}_m = U_m D_m V_m^\top = Z_m V_m^\top, \quad Z_m = X V_m = U_m D_m.$$

This is the rank- m PCA reconstruction of X . It first builds m new derived features,

$$Z_m = (Xv_1, \dots, Xv_m),$$

then maps those m score columns back to the original feature space. In this sense, PCA compression replaces the original p features by m carefully chosen feature combinations.

The important optimality result is:

$$\hat{X}_m = \operatorname{argmin}_{A \in \mathbb{R}^{n \times p}, \operatorname{rank}(A) \leq m} \|X - A\|_F^2.$$

Here A is any candidate rank- m approximation to X , and $\|\cdot\|_F$ is the Frobenius norm:

$$\|X - A\|_F^2 = \sum_{i=1}^n \sum_{j=1}^p (x_{ij} - a_{ij})^2.$$

Then the reconstruction error is

$$\|X - \hat{X}_m\|_F^2 = \sum_{j=m+1}^r d_j^2.$$

This is the unsupervised meaning of PCA: among all m -dimensional linear representations, PCA gives the best squared-error reconstruction of the centered data matrix.

! PCA optimizes reconstruction, not every downstream task

PCA preserves directions with large variance because those directions minimize squared reconstruction error. This does not guarantee that the leading components preserve class separation, rare groups, causal structure, or predictive signal.

If we use the PCA scores as predictors in a linear regression, we obtain principal component regression (PCR). PCR adds a supervised prediction step, but the PCA directions themselves are still chosen without using y .

13.2.1 A Two-Dimensional Reconstruction Example

The dataset is generated in the following way. To make the later PCA workflow simpler, we center the data from the beginning.

```
import numpy as np

rng = np.random.default_rng(1)

Sigma = np.array([
    [0.5, -0.65],
    [-0.65, 1.0],
])

X_original = rng.multivariate_normal(mean=[1, 2], cov=Sigma, size=100)
X_centered = X_original - X_original.mean(axis=0) ①
```

① `StandardScaler(with_std=False)` also works.

We perform PCA and inverse-transform the one-dimensional scores to reconstruct the data in the original feature space.

```

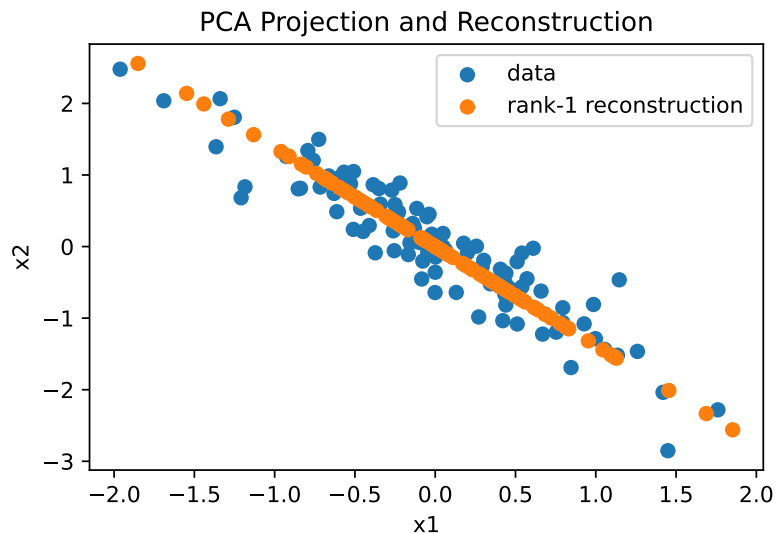
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

pca_1d = PCA(n_components=1)
score_1d = pca_1d.fit_transform(X_centered)
X_hat_1d = pca_1d.inverse_transform(score_1d)

plt.scatter(X_centered[:, 0], X_centered[:, 1], label="data")
plt.scatter(X_hat_1d[:, 0], X_hat_1d[:, 1], label="rank-1 reconstruction")

plt.xlabel("x1")
plt.ylabel("x2")
plt.title("PCA Projection and Reconstruction")
plt.legend();

```



The orange points are not new observations. They are the rank-1 reconstructions of the original observations obtained by projecting the data onto the first principal component and then mapping them back to the original space. Under the squared-error criterion, this gives the best rank-1 approximation.

13.2.2 Centering, Scaling, and What PCA Preserves

PCA depends on the geometry of the input matrix.

Centering is usually essential. Without centering, the first principal component may partly reflect the location of the data cloud relative to the origin rather than the variation around the mean. This is the same phenomenon illustrated in Example 5.2.

Scaling is a modeling choice. `sklearn.decomposition.PCA` centers features automatically, but it does not scale them to unit variance. Because the loading vector assigns weights to the input variables, variables with larger scales contribute more to the covariance matrix and therefore tend to have greater influence on the principal components. In practice, the following guidelines are commonly used:

- Scale the variables when their units are not comparable.
- Do not scale them when the original magnitude of variation is meaningful.
- State the choice explicitly, because it changes the covariance matrix and therefore the principal components.

Example 13.4. We first create a version of the same data in which `x2` is measured on a scale ten times larger than `x1`.

```
X_modified = X_original * np.array([1, 10])
```

Now compare PCA with and without standardization.

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import pandas as pd

X_scaled = StandardScaler().fit_transform(X_modified)

pca_centered = PCA(n_components=2).fit(X_modified)
pca_scaled = PCA(n_components=2).fit(X_scaled)

pd.DataFrame(
    {
        "centered only": pca_centered.explained_variance_ratio_,
        "centered and scaled": pca_scaled.explained_variance_ratio_,
    },
    index=["PC1", "PC2"],
)
```

	centered only	centered and scaled
PC1	0.999259	0.964569
PC2	0.000741	0.035431

The two analyses use the same observations, but the second feature has been measured on a scale ten times larger. Without scaling, PCA treats that larger numerical variation as geometrically important, and therefore PC1 dominates. After standardizing, the comparison is based on the correlation structure instead of the raw units. PC2 can then explain a nontrivial share of the standardized variance, because PCA is no longer dominated by the raw numerical scale of the second feature.

Preprocessing defines the geometry

PCA does not know which units are scientifically meaningful. Changing units, transforming variables, or standardizing features changes the geometry that PCA tries to preserve.

13.2.3 Choosing a Dimension Without y

In PCR, the number of components is chosen by prediction error. Without y , the choice must be tied to the representation goal.

Common criteria include:

- cumulative explained variance;
- reconstruction error;
- visual interpretability in two or three dimensions;
- denoising performance;
- stability of a downstream unsupervised method such as clustering;
- computational constraints.

The right criterion depends on the goal of the analysis. As one common example, we can choose m by setting a threshold for the cumulative explained variance ratio.

Example 13.5. We use the digits dataset from `sklearn` to demonstrate the example. This is about handwritten digits. The `dataset` is from `sklearn` that is partially copied from [UCI machine learning repository](#).

The dataset contains 1797 8x8 images. The data flattens the images into 1D arrays, and therefore the dataset is a (1797, 64) numpy array.

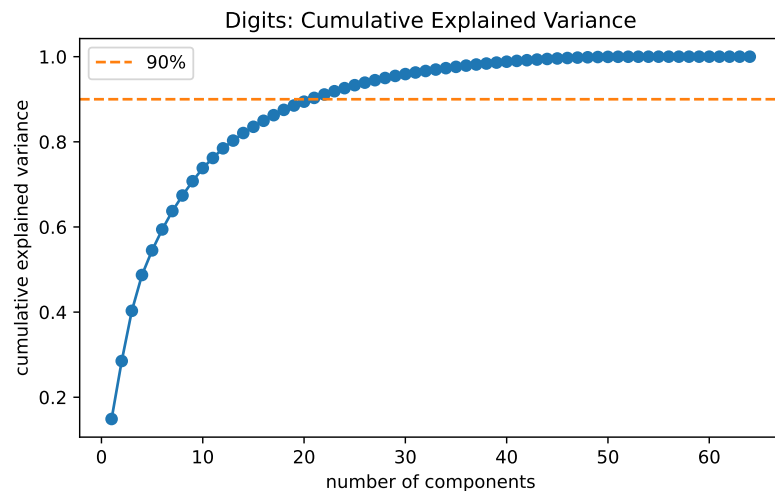
```
from sklearn.datasets import load_digits

X_digits = load_digits().data
```

We use `np.cumsum()` to find the total variance explained of the first several components. In this example, we decide to use 90% as the threshold to determine the number of components. This 90% threshold is only for illustration; there is no special reason that this example must use 90%.

```
digits_pca_full = PCA().fit(X_digits)
cum_ratio = np.cumsum(digits_pca_full.explained_variance_ratio_)

fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(np.arange(1, len(cum_ratio) + 1), cum_ratio, marker='o')
ax.axhline(0.90, color="C1", linestyle="--", label="90%")
ax.set_xlabel("number of components")
ax.set_ylabel("cumulative explained variance")
ax.set_title("Digits: Cumulative Explained Variance")
ax.legend();
```



If it is not easy to read from the plot, we can also use `np.argmax` to find the first term that `cum_ratio` exceeds `.90`.

```
np.argmax(cum_ratio >= 0.90) + 1
```

- ① `np.argmax(cum_ratio >= 0.90)` returns the first 0-based index where the cumulative ratio reaches 0.90. The `+1` converts that index to the number of components.

```
np.int64(21)
```

This number answers a reconstruction-style question: how many components are needed to explain 90% of the sample variance? It does not automatically answer how many components are best for clustering, classification, or scientific interpretation.

i Choosing an explained-variance threshold

In practice, the threshold should be justified for the specific goal of the analysis. There is no universal threshold that works for all PCA applications.

Common values such as 90%, 95%, or 99% are heuristics for reconstruction or compression. A higher threshold keeps more detail but gives less dimension reduction; a lower threshold gives stronger compression but may lose

important structure.

For visualization, we usually choose 2 or 3 components directly. For denoising, an elbow in the scree plot or visible reconstruction quality may be more useful than a fixed percentage. For clustering or other downstream unsupervised tasks, try several values of m and check whether the downstream result is stable.

13.3 More Applications of PCA

Different goals lead to different choices of m . We now look at three common uses of PCA: compression, denoising, and preprocessing.

13.3.1 Compression

Low-rank reconstruction can compress data by storing the scores Z_m and loadings V_m instead of all of X . The scores are the values of m derived features; the loadings say how those derived features relate back to the original p features.

There are two storage costs: shared quantities, such as the loading matrix and feature means, and per-observation quantities, namely the m scores. PCA compression is therefore most useful when many observations share the same fitted PCA basis.

For images, reconstruction makes the compression tradeoff visible.

We use the digits dataset introduced above to illustrate the idea.

```
from sklearn.datasets import load_digits

X_digits = load_digits().data
```

We fit PCA on the full digits dataset, and then visualize the reconstruction of the image at index 0.


```
x0 = X_digits[[0]]
```

Once the flattened array is reshaped into an 8x8 matrix, we may use `imshow()` to display the image. We then apply PCA and display the reconstructed images.

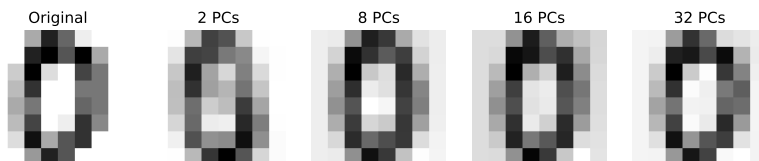
```
from sklearn.decomposition import PCA

components_to_show = [2, 8, 16, 32]

fig, axs = plt.subplots(1, len(components_to_show) + 1, figsize=(11, 2.3))

axs[0].imshow(x0.reshape(8, 8), cmap="gray_r")
axs[0].set_title("Original")
axs[0].axis("off")

for ax, m in zip(axs[1:], components_to_show):
    pca_m = PCA(n_components=m).fit(X_digits)
    x_hat = pca_m.inverse_transform(pca_m.transform(x0))
    ax.imshow(x_hat.reshape(8, 8), cmap="gray_r")
    ax.set_title(f"{m} PCs")
    ax.axis("off");
```



Small m produces a rough reconstruction because many components have been discarded. Larger m restores more detail, but it also requires storing more scores and loadings. Because this digit is relatively simple, the visual differences between some reconstructions may be subtle.

We can measure the per-image storage after the PCA loading matrix has been learned and shared across the dataset. Each original image stores 64 pixel values. With m components, each image is represented by m scores. Ignoring the shared loading matrix and feature means, the theoretical per-image storage ratios are:

	scores per image	original values per image	relative storage per image
2 PCs	2	64	0.03125
8 PCs	8	64	0.12500
16 PCs	16	64	0.25000
32 PCs	32	64	0.50000

We can also check the actual memory used by the NumPy arrays with `.nbytes`. Since `sklearn.PCA` centers the data, the compressed representation must store the scores, the loading matrix, and the feature means needed for reconstruction.

```
original_bytes = X_digits.nbytes

storage_rows = []
for m in components_to_show:
    pca_m = PCA(n_components=m).fit(X_digits)
    Z_m = pca_m.transform(X_digits)

    compressed_bytes = Z_m.nbytes + pca_m.components_.nbytes + pca_m.mean_.nbytes
    storage_rows.append(
        {
            "components": m,
            "original KB": original_bytes / 1024,
            "compressed KB": compressed_bytes / 1024,
            "compressed / original": compressed_bytes / original_bytes,
        }
    )

pd.DataFrame(storage_rows)
```

	components	original KB	compressed KB	compressed / original
0	2	898.5	29.578125	0.032919
1	8	898.5	116.812500	0.130008
2	16	898.5	233.125000	0.259460
3	32	898.5	465.750000	0.518364

These measured ratios are slightly larger because they include the shared quantities, not only the score values for each image.

For a single image alone, PCA is not necessarily a compression method. The benefit comes when many observations share the same loading matrix and each observation only needs its own low-dimensional score vector.

13.3.2 Denoising

PCA can also denoise data when noise is spread across many low-variance directions and signal is concentrated in the leading components. In that case, reconstructing with a moderate number of components can keep the main structure while discarding some noise.

We illustrate the idea by adding noise to the `x0` image from the digits dataset.

```
from sklearn.datasets import load_digits

X_digits = load_digits().data
x0 = X_digits[[0]]

rng = np.random.default_rng(3)
x_noisy = x0 + rng.normal(scale=4, size=x0.shape)
```

We learn the PCA basis from the full digits dataset, not from this single noisy image. The leading directions therefore represent common digit-shaped variation, while the added pixel noise is less aligned with those directions.

For denoising, m is often chosen by inspecting reconstruction quality or by looking for an elbow in the explained-variance curve, rather than by using a fixed threshold. Here we use 16 components as an illustrative middle ground: enough to preserve the main digit shape, but not so many that the reconstruction simply keeps all of the noisy variation.

```
from sklearn.decomposition import PCA

m_denoise = 16
pca_denoise = PCA(n_components=m_denoise).fit(X_digits)
x_denoised = pca_denoise.inverse_transform(pca_denoise.transform(x_noisy))
```

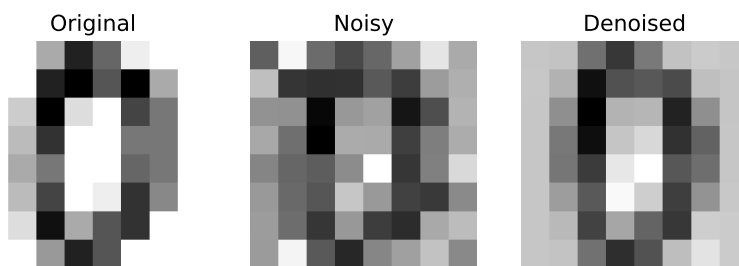
```
import matplotlib.pyplot as plt

fig, axs = plt.subplots(1, 3, figsize=(7.5, 2.5))

axs[0].imshow(x0.reshape(8, 8), cmap="gray_r")
axs[0].set_title("Original")
axs[0].axis("off")

axs[1].imshow(x_noisy.reshape(8, 8), cmap="gray_r")
axs[1].set_title("Noisy")
axs[1].axis("off")

axs[2].imshow(x_denoised.reshape(8, 8), cmap="gray_r")
axs[2].set_title("Denoised")
axs[2].axis("off");
```



The denoised image is still a low-rank reconstruction, but the goal is different from compression. Instead of using as few numbers as possible, we choose enough components to preserve the digit shape while smoothing out variation that does not align with the leading PCA directions.

13.3.3 PCA as Preprocessing

PCA is not only used inside PCR. It can also be used as a preprocessing step before clustering, visualization, anomaly detection, nearest-neighbor methods, or other downstream analyses. This step is optional: whether PCA helps depends on the geometry of the data and the goal of the analysis.

For clustering, a common workflow is:

1. preprocess the features;
2. fit PCA;
3. retain m component features;
4. cluster the rows of the score matrix.

This can remove noise, reduce computation, and stabilize distances. But the clustering algorithm now sees the derived features $Z_m = XV_m$, not the original columns of X . Since PCA preserves high-variance directions rather than cluster separation directly, it should be treated as a modeling choice and checked against the clustering goal.

The following synthetic example shows a case where PCA helps clustering. The true cluster structure is two-dimensional, but the observed data also contains many noisy features. K-means in the full noisy feature space is partly distracted by the noisy coordinates. After PCA reduces the data to two component features, K-means works better.

We first generate a dataset with only two signal features, then append 200 noise features to form the observed data matrix.

```
import numpy as np

rng = np.random.default_rng(12)
n_per_cluster = 50

centers = np.array([[-2, 0], [2, 0], [0, 2.5]])
y_cluster = np.repeat([0, 1, 2], n_per_cluster)

signal = np.vstack(
    [rng.normal(loc=center, scale=0.8, size=(n_per_cluster, 2)) for center in centers]
)
noise = rng.normal(scale=0.8, size=(len(y_cluster), 200))

X_cluster = np.column_stack([signal, noise])
```

We use the adjusted Rand index to compare the clustering results. The true labels are used only for evaluation in this constructed example. The PCA step and K-means clustering do not use the labels.

```

from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

scores_cluster = PCA(n_components=2).fit_transform(X_cluster)

labels_raw = KMeans(n_clusters=3, n_init=20, random_state=1).fit_predict(X_cluster)
labels_pca = KMeans(n_clusters=3, n_init=20, random_state=1).fit_predict(scores_cluster)

from sklearn.metrics import adjusted_rand_score
import pandas as pd

pd.DataFrame(
    {
        "adjusted Rand index": [
            adjusted_rand_score(y_cluster, labels_raw),
            adjusted_rand_score(y_cluster, labels_pca),
        ],
    },
    index=["K-means on original features", "PCA + K-means"],
)

```

	adjusted Rand index
K-means on original features	0.337112
PCA + K-means	0.739882

The clustering results are displayed below. The first panel shows the true labels for reference.

```

import matplotlib.pyplot as plt

fig, axs = plt.subplots(1, 3, figsize=(12, 3.8))

axs[0].scatter(signal[:, 0], signal[:, 1], c=y_cluster, s=18)
axs[0].set_xlabel("signal feature 1")
axs[0].set_ylabel("signal feature 2")
axs[0].set_title("True Labels")

axs[1].scatter(signal[:, 0], signal[:, 1], c=labels_raw, s=18)

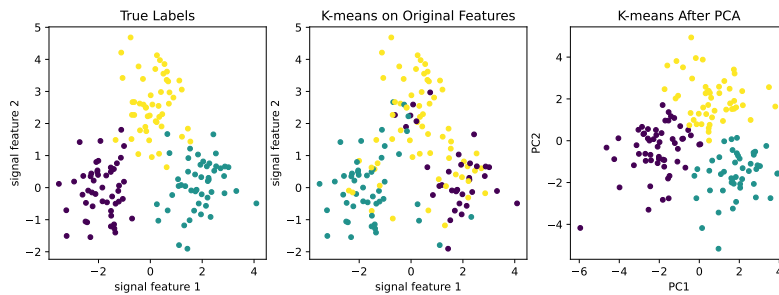
```

```

axs[1].set_xlabel("signal feature 1")
axs[1].set_ylabel("signal feature 2")
axs[1].set_title("K-means on Original Features")

axs[2].scatter(scores_cluster[:, 0], scores_cluster[:, 1], c=labels_pca, s=18)
axs[2].set_xlabel("PC1")
axs[2].set_ylabel("PC2")
axs[2].set_title("K-means After PCA");

```



This example works because the cluster signal is recoverable from the leading PCA directions, while many noisy coordinates are reduced to a smaller score representation.

i What the plot does not show

The plots show only the two signal features so that the cluster structure is visible. However, K-means on the original features is applied to the full matrix `X_cluster`, which also contains 200 noisy features. Those noisy coordinates affect the Euclidean distances used by K-means, even though they are not drawn in the figure. After PCA, the data are represented by two score columns, so much of the noise influence is reduced before clustering.

In practice, PCA as preprocessing should still be checked against the downstream goal. If the cluster separation lies in a low-variance direction, PCA may remove the relevant structure instead.

14 Embedding and Visualization

The previous chapter treated PCA as a low-rank representation of a data matrix. PCA constructs new coordinates by multiplying the data matrix by loading vectors, $Z = XV_m$.

In this chapter, an **embedding** means a lower-dimensional coordinate representation of the observations:

$$x_i \in \mathbb{R}^p \longrightarrow z_i \in \mathbb{R}^m, \quad m \ll p.$$

i Three uses of the word embedding

- In this chapter, an embedding is a lower-dimensional representation of high-dimensional observations.
- In mathematics, an embedding usually means an injective map that preserves the relevant structure.
- In language models, an embedding is a vector representation of words, sentences, or documents, usually designed so that similar meanings have nearby vectors.

These uses are related in spirit, but they are not the same mathematical object. In this chapter, injectivity or surjectivity is not the main issue. What matters is which relationships among observations the embedding tries to preserve.

PCA is a linear embedding: each coordinate is a feature combination, and the axes can be interpreted through loadings. The methods in this chapter usually build coordinates from distances, neighborhoods, or graphs instead. Their axes are

useful for visualization, but they usually do not have direct feature interpretations.

We focus on three methods:

- spectral embedding, which builds coordinates from a similarity graph;
- t-SNE, which matches high-dimensional and low-dimensional neighborhood probabilities;
- UMAP, which builds and matches nearest-neighbor graphs.

The shared goal is not to reconstruct X feature by feature. The goal is to make selected relationships among observations visible in a small number of coordinates, usually two.

Embeddings are task-dependent

In this chapter, embeddings are mainly used for visualization and sometimes for clustering. They can also be useful for other tasks, but not every embedding is useful for every task.

The reason is that different methods preserve different relationships. PCA preserves variance and optimizes linear reconstruction. Spectral embedding, t-SNE, and UMAP preserve neighborhoods or graph structure in different ways. For example, t-SNE is useful for visualizing local neighborhoods, but distances in a t-SNE plot are not reliable enough to automatically use as input to a distance-based method such as KNN. A good embedding plot is therefore a guide for exploration, not a guarantee that the coordinates are appropriate for clustering, prediction, compression, or denoising. We will discuss these later in more details.

Because an embedding is built from the rows of X , two choices come before the algorithm itself:

- how each observation is represented as a row vector;
- how similarity or distance between rows is defined.

Thus a common workflow is

$X \rightarrow \text{row representation} \rightarrow \text{similarity or neighborhood structure} \rightarrow \text{low-dimensional embedding}.$

14.1 Spectral Embedding

Spectral embedding starts by replacing the original data matrix with a graph. The rows $x_i \in \mathbb{R}^p$ are observations in feature space. Spectral embedding keeps the same observations, but represents their relationships by a graph: the vertices are observations, and edges connect observations that are similar, often nearest neighbors.

The main idea is:

$X \rightarrow \text{similarity graph} \rightarrow \text{graph Laplacian} \rightarrow \text{observation-space eigenvector coordinates}.$

This is useful when the structure is better described by connectivity than by distance to a single center. Spectral clustering uses the same embedding and then applies a label-assignment method such as K-means.

14.1.1 Mathematical Idea

Let $W = [W_{ij}] \in \mathbb{R}^{n \times n}$ be a symmetric weighted adjacency matrix. A large W_{ij} means observations i and j are strongly connected. Define the degree matrix

$$D = \text{diag}(d_1, \dots, d_n), \quad d_i = \sum_j W_{ij}.$$

The unnormalized graph Laplacian is

$$L = D - W.$$

Both W and L are $n \times n$ matrices, so they act on vectors in observation space $\mathbb{R}^n$. A vector

$$y = [y_1, \dots, y_n]^\top \in \mathbb{R}^n$$

assigns one coordinate value to every observation. The entry y_i is the coordinate assigned to observation i .

i What does L do?

The matrix L compares each observation with its neighbors in the graph. If $y \in \mathbb{R}^n$ assigns one number to each observation, then the i -th entry of Ly is

$$(Ly)_i = d_i y_i - \sum_j W_{ij} y_j = \sum_j W_{ij} (y_i - y_j).$$

Thus $(Ly)_i$ is a weighted difference between the coordinate of observation i and the coordinates of its neighbors.

- If y_i is close to the coordinates of strongly connected neighbors, then $(Ly)_i$ is small.
- If y_i is very different from strongly connected neighbors, then $(Ly)_i$ is large.

In this sense, L measures how rough or unsmooth the coordinate vector y is over the graph. Smooth coordinates change slowly across strong edges.

The **graph roughness** of an observation-space coordinate vector y is

$$\text{roughness}_G(y) = y^\top Ly = \frac{1}{2} \sum_{i,j} W_{ij} (y_i - y_j)^2.$$

It measures how much the coordinate values change across graph edges, with larger penalties on stronger edges. This expression is small when strongly connected observations receive similar coordinates. This is exactly the behavior we want from a graph embedding. If W_{ij} is large, then observations i and j are strongly connected, so we want their embedded coordinates to be close:

$$y_i \approx y_j.$$

The term $W_{ij}(y_i - y_j)^2$ penalizes the opposite situation: a strong edge with very different coordinates. Therefore minimizing $y^\top Ly$ means finding an observation-space coordinate column that keeps strongly connected observations close together. In other words, we need the eigenvectors corresponding to the smallest nonzero eigenvalues of L .

Click to expand.

More explicitly, a one-dimensional observation-space coordinate can be described as trying to solve

$$\min_y y^\top Ly \quad \text{subject to } \|y\|^2 = 1 \text{ and } y \perp \mathbf{1}.$$

The constraints remove trivial solutions. Without them, $y = 0$ minimizes the objective. Even with a nonzero vector, the constant coordinate $y_1 = \dots = y_n$ makes every coordinate difference zero, so it does not separate observations. The constraint $\|y\|^2 = 1$ fixes the scale, and $y \perp \mathbf{1}$ removes the constant coordinate.

This optimization problem leads to eigenvectors because L is symmetric. Write its orthonormal eigen-decomposition as

$$L = V\Lambda V^\top.$$

If $v_1, \dots, v_n$ are the eigenvectors of L , then any unit vector y can be expanded in this eigenbasis:

$$y = a_1 v_1 + \dots + a_n v_n, \quad \sum_{k=1}^n a_k^2 = 1.$$

Then

$$y^\top Ly = \sum_{k=1}^n \lambda_k a_k^2.$$

Thus $y^\top Ly$ is a weighted average of the eigenvalues. To make it small, we put weight on eigenvectors with small eigenvalues. For a graph Laplacian, the smallest eigenvalue is usually 0, with a constant eigenvector on each connected component. That eigenvector has zero graph roughness, but it gives every observation the same coordinate. After removing this constant direction, the first useful one-dimensional spectral coordinate is the eigenvector corresponding to the smallest positive eigenvalue:

$$Lv = \lambda v, \quad \lambda = \min\{\lambda_k : \lambda_k > 0\}.$$

This eigenvector $v = [v_1, \dots, v_n]^\top$ assigns one coordinate value to each observation: the entry v_i becomes the new coordinate of observation i . Since

$$v^\top Lv = \lambda v^\top v,$$

the smallest positive eigenvalue gives the nonconstant coordinate with the smallest graph roughness. Therefore strongly connected observations tend to receive similar values. Additional spectral coordinates come from the next smallest positive eigenvalues.

Consider the eigenvectors of the graph Laplacian $v_1, \dots, v_n$, ordered by ascending eigenvalues. Each selected eigenvector gives one coordinate column for all observations. For an m -dimensional spectral embedding, we place the first m nonconstant eigenvectors side by side:

$$Z = \begin{bmatrix} v_2 & v_3 & \cdots & v_{m+1} \end{bmatrix} \in \mathbb{R}^{n \times m}.$$

The embedding is then obtained by reading across rows of this matrix. The i -th row of Z gives the embedded coordinate of observation i :

$$z_i^\top = [v_2(i) \quad v_3(i) \quad \cdots \quad v_{m+1}(i)],$$

where $v_k(i)$ is the i -th entry of the k -th Laplacian eigenvector. In words, observation i gets its first coordinate from the i -th entry of v_2 , its second coordinate from the i -th entry of v_3 , and so on.

Example 14.1 (A four-point graph). Suppose the original dataset has four observations and any number of predictors:

$$X \in \mathbb{R}^{4 \times p}.$$

The value of p does not matter for this example. The spectral embedding starts after those observations have been converted into a similarity matrix. Suppose observations 1 and 2 are strongly connected, observations 3 and 4 are strongly connected, and observations 2 and 3 are weakly connected:

$$W = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0.2 & 0 \\ 0 & 0.2 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

This matrix says that the graph has two strong pairs, $(1, 2)$ and $(3, 4)$, with a weak bridge between observations 2 and 3. From this matrix we form the graph Laplacian $L = D - W$ and compute the eigenvectors. A small nonconstant Laplacian eigenvector is approximately

$$v_2 \approx \begin{bmatrix} 0.547 \\ 0.448 \\ -0.448 \\ -0.547 \end{bmatrix}.$$

Using v_2 as a one-dimensional embedding means the output has dimension

$$Z \in \mathbb{R}^{4 \times 1}.$$

The embedding is obtained by reading the entries of v_2 :

$$Z = v_2 \approx \begin{bmatrix} 0.547 \\ 0.448 \\ -0.448 \\ -0.547 \end{bmatrix}.$$

Thus the spectral embedding maps the original observations as follows:

$$x_1 \mapsto 0.547, \quad x_2 \mapsto 0.448, \quad x_3 \mapsto -0.448, \quad x_4 \mapsto -0.547.$$

The result tells us that observations 1 and 2 are close in the embedding, observations 3 and 4 are close in the embedding, and there is a large gap between the two pairs. This reflects the graph structure encoded in W : two strong within-pair connections and one weak bridge. The coordinate is not a feature combination like Xv in PCA, where $v \in \mathbb{R}^p$ is a loading vector. It is a coordinate read from a Laplacian eigenvector, so its values are determined by graph connectivity.

14.1.2 Spectral Embedding in Python

We can use `SpectralEmbedding` from `sklearn.manifold` to compute a spectral embedding. The first hyperparameter, `n_components`, specifies the dimension of the embedding. However, it is usually not the most important hyperparameter.

As discussed earlier, spectral embedding starts by constructing a local connectivity graph. There are several ways to construct the connectivity matrix W . One of the most common approaches is the nearest neighbors method:

1. Choose the number of nearest neighbors, `n_neighbors`.
2. For each point, find its `n_neighbors` nearest neighbors.
3. Set $W_{ij} = 1$ if points i and j are connected, and $W_{ij} = 0$ otherwise.

This produces a sparse connectivity matrix W , which is then used to compute the graph Laplacian and the spectral embedding.

In `SpectralEmbedding`, this construction is specified by `affinity="nearest_neighbors"` together with the parameter `n_neighbors`. In practice, choosing an appropriate neighborhood size or a different way of constructing the connectivity graph is often more important than choosing the embedding dimension.

The formulas above use the unnormalized graph Laplacian $L = D - W$ for exposition. The `sklearn` implementation uses a normalized graph Laplacian. The interpretation is the same at this level: the embedding favors coordinates that vary smoothly over the graph, but the exact matrix used in the computation is different.

SpectralClustering

A common use of spectral embedding is to apply a clustering algorithm, such as `KMeans`, to the embedded coordinates. You can do this manually by combining `SpectralEmbedding` with `KMeans`, or you can use `SpectralClustering` from `sklearn.cluster`, which performs these two steps automatically.

If you want to visualize the embedding itself, however, you should use `SpectralEmbedding` directly and then apply the clustering algorithm manually.

14.1.2.1 Example

We generate a dataset consisting of two concentric circles and add a small number of bridge points connecting them. The outer circle has uneven density, meaning that the point density varies along its perimeter.

Although the bridge points connect the two circles geometrically, we still regard the data as containing two clusters: the inner circle and the outer circle. The bridge points are treated as

noise rather than evidence that the two circles belong to the same cluster.

The dataset is generated manually using NumPy because `sklearn.datasets.make_circles` cannot generate circles with non-uniform point density. The dataset is visualized after generating.

```
import numpy as np
import matplotlib.pyplot as plt

n_samples = 900
rng = np.random.default_rng(0)

n_inner = n_samples // 2
n_outer = n_samples - n_inner

# Inner circle
theta = rng.uniform(0, 2 * np.pi, n_inner)
r = rng.normal(1.0, 0.035, n_inner)
x_in = r * np.cos(theta)
y_in = r * np.sin(theta)

# Outer circle with uneven density
theta_out = []
while len(theta_out) < n_outer:
    theta = rng.uniform(0, 2 * np.pi, 6 * n_outer)
    prob = 0.08 + 0.92 * np.sin(2.5 * theta) ** 2
    keep = rng.random(len(theta)) < prob
    theta_out.extend(theta[keep])

theta_out = np.array(theta_out[:n_outer])

r = rng.normal(2.6, 0.04, n_outer)
x_out = r * np.cos(theta_out)
y_out = r * np.sin(theta_out)

X = np.vstack([
    np.column_stack([x_in, y_in]),
    np.column_stack([x_out, y_out]),
])
```

```

y = np.array([0] * n_inner + [1] * n_outer)

# Bridge points
n_bridge = 10
theta = rng.normal(0.4, 0.02, n_bridge)
r = np.linspace(1.05, 2.55, n_bridge)
r += rng.normal(0, 0.02, n_bridge)

x = r * np.cos(theta)
y_bridge = r * np.sin(theta)

X_bridge = np.column_stack([x, y_bridge])

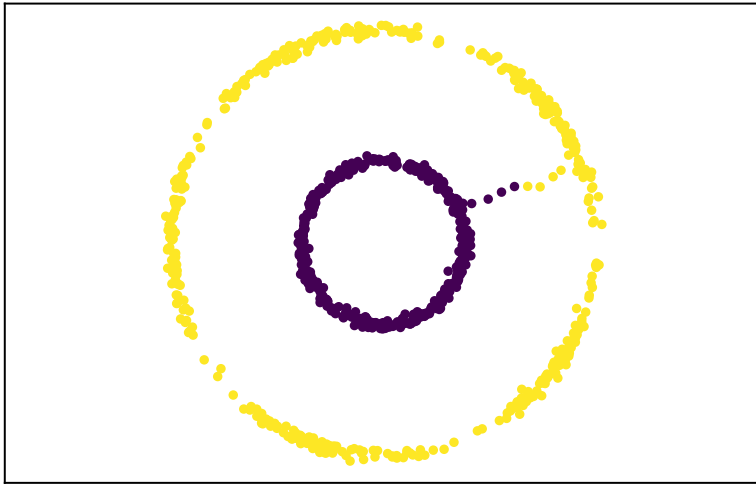
# Assign each bridge point to the nearest circle
labels = (np.abs(r - 1.0) > np.abs(r - 2.6)).astype(int)

X = np.vstack([X, X_bridge])
y = np.concatenate([y, labels])

plt.scatter(X[:, 0], X[:, 1], c=y, s=8)
plt.title("Ground Truth")
plt.xticks([])
plt.yticks([])
plt.axis("equal");

```

Ground Truth



Now we apply the four clustering methods and compare their results. Since the true labels are available, we use the adjusted Rand index (ARI) to evaluate the clustering performance.

For all methods, the hyperparameters were only tuned lightly rather than exhaustively. Better performance may be achievable with more careful tuning, but the overall conclusions are unlikely to change. The purpose of this example is not to compare the best possible performance of each algorithm, but to illustrate their typical behavior on this dataset.

```
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.manifold import SpectralEmbedding

kmeans_labels = KMeans(n_clusters=2, n_init=10, random_state=0).fit_predict(X)

ha_labels = AgglomerativeClustering(n_clusters=2, linkage="ward").fit_predict(X)

dbscan_labels = DBSCAN(eps=0.5, min_samples=10).fit_predict(X)

Z = SpectralEmbedding(n_components=2, n_neighbors=15, random_state=0).fit_transform(X)
spectral_labels = KMeans(n_clusters=2, random_state=0, n_init=10).fit_predict(Z)

from sklearn.metrics import adjusted_rand_score
```

```

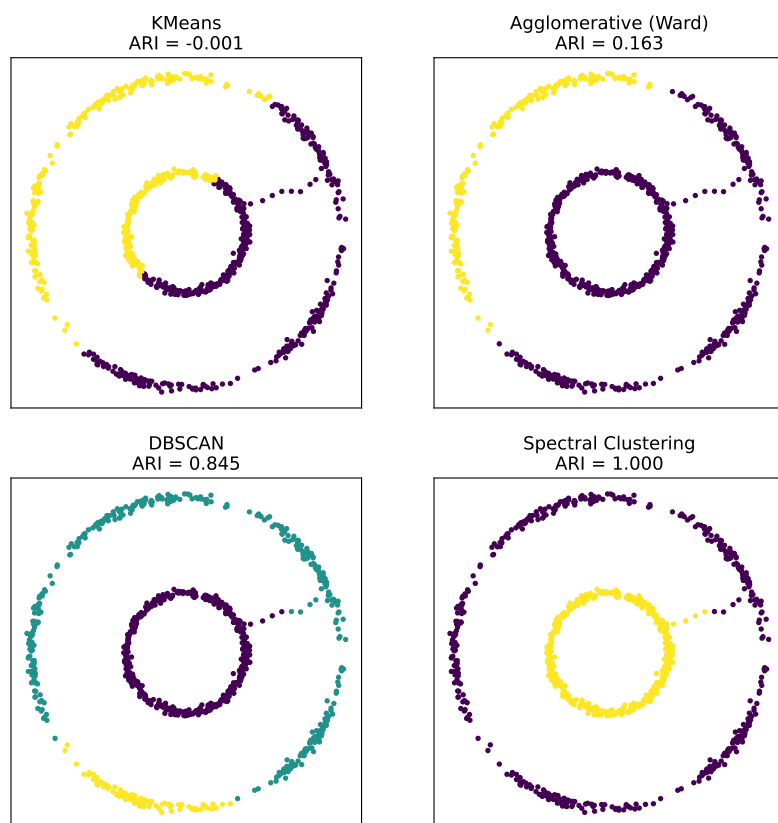
labels = {
    "KMeans": kmeans_labels,
    "Agglomerative (Ward)": ha_labels,
    "DBSCAN": dbscan_labels,
    "Spectral Clustering": spectral_labels,
}
results = {
    name: (label, adjusted_rand_score(y, label)) for name, label in labels.items()
}

fig, axes = plt.subplots(2, 2, figsize=(10, 10))
axes = axes.ravel()

for ax, (name, (labels, ari)) in zip(axes, results.items()):
    ax.scatter(X[:, 0], X[:, 1], c=labels, s=8)
    ax.set_title(f"{name}\nARI = {ari:.3f}", fontsize=13)

    ax.set_xticks([])
    ax.set_yticks([])
    ax.set_aspect("equal");

```



The results illustrate the different assumptions made by the clustering algorithms.

- **KMeans:** Fails. KMeans partitions the data according to the distance to cluster centroids. Since the two circles share the same center, the algorithm cannot separate them into inner and outer circles. Instead, it splits the data into two roughly equal halves.
- **Agglomerative (Ward):** Performs slightly better but still fails. Ward linkage merges clusters to minimize the increase in within-cluster variance. This favors compact, approximately spherical clusters rather than ring-shaped clusters, so it also tends to cut across the circles instead of preserving their circular structure.
- **DBSCAN:** Performs reasonably well but is affected by the bridge points. Since DBSCAN defines clusters through density connectivity, the bridge creates a path between

the two circles, causing some points to be merged incorrectly. Furthermore, because DBSCAN uses a single density threshold, it struggles with the uneven density of the outer circle and fails to identify it as one cluster.

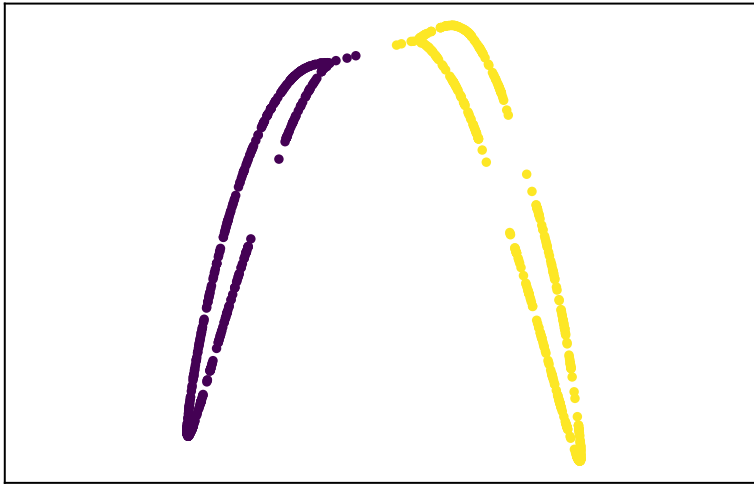
- Spectral Clustering: Correctly recovers the two circles. Instead of clustering the original coordinates directly, it first constructs a similarity graph and computes a spectral embedding. In this new representation, points belonging to the same circle become much closer, while the weak bridge has little influence. A simple clustering algorithm such as KMeans can then easily separate the two clusters.

The key idea is that spectral clustering does not rely on Euclidean geometry in the original feature space. Instead, it transforms the data into a new space where the cluster structure becomes much easier to identify.

The figure below shows the two-dimensional spectral embedding. Although the circles are not linearly separable in the original space, they become well separated after the embedding, making KMeans sufficient to recover the correct clusters.

```
plt.scatter(Z[:, 0], Z[:, 1], c=y, s=8)
plt.title("Spectral Embedding")
plt.xticks([])
plt.yticks([])
plt.axis("equal");
```

Spectral Embedding



14.2 t-SNE

t-SNE stands for t-distributed Stochastic Neighbor Embedding. It is designed mainly for visualization. It does not try to reconstruct the original features and does not produce interpretable axes. Instead, it tries to preserve local neighborhoods.

The main idea is:

$X \rightarrow$ high-dimensional neighbor probabilities $P \rightarrow$ low-dimensional coordinates with probabilities Q .

14.2.1 Mathematical Idea

The core algorithm is simple:

1. Compute pairwise similarities among the original observations $x_1, \dots, x_n$. These similarities are converted into a probability matrix $P = (p_{ij})$, where larger p_{ij} means that observations i and j are close neighbors in the original space.

2. Introduce low-dimensional coordinates $z_1, \dots, z_n$. These coordinates are the unknown variables that t-SNE must solve for.
3. From the distances among the current z_i 's, compute another probability matrix $Q = (q_{ij})$. Larger q_{ij} means that points i and j are close in the low-dimensional display. Note that Q is a function of $Z = (z_1, \dots, z_n)$.
4. Adjust all coordinates $z_1, \dots, z_n$ so that Q is close to P in the sense of Kullback-Leibler divergence.

Thus t-SNE chooses the plotted coordinates by solving

$$\min_{z_1, \dots, z_n} \text{KL}(P \| Q(z_1, \dots, z_n)),$$

where

$$\text{KL}(P \| Q) = \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right).$$

In practice, the coordinates are initialized, often randomly or from PCA, and then updated by gradient descent. The next formulas define the two probability matrices P and Q in detail.

i How t-SNE builds P

For each observation i , t-SNE defines the probability that i chooses observation j as a neighbor:

$$p_{j|i} = \frac{\exp \left(-\|x_i - x_j\|^2 / (2\sigma_i^2) \right)}{\sum_{k \neq i} \exp \left(-\|x_i - x_k\|^2 / (2\sigma_i^2) \right)}, \quad j \neq i.$$

The conditional probabilities are then symmetrized to produce p_{ij} , where n is the number of observations:

$$p_{ij} = \frac{1}{2n} (p_{i|j} + p_{j|i}).$$

Geometrically, p_{ij} summarizes the relationship between x_i and x_j in the original high-dimensional space. If x_i and x_j are close relative to the local scale around them, then p_{ij} is large.

i How t-SNE builds Q

For low-dimensional coordinates z_i , t-SNE defines similarities using a heavy-tailed Student- t kernel:

$$q_{ij} = \frac{(1 + \|z_i - z_j\|^2)^{-1}}{\sum_{k \neq \ell} (1 + \|z_k - z_\ell\|^2)^{-1}}.$$

Geometrically, q_{ij} summarizes the relationship between the plotted points z_i and z_j . If the two plotted points are close, then q_{ij} is large; if they are far apart, then q_{ij} is small.

i Perplexity

perplexity can be read as an effective neighborhood size. It controls how many neighbors each point effectively pays attention to.

- Larger **perplexity** means a broader neighborhood. The probability distribution $p_{\cdot|i}$ is flatter and spreads mass over more nearby observations. This corresponds to a larger σ_i .
- Smaller **perplexity** means a more local neighborhood. The probability distribution $p_{\cdot|i}$ concentrates more strongly on the closest observations. This corresponds to a smaller σ_i .

More precisely, for each point i , t-SNE chooses σ_i so that the entropy of the conditional distribution $p_{\cdot|i}$ matches the requested perplexity:

$$\text{Perp}(p_{\cdot|i}) = 2^{H(p_{\cdot|i})}, \quad H(p_{\cdot|i}) = - \sum_{j \neq i} p_{j|i} \log_2 p_{j|i}.$$

Thus the user fixes one common **perplexity** value, while t-SNE solves a separate σ_i for each observation. The σ_i values can differ, allowing dense and sparse regions to use different local distance scales while keeping the same effective neighborhood size.

14.2.2 What structure does t-SNE preserve?

t-SNE converts geometry into probabilities.

- The matrix $P = (p_{ij})$ describes neighborhood relationships in the original high-dimensional space: nearby observations have larger p_{ij} , and faraway observations have smaller p_{ij} .
- The matrix $Q = (q_{ij})$ describes neighborhood relationships in the low-dimensional plot in the same way.

Minimizing $\text{KL}(P\|Q)$ tries to make these two probability patterns match.

The direction of the KL divergence matters.

- The term $p_{ij} \log(p_{ij}/q_{ij})$ strongly penalizes the case where p_{ij} is large but q_{ij} is small. Thus, if two observations are close neighbors in the original space, t-SNE tries hard to keep them close in the plot.
- The converse is weaker: if two points appear close in the t-SNE plot, they were not necessarily close in the original space. Pairs with very small p_{ij} contribute little directly to the objective, so t-SNE is less strict about the exact arrangement of points that were already far apart.

Thus t-SNE preserves local neighborhood probabilities, not the original coordinates, not the original Euclidean distances, and not the meanings of the original features. In plain language:

- Original-space neighbors are likely to remain close in the t-SNE plot.
- Points that are close in the t-SNE plot are not guaranteed to have been close in the original space.
- A visible t-SNE patch may contain several real subgroups.
- Distances between separated patches, patch sizes, empty space, orientation, and apparent density should not be used as quantitative evidence.

So t-SNE answers a local question: “which observations are neighbors of which other observations?” It does not faithfully answer global metric questions such as “how far apart are these two groups?” or “which visual cluster is larger or denser?”

i t-SNE as a diagnostic plot

The main use of t-SNE is visualization after dimension reduction. It is not usually used as a feature representation for a downstream model. Instead, it helps us inspect whether patterns found by other methods are visible in local neighborhood structure.

For example, after fitting a clustering algorithm in the original feature space, we can color a t-SNE plot by the cluster labels. If the colors form well-separated local patches, then the clustering is consistent with the neighborhood structure shown by t-SNE. If the colors are heavily mixed, then the clustering may be failing to capture local structure, or the data may not contain clearly separated groups.

This is a diagnostic check, not a proof. t-SNE can make separation look stronger than it really is, and the apparent distance between colored patches is not a quantitative cluster separation. Quantitative measures such as nearest-neighbor overlap, trustworthiness, silhouette score, or adjusted Rand index should be used when a formal comparison is needed. The plot is useful because it makes possible problems visible quickly.

14.2.3 t-SNE in Python

We may use `TSNE` from `sklearn.manifold` to compute a t-SNE embedding. Unlike PCA or many other `sklearn` transformers, standard `sklearn` t-SNE does not provide a `.transform()` method for new observations. The usual workflow is to fit the embedding and return the plotted coordinates directly with `.fit_transform()`.

```
from sklearn.manifold import TSNE

tsne = TSNE(
    n_components=2,
    perplexity=30,
    init="pca",
    learning_rate="auto",
```

```
    random_state=1
)
Z = tsne.fit_transform(X)
```

The parameter `n_components=2` asks for a two-dimensional display. `perplexity=30` sets the effective neighborhood size used when constructing the high-dimensional probabilities. The argument `init="pca"` specifies how the optimization is initialized. Since t-SNE minimizes its objective by an iterative optimization method, the low-dimensional coordinates must be assigned initial values before the optimization begins. Two common choices are random initialization (`init="random"`) and PCA initialization (`init="pca"`). We set `init="pca"` explicitly here.

With PCA initialization, the optimization starts from the coordinates obtained by projecting the data onto the first `n_components` principal components. In other words, the PCA embedding serves as the initial low-dimensional representation, and t-SNE then refines these coordinates to minimize its own objective. PCA initialization often converges faster and gives more stable results than random initialization. It can also make the early layout reflect some broad linear structure. However, it does not guarantee a better final embedding.

The initialization is not the same as the objective. t-SNE still optimizes the t-SNE loss after initialization. However, because the optimization is nonconvex, different initializations can lead to different local solutions. This is one reason to check whether visible patterns are stable across reasonable settings.

i PCA initialization vs. PCA preprocessing

Do not confuse `init="pca"` with applying PCA before t-SNE.

- `init="pca"` uses a PCA embedding only as the initial values for gradient descent. The neighborhood probabilities are still computed from the data passed to TSNE.
- Applying PCA before t-SNE replaces the original data with a lower-dimensional representation (for

example, 50 principal components). In this case, t-SNE computes neighborhood probabilities from the PCA-transformed data.

The latter is often recommended for high-dimensional datasets because it can reduce noise and speed up computation. The idea is to use PCA as a preliminary compression step, often to 30 or 50 dimensions, and then let t-SNE perform the main nonlinear embedding from that cleaner representation.

14.2.4 Digits example

We use the handwritten digits data from `sklearn` as an example. The data have 64 pixel features. We standardize the features, reduce them to 30 PCA coordinates, and then run t-SNE. The labels are used only for coloring the final plot.

```
from sklearn.datasets import load_digits
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE

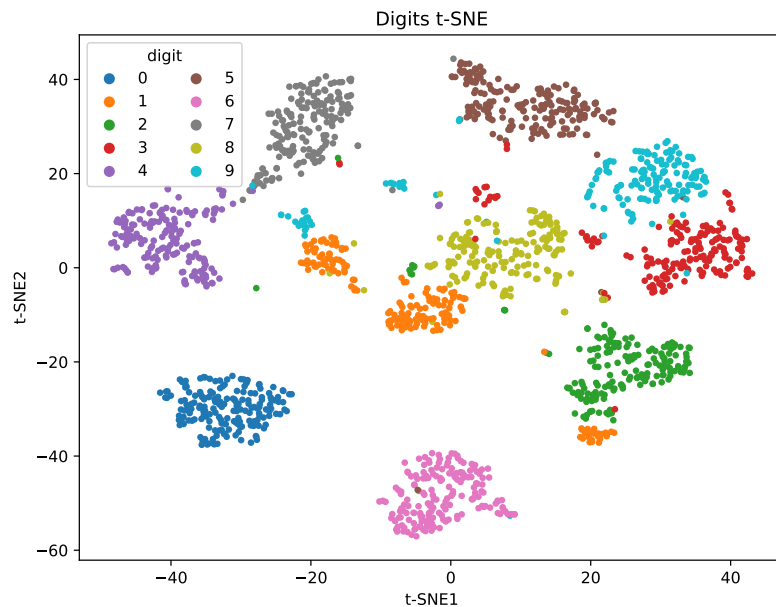
digits = load_digits()
X_digits = StandardScaler().fit_transform(digits.data)
y_digits = digits.target

X_digits_pca = PCA(n_components=30, random_state=1).fit_transform(X_digits)

tsne = TSNE(
    n_components=2,
    perplexity=30,
    init="pca",
    learning_rate="auto",
    random_state=1,
)
Z_tsne = tsne.fit_transform(X_digits_pca)
```

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(8, 6))
scatter = ax.scatter(Z_tsne[:, 0], Z_tsne[:, 1], c=y_digits, cmap="tab10", s=10)
ax.set_xlabel("t-SNE1")
ax.set_ylabel("t-SNE2")
ax.set_title("Digits t-SNE")
ax.legend(*scatter.legend_elements(), title="digit", ncol=2)
```



In this example, we use `perplexity=30`, a common starting point for a moderate-size dataset. This value should be treated as a tuning choice rather than a fixed default.

In practice, choose `perplexity` by comparing several reasonable values. For a moderate-size dataset, values such as 5, 10, 30, and 50 are common candidates. Use smaller values when you want to emphasize very local neighborhoods, and larger values when you want a smoother display based on broader neighborhoods. The chosen `perplexity` should be well below the sample size. The result can also depend on initialization and random seed, so a pattern is more credible when it persists across several reasonable `perplexity` values and random seeds.

14.2.5 More about perplexity

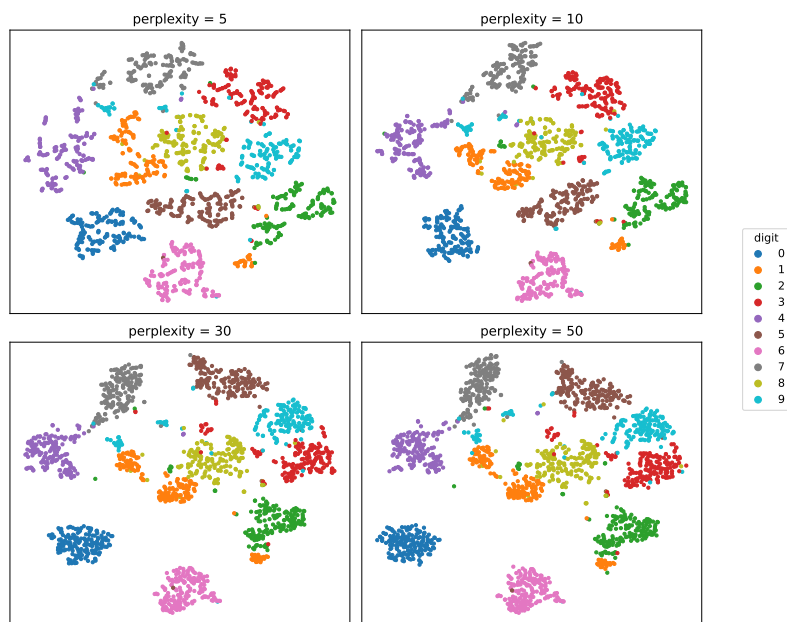
The plots below use the same preprocessed digits data and the same random seed, but change **perplexity**.

```
perplexities = [5, 10, 30, 50]

fig, axs = plt.subplots(2, 2, figsize=(10, 8))
scatter = None

for ax, perplexity in zip(axs.ravel(), perplexities):
    model = TSNE(
        n_components=2,
        perplexity=perplexity,
        init="pca",
        learning_rate="auto",
        random_state=1,
    )
    Z = model.fit_transform(X_digits_pca)
    scatter = ax.scatter(Z[:, 0], Z[:, 1], c=y_digits, cmap="tab10", s=8)
    ax.set_title(f"perplexity = {perplexity}")
    ax.set_xticks([])
    ax.set_yticks([])

fig.legend(
    *scatter.legend_elements(),
    title="digit",
    loc="center right",
)
fig.tight_layout(rect=[0, 0, 0.9, 1]);
```



The figure illustrates how **perplexity** changes the neighborhood scale used by t-SNE. Smaller values emphasize fine local structure, often producing many tight or fragmented clusters. Larger values consider broader neighborhoods, leading to smoother and more stable embeddings.

Some observations from the plots are:

- All **perplexity** values separate most digit classes, indicating that the handwritten digits have meaningful local neighborhood structure in the original high-dimensional space.
- With small **perplexity** (5–10), the embedding focuses on very local neighborhoods. Clusters are more fragmented, and some digit classes appear as multiple disconnected regions. This suggests that some digit classes contain multiple local neighborhoods in the original feature space, although these disconnected regions should not automatically be interpreted as genuine subtypes.
- In this example, increasing **perplexity** (30–50) produces more compact and stable clusters. Samples from the same digit are grouped more consistently, resulting in a cleaner overall layout.

- The embeddings for **perplexity** 30 and 50 are visually similar, suggesting that the recovered local neighborhood structure is relatively stable over this range.
- Some overlap between visually similar digits persists under all settings, possibly reflecting genuine ambiguity in the handwritten digits rather than an inappropriate choice of **perplexity**.
- Although local neighborhoods are largely preserved, the relative positions, sizes, and distances between clusters change noticeably across **perplexity** values. Therefore, these global geometric relationships should not be interpreted quantitatively.

Overall, **perplexity** controls the trade-off between emphasizing fine local neighborhoods and producing a smoother representation of the underlying neighborhood structure, rather than determining the number of clusters.

A useful **perplexity** is one that produces stable local neighborhoods across a reasonable range of parameter values rather than one that gives the most visually appealing plot. Comparing several **perplexity** values is therefore often more informative than selecting a single “best” visualization. When class labels are available, they should be used only after fitting to verify whether nearby points correspond to similar observations. The embedding can then guide further exploration of the original data. For example, overlapping digit classes can be examined in the original images to determine whether the ambiguity reflects genuine variation in handwriting rather than an artifact of the visualization.

14.3 UMAP

UMAP stands for Uniform Manifold Approximation and Projection. Like t-SNE, it is a nonlinear embedding method based on local neighborhoods and is often used for two-dimensional visualization. The main difference is that UMAP builds a weighted nearest-neighbor graph in the original space and then tries to reproduce that graph in a low-dimensional display.

Unlike standard `sklearn` t-SNE, the common Python implementation of UMAP can also transform new observations after fitting. This makes UMAP somewhat easier to reuse in a later workflow, although the first purpose here is still visualization and diagnosis.

The main idea is:

$X \longrightarrow$ high-dimensional nearest-neighbor graph $W \longrightarrow$ low-dimensional coordinates with a similar graph.

14.3.1 Mathematical Idea

The core algorithm is similar in spirit to t-SNE:

1. For each observation, find its nearest neighbors in the original space.
2. Turn these local neighborhoods into a weighted graph $W = (w_{ij})$. A larger w_{ij} means that observations i and j are strongly connected in the original representation.
3. Introduce low-dimensional coordinates $z_1, \dots, z_n$.
4. From the distances among the z_i 's, build a low-dimensional graph $R(Z) = (r_{ij})$.
5. Move the z_i 's so that the low-dimensional graph resembles the original graph.

In other words, UMAP chooses the plotted coordinates by minimizing a graph cross-entropy loss between the original graph W and the low-dimensional graph $R(Z)$:

$$\min_{z_1, \dots, z_n} \text{CE}(W, R(Z)).$$

Thus UMAP does not try to reconstruct the original features. It tries to preserve a local nearest-neighbor graph. The details use fuzzy-set language, but the practical interpretation is simple: observations that are strongly connected in the original graph should stay close in the display, while weakly connected observations may be placed farther apart.

i How UMAP builds W

UMAP first finds the nearest neighbors of each observation according to the chosen distance metric. The parameter **n_neighbors** controls the size of these local neighborhoods. For each point x_i , UMAP uses only its selected nearest neighbors as candidate directed edges from i . Points outside this selected neighbor set usually have directed weight $a_{ij} = 0$ from the point of view of i .

After the candidate neighbors are selected, UMAP assigns a local distance offset ρ_i and a local scale σ_i for each point. The offset ρ_i is usually the distance from x_i to its nearest neighbor. This makes the nearest neighbor receive weight 1. The scale σ_i controls how quickly the remaining selected neighbors lose weight as distance increases. Both ρ_i and σ_i are chosen automatically by the algorithm for each point; they are not tuning parameters set by the user.

A simplified version of the directed neighbor weight from i to j is

$$a_{ij} = \exp \left(-\frac{\max\{0, d(x_i, x_j) - \rho_i\}}{\sigma_i} \right).$$

Thus the construction is

$$\mathbf{n_neighbors} \longrightarrow \text{selected neighbor set} \longrightarrow (\rho_i, \sigma_i) \longrightarrow a_{ij} \longrightarrow W.$$

The parameter **n_neighbors** decides which points are considered local neighbors. The parameters ρ_i and σ_i then decide how strongly those selected neighbors are connected. Thus UMAP, like t-SNE, uses local distance scales instead of one global bandwidth for the whole dataset.

The formula has a simple piecewise interpretation. If x_j is within the local offset of x_i , so that

$$d(x_i, x_j) \leq \rho_i,$$

then $\max\{0, d(x_i, x_j) - \rho_i\} = 0$ and therefore $a_{ij} = \exp(0) = 1$. If x_j is farther away than ρ_i , then

$$a_{ij} = \exp \left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i} \right),$$

so the directed connection strength decays exponentially with distance. For the directed weights a_{ij} , the nearest neighbor has weight approximately 1, selected but farther neighbors receive exponential weights between 0 and 1, and points outside the selected `n_neighbors` set have weight 0 from i to j .

The directed weights are then combined into a symmetric graph:

$$w_{ij} = a_{ij} + a_{ji} - a_{ij}a_{ji}.$$

This produces the high-dimensional graph $W = (w_{ij})$. Formally, W is an $n \times n$ weighted adjacency matrix: w_{ij} measures the strength of the local connection between observations i and j in the original space. It is not a distance matrix. Larger w_{ij} means a stronger neighbor relationship, and most entries are zero or very small because UMAP is based on nearest neighbors.

The final weight w_{ij} is symmetric, so it combines both directions. Even if x_j is not selected as a neighbor of x_i , the final w_{ij} can still be positive if x_i is selected as a neighbor of x_j .

i How UMAP builds $R(Z)$ and the cross-entropy

After low-dimensional coordinates $z_1, \dots, z_n$ are introduced, UMAP defines a low-dimensional graph $R(Z) = (r_{ij})$ from pairwise distances in the plot. A common form is

$$r_{ij} = \frac{1}{1 + \alpha \|z_i - z_j\|^{2\beta}},$$

Here α and β are not usually chosen directly by the user. They are fitted internally from UMAP's display parameters, especially `min_dist`. If two plotted points are close, r_{ij} is large; if they are far apart, r_{ij} is small.

UMAP then moves the coordinates so that $R(Z)$ resembles W . This is done by minimizing a cross-entropy loss of the form

$$\text{CE}(W, R) = - \sum_{i \neq j} [w_{ij} \log r_{ij} + (1 - w_{ij}) \log(1 - r_{ij})] .$$

The first term encourages pairs with large w_{ij} to stay close in the plot. The second term discourages weakly connected pairs from all collapsing into the same place. In practice, UMAP optimizes the low-dimensional coordinates by stochastic gradient descent: sampled edges from W pull connected points together, while randomly sampled non-edges push unrelated points apart.

i What `min_dist` controls

The parameter `min_dist` is the effective minimum distance between nearby points in the low-dimensional embedding. It is not a hard constraint that all plotted points must be at least this far apart. Instead, it defines the distance scale at which the low-dimensional similarity should begin to decay.

UMAP does not usually ask the user to choose α and β directly. Instead, it fits α and β from `min_dist` so that

$$\frac{1}{1 + \alpha d^{2\beta}}$$

has a shape similar to a target curve: the curve stays near 1 for distances below `min_dist`, then decreases after that. Thus the tuning chain is

$$\text{min_dist} \longrightarrow (\alpha, \beta) \longrightarrow r_{ij} = \frac{1}{1 + \alpha \|z_i - z_j\|^{2\beta}} \longrightarrow R(Z).$$

The practical interpretation is:

- `min_dist` does not define the original nearest-neighbor graph W . That graph is mainly determined by `metric` and `n_neighbors`.
- `min_dist` affects the low-dimensional graph $R(Z)$ through the fitted values of α and β .

- Smaller `min_dist` allows points that are strongly connected in W to be placed closer together in the low-dimensional embedding. This usually produces tighter visual clusters.
- Larger `min_dist` keeps strongly connected points more spread out. This can make continuous structure and within-group variation easier to see, but the plot may look less clustered.
- A tight island in a UMAP plot does not automatically mean that the original data have high density there. It may simply reflect a small `min_dist`.

14.3.2 What structure does UMAP preserve?

UMAP preserves local nearest-neighbor graph structure, not the original coordinates, not the original feature meanings, and not exact Euclidean distances. The input graph depends on the chosen distance metric and on the neighborhood size `n_neighbors`. These determine the original graph W that UMAP tries to preserve. The parameter `min_dist` does not change W ; it changes how tightly connected points may be packed in the low-dimensional display.

In plain language:

- If two observations are nearest neighbors in the original representation, UMAP tries to keep them close or connected in the plot.
- If two groups are connected by many nearest-neighbor edges, UMAP may place them near each other even if the original geometry is curved or nonlinear.
- Distances between faraway islands, island sizes, axis directions, and visual densities should not be treated as exact quantitative information.
- Compared with t-SNE, UMAP often keeps a little more broad structure, but this depends strongly on `n_neighbors` and should still be checked.

i UMAP as a diagnostic plot

UMAP is useful when we want to inspect whether local nearest-neighbor relationships reveal structure that a linear plot such as PCA misses. Like t-SNE, it is mainly a visualization and diagnostic tool: visible groups are hypotheses to investigate, not automatic cluster definitions.

UMAP can also be useful after clustering. If cluster labels form coherent local regions in a UMAP plot, then the clustering is consistent with the nearest-neighbor graph shown by UMAP. If labels are heavily mixed, the clustering may not match local structure, or the data may not contain clearly separated groups. Formal evaluation should still use criteria tied to the analysis goal, such as nearest-neighbor overlap, silhouette score, or adjusted Rand index when labels are available.

14.3.3 UMAP in Python

The standard implementation is provided by `umap-learn`. If it is not installed, you may install it directly.

```
uv add umap-learn
```

We use the same standardized digits data and the same 30-dimensional PCA preprocessing as in the t-SNE example.

```
import umap

umap_model = umap.UMAP(
    n_neighbors=15,
    min_dist=0.1,
    n_components=2,
    metric="euclidean",
    random_state=1,
)

Z_umap = umap_model.fit_transform(X_digits_pca)
```

In this example, `n_neighbors=15` sets a moderate local neighborhood size, and `min_dist=0.1` allows fairly compact visual groups. These are tuning choices, not universal defaults.

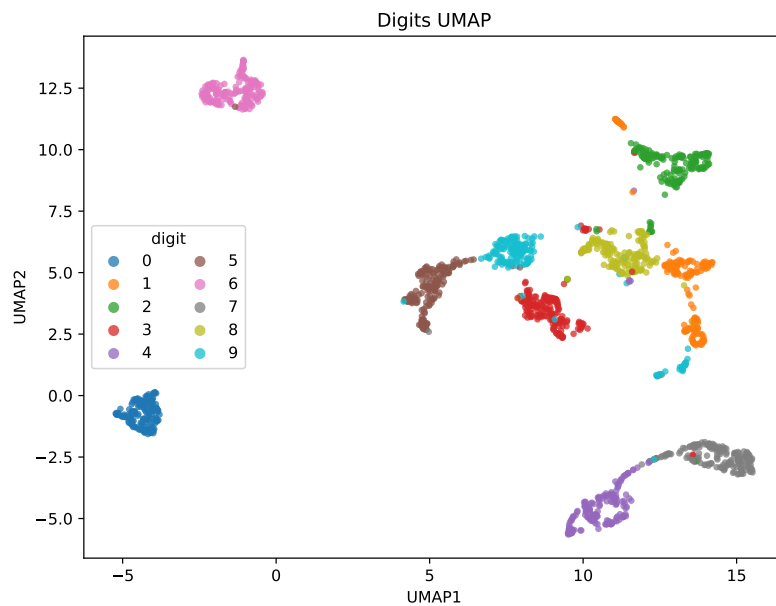
- The parameter `n_neighbors` controls the construction of the high-dimensional graph W : it determines how many nearby observations each point uses when UMAP builds the weighted nearest-neighbor graph. A small value, such as 5 or 10, makes UMAP focus on very local neighborhoods. This can reveal fine structure, but it can also break continuous structure into many small islands. A larger value, such as 30 or 50, asks UMAP to use broader neighborhoods. This often gives a smoother layout and may preserve more large-scale relationships, but it can hide small local differences.
- The parameter `metric` defines distance in the original feature space, so it determines which observations count as nearest neighbors before UMAP does any embedding. For standardized numeric features, "euclidean" is a common starting point. For sparse counts or text-like features, cosine distance is often more appropriate. Changing `metric` can change the graph W itself, so it can change the embedding more fundamentally than changing only the visual display parameters.
- The parameter `min_dist` affects the low-dimensional graph $R(Z)$ rather than the original graph W . Smaller values allow tighter visual groups, while larger values keep connected points more spread out. In practice, try several combinations of `n_neighbors` and `min_dist`. Patterns that remain visible across reasonable settings are more credible than patterns that appear only under one parameter choice.

i Initialization in UMAP

UMAP can use PCA initialization, but it usually does not rely on it. Since UMAP first builds a nearest-neighbor graph W , the default initialization is typically graph-based, using spectral initialization from that graph. PCA initialization is possible, but it is not the central idea.

We plot the result from the above code.

```
fig, ax = plt.subplots(figsize=(8, 6))
scatter = ax.scatter(
    Z_umap[:, 0],
    Z_umap[:, 1],
    c=y_digits,
    cmap="tab10",
    s=10,
    alpha=0.75,
)
ax.set_xlabel("UMAP1")
ax.set_ylabel("UMAP2")
ax.set_title("Digits UMAP")
ax.legend(*scatter.legend_elements(), title="digit", ncol=2);
```



We will try different combinations of `n_neighbors` and `min_dist`.

```
n_neighbors_values = [5, 15, 50]
min_dist_values = [0.05, 0.10, 0.50]

fig, axes = plt.subplots(3, 3, figsize=(12, 10))
```

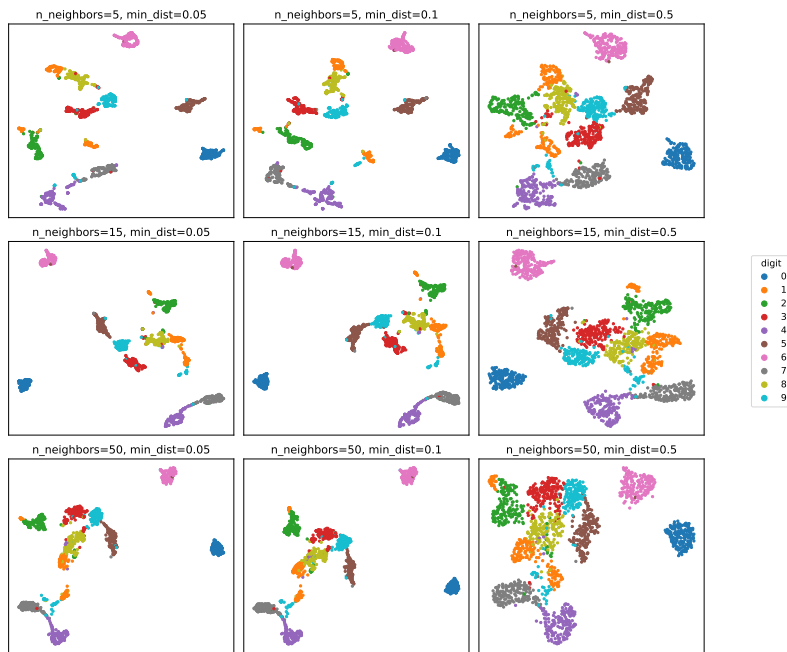
```

scatter = None

for row, n_neighbors in enumerate(n_neighbors_values):
    for col, min_dist in enumerate(min_dist_values):
        ax = axs[row, col]
        model = umap.UMAP(
            n_neighbors=n_neighbors,
            min_dist=min_dist,
            random_state=1,
        )
        embedding = model.fit_transform(X_digits_pca)
        scatter = ax.scatter(
            embedding[:, 0],
            embedding[:, 1],
            c=y_digits,
            cmap="tab10",
            s=6,
        )
        ax.set_title(f"n_neighbors={n_neighbors}, min_dist={min_dist}")
        ax.set_xticks([])
        ax.set_yticks([])

fig.legend(
    *scatter.legend_elements(),
    title="digit",
    loc="center right",
)
fig.tight_layout(rect=[0, 0, 0.9, 1]);

```



Some observations from the plots are:

- The overall class structure is relatively stable across all parameter settings. Most digit classes remain recognizable, indicating that the underlying local neighborhood structure is robust.
- Digits 0 and 6 remain well separated from most other classes in all embeddings, suggesting that these digits have distinctive local pixel patterns.
- Several groups of digits remain in neighboring regions across different parameter settings, although their relative positions change as the hyperparameters vary.
- Some digit classes appear as multiple disconnected regions when `n_neighbors` is small. This suggests that the class may contain several distinct local neighborhoods in the original feature space, although these should not automatically be interpreted as true subtypes.
- Increasing `min_dist` primarily changes the compactness of the embedding. Smaller values produce tighter clusters, whereas larger values spread nearby points farther apart.
- Increasing `n_neighbors` generally produces a smoother and more stable overall layout by incorporating broader

neighborhood information during graph construction.

- Although the global layout changes noticeably across parameter settings, the major local neighborhoods remain largely consistent.

These patterns can also guide inspection of the original data. For example, if several digit classes remain connected or neighboring in the UMAP plot, we can examine those images to understand which strokes or handwriting styles make the classes similar.

14.4 Comparing the Methods

The unsupervised methods in this chapter and the previous chapter are related because they all produce lower-dimensional representations of high-dimensional data. For that reason, they may appear to be interchangeable: PCA, spectral embedding, t-SNE, and UMAP can all be used to make a two-dimensional plot.

However, they are not trying to preserve the same structure. PCA preserves linear variance. Spectral embedding preserves connectivity in a chosen graph. t-SNE attempts to preserve neighborhood probabilities. UMAP attempts to preserve the weighted nearest-neighbor graph structure. Therefore, the right method depends on the question we want the visualization to answer.

This section compares the methods in two steps. First, we compare t-SNE and UMAP directly, since both are nonlinear visualization methods based on local neighborhoods. Then we compare all four methods together to show how linear variance, graph connectivity, local neighbor probabilities, and nearest-neighbor graph structure lead to different visual conclusions.

14.4.1 t-SNE vs UMAP

t-SNE and UMAP are both nonlinear visualization methods based on local relationships. They both start from high-dimensional distances, but they preserve different objects.

For t-SNE, distances are converted into neighbor probabilities. The high-dimensional probability distribution is $P = (p_{ij})$, the low-dimensional probability distribution is $Q = (q_{ij})$, and t-SNE moves the plotted points so that Q matches P . The main tuning parameter is **perplexity**, which controls the effective neighborhood size used to construct P .

For UMAP, distances are converted into a weighted nearest-neighbor graph. The high-dimensional graph is $W = (w_{ij})$, the low-dimensional graph is $R(Z) = (r_{ij})$, and UMAP moves the plotted points so that $R(Z)$ resembles W . The parameter **n_neighbors** controls the neighborhood scale used to build W , while **min_dist** controls how tightly connected points may be packed in the display.

This difference affects how we read the plots. In a t-SNE plot, a continuous bridge made of many nearby points is meaningful: it suggests that the original data may contain a chain of local neighbor relationships connecting the two regions. But a large empty distance between two t-SNE islands should not be interpreted as a reliable measurement of how far apart the groups are in the original space. UMAP has a more direct graph interpretation: connected regions in the plot often reflect connections in the nearest-neighbor graph W , although that graph still depends on the chosen **metric** and **n_neighbors**.

So t-SNE is especially useful for inspecting very local neighborhoods. UMAP is especially useful for inspecting nearest-neighbor graph structure, comparing neighborhood scales, and sometimes reusing a fitted embedding for new observations. Neither method should be used to make strict claims about global distances, island sizes, densities, or axis meanings.

14.4.1.1 MNIST

We compare the methods on the famous [MNIST handwritten digit](#) dataset. MNIST has 70,000 images, and each image is represented by $28 \times 28 = 784$ pixel features. Since the original data are 784-dimensional, we cannot directly visualize the original feature space as a scatter plot. Instead, we first inspect a few raw digit images, and then visualize two-dimensional embeddings.

We use `fetch_openml("mnist_784")` to load the dataset. For speed, the embedding examples below use a fixed random subset of 3000 images.

```
from sklearn.datasets import fetch_openml
from sklearn.preprocessing import StandardScaler
import numpy as np

mnist = fetch_openml("mnist_784", version=1, as_frame=False)

rng = np.random.default_rng(1)
sample_index = rng.choice(mnist.data.shape[0], size=3000, replace=False)

X_mnist_raw = mnist.data[sample_index]
y_mnist = mnist.target[sample_index].astype(int)

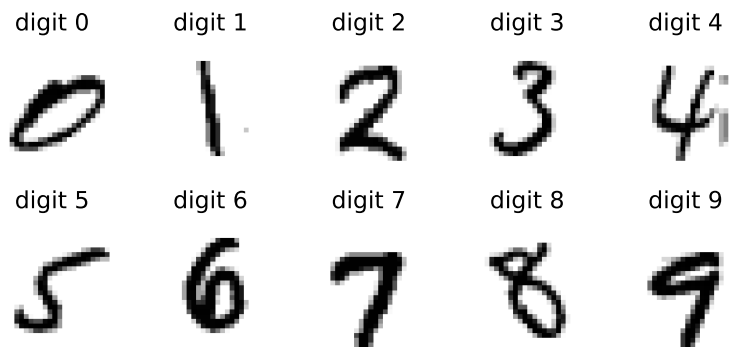
X_mnist = StandardScaler().fit_transform(X_mnist_raw)
```

Here are some sample images from MNIST. As with the `digits` dataset, we use `imshow()` to display each digit as a matrix of pixel intensities.

```
import matplotlib.pyplot as plt

fig, axs = plt.subplots(2, 5, figsize=(7, 3))

for ax, digit in zip(axs.ravel(), range(10)):
    image = X_mnist_raw[y_mnist == digit][0].reshape(28, 28)
    ax.imshow(image, cmap="gray_r")
    ax.set_title(f"digit {digit}")
    ax.axis("off");
```



Now we compare t-SNE and UMAP in two ways on this subset. The first row uses the standardized 784-dimensional pixel features directly. The second row first reduces the data to 50 principal components and then applies t-SNE or UMAP. The colors are the digit labels, used only after fitting to interpret the plots.

```
from sklearn.manifold import TSNE
from sklearn.decomposition import PCA
import umap

X_mnist_pca50 = PCA(n_components=50, random_state=1).fit_transform(X_mnist)

Z_tsne_raw = TSNE(
    n_components=2, perplexity=50, init="pca", learning_rate="auto", random_state=1
).fit_transform(X_mnist)

Z_umap_raw = umap.UMAP(
    n_components=2, n_neighbors=15, min_dist=0.1, metric="euclidean", random_state=1
).fit_transform(X_mnist)

Z_tsne_compare = TSNE(
    n_components=2, perplexity=50, init="pca", learning_rate="auto", random_state=1
).fit_transform(X_mnist_pca50)

Z_umap_compare = umap.UMAP(
    n_components=2, n_neighbors=15, min_dist=0.1, metric="euclidean", random_state=1
).fit_transform(X_mnist_pca50)

embedding_grid = {
```

```

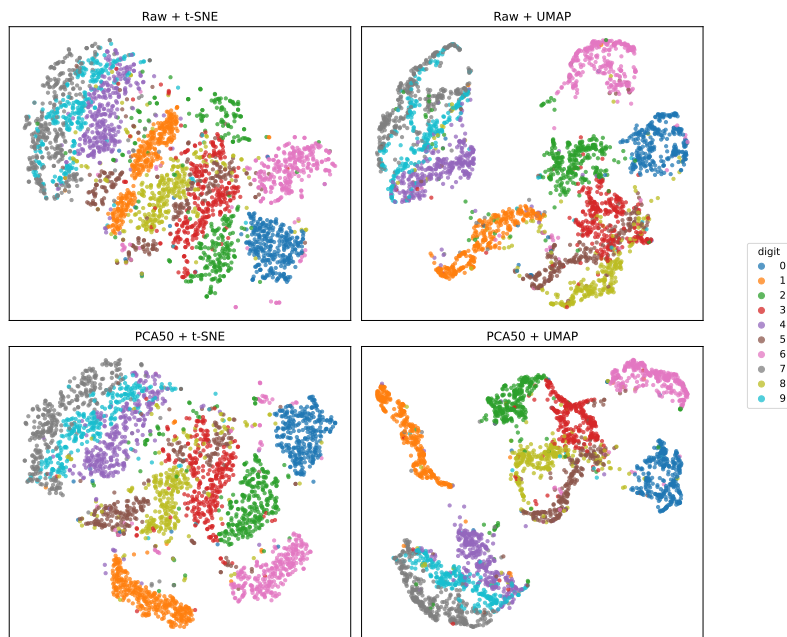
"Raw + t-SNE": Z_tsne_raw,
"Raw + UMAP": Z_umap_raw,
"PCA50 + t-SNE": Z_tsne_compare,
"PCA50 + UMAP": Z_umap_compare,
}

fig, axs = plt.subplots(2, 2, figsize=(11, 9))

scatter = None
for ax, (name, Z_method) in zip(axs.ravel(), embedding_grid.items()):
    scatter = ax.scatter(
        Z_method[:, 0], Z_method[:, 1], c=y_mnist, cmap="tab10", s=10, alpha=0.75
    )
    ax.set_title(name)
    ax.set_xticks([])
    ax.set_yticks([])

fig.legend(*scatter.legend_elements(), title="digit", loc="center right")
fig.tight_layout(rect=[0, 0, 0.9, 1]);

```



Since neither method aims to preserve global geometry, we focus on the stability of local neighborhoods and on the qualitative

differences between the resulting embeddings.

The figure highlights two main comparisons: the effect of PCA preprocessing and the differences between t-SNE and UMAP.

Raw versus PCA50

- In this MNIST subset, applying PCA before t-SNE or UMAP changes the exact layout but preserves much of the local neighborhood structure. The PCA50 embeddings therefore produce similar digit neighborhoods while reducing the dimensionality from 784 to 50.
- In this example, the PCA50 embeddings appear slightly cleaner, with fewer scattered points and more coherent local neighborhoods. In practice, PCA preprocessing can improve computational efficiency and reduce low-variance noise while retaining much of the information needed for nonlinear visualization.

t-SNE versus UMAP

- Both methods recover clear local neighborhoods for the handwritten digits. Images with the same label tend to occupy nearby regions, although several visually similar digit classes still overlap.
- With these settings, the UMAP embeddings show several groups of similar digits as relatively well-separated neighboring regions. For example, groups of visually similar digits, such as 3, 5, and 8 or 4, 7, and 9, tend to occupy neighboring regions, reflecting similarities in their handwritten shapes.
- With these settings, the t-SNE embeddings connect these similar digit classes through more gradual transitions. Instead of forming clearly separated islands, neighboring classes blend through intermediate handwriting styles, giving the overall layout a more continuous appearance.

These differences should be read as differences in the visualization objectives, not as proof that one method has found the true geometry. UMAP’s clearer separation and t-SNE’s smoother transitions are both diagnostic patterns to investigate, not final cluster definitions.

14.4.2 Four-method comparison

PCA, spectral embedding, t-SNE, and UMAP all create low-dimensional coordinates, but they answer different diagnostic questions. PCA is the linear baseline. Spectral embedding is graph-based and focuses on connectivity. t-SNE focuses on local neighbor probabilities. UMAP focuses on a weighted nearest-neighbor graph.

Method	Mainly preserves	Does not preserve well	Best for checking
PCA	linear variance and large-scale linear directions	nonlinear neighborhoods and curved manifolds	whether a simple linear projection already explains the main variation
Spectral embedding	connectivity in a chosen graph	original feature axes and exact Euclidean distances	whether a nearest-neighbor graph separates groups
PCA50 + t-SNE	local neighborhood probabilities after PCA preprocessing	global distances, island sizes, densities, axis directions	whether local neighborhoods or clustering labels are visually mixed

Method	Mainly preserves	Does not preserve well	Best for checking
PCA50 + UMAP	local nearest-neighbor graph after PCA preprocessing, with some broader structure depending on <code>n_neighbors</code>	exact distances, densities, and axis meanings	nonlinear local visualization and graph-based neighborhood structure

We now apply all four methods to the same MNIST subset. PCA and spectral embedding are applied to the standardized pixel features, while t-SNE and UMAP are applied after PCA preprocessing to 50 dimensions.

```
from sklearn.manifold import SpectralEmbedding
Z_pca2 = PCA(n_components=2, random_state=1).fit_transform(X_mnist)

Z_spectral_mnist = SpectralEmbedding(
    n_components=2,
    affinity="nearest_neighbors",
    n_neighbors=15,
    random_state=1,
).fit_transform(X_mnist)

embeddings = {
    "PCA": (Z_pca2, "PC1", "PC2"),
    "Spectral embedding": (Z_spectral_mnist, "SE1", "SE2"),
    "PCA50 + t-SNE": (Z_tsne_compare, "t-SNE1", "t-SNE2"),
    "PCA50 + UMAP": (Z_umap_compare, "UMAP1", "UMAP2"),
}

fig, axs = plt.subplots(2, 2, figsize=(11, 9))

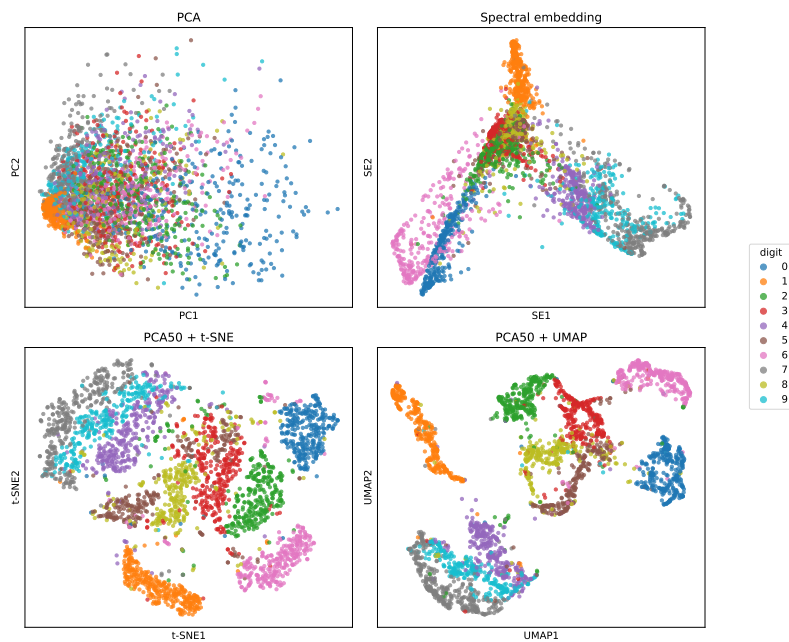
scatter = None
for ax, (name, (Z_method, xlabel, ylabel)) in zip(axs.ravel(), embeddings.items()):
```

```

scatter = ax.scatter(
    Z_method[:, 0],
    Z_method[:, 1],
    c=y_mnist,
    cmap="tab10",
    s=10,
    alpha=0.75,
)
ax.set_title(name)
ax.set_xlabel(xlabel)
ax.set_ylabel(ylabel)
ax.set_xticks([])
ax.set_yticks([])

fig.legend(*scatter.legend_elements(), title="digit", loc="center right")
fig.tight_layout(rect=[0, 0, 0.9, 1]);

```



The four methods provide different views of the same high-dimensional image subset because they preserve different aspects of the original data. Some observations from the comparison are:

- PCA provides the linear reference view by projecting the images onto two directions of maximum variance. It captures the dominant linear variation but leaves substantial overlap because the digit classes are not linearly separable.
- Spectral embedding reveals nonlinear organization that is not visible in the PCA projection. Because the embedding is constructed from a nearest-neighbor graph, the resulting layout depends strongly on the chosen graph.
- PCA50 + t-SNE and PCA50 + UMAP recover many of the same local neighborhoods, but organize them differently because they optimize different objective functions.

The four methods should therefore be interpreted differently because each is designed to preserve a different aspect of the original data. PCA asks whether a low-dimensional linear projection already captures the dominant variation. Spectral embedding asks whether a neighborhood graph reveals meaningful connected structure. t-SNE asks whether local neighborhoods remain coherent after embedding. UMAP asks whether the nearest-neighbor graph exhibits stable graph-connected regions. None of these methods provides a geometrically faithful map of the original feature space. Their interpretation should therefore always be based on the structure each method is designed to preserve.

i Using all 70,000 MNIST images

The examples above use a fixed subset of 3,000 images for speed. The next two figures repeat the previous comparisons using the full MNIST dataset with 70,000 images. The first figure compares t-SNE and UMAP, with or without PCA, and the second figure compares PCA, spectral embedding, PCA50 + t-SNE, and PCA50 + UMAP.

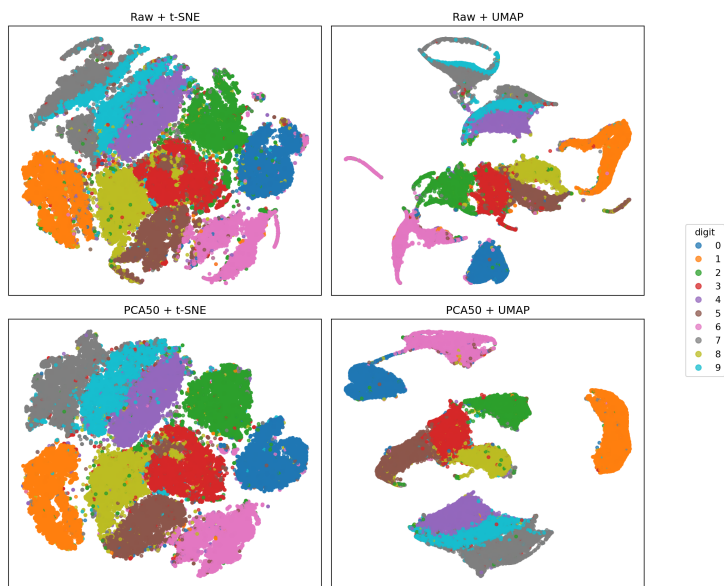


Figure 14.1: t-SNE and UMAP comparison on all 70,000 MNIST images

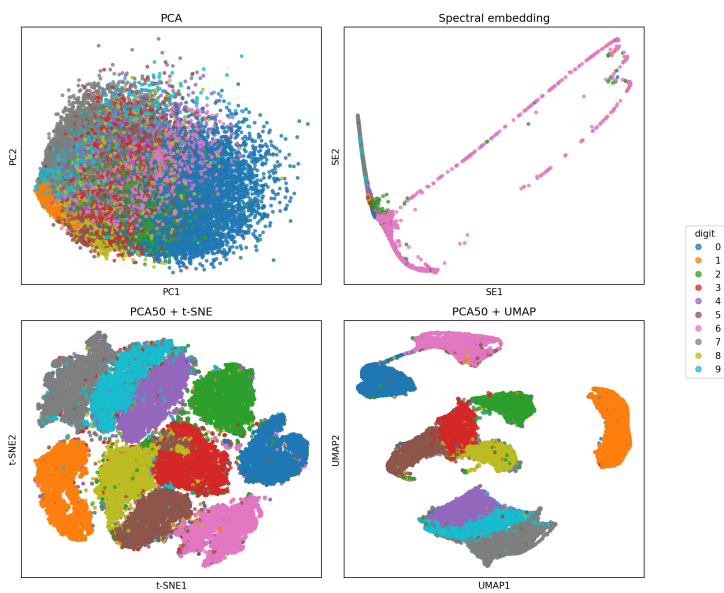


Figure 14.2: Four-method comparison on all 70,000 MNIST images

Compared with the 3,000-image subset:

- The main qualitative conclusions are broadly similar. The 3,000-image subset already captures many of the important neighborhood relationships among the handwritten digits.
- The plots become visually denser with 70,000 images because many more intermediate handwriting styles are included.
- PCA changes the least: the dominant linear directions remain similar, and the larger dataset mainly increases sampling density.
- Spectral embedding becomes smoother because the nearest-neighbor graph is constructed from many more observations, making graph branches more complete and less affected by sampling variability.
- The t-SNE embedding becomes substantially denser. Many apparent gaps between neighboring digit groups are filled by intermediate observations, producing smoother local transitions.
- The UMAP embedding also becomes denser while retaining more clearly segmented graph-connected regions.
- Increasing the sample size improves the completeness and stability of the visualizations, but it does not make global distances, region sizes, or visual densities quantitatively reliable.
- Running t-SNE and UMAP on all 70,000 images is substantially more computationally expensive than using a representative 3,000-image subset.

Apparent clusters are not proof of clusters

Visible separation in an embedding is a diagnostic clue, not a cluster definition by itself. A gap, island, or bridge should be checked against the structure the method is designed to preserve and, when needed, against quantitative criteria tied to the analysis goal.

A Python IDE Setup

There are many ways to set up a Python development environment, and each has its own advantages and disadvantages. In this tutorial, we will cover one of the most popular current setups: VS Code + uv.

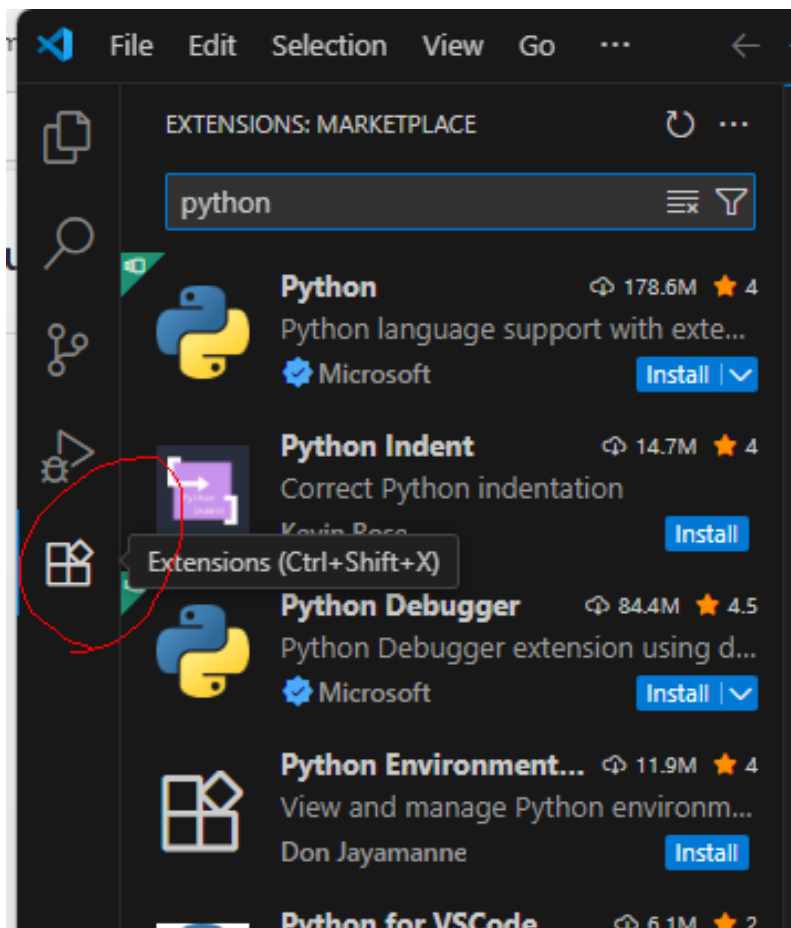
For now, we will focus only on the practical setup steps. The underlying concepts will be discussed in the later appendices.

This tutorial uses Windows 11. If you use another operating system, the steps are very similar, but you may need to make some minor but straightforward modifications. If you encounter major issues, please contact me.

A.1 VS Code

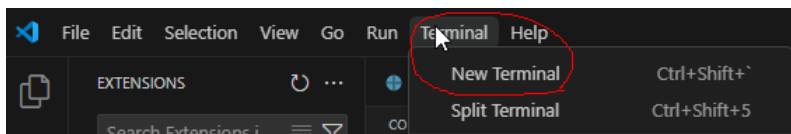
We choose VS Code as our IDE for Python. The installation is straightforward starting from the [Official website](#). After installation, we need to do a few configurations to make the experience better.

1. For development of Python, the essential plugins are the Python plugin and Jupyter plugin. You only need to install the two main ones, and all related plugins will be installed automatically.

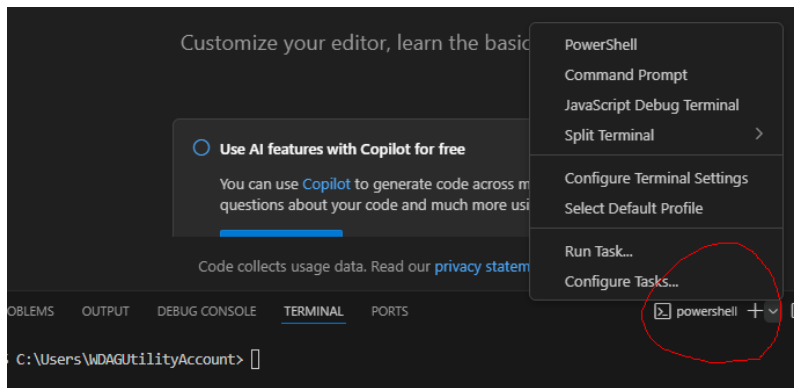


You may either search the extensions directly from VS Code extension tab (which is usually located on the left side bar), or you may install it from the link [python](#) and [jupyter](#).

2. Sometimes you may need to use the terminal inside VS Code. You may start the terminal from top menu (whose hotkey is by default `ctrl+shift+``).



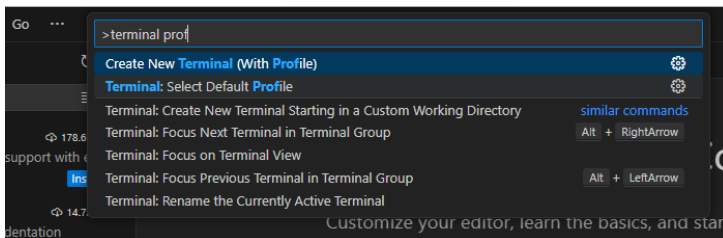
Note that there are many different choices of terminals. After you start the first default terminal, you may start any new terminals using the small + and the v mark on the right.



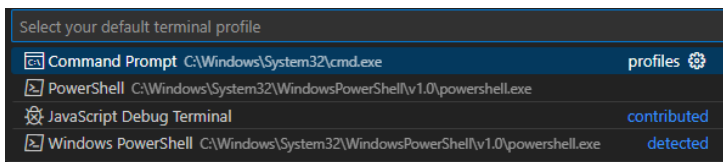
🔥 Change Terminal default profile

In Windows the default terminal is Powershell when you first install VS Code. In the past it had bugs with `conda` (and I don't know whether it is fixed now). In order to avoid headaches due to the bugs, we simply switch to other terminals, like Command Prompt. You may change the default terminal by

1. Press `Ctrl+Shift+P` or `F1` to open the top prompt menu.
2. Find the `Terminal: Select Default Profile` command.



3. Select the desired terminal as the default one.



A.2 Python Virtual Environment installation

There are many tools available for managing Python environments and packages. As of Summer 2026, `uv` has become one of the most popular choices.

`uv` is an all-in-one tool for managing Python installations, virtual environments, and project dependencies. With `uv`, you do not need to install Python separately, as it can download and manage Python versions for you. You may visit the [official documentation](#) to install `uv`.

Windows headache

There is a possibility that Smart App Control (SAC) or other Application Control policies in Windows 11 may block the execution of `uv.exe`. In this case, you may see a message similar to

```
'C:\Users\WDAGUtilityAccount\.local\bin\uv.exe'  
was blocked by your organization's Device Guard policy.  
Contact your support person for more info.
```

This may occur because Windows does not yet consider the executable sufficiently trusted under its application-control policies.

If this happens, note that the exact cause and solution can vary across machines, Windows versions, and security configurations. In addition, both Windows and `uv` are evolving rapidly, so solutions that work today may change in the future.

You are encouraged to search for the solution that best fits your situation. For this course, one workaround is to use WSL (see Section [A.4](#)).

After installation (and you may need to restart VS Code or open a new terminal window so that the `uv` command becomes available), you can use the following commands to verify that `uv` is working correctly.

Command	Purpose	Example Output
<code>uv --version</code>	Display the installed <code>uv</code> version. Useful for verifying that <code>uv</code> is installed correctly.	<code>uv 0.11.17</code>
<code>uv self update</code>	Update <code>uv</code> itself to the latest available version. Similar to updating package managers such as <code>pip</code> or <code>cargo</code> .	Upgraded <code>uv</code> from <code>v0.11.4</code> to <code>v0.11.17</code> !
<code>uv python list</code>	Show all Python versions available locally and those that can be installed through <code>uv</code> .	Lists installed and downloadable Python versions.

A.2.1 Install the first Python

Since VS Code often expects to discover at least one Python interpreter before it can fully initialize its Python and Jupyter services, we first install a Python version managed by `uv`.

```
uv python install
```

This Python installation is managed entirely by `uv`; it is not intended to be used directly for course work. Instead, it serves as a bootstrap interpreter that allows VS Code to detect Python correctly and manage project-specific virtual environments more reliably. You can verify the installed Python versions using `uv python list` as mentioned above.

If you skip this step and directly jump into the virtual environment, there is a possibility that Python in VS Code doesn't work well.

A.2.2 Setup the environment for the course

To set up the environment for this course, we provide the [toml file](#) and [python version file](#) for this course.

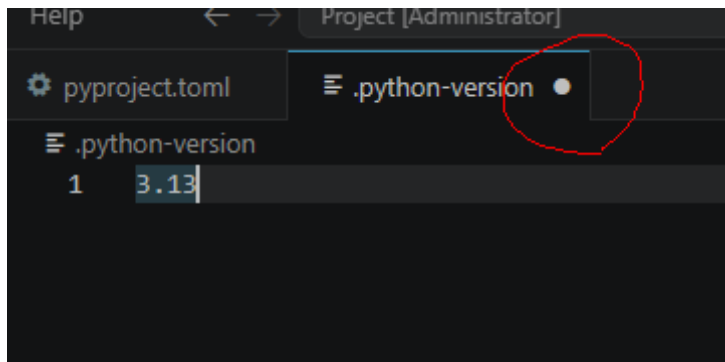
Alternatively, you may create a file named `pyproject.toml` and add the following content:

```
[project]
name = "STAT432"
version = "0.1.0"
requires-python = ">=3.13"
dependencies = [
    "jupyter>=1.1.1",
    "matplotlib>=3.10.9",
    "pandas>=3.0.2",
    "scikit-learn>=1.8.0",
    "seaborn>=0.13.2",
    "statsmodels>=0.14.6",
    "xgboost>=3.2.0",
]
```

The python version file only contains the version number `3.13` inside since this is the version of Python I use when I wrote the notes. You may either download the file and put it inside the folder or create a file with the name `.python-version` and write `3.13` inside.

Autosave

Before proceeding, make sure the file is saved.
By default, VS Code does not enable Auto Save. If you see a small dot next to the file name, the file contains unsaved changes.



Many beginner problems occur because they modify a file but forget to save it before running commands.

1. Create a new folder. The folder name is irrelevant. Put the files `pyproject.toml` and `.python-version` in the folder.
2. Open a terminal (no matter whether within VS Code or not) and enter the folder. Use the command `uv sync`.

After the command finishes, `uv` creates a virtual environment named `.venv` and installs all required packages specified in `pyproject.toml`. The resulting environment should closely match the one used to prepare these notes.

Now it is possible to start using the virtual environment for the course. In case that you need to modify the virtual environment, please continue to read the following optional part.

i Modifying the Environment

To install a package and update the project configuration, use `uv add <package-name>`.

For example, you may need `openpyxl` to read and write Excel files:

```
uv add openpyxl
```

This command updates the project configuration in `pyproject.toml` and installs the package into the project's virtual environment `.venv`.

Most package management tasks can be handled through

```
uv add.
```

i Jupyter kernels and modifying venv

A Jupyter kernel is the Python environment used by a notebook. When a notebook cell is executed, the code runs inside the selected kernel. Therefore, it is important to connect the notebook to the kernel associated with your course environment.

Modern versions of VS Code can usually detect a project managed by `uv` and use it directly as a notebook kernel. In many cases, no manual kernel registration is required. If VS Code does not automatically detect the environment, you can create a dedicated Jupyter kernel:

```
uv run python -m ipykernel install --user --name stat432 --display-name "(STAT432)"
```

The option `--name stat432` specifies the kernel identifier. In this example the identifier is `stat432`, but you may choose any name that is easy for you to recognize. This identifier is used internally by Jupyter and should be unique on your system.

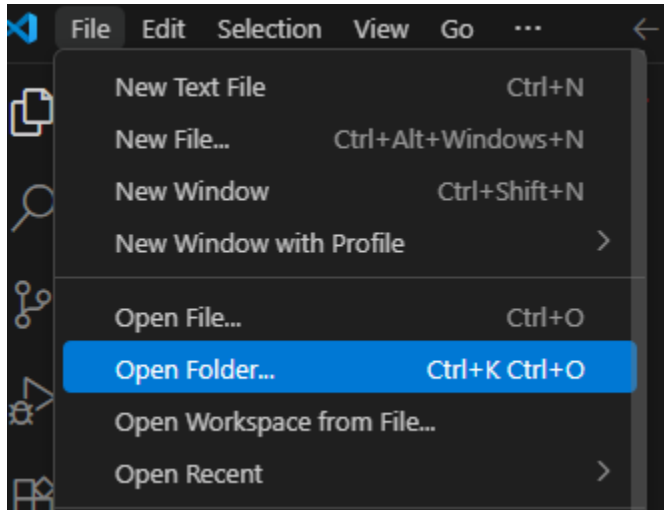
The option `--display-name "(STAT432)"` specifies the name shown in VS Code and Jupyter when selecting a kernel. Unlike the identifier, the display name does not need to be unique, although a descriptive name is recommended.

A.3 Back to VS Code and start Jupyter notebook

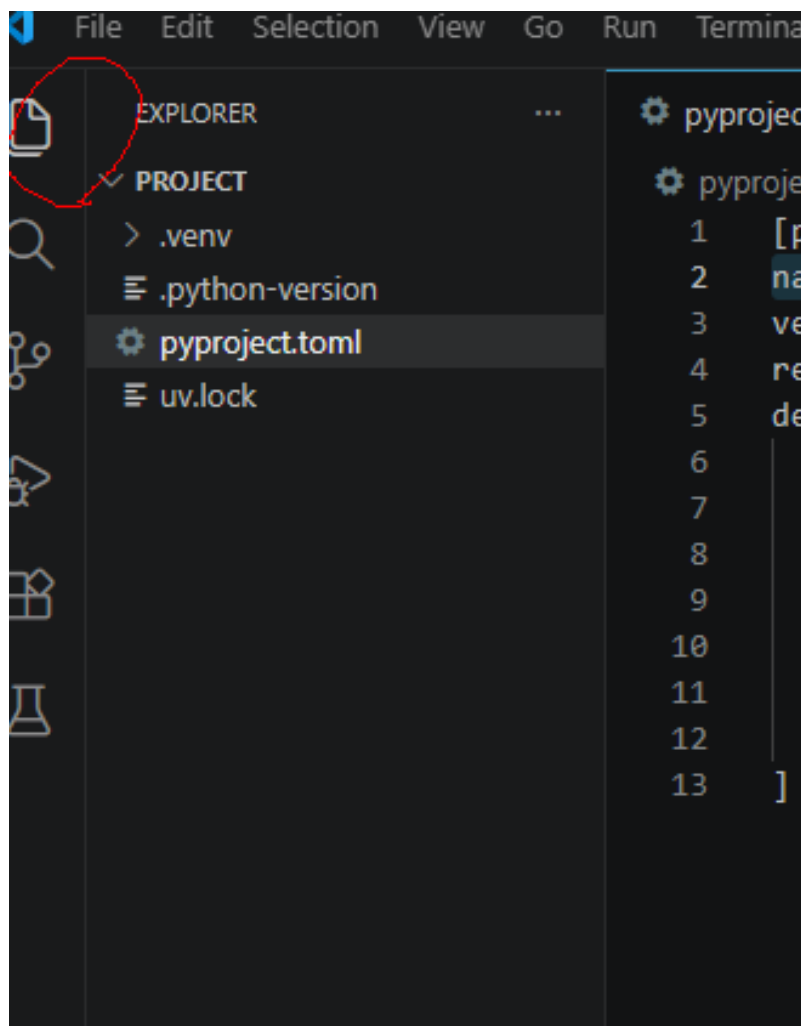
After Python is installed and virtual environment is set up, we could start to do a Hello World project.

The way VS Code organizes projects is through folders by default. So we first start a new folder (called the working folder) and all project related files should be put inside this folder. (There are ways to work with files outside the working folder, but you don't need it in most cases.)

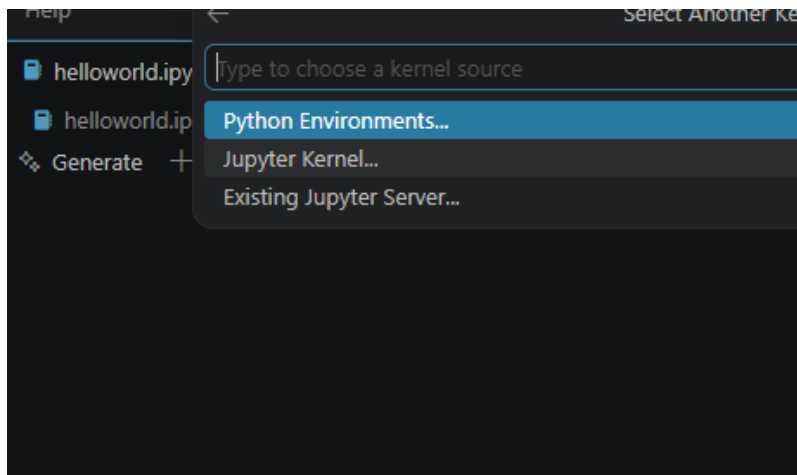
Since we already set up a folder (with a virtual environment), we open it as our working folder.



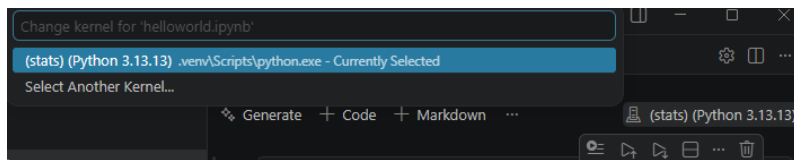
The folder already contains the project files and the environment files. You may see them from the file explorer.



Then you may create new files/folders or copy files/folders into this working folder. For demonstration a new file called `helloworld.ipynb` is created.

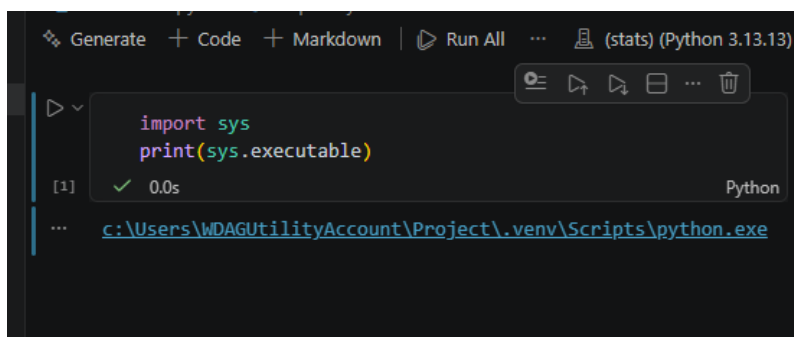


- If you created a new Jupyter kernel, you can find it from **Jupyter kernel**....
- If not, you can find your virtual environment in the working folder from **Python Environments**....



After the kernel is selected, we could start using the notebook. I prefer to use the following code as hello world. It can also tell us whether the correct python is used.

```
import sys
print(sys.executable)
```



A.3.1 Output to html and pdf

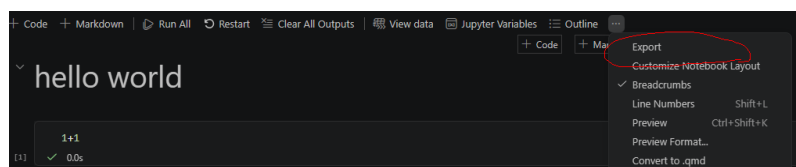
After finishing your Jupyter Notebook, you need to generate a PDF version for assignment submission.

Since we do not want to spend too much time on document formatting, we will use a simple workflow. Although there are many more sophisticated tools available (such as Pandoc, Quarto, and LaTeX-based workflows), the approach below is completely sufficient for this course.

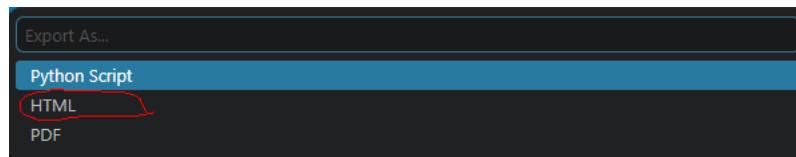
Our workflow is:

0. Restart the kernel and run all.
1. Export the notebook to HTML using VS Code.
2. Open the HTML file in a web browser.
3. Print the page to a PDF file.

First, locate the notebook toolbar and click the `...` button on the right side. From the expanded menu, select **Export**.



Then choose HTML.



This will create an HTML file. Open the file in your preferred web browser and print it to PDF. Most modern browsers (such as Chrome, Edge, and Firefox) support printing directly to PDF.

For assignment submission, submit both the PDF file and the notebook file. Grading will be based primarily on the PDF file, while the notebook file serves as a backup source in case I need to inspect the code or verify specific parts of the submission.

Restart and Run All

A Jupyter notebook keeps its internal state in memory while you work. As a result, cells can be executed in different orders, making it possible for the notebook contents and the actual execution state to become inconsistent.

For example, a variable may exist because it was created in an earlier session, even though the cell that creates it has been deleted and no longer appears before its first use in the notebook. Such notebooks may appear to run correctly during editing but fail when executed from scratch.

To avoid these issues, you are required to restart the kernel and run all cells before creating your submission PDF. Any errors that occur during this process must be fixed before submission.

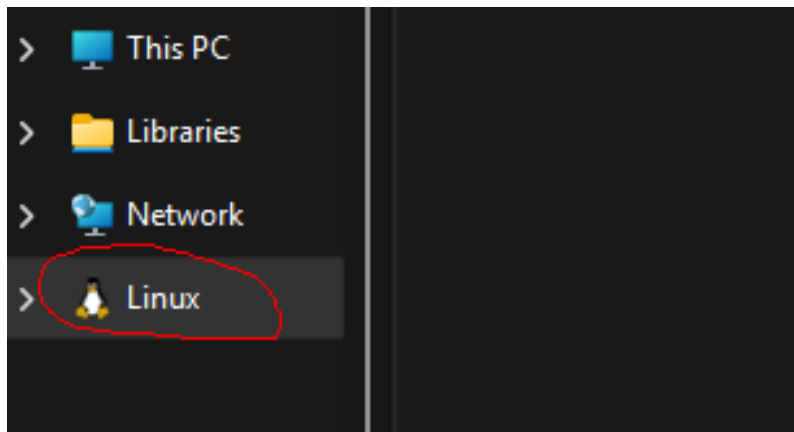
Although notebook workflows have several well-known limitations, Jupyter notebooks remain the dominant platform for data science, machine learning, and scientific computing. New alternatives such as `marimo` have attracted considerable attention in recent years, but they are still relatively young and may require several more years to reach the level of adoption enjoyed by Jupyter — if they ever do.

A.4 WSL

WSL stands for Windows subsystem for Linux. It allows you to run a Linux environment directly inside Windows without installing a separate operating system.

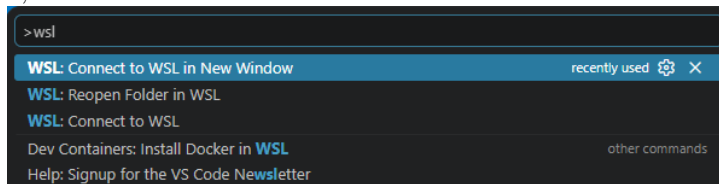
To install WSL, please follow the [official Microsoft guide](#).

After installation, you will have a Linux system (usually Ubuntu) running alongside Windows. The Linux file system is separate from the Windows file system, although you can access it through File Explorer to copy / move files back and forth.



VS Code provides excellent support for WSL through the Remote WSL extension. To open a WSL window, you may:

- Click the button in the lower-left corner of VS Code 
- Or open the Command Palette (**Ctrl+Shift+P** or **F1**), search for **WSL**, and select one of the available options. For most situations, **Connect to WSL in New Window** is the



easiest choice.

Once connected to WSL, you are working inside a Linux environment rather than Windows.

Because WSL is a separate system, you will need to:

- Open your project folder again.
- Install **uv**, Python and other tools inside WSL.
- Create virtual environments inside WSL.
- Install packages inside WSL.

You may follow the same instructions presented earlier to install **uv** and set up your Python environment. The overall process is the same, although the commands will be the Linux versions rather than the Windows versions.

Note that if you install `uv` in Windows, it is not automatically available inside WSL. Similarly, software installed inside WSL is generally not available in Windows.

In practice, it is usually easiest to keep an entire project inside WSL and perform all Python-related work there.

 Tip

Avoid storing virtual environments inside the Windows file system when working primarily in WSL. For best performance and compatibility, create your project folders inside your Linux home directory (for example, `~/Project`) rather than in mounted Windows directories (for example, `/mnt/c/...`).

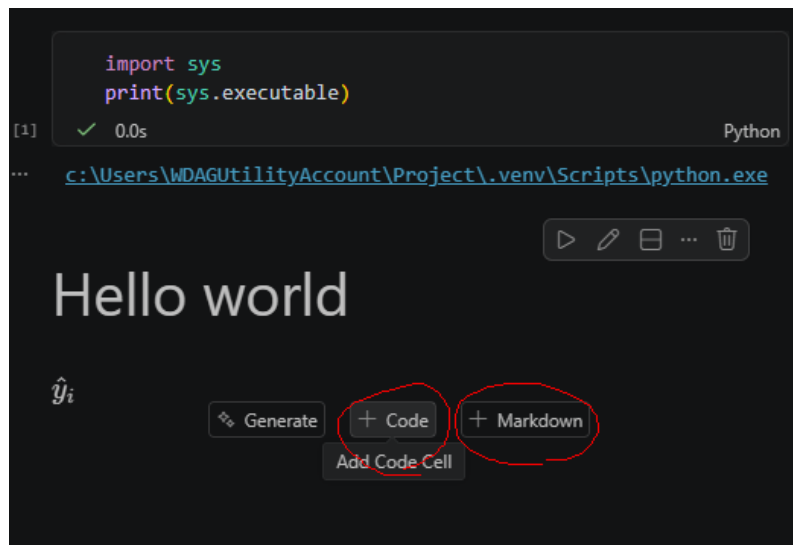
B Jupyter Notebook and Markdown

One of the best features of a Jupyter Notebook is that it can combine code, narrative text, and output in a single file.

There are two types of cells in a notebook: Code cells and Markdown cells.

- Code cells are used to run Python code.
- Markdown cells are used to write text, explanations, equations, and documentation.

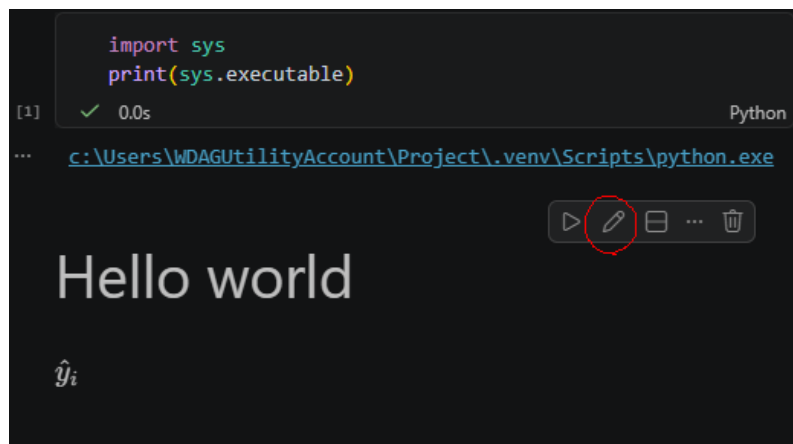
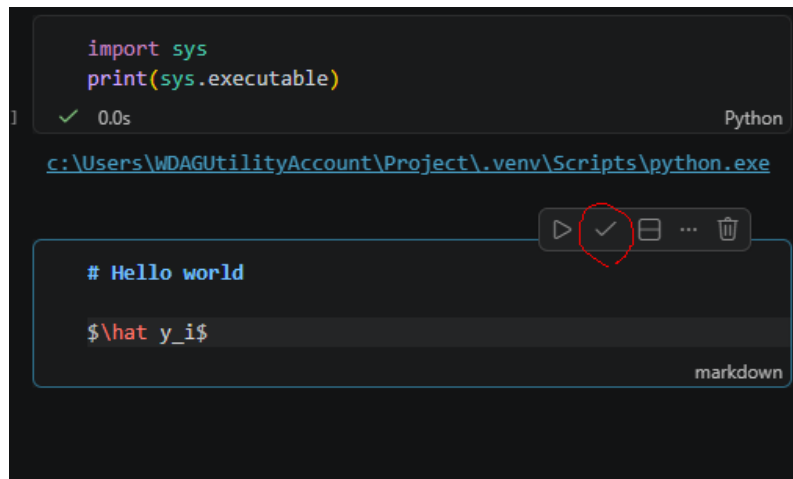
In statistical learning, code cells are typically used for data analysis, model fitting, and computations, while Markdown cells are used to explain the methods, interpret the results, and document the workflow.



A notebook combines code, output, and narrative in a single document, making it an ideal environment for data science and machine learning.

 Tip

You can always click the small **Edit / Stop Editing** button to preview how your Markdown text will be rendered.



Markdown is a lightweight text format. In this course, we will use only the most basic Markdown syntax.

B.1 Text Formatting

Markdown Syntax	Output
<code>*italics*, **bold**, ***bold italics***</code>	<i>italics</i> , bold , <i>bold italics</i>
<code>~~strikethrough~~</code>	strikethrough
<code>`verbatim code`</code>	verbatim code
<code>```\nmultiline\ncode\n```</code>	multiple code

B.2 Headings

Markdown Syntax	Output
<code># Heading 1</code>	Heading 1
<code>## Heading 2</code>	Heading 2
<code>### Heading 3</code>	Heading 3

Tip

Note that for heading, there is a space after #.

B.3 Lists

Markdown Syntax	Output
<code>* unordered list</code>	• unordered list
<code>+ sub-item 1</code>	– sub-item 1
<code>+ sub-item 2</code>	– sub-item 2
<code>- sub-sub-item 1</code>	* sub-sub-item 1

Markdown Syntax	Output
<pre>1. ordered list 2. item 2 i) sub-item 1 A. sub-sub-item 1</pre>	<pre>1. ordered list 2. item 2 i) sub-item 1 A. sub-sub-item 1</pre>



Tip

Note that for lists, there is a space after ..

B.4 Essential LaTeX

Use single dollar signs for inline math:

The fitted value is $\hat{y}_i$.

The fitted value is $\hat{y}_i$.

Use double dollar signs for displayed equations:

```
$$
Y = \beta_0 + \beta_1 X + \epsilon.
$$
```

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

Common symbols:

$\hat{y}$	predicted value
$\bar{x}$	sample mean
β_0	parameter
X^T	transpose
$\sum_{i=1}^n$	summation

$\hat{y}$: predicted value

$\bar{x}$: sample mean

β_0 : parameter

$X^\top X$: transpose

$\sum_{i=1}^n$: summation

LaTeX supports a wide range of mathematical notation, but the basic symbols introduced here will be enough for all of the work in this course.

C Python Basics for sklearn

This note introduces only the Python syntax needed for the rest of these notes. The goal is not to cover Python as a full programming language. The goal is to make code using `numpy`, `matplotlib`, and `sklearn` readable.

C.1 Running Python Code

A Python code chunk looks like this:

```
# This is a comment

x = 10      # assignment uses =

a, b = 2, 3
a + b      # last expression is automatically displayed
```

5

Tip

In Python power operation is `**`. Therefore `3**2` is 9.

Notebook

1. When a code cell is executed, the value of the last expression is automatically displayed. Any explicit output commands, such as `print()`, will also produce output.
2. Use `#` comments only for short notes that help explain the code. Detailed explanations and discussion

should be written in Markdown cells, where they are easier to read and maintain.

C.1.1 Indentation

Python uses indentation to mark code blocks. R uses braces `{}`; Python uses indentation.

For example, a `for` loop:

```
for k in [1, 3, 5]:  
    print(k)
```

```
1  
3  
5
```

The indented block belongs to the loop. These notes use loops only when they are helpful for plotting or comparing models.

C.1.2 `if` and `for`

The logic of `if` statements and `for` loops is straightforward. When using them, pay attention to indentation. A colon `:` indicates the beginning of an indented code block.

```
x = 1  
if x == 2:  
    print('good')  
else:  
    print('not good')
```

```
not good
```

```
for k in range(4):  
    print(k)
```

```
0
1
2
3
```

```
for val in ['a', 'b', 1, 3]:
    print(val)
```

```
a
b
1
3
```

C.1.3 Functions and Objects

This is not an object-oriented programming course, so don't worry too much about the details. The important idea is the distinction between a function, a class, and an object created from a class.

- Function: performs an action and returns a result. Called with parentheses: `round(3.14, 2)`
- Class: a blueprint for creating objects.
- Object: stores information and provides methods. Accessed with a dot: `object.method()`

Example in scikit-learn:

```
model = LinearRegression()
```

Here

- `LinearRegression` is the class;
- `LinearRegression()` creates a linear regression model object;
- the object is assigned to the variable `model`.

```
model.fit(X_train, y_train)
coef = model.coef_
```

Here

- `fit()` is method;
- `coef_` stores information about the fitted model;
- methods are called using `object.method(...)`;
- stored information is accessed using `object.attribute`.

In almost all **scikit-learn** code, you should work with objects rather than classes. A common mistake is to pass a class where an object is required.

```
pipe = make_pipeline(StandardScaler, LinearRegression) # wrong
pipe = make_pipeline(StandardScaler(), LinearRegression()) # correct
```

The same idea applies to most **scikit-learn** components, including transformers, models, and pipelines. In general, if something is used as a step in a pipeline, you should create an object by adding `()`.

. (Dot)

The meaning of `.` is different in Python and R.

- In Python, `.` is used to access methods and attributes of an object.

```
model.fit(X, y)
model.coef_
```

- In R, `.` is usually just part of a variable or function name.

```
my.variable <- 1
```

Do not interpret dots in R names as object access.

C.1.4 Importing Packages

Python packages are loaded with `import`.

The usual imports in these notes are:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

The names on the right are aliases:

- `numpy` is used as `np`;
- `pandas` is used as `pd`;
- `matplotlib.pyplot` is used as `plt`.

After importing a package this way, you access its functions using the package name, e.g.

- `np.array([1, 2, 3])`
- `pd.DataFrame({"x": [1, 2, 3]})`
- `plt.plot([1, 2, 3], [4, 5, 6])`

i Note

When a package is imported with `import ... as ...`, you must use the alias to access its functions and classes instead of the original name.

Sometimes only specific functions or classes are imported.

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

These objects can then be used directly.

```
model = LinearRegression()
```

i Note

After `from sklearn.linear_model import LinearRegression`, you may use `LinearRegression()` directly, but other objects from `sklearn.linear_model` are not automatically imported.

C.2 Data structure

C.2.1 Basic Objects

Python has several basic data types.

```
x = 3          # integer
y = 2.5        # float
name = "tree"  # string
flag = True    # Boolean
```

Use `type()` to check the type of an object.

```
type(y)
```

float

Python is case-sensitive. `x` and `X` are different objects.

C.2.2 Lists

A list is an **ordered collection** of objects. Lists are commonly used to store variable names, column names, or collections of parameter values.

Purpose	Example	Explanation
Create a list	<code>features = ["age", "income", "education"]</code>	Creates a list containing three strings.
Access the first element	<code>features[0]</code>	Returns "age". Python indexing starts at 0.
Access the last element	<code>features[-1]</code>	Returns "education". Negative indices count from the end.

Purpose	Example	Explanation
Select several elements	<code>features[0:2]</code>	Returns <code>["age", "income"]</code> . The stop index is not included.
Add an element	<code>features.append("gender")</code>	Adds "gender" to the end of the list.
Count elements	<code>len(features)</code>	Returns the number of elements in the list.

Note

Unlike R, Python uses zero-based indexing, so the first element has index 0.

C.2.3 Dictionaries

A dictionary stores **key-value pairs**. It is similar to a named list in R and is commonly used to store settings, options, and model parameters.

Purpose	Example	Explanation
Create a dictionary	<code>params = {"n_neighbors": 5, "weights": "uniform"}</code>	Creates a dictionary with two keys and their values.
Access a value	<code>params["n_neighbors"]</code>	Returns 5.
Add or update a value	<code>params["metric"] = "euclidean"</code>	Adds a new key-value pair or updates an existing one.
Count entries	<code>len(params)</code>	Returns the number of key-value pairs.
Store model settings	<code>{"max_depth": 3, "min_samples_leaf": 5}</code>	A common use in machine learning.
Store parameter grids	<code>{"max_depth": [2,3,4]}</code>	Often used when tuning hyperparameters.

A typical parameter grid in `scikit-learn` looks like:

```
param_grid = {
    "max_depth": [2, 3, 4],
    "min_samples_leaf": [1, 5, 10],
}
```

For pipelines, parameter names often use two underscores:

```
param_grid = {
    "model__C": [0.1, 1, 10],
}
```

i Note

In a pipeline, a parameter name of the form `step__parameter` refers to a parameter inside a specific pipeline step. For example, `model__C` means the parameter `C` of the pipeline step named `"model"`. More terminologies will be discussed in related sections.

C.3 `numpy.array`

The most important data structure for numerical computation in Python is `numpy.array`.

```
import numpy as np

x = np.array([1, 2, 3, 4])
x
```

```
array([1, 2, 3, 4])
```

Arrays have a shape and a dimension.

```
print(x.shape)
print(x.ndim)
```

```
(4,)
1
```

So `x` is a 1D array containing 4 elements.

C.3.1 2D Arrays

A 2D array is similar to a matrix.

```
X = np.array([
    [1, 2],
    [3, 4],
    [5, 6]
])
X
```

```
array([[1, 2],
       [3, 4],
       [5, 6]])
```

```
print(X.shape)
print(X.ndim)
```

```
(3, 2)
2
```

Here `X` is a 2D array with 3 rows and 2 columns.

C.3.2 Indexing

Use `[row, column]` indexing.

```
X[0, 1]    # row 0, column 1
X[:, 0]    # first column
X[1, :]    # second row
```

The symbol `:` means “all indices” along that axis.

Slicing can select ranges.

```
X[0:2, 1]      # first 2 rows, column 1
X[0, 0:2]      # row 0, first 2 columns
X[0:2, 0:2]    # upper-left 2x2 block
```

i Note

`X[:, 0]` returns a 1D array with shape `(n,)`, not a column vector.

To obtain a column vector, use

```
X[:, [0]]
# or
X[:, 0].reshape(-1, 1)
```

which both have shape `(n, 1)`.

C.3.3 Common Operations

Arrays support vectorized arithmetic.

```
x + 1
x * 2
np.mean(x)
```

Operations are applied elementwise.

There are two special operations that is worth mentioning.

i Matrix multiplication

There are several ways to perform matrix multiplication in NumPy. Here we introduce the `@` operator because it has the simplest syntax.

```
import numpy as np
```

```
X = np.array([[1, 2], [3, 4], [5, 6]])  
y = np.array([1, 2])  
X @ y
```

```
array([ 5, 11, 17])
```

```
X = np.array([[1, 2], [3, 4], [5, 6]])  
y = np.array([1, 2]).reshape(-1, 1)  
X @ y
```

```
array([[ 5],  
       [11],  
       [17]])
```

Note that the result depends on the dimensions of the inputs. NumPy automatically applies the appropriate matrix multiplication rule based on the shapes of the arrays.

- In the first example, `y` is a one-dimensional array with shape `(2,)`, so the result is also one-dimensional with shape `(3,)`.
- In the second example, `y` is a column vector with shape `(2,1)`, so the result is a two-dimensional array with shape `(3,1)`.

Broadcasting

Broadcasting is a NumPy mechanism that allows arithmetic operations between arrays of different shapes. Instead of explicitly copying data, NumPy automatically expands arrays along dimensions of size 1 when the shapes are compatible. Two dimensions are compatible if they are equal or if one of them is 1.

A common example is arithmetic between an array and a scalar:

```
import numpy as np

X = np.array([[1, 2], [3, 4], [5, 6]])
X + 10

array([[11, 12],
       [13, 14],
       [15, 16]])
```

The scalar is automatically applied to every element of the array.

Broadcasting can also occur between higher-dimensional arrays. For example,

```
a = np.array([[1, 2, 3]])
b = np.array([[10], [20]])
a + b

array([[11, 12, 13],
       [21, 22, 23]])
```

Here **a** has shape (1,3) and **b** has shape (2,1). These shapes are compatible because, along each axis, either the dimensions are equal or one of them is 1.

Broadcasting expands dimensions of size 1 to match the corresponding dimension of the other array. In this example, the single row of **a** behaves as if it were repeated twice, while the single column of **b** behaves as if it were repeated three times, resulting in arrays of shape (2,3). Broadcasting can also be used to compute pairwise differences efficiently.

```
import numpy as np

X1 = np.array([[1, 2],
               [3, 4]])

X2 = np.array([[1, 2],
               [3, 4]])

d = X1[:, None] - X2[None, :]
d
```



```
array([[[ 0,  0],
        [-2, -2]],

       [[ 2,  2],
        [ 0,  0]]])
```

`None` inserts a new dimension at the specified location. Therefore `X1[:, None]` has shape (2,1,2) while `X2[None, :]` has shape (1,2,2).

NumPy automatically broadcasts these arrays to shape (2, 2, 2) and computes all pairwise differences at once. In this example, each row of `X1` is subtracted from each row of `X2`. The result satisfies $d[i, j] = X1[i] - X2[j]$ for all valid indices i and j .

This technique is frequently used when computing distance matrices in machine learning algorithms.

C.3.4 Axis-wise

Many functions accept an `axis=` argument.

```
X # show X
```

```
array([[1, 2],
       [3, 4],
       [5, 6]])
```

Computes the sum across rows:

```
X.sum(axis=0)
```

```
array([ 9, 12])
```

Computes the mean across columns:

```
X.mean(axis=1)
```

```
array([1.5, 3.5, 5.5])
```

A useful rule:

- `axis=0`: operate down rows (one result per column)
- `axis=1`: operate across columns (one result per row)

C.3.5 Reshaping

A 1D array can be converted into a matrix. Consider a 1D array

```
x = np.array([1, 2, 3, 4])
```

To create a column vector:

```
x.reshape(-1, 1)
```

```
array([[1],  
       [2],  
       [3],  
       [4]])
```

To create a row vector:

```
x.reshape(1, -1)
```

```
array([[1, 2, 3, 4]])
```

The value `-1` means “infer this dimension automatically.”

A common pattern is

```
test_x = np.linspace(0, 1, 101).reshape(-1, 1)
```

On the other side, to flatten an array back to 1D:

```
X.reshape(-1)
```

```
array([1, 2, 3, 4, 5, 6])
```

NumPy reads entries row by row and produces a one-dimensional array.

Tip

Other flattening methods include `.ravel()` and `.flatten()`. The difference is about how copy-and-view is handled. We mainly use `.reshape(-1)` because it makes the resulting shape explicit.

Note

One common source of bugs in statistical learning is using arrays with incorrect shapes.

In `scikit-learn`, predictors `X` are usually stored as a 2D array and responses `y` as a 1D array. Other libraries may use different conventions, and the required shapes often depend on the specific model or loss function.

When shape-related errors occur, always check the documentation of the library you are using.

C.3.6 Arrays versus Lists

Lists are general-purpose containers, and its addition concatenates. For example

```
[1, 2, 3] + [4, 5, 6]
```

```
[1, 2, 3, 4, 5, 6]
```

Arrays are designed for numerical computation. Its operations are align with the regular math operations. Therefore for numerical work and machine learning, use `numpy.array` whenever possible.

C.4 Random Numbers

These notes often use `numpy` random number generators for simulation.

```
rng = np.random.default_rng(1)

x = rng.normal(size=5)
x
```

```
array([ 0.34558419,  0.82161814,  0.33043708, -
 1.30315723,  0.90535587])
```

Note

The number 1 is a seed. It makes the random output reproducible.

On a single machine, if the seed is fixed the results are supposed to be the same.

You may choose any number as the seed. You may use the same seed when you want to see the same result.

Some common random functions are:

```
rng.normal(loc=0, scale=1, size=5)
```

```
array([ 0.44637457, -0.53695324,  0.5811181 ,  0.3645724 ,  0.2941325 ])
```

```
rng.uniform(0, 1, size=5)
```

```
array([0.75351311, 0.53814331, 0.32973172, 0.7884287 , 0.30319483])
```

For a two-dimensional predictor matrix:

```
X = rng.normal(size=(10, 3))
X.shape
```

```
(10, 3)
```

C.5 pandas

pandas can be treated as a wrapper of `numpy.array` which make it more readable. The complete pandas is very complicated. Here we only mention a few very important concepts.

To import pandas, in most cases we set `pd` as its alias.

```
import pandas as pd
```

C.5.1 DataFrame and Series

In pandas, `DataFrame` is the 2D data table, and `Series` is the 1D column. There are many ways to create a `DataFrame`. Here we focus on one of them, that is to convert a `dict` into a `DataFrame`.

```
results = {
    'round': [1, 2, 3, 4, 5],
    'score': [0.22, 0.33, 0.44, 0.55, 0.66],
    'mse': [0.12, 0.11, 0.10, 0.09, 0.08]
}
df_res = pd.DataFrame(results)
df_res
```

	round	score	mse
0	1	0.22	0.12
1	2	0.33	0.11
2	3	0.44	0.10
3	4	0.55	0.09
4	5	0.66	0.08

A single column is returned as a `Series`.

```
df_res['score']
```

```
0    0.22
1    0.33
2    0.44
3    0.55
4    0.66
Name: score, dtype: float64
```

Notice that this is similar to `numpy` indexing. Selecting a single column by name returns a 1D object. If we instead pass a list of column names, even if the list contains only one column (e.g. `['score']` below), the result is a 2D `DataFrame`.

```
df_res[['score']]
```

	score
0	0.22
1	0.33
2	0.44
3	0.55
4	0.66

This behavior is analogous to the difference between `X[:, 0]` and `X[:, [0]]` for a `numpy` array:

- `df_res["score"]` returns a `Series` (1D)
- `df_res[["score"]]` returns a `DataFrame` (2D)
- `X[:, 0]` returns a 1D array
- `X[:, [0]]` returns a 2D array

C.5.2 Read and write files

In `scikit-learn`, predictors and responses can be stored either as `numpy` arrays or as `pandas` objects. Throughout this course, many examples use `numpy`, but most code works equally well with `pandas` data structures.

When working with external datasets, a common workflow is:

1. Read the data into a `DataFrame`.

2. Perform data cleaning and preprocessing.
3. Extract predictors and responses.
4. Fit a statistical learning model.

Two of the most common tabular file formats are CSV files and Excel spreadsheets.

i CSV

CSV stands for comma-separated values. A CSV file is a plain-text file that stores tabular data using a separator between columns.

Common separators include:

- , (comma)
- ; (semicolon)
- \t (tab)

Consider the following file [example.csv](#).

```
round    score    mse
1    0.220000    0.120000
2    0.330000    0.110000
3    0.440000    0.100000
4    0.550000    0.090000
5    0.660000    0.080000
```

In Python we could use `pd.read_csv()` to read it.

```
import pandas as pd
df = pd.read_csv('example.csv')
df
```

	round\t	score\t	mse
0	1\t	0.220000\t	0.120000
1	2\t	0.330000\t	0.110000
2	3\t	0.440000\t	0.100000
3	4\t	0.550000\t	0.090000
4	5\t	0.660000\t	0.080000

The result is incorrect because this file uses tab characters (`\t`) rather than commas as separators. Specify the separator explicitly:

```
df = pd.read_csv('example.csv', sep='\t')
df
```

	round	score	mse
0	1	0.22	0.12
1	2	0.33	0.11
2	3	0.44	0.10
3	4	0.55	0.09
4	5	0.66	0.08

Excel

Excel spreadsheets can be read using `pd.read_excel()`. We provide the `example.xlsx` as an example.

```
import pandas as pd
df = pd.read_excel('example.xlsx')
df
```

	round	score	mse
0	1	0.22	0.12
1	2	0.33	0.11
2	3	0.44	0.10
3	4	0.55	0.09
4	5	0.66	0.08

Unlike CSV files, Excel files are stored in a binary format. Therefore, `pandas` requires an additional package called an engine to read and write them. The most commonly used engine is `openpyxl`.

```
uv add openpyxl
```

Although not provided in the toml file, `openpyxl` was added to our environment as an example in the introduction. If you didn't add it at that time you may add it now.

Writing files is similar to reading files. The main difference is that writing starts from an existing `DataFrame` instead of starting from `pd`.

```
df.to_csv('example2.csv')  
df.to_excel('example2.xlsx')
```

C.5.3 Relative paths

In this course, always use relative paths when reading or writing files.

A relative path starts from the current working directory rather than from the root of the file system.

Suppose your working directory is

```
C:/Users/WDAGUtilityAccount/Project/
```

and you have the following structure:

```
Project/  
  example.csv  
  foldername/
```

Then you can read and write files using:

```
df = pd.read_csv("example.csv")
df.to_csv("foldername/result.csv")
```

Notice that neither path begins with a drive name such as C: or D:.

If your files are stored outside the working directory, **copy them into the working directory** before reading them.

Note

Although absolute paths such as

```
pd.read_csv("D:/Files/example.csv")
```

often work on your own computer, they may fail on another computer because the file locations are different. Using relative paths makes your code more portable and easier to share.

C.6 matplotlib Basics

These notes use only basic matplotlib. The standard import is:

```
import matplotlib.pyplot as plt
```

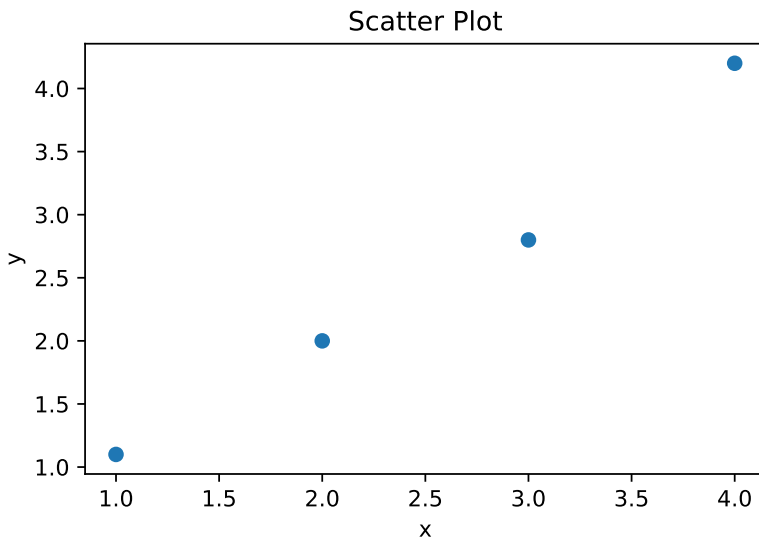
We provide some code examples here. Most of the commands are straightforward. Please check the [official document](#) if you need more details.

C.6.1 Scatter Plot

```
x = np.array([1, 2, 3, 4])
y = np.array([1.1, 2.0, 2.8, 4.2])

plt.scatter(x, y)
```

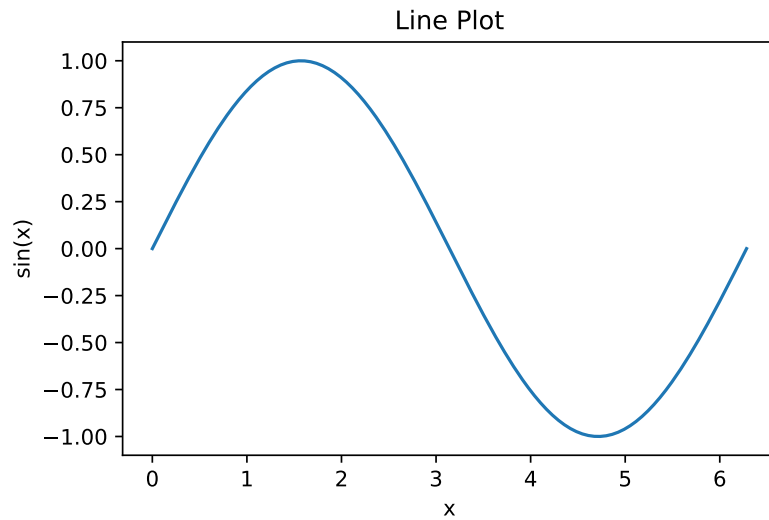
```
plt.xlabel("x")
plt.ylabel("y")
plt.title("Scatter Plot");
```



C.6.2 Line Plot

```
x_grid = np.linspace(0, 2 * np.pi, 200)
y_grid = np.sin(x_grid)

plt.plot(x_grid, y_grid)
plt.xlabel("x")
plt.ylabel("sin(x)")
plt.title("Line Plot");
```

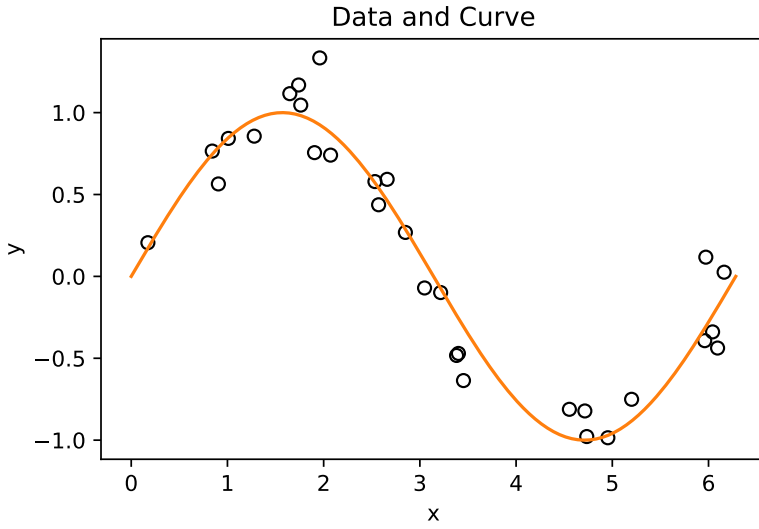


C.6.3 Scatter and Line Together

```
rng = np.random.default_rng(1)

x = rng.uniform(0, 2 * np.pi, size=30)
y = np.sin(x) + rng.normal(scale=0.2, size=30)
x_grid = np.linspace(0, 2 * np.pi, 200)

plt.scatter(x, y, facecolors="none", edgecolors="black")
plt.plot(x_grid, np.sin(x_grid), color="C1")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Data and Curve");
```



This pattern appears often when we plot data points and a fitted curve.

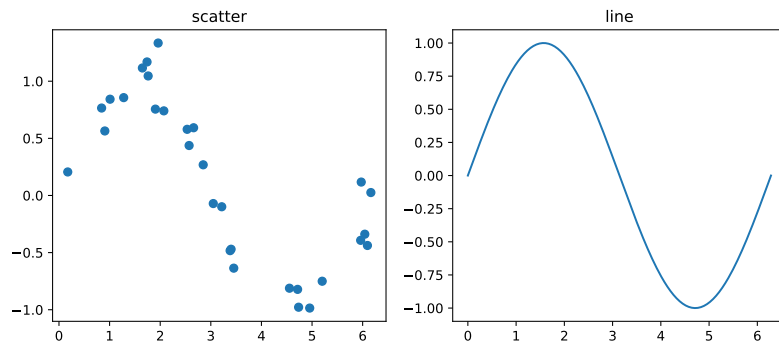
C.6.4 Multiple Plots

Use `plt.subplots` for multiple panels.

```
fig, axs = plt.subplots(1, 2, figsize=(10, 4))

axs[0].scatter(x, y)
axs[0].set_title("scatter")

axs[1].plot(x_grid, np.sin(x_grid))
axs[1].set_title("line");
```



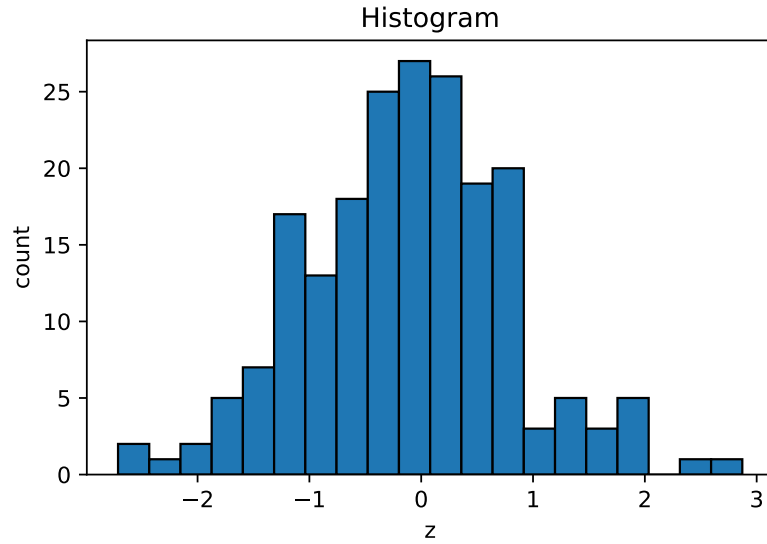
i Note

Notice that when using an axes object, many plotting commands have different names. For example, `plt.title()` becomes `ax.set_title()`. Therefore, you cannot always replace `plt` with `ax` directly. For details, consult the official documentation.

C.6.5 Histograms

```
z = rng.normal(size=200)

plt.hist(z, bins=20, edgecolor="black")
plt.xlabel("z")
plt.ylabel("count")
plt.title("Histogram");
```



C.7 Common Errors

C.7.1 One-Dimensional X

The most common error is passing a one-dimensional predictor array to `sklearn`.

Bad	Good
<code>x = np.array([1, 2, 3, 4])model.fit(x, y)</code>	<code>X = x.reshape(-1, 1)model.fit(X, y)</code>

C.7.2 Class Name versus Model Object

Another common mistake is confusing an `sklearn` class with an object created from that class.

For example, `DecisionTreeClassifier` is the class itself.

Wrong	Right
<pre>model = DecisionTreeClassifier(model_fit(X_train, y_train)</pre>	<pre>model = DecisionTreeClassifier()model.fit(X_train, y_train)</pre>

C.7.3 Mixing Lists and Arrays

Use arrays for elementwise numerical operations.

```
[1, 2, 3] * 2
```

```
[1, 2, 3, 1, 2, 3]
```

Plain lists do not always behave like numerical vectors.

```
import numpy as np
np.array([1, 2, 3]) * 2
```

```
array([2, 4, 6])
```


D Some Python concepts

D.1 Language

Python is a programming language. In other words, it is a collection of syntaxes.

D.2 Interpreters

Python needs to be interpreted into codes that computers can understand. Therefore there should be some programs that translate Python scripts. These programs are called *interpreters*.

CPython is the reference implementation of Python and the implementation most people use. It is written mainly in C and Python. New Python language features are normally developed and tested first in CPython.

Other implementations include [PyPy](#), [Jython](#) and [IronPython](#). They are useful in specific contexts, but they do not always support the same Python versions or the same package ecosystem as CPython. For example, PyPy focuses on speed through a JIT compiler, Jython integrates with the JVM but its current stable release is Python 2.7-based, and IronPython integrates with .NET but lags behind modern CPython versions.

Since **CPython** is the most widely used and best-supported implementation, it is the best choice, at least for beginners. And actually, if you have no idea about this topic, but you use Python, it is highly possible that you are using **CPython**.

Note

We mentioned “interpreter” here. There are mainly two types of implementations of programming languages: interpreters and compilers. There are also some additional types like just-in-time compilers which can be treated as combinations of the two.

Python is often described as an *interpreted* language, but CPython first compiles source code to bytecode and then executes that bytecode in the Python virtual machine. For beginners, the practical point is that Python supports an interactive workflow and does not require a separate manual compile step.

D.3 REPL

There are two ways to use Python interpreter. One way is to execute a Python script stored in a file. The second way is called *the interactive shell*, that Python interpreter read the input from user directly, and print the result immediately. The model is like code example: prompt the user for some code, and when they’ve entered it, execute it in the same process. This model is often called a [REPL](#), or *Read-Eval-Print-Loop*.

Shell, *terminal*, *console* have different meanings in their original contexts. However, nowadays, especially when talking about Python interactive shell, these terminologies are used interchangeably. They are referred to the frontend of the system. In other words, the main task for the Python interactive shell is to handle the user inputs and communicate with the backend, which is also called a *kernel*. We won’t distinguish the real differences between these terminologies. The kernel will be discussed in the next section.

The standard Python REPL can usually be started with `python` or `python3`. On Windows, `py` may also be available. To quit, use `quit()`, `exit()`, `Ctrl+D` on macOS/Linux, or `Ctrl+Z` then `Enter` on Windows.

i Note

In the REPL model, the backend (evaluation) is basically handled by the Python interpreter. The frontend is dealing with the user interface. Some typically tasks include the *primary/secondary prompt* and multi-line commands. The original REPL is very limited.

D.4 IPython

IPython was initially designed as an Enhanced interactive Python shell. However after many year's development, the whole IPython project becomes too big to maintain as one single project. Therefore it is now split into many smaller projects. The two most popular projects are IPython and Jupyter. This is called [the Big Split](#).

The current IPython play two fundamental roles:

- Terminal IPython as the familiar REPL;
- The IPython kernel (which is defined below) that provides computation and communication with the frontend interfaces, like the notebook.

The core idea in the design of IPython is to abstract and extend the notion of a traditional REPL environment by decoupling the evaluation into its own process. We call this process a *kernel*: it receives execution instructions from clients and communicates the results back to them.

This decoupling allows us to have several clients connected to the same kernel, and even allows clients and kernels to live on different machines. This two-process model is now used by most of the Jupyter project.

You can launch the IPython shell on the command line with the `ipython` command (which similar to `python` case requires `PATH` configuration), and quit the shell with `exit/exit()/quit/quit()` commands.

The reference Python kernel provided by IPython is called `ipykernel`. With `ipykernel` you may create and maintain multiple kernels.

D.5 Jupyter

Jupyter is an ecosystem for interactive computing. It originated from the IPython Notebook project and later evolved into a language-independent platform supporting many programming languages. The name **Jupyter** is inspired by **Julia**, **Python**, and **R**, the three languages that were central to many early data science workflows.

Today, Jupyter includes several user interfaces, such as Jupyter-Lab, Jupyter Notebook, Jupyter Console, and QtConsole. In this course, however, the important idea is not a particular interface, but the relationship among a notebook frontend, a kernel, and the notebook file stored on disk.

A Jupyter notebook is saved as a file with extension `.ipynb`. Internally, the file uses a structured JSON format that stores:

- code cells;
- Markdown cells;
- outputs;
- notebook metadata.

When you run code in a notebook, the frontend sends the code to a kernel, which executes the code and returns the results. In a Python notebook, a kernel can be viewed as a Python process running in the background. It executes code, stores variables, and communicates with the notebook interface.

Different programming languages can be supported by different kernels. For example:

- `ipykernel` for Python;
- `IRkernel` for R;
- `IJulia` for Julia.

Kernel

A kernel is associated with a particular Python environment, but it is not the environment itself. For example, multiple kernels may use the same Python environment while maintaining completely separate variables and memory. Conversely, multiple notebook windows may connect to the same kernel and therefore share the same variables and computational state.

For this course, you can think of the notebook as the user interface, the kernel as the Python process that runs your code, and the `.ipynb` file as the notebook document saved on disk.

D.5.1 Multi-kernels setup

This section is mainly following the [official document](#).

To install one IPython kernel, you may install `ipykernel` in the environment. If you want to have multiple IPython kernels for different `venvs`, you will need to specify unique names for the kernelspecs.

1. Activate the environment you want.
2. Install the kernel in the environment.

```
uv add ipykernel
uv run python -m ipykernel install --user --name myenv --display-name "Python (myenv)"
```

`--user` means that the kernel is installed in the user's folder instead of a system folder, and it can be removed. The `--name` value (in this case it is `myenv`) is used by Jupyter internally. These commands will overwrite any existing kernel with the same name. `--display-name` is what you see in the notebook menus.

3. You could use the command to find all kernels installed in your system.

```
uv run jupyter kernelspec list
```

Available kernels are shown, as well as the path to the kernel configuration file `kernel.json`. The most important configuration is the path to the Python interpreter executable file.

Listing D.1 `kernel.json`

```
{
  "argv": [
    "C:\\Users\\WDAGUtilityAccount\\Project\\.venv\\Scripts\\python.exe",
    "-Xfrozen_modules=off",
    "-m",
    "ipykernel_launcher",
    "-f",
    "{connection_file}"
  ],
  "display_name": "(myenv)",
  "language": "python",
  "metadata": {
    "debugger": true
  },
  "kernel_protocol_version": "5.5"
}
```

Notice that the path in `argv` points to the Python interpreter used by this kernel. Different kernels may point to different Python environments, or multiple kernels may point to the same environment.